Loop Walking

A Field Guide to Hacking the Universe with Holonomy

The reader’s edition

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Preprint. This guide has not been peer reviewed, and its search of the literature is limited. It comes from a part-time research team that works in the open and gives its tools away. The reader’s edition of version 7, 2026-11-10, with the work through Question the Substrate and Two Arrows, made by script from the full edition (concept DOI 10.5281/zenodo.21811588), whose row numbers it keeps. *CC BY 4.0.*

About this edition

This is the reader’s edition of *Loop Walking*. It is made by script from version 7’s full edition at every build and keeps its row numbers. Every row has a short entry here: the row’s rule in its title, one to three sentences drafted from the row’s own words, with every number, run name and quoted phrase in them checked against the row, and two lines generated from the full edition’s own tables. The full edition is the encyclopedia on the shelf. Every run, number and cost behind an entry stands in its row there, with the appendices that list every arrival by family, every row by domain and every transfer of a lesson from one setting to another.

The lines are an honest map. *Holds* says where a rule has been carried to another system or setting and how it fared there. The full edition records 519 such carryings on 322 rows: 222 held, 79 held weaker and 218 failed, and each failure names whose break it was (the ruler’s 62, the substrate’s 58, a seam’s 48, a knob’s 41, a wall’s 28, not stated 60). A lesson fails to carry about as often as it carries. 562 rows have not yet been carried anywhere, and their entries say so: they are single reports until another system tries them. *Met in* names the row’s domains, from the sixteen the guide uses. *Counted* names the families of the ranking whose arrivals land at the row, with how many. *Shown* names the projects that show the row, by the short names in the key below, and the row’s number in the full edition.

In the lines	Project or paper	Its record
version 6	Version 6 of this guide, and the papers it drew on	concept DOI 10.5281/zenodo.21811588
Loop Atlas	The Loop Atlas sessions, *The Loop Atlas*	concept DOI 10.5281/zenodo.24100672
R2W	R2W, *A Rider in Two Waves* (the RF Spoke in its early records)	concept DOI 10.5281/zenodo.23999853
triple-star	The triple-star sessions, *Held, Rocking, Locked*	concept DOI 10.5281/zenodo.23755473
Dancing on the Sun	*Dancing on the Sun*	concept DOI 10.5281/zenodo.23067245
Latch Atlas	*The Latch Atlas*	concept DOI 10.5281/zenodo.23068382
Jellyfish	*Jellyfish: the Holonomist of the Sea*	concept DOI 10.5281/zenodo.22908909
Linking and Magnetic Forest	*Linking Is the Latch* and *Instruments for the Magnetic Forest*	concept DOIs 10.5281/zenodo.22906667 and 10.5281/zenodo.22906780
Question the Substrate	*Question the Substrate* (the two-voices series, Paper 1)	concept DOI 10.5281/zenodo.24094163
Two Arrows	*Two Arrows* (the two-voices series, Paper 2)	concept DOI 10.5281/zenodo.24120415
Field Guide sessions	The Field Guide's own working sessions	the full edition's record

Abstract

Loop Walking is a field guide to reading systems by holonomy: what a quantity carried round a closed path comes home changed by. Version 7 gathers the lessons of three projects, registered from 29 September to 2 November 2026, a triple-star toy, a driven rider in two waves and the Loop Atlas of toy machines, and of the two-voices papers on learning minds, *Question the Substrate* and *Two Arrows*, published on 9 and 10 November, into 884 register rows, 344 of them new since version 6, tagged with 16 domains. It reports what the method achieved as well as how to use it. Across rows 1 to 847 of the register, 519 times a lesson was carried from the setting where it was learned to another; 222 held, 79 held weaker and 218 failed, and the 222 that held sit in rows spanning all 16 domains. The three projects watch one object, an angle carried round a circle by a flow and read once per tick of an outside clock, and one fork decides most of what follows: the area-keeping triple star's class doubles with the constant 8.72 (its own island doubled once, then lost its stars), the dissipative rider's with 4.669, and a lightly leaky map crosses over once a trip keeps less than about half its area (0.440 to 0.524 across five leaks). A reading of pictures made after seeing the data was tested blind: chaos on the borders between treads, read at 84 per cent on a map already seen, held at 97 per cent on a fresh grid. A power law registered with two rivals held blind, 8 of 8 and then 4 of 4, and both rivals failed. The 582 arrivals of lessons gathered from 6 to 23 October were counted into 13 families, two of them defined on 23 October, 446 of the arrivals by fresh readers, and a blind re-sort of a sample agreed 85 per cent of the time (Cohen's kappa 0.67). Two falling cats that look like strangers, one twisting at its waist and one twist-free, turn out to be one cat under two rules, alike within 2.842×10^{-13} degrees on nine swelling paths. New in this version is a register of results beside the record of misses, so that what holds reaches the front page: the triple-star thread landed 236 of 490 predictions registered before their runs (to its paper's publication, 28 October), the Loop Atlas held 706 of 1,104 judged (at its book's publication on 9 November, concept DOI 10.5281/zenodo.24100672), and R2W, whose book *A Rider in Two Waves* was published on 5 November (concept DOI 10.5281/zenodo.23999853), landed at least one prediction in 448 of 527 graded tests. The two-voices papers held 122 of 212 and 87 of 170 predictions registered before their runs: a learning mind cannot grade its own homework, since a self-training chain that keeps the half of its labels it is surest of lost 25.3 points where the same cut ranked by a frozen first network lost 2.1, and it heals its answers long before its self, keeping 0.0167 of a loop's change in its answers and 0.3049 in its hidden layer. The largest results include the Loop Atlas's 51 matches with the published record, the sign slip behind a widely copied table's odd-order doubling constants, and the triple-star paper's one-sided door (163 of 166 escapes). The guide is a preprint, not peer reviewed, with a limited literature pass.

What this guide achieved

1. **The method travels, and the guide counts where.** In rows 1 to 847, 519 times a lesson was carried from where it was learned to another setting, on 322 of the 840 rows read; 222 held, 79 held weaker and 218 failed, and the rows holding the 222 that held span all 16 of the guide's domains (appendix "Transfers, row by row"); the 44 rows from 848 to 891 are not yet read for transfers, so each count is a floor.
2. **One fork decides most of what follows.** The triple star keeps area, and its class's doubling constant, 8.72, was matched on an area-keeping test map (8.721082), while the triple's own island doubled once and every star round it then escaped within 1,600 passes; the rider loses energy, and its class's constant, 4.669, was matched on a damped test map (4.669197); both constants are benchmarks, known and matched here; a lightly leaky map reads the frictionless constant for its first doublings and crosses over once one trip round its orbit keeps less than about half its area, 0.440 to 0.524 across five leaks, a crossover known in kind whose threshold is measured here (row 680; Two Readers, section 8).
3. **A slip time forecast from an island's width.** At a lock's clean edge the time to slip grows like a pendulum's logarithm: 7.57 and 7.37 passes per decade of distance at two clean rims, against 7.29 and 7.83 forecast from the island's own width with nothing fitted, where a square-root law would have grown the slip period 4.5 times over the span and it grew 2.4; the logarithm is known in kind, and its forecast from the width is measured here (row 675; Two Readers, section 8).
4. **A law read after the data, then tested blind.** A power law seen after one table ($\delta_2 = \delta_1^2$) was registered with two rivals before the next rows were opened; it held blind, 8 of 8 and then 4 of 4 in the junction, and both rivals failed (runs TOY-SIDE-3g, TOY-SIDE-4g and JJ-SMALL-1c; row 79).
5. **The eye, tested blind.** Chaos on the borders between treads, read after seeing the data at 84 per cent on a map already seen, held at 97 per cent on a fresh grid and at three chaos thresholds, so it is structure, not binning (row 655); a second reading, a staircase's turns marked by a change of word, failed blind and stays on the record (row 369). Reads by eye that named a relation held 19 times in 30, against 11 in 28 for reads that named a shape (rows 100 and 714; Two Readers, section 9), and the four straight lines Bear drew over a converter's chaos are each a real edge of it, within 0.12 A (row 873).
6. **Counts survive a change of machine.** Four riders run on a laptop and again in the container agreed on their slips, and on their stretch rates only within 0.012 (row 103). The triple-star thread's long runs, made again from nothing on a second machine, gave 103 lines: 83 the same, 20 the same verdict with other digits and none different; verdicts carry across machines, last digits do not (run XC1; row 810).
7. **A count reads through an instrument's turning.** A turning of 5.5 degrees put in on purpose across 38 sunspot maps moved 31 of 154,267 readable loop counts; the offset found in every sunspot's current behaves like exactly such a turning, and it survived blind tests on 24 of 26 and 45 of 50 fresh regions (row 567; from *Dancing on the Sun*, concept DOI 10.5281/zenodo.23067245).
8. **Four tests tell a true nothing from two hands cancelling when a knob has no sign.** A symmetry's nothing sits at the arithmetic floor, 6.994×10^{-15} against one rider's 2.599775×10^{-2}; a balance was placed blind within 0.04 and 0.006 per cent. The size sweep alone is not enough when both hands share a factor of the knob: on the Loop Atlas's crest-firer it came out flat within 1.5 per cent at every level, and the other three tests found the two hands (row 805).
9. **What keeps happening, counted rather than remembered.** 582 arrivals of lessons, gathered from 6 to 23 October, sorted into 13 families, 446 of them by fresh readers and 136 by the session that gathered version 7; a blind re-sort of 60 of the first 801 units, made before the two families of 23 October were defined, agreed 85 per cent of the time (Cohen's kappa 0.67) and named the same family 16 times in 16 where both saw a lesson; the commonest move, making every check able to fail, arrived 131 times (the Register's opening).
10. **A project's record of misses finds what its summaries drop.** Of the Loop Atlas's misses 1 to 457, 85 (18.6 per cent: 63 of the first 288 and 22 of the next 169) carried a lesson that had never reached the guide, whole or in part, through the project's own list of items; for misses 219 to 288 that count was forecast before it was made, at about 15 of 70, and came to 16. The triple-star thread's 268 missed and unread readings, read the same way on 23 October, gave 13 carried only in part and none missing (rows 784 to 789, 794, 806 and 834 to 837, and additions elsewhere).
11. **Two falling cats are one cat under two rules.** A cat twisting at its waist and a thinner twist-free cat bent to the mirror angle turn alike at every shape and on every path, within 2.842×10^{-13} degrees on nine swelling paths and with the rates within 9.3×10^{-15} at 2,000 random shapes; the pairing was not found in six sources read in full, among them Rademaker and ter Braak 1935, Kane and Scher 1969, Montgomery 1993, Gbur 2019 and Batterman 2003, nor in two more read on 22 October, when the Loop Atlas claimed it as new on a limited, noncommercial review, the claim held until advised otherwise; across Iwai's general model of 2010, two cylinders on a joint that allows any twist, the mirror holds only with the hinge at the tubes' centres, and his book of 2021 is the likeliest place left for it (rows 820, 742 and 511).

12. **A widely copied table's odd entries, and the one slip that makes them.** Briggs's table of doubling constants (1991; copied by MathWorld and by Finch's Mathematical Constants, 2003) is right in α at every order from 2 to 12 and in δ at every even order, and off in δ at every odd one, 5.9679687038 for 6.0846890743 at order 3; one sign slip in the doubling operator reproduces every odd entry, the printed δ(3) to 2×10^{-29}, and a fresh helper given no number re-derived the map's values and the printed one with code of its own, all five of its registered predictions holding. McCartney (2020) found the odd orders wrong first, and the corrected values are his; the cause is new as far as a limited pass shows, a pointer for a specialist (rows 874 and 875, and the Loop Atlas's note, *A Mirror in Feigenbaum's Table*, concept DOI 10.5281/zenodo.24101154).
13. **The toys meet the published record 51 times.** At its book's publication on 9 November the Loop Atlas counted 51 benchmarks against the published record, among them Feigenbaum's δ measured at 4.66920159 for a rounded top (row 771), Adler's slip period met exact to 1.1×10^{-8} (row 675), the needle's slips meeting the exact escape rate at 0.9885 ± 0.0122 where Kramers' formula runs 9 per cent high (row 725), and the pull between distant vortex rings falling as the cube of their distance, read from a chaotic train two ways, 3.17 and 2.97 (row 865). Beside them stand 16 found-here candidates, one claimed (item 11), and a corner ratio, −1.9644458, on which the logistic and sine maps agree to the ninth figure and which a limited pass did not find named (row 877).
14. **One toy's road forecasts another's.** A rule R2W read in its rider's locked step, each gap to 4.669 about a quarter of the last, forecast the Loop Atlas needle's fourth band ratio at 4.6585 and 4.6591; the needle gave 4.6599 and 4.6604, and its fifth, forecast at 4.6672 and 4.6673, came in at 4.6673 from above and 4.6672 from below (row 872).
15. **One resonance measured from the inside, and published.** *Held, Rocking, Locked* (28 October; concept DOI 10.5281/zenodo.23755473) follows a body a millionth of either star's mass round an eccentric pair of equal stars, a circumbinary planet's setting. Mardling's resonance pendulum, tested piece by piece, holds in its relation between the rocking rate and the island's width, within 1.1 per cent at the pair's eccentricity 0.10, while her widths run 20 to 31 per cent wide (rows 506 and 678); and past the stability limit the island loses its stars by one side: with the star's eccentricity at 0.20, 163 of 166 escapes leave toward the pair, where the published stability rules smooth the resonance over (rows 599 and 863; its register in the part on the headline register).
16. **The arithmetic can fake resonance physics.** At a tolerance that looks tight, a general-purpose integrator dragged every unlocked rocking rate of the triple star down by about 2.8×10^{-8} cycles a pass in every pass, enough to fake a lock's slide and part of its wear, and a tolerance 100 times tighter left 0.0093 of it (rows 599 and 741); the energy's drift falls by the same powers per tenfold tightening as the creep left in the rates, 1.46 and 1.92 against 1.41 and about 2.0, so that creep is the integrator's own (row 858).
17. **A law with nothing fitted.** When one extra tone is added to R2W's drive, the dancing pair moves by a closed form with no free number in it; rewritten on the passing matrix, it reproduced all 56 measured moves, harmonics 1 to 8 at seven dances, within 0.00004 of a turn (run KERNEL-1; row 699).
18. **Every project counted what landed, not only what missed.** By its paper's publication on 28 October the triple-star thread had registered 490 predictions before their runs: 236 landed, 211 missed and 43 could not be read, and 35 of its 92 redesigns of earlier misses landed; at its book's publication on 9 November the Loop Atlas had judged 1,104 registered predictions, 706 held, 369 missed and 29 undecided, with 572 planted checks beside them, 555 held; and at its book's publication on 5 November R2W had graded 527 tests against predictions written first, 448 with at least one prediction landed (223 with every prediction and 30 only through the named rival) and 79 missed outright (rows 646, 661, 854 and 882; the part on the headline register).
19. **A learning mind cannot grade its own homework, and heals its answers long before its self.** *Question the Substrate* (9 November; concept DOI 10.5281/zenodo.24094163) registered 212 predictions before their runs, 122 held, 89 missed and 1 held in part: a self-training chain that keeps the half of its own labels it is surest of lost 25.3 points and let an answer holding 11 per cent of the data die out, while the same cut ranked by the frozen first network lost 2.1 and kept it alive (survival 0.83 against 0.02); every failing rule ranked or copied by the chain itself, and every cure was a fixed reference from outside (rows 77 and 883). *Two Arrows* (10 November; concept DOI 10.5281/zenodo.24120415) registered 170, 87 held and 83 missed: after a closed loop of tasks a learner's answers keep 0.0167 of their change and its hidden layer 0.3049, and only a pull toward a record of the hidden layer taken before the loop brings it home, to 0.0003 (rows 888 and 889; the part on the headline register).

About this guide

The cosmos is an amazingly interconnected place. Every relationship between any two or more points in spacetime — from the way a DNA strand twists, to the way a fault line accumulates stress, to the way starlight curves around mass — is a fiber bundle waiting to be read. The bundles are always there. What varies is whether we have the sensor data and the geometry to reach them.

Loop walking is the method for reaching them. You choose a path that comes back to where it started. You walk it. You measure what fails to close when you return. That gap — the holonomy — is the geometry telling you something. Not a size. A rotation, a winding, a count: something written into the structure of the substrate that no smooth push

can erase and no noise can manufacture.

A word about what this is. This is a working guide, kept up to date as the work goes. It records best practices even when we choose to ignore them, and bad results as eagerly as good ones. Do not be afraid to fork and pivot; just maintain full documentation and disclosure practices. The failures are as important as the findings, and in this work they usually arrive first.

Think of this guide as a holiday tree. The four limbs are the branches. The ornaments hanging from them are treasured lessons, each one earned on a specific project and priced so it travels to the next one. The tree keeps growing — limbs can be differentiated or added as the method finds new substrates. This document is the tree at version 7.

Why it exists. Two reasons, and the selfish one is worth naming. When I run an agentic AI research partner on these methods, it needs to be spun up: it needs the vocabulary, the lessons, the rule system built over these months, and the knowledge that the rules help but are not sacred. They keep results legitimate and save enormous time, but they are tuned down carefully depending on the project — because the rules teach through failure, and that is their whole point. This guide is that spin-up document. It is also what I wish had existed when I started. And it was not built alone — it is the record of a collaboration: the researcher, an AI assistant, and the subconscious that kept surfacing connections at 3 a.m. The ‘we’ and ‘our’ throughout are real.

How version 7 was gathered. We opened a session on 23 September 2026 to hold the guide’s next upgrade after version 6. The lessons we learned formed its structure, and the papers themselves filled out the limbs of the tree as we recorded the questions that grew out of the data. Reading loops became writing loops in the Latch Atlas. Loops that needed walls, or each other, for a push became loops without history or walls in the Loop Atlas. The Loop Atlas spun off A Rider in Two Waves (R2W, called the RF Spoke in its early records), to work out how quadrature and timing make gearboxes and drive trains. We engineered with and through more chaos and binning in these weeks than in the six weeks before them combined. We watched the discovery tests and the behaviours on the pictures cross the projects like a ripple: driven pendulums turned up in all three projects in the same window, damped in two of them and frictionless in the triple star, and these little patterns clustered loudly. At moments it was a little eerie; I would lose track of which project’s results I was reviewing, because the language was the same in all three. Each paper still had its own views and branches: fast and slow chaos in the three-body work; R2W sharing its depth-axis explorer, a toy with knobs, with the Loop Atlas, which reshaped it into the stretch-and-squeeze explorer, the same ideas under different names; and R2W mapping its own toy’s atlas of where and how we can grip RF and use it as an engine and a gearbox. I am still learning the language for what is in my head, because some part of me still expects to walk into a dark room, have the lights snap on, and hear everyone shout “Surprise! It was all an elaborate joke.” That is how it feels to play loop walker with holonomy every night at my dining table, in a way that connects with so many cool spaces and geometries.

What version 7 includes. This version includes the work through the two-voices papers, *Question the Substrate* (published on 9 November 2026, concept DOI 10.5281/zenodo.24094163) and *Two Arrows* (published on 10 November 2026, concept DOI 10.5281/zenodo.24120415). Their lessons are rows 883 to 891 and additions to 20 earlier rows, each naming its paper.

Quotations. Bear’s words from working conversations, and the labels he wrote on pictures, are quoted in his own words, with typing slips and capitals set right; the projects’ own records keep them as typed.

The glasses

I want to describe how I see things, because it will show up everywhere in this guide and the research papers that companion it, and knowing it in advance may save you some confusion.

I experience the universe as mostly ternary. Slopes go up, sit flat, or fall: −1, 0, +1. Curvature turns left, keeps straight, or turns right. A ternary operator is not a simplification — it is the native language of the thing changing direction. Rates of change, rates of change of those rates, and rates of change of those — expressed in three signs — describe almost everything this method reads. The hierarchy goes: value, slope, curvature, and beyond. Each level is ternary. When I find a substrate that resists the ternary frame, I pay attention to that resistance.

I see fiber bundles everywhere. Every pair of points in space has a relationship. The collection of all those relationships is the fiber bundle network. It is always there — the instrument does not create it, only exposes it. The instrument’s job is to find a seam or a junction where the geometry is readable, walk a loop around it, and report what the geometry accumulated. Some substrates offer that seam readily. Others hide it behind instrumentation limits. But the bundle is there regardless.

I treat every failure as a lesson that has to be extracted deliberately. The method delivers failure routinely, and that is the point. A null result, read correctly, is more valuable than a positive result read carelessly, because a null with a cause is a specification: it tells you exactly where the instrument ends and the substrate begins.

A vocabulary for this guide

These words appear throughout, often without warning. The definitions below are how this corpus uses them — not necessarily how a textbook would, though the textbook usually agrees.

Invariant. Something that stays the same even when everything around it changes. An invariant is not a coincidence — it is a consequence: the geometry forces it constant, and no noise, calibration drift, or perturbation can move it without crossing a threshold that has a name and a measurement. In this corpus we hunt non-trivial invariants: quantities that topology makes permanent. When you find one, you have found a direct readout of the substrate's geometry. An invariant is not a bad thing — it is the whole point.

Loop walking. You pick a closed path — a route that returns to its starting point. You carry some quantity along that path and observe how it has changed when you return. The change is the reading. This is the entire method. Every technique in this guide is a way to choose the loop, carry the quantity, or interpret the reading.

Holonomy. The change that accumulates when you transport something around a closed loop. A compass carried around a mountain may point in a different direction when it arrives back at the start — that difference is the holonomy of that path in that field. This guide reads holonomy not as a curiosity but as an instrument: the holonomy reports what the geometry did to whatever you were carrying. Not every leftover is a holonomy: run the loop out and back along one line, enclosing no area, and a holonomy leaves nothing there, while a knee (hysteresis) leaves its band (row 116).

Fiber bundle. Imagine every point in space has a small room attached to it. The space you are moving through is the base. The rooms are the fibers. The rules for how you move from one room to the next as you travel are the connection. The whole system — space, rooms, and rules — is the fiber bundle. The key fact in this corpus: the bundles are always there. What varies is our ability to access them, which depends on what sensors we have and what the geometry looks like at the seams and junctions.

Differential geometry. The mathematics of curved spaces. This guide uses it as a gateway into the geometry of the substrate, not as the destination. You do not need to compute it — you need to know that it organizes which loops are equivalent, which gaps are meaningful, and why the winding number is protected.

Winding number and integer channels. How many times something wraps around. One full wrap is winding number 1. Two is 2. The crucial property: it cannot change by fractions. It cannot drift between checks. White noise cannot push it from 3 to 3.7 — it can only jump by a whole step, and that jump requires a crossing event with a number on it. This is the protection this method exploits.

Residual. What remains after the integer part is extracted. The continuous signal that rides alongside the winding count. The residual tells you about the substrate's smooth properties; the winding tells you about its topology.

Amplitude blindness. This instrument does not care how large the signal is, only about its topological structure. Magnitude is deliberately discarded. That is not a limitation — it is the design. Amplitude-blindness is what makes the read immune to calibration drift, to scaling errors, to the difference between a big version and a small version of the same geometry.

The competence surface. Not "the instrument works" — but "here is where it works, here is where it goes dark, and here is the measurement that tells them apart." Every instrument this guide describes ships with its own visibility matrix: the list of what it can see, at what aperture, and what it cannot see and why. A limitation with a known cause and a named price is a design parameter, not a failure. The surface is drawn from the ruler (defined below, after the floor): wherever the effect outgrows the ruler's step, scatter and bias together, the instrument sees, and wherever it does not, the instrument goes dark, however good the ruler is elsewhere.

Common mode. When the same signal arrives at all your sensors simultaneously. Common-mode signals are indistinguishable from reality — every sensor agrees, and that agreement is the trap. An instrument that cannot reject common mode cannot separate a global perturbation from a local one. Common-mode rejection is the art of building readings that depend on *differences* between sensors rather than their shared level.

The floor. How much your measurement moves when nothing is happening. Before any reading means anything, you measure the floor: the amplitude of your quantity across an ensemble where nothing is wrong. If the effect you are looking for is smaller than the floor, no processing downstream can recover it. That comparison — effect divided by floor — is one division, and it costs about a minute, and it kills more bad projects than any other check in this guide.

The ruler. Whatever turns a claim into a number: a count, a fit, a grid, a reader, a picture's pixels read on its own axes. In this guide a person's eye and a blind reader propose and the ruler decides, but the ruler has limits of its own, and each leaves a mark that can be told apart from the others. Its *step* is the smallest change it can see: a reading on a grid is good only to the grid, a number carried to too few digits is good only to its digits, and a system that will not commit to one side of a step's edge makes signal out of the edge (chaos and binning; rows 686 and 738). Its *scatter* is how far it wanders when nothing is happening: scatter changes sign from run to run and shrinks when readings are averaged (the floor). Its *bias* is a lean it adds to every reading: small, steady, the same everywhere, and it does not

average away, so an effect with those three properties is the ruler's fingerprint until a known answer says otherwise (row 741). Its *frame* is what it measures against: a ratio's partner, a zero that can move, a coordinate that bunches its marks in one place; when the frame moves, the reading moves with it (rows 308 and 730). Each mark has its own test. Scatter shrinks with averaging and changes with the seed. Bias survives averaging and changes when the ruler is changed. A step's edge moves when the grid is shifted by half a step. A frame's motion disappears when both ends are measured. The ruler is not the competence surface: the ruler is the instrument in the hand, and the competence surface is its map.

Chaos and binning. A recurring partner in this work, named as a family in version 7, which counted 63 arrivals of it, 3 of them from sessions that had not yet read the guide (the Register, "What kept arriving"); a test at three chaos thresholds can tell it from structure (row 655). When a chaotic or near-chaotic system sits near the boundary between two measurement bins, it flickers back and forth across the edge, producing apparent signal that is really the bin boundary, not the physics. The formal name shifts by domain — Arnold tongues in oscillators, level repulsion in quantum chaos, resolution walk in numerical analysis — but the structure is the same: discretisation meets a system that would rather not commit to one side. Recognising this pattern is one of the most transferable skills in this corpus.

Memory reading. A per-cycle measurement that changes by its own size when the clock shifts a tenth of a period — reported as a slope and a crossing, not a value. Before quoting any per-cycle number from a medium with memory, sweep at least five phase points and report the line.

Floor (instrument zero). The instrument's own answer where the true answer is known to be zero. For a synthetic instrument, this floor is stated on the first page of results. A result inside the floor has a magnitude and no sign.

Station-keeping within the floor. The honest verdict for a body or measurement whose motion is smaller than the instrument's floor. The sign is not resolved — this is not a failure, it is a precisely stated result.

Two clocks. The verdict when two time scales of one process are shown not to scale together, with the predicted ratio committed at a band before the runs. A falsified ratio is a finding about the medium, not a failure.

Duct. A mechanism the instrument found that the object being modelled does not possess. Kept, fenced as not a design, pointed at a specialist.

Latch. A whole number that costs energy to hold and sits behind a wall higher than heat's jiggle. It holds nothing itself; it forbids the store beneath it from emptying below a floor. The Latch Atlas prices one in each of thirteen media.

Label. A whole number that travels and holds nothing: twisted light, radio, sound. It becomes a latch only when it lands on matter whose phase must close on itself.

Powered lock. A lock that holds its count only while it is driven, like a voltage standard's steps, which vanish when the tone stops.

Undecided. A verdict, not a failure to reach one: a pre-registered test whose result landed in the gap its own rule left, reported as written. It differs from HELD, which waits on a missing artifact.

Tread. A flat step in a two-rhythm rider's staircase, where its slip holds a fixed ratio across a range of pace. Treads appear only where phase is carried forward from one cycle to the next (row 605), and one is accepted only with flat shoulders and after the null case has run (row 603). It is not the Pythagorean comma, which is a closure gap (row 202).

Memory. A word with four senses in this project, each right in its own room, so a sentence that carries it across rooms says which. A medium that remembers, so that a per-cycle number moves with the clock: the "Memory reading" above, whose five-point sweep proved itself on the Loop Atlas's own rows where the line was steep and was not needed where it was flat. A rider that does not forget its start, read as a largest Lyapunov exponent near zero (row 605). Route-dependence, where two locks overlap and the state you hold depends on the road you came by, which needs mass (row 670). And a dial in the Loop Atlas's needle called "memory", which is its inertia (row 643). The needle's equation sets its drag to 1, so that dial is the needle's mass over its drag, the time it takes to forget its speed, and in the toys memory and mass are one number. A method that erases memory, such as heavy drag or a reset of the speed, removes the mass's coasting from the motion. Whether every effect of mass goes with it, its weight and its stored energy among them, is an open question this guide carries for a future project.

Anomalistic lag. A binary's periapsis clock running behind its energy clock, as the Moon's anomalistic month, perigee to perigee, runs longer than its sidereal month. The name the triple-star sessions gave a lag of this kind in their runs. "Two clocks" keeps its narrow sense, two time scales of one process shown not to scale together, and two clocks tied to each other by their events are coupled clocks.

Coupled clocks. Two clocks that each keep time by their own events and are tied to each other. They come home together after the denominators of the continued fraction of their ratio, and a lock is where they agree. "Two clocks" itself keeps only its narrow sense: the verdict when two time scales of one process are shown not to scale together.

Outside tick. In a strobed picture, the tick set from outside that never feels the rider: the passing of a ripple, a star's closest approach, a drive's period. Its partner is the rider's own pace, set by the rider's slope and its drag; a lock is the rider's own pace giving in to the outside tick where the two are already close. A third timing reads neither: the event, which opens a gate when something arrives. A motion that keeps its rhythm between kicks, set against chaos, is called regular, not a clock.

How to read this guide

If you are a returning session or an AI partner: start at Station 0 of *The Walk*. That station tells you what has already been tried, what is on hold, and what has been closed. Then walk forward from there. Load limb sections when a station raises a question you cannot answer from the prose.

If you are new to this: start at Station 1. Skip Station 0. Walk the ten stations, which are written to be followed in order on the first trip. When a station points you at a row number, that number is an entry in the register at the back. The register is the dictionary; the walk is the phrasebook.

If you are reading for the pleasure of it: this guide is also written for people who do not know anything about this yet. There is a great deal left to discover, and no certificate is required to start looking. Holonomy is in no hurry.

If you are reading a picture of data: the part after the walk, *Two Readers and a Ruler*, is an order of operations for it. A person's eye and a blind second reader propose, and a ruler decides. Each of its lessons is also a row in the register, and it cites them by number.

The Walk's four figures are embedded at the stations that need them. *Two Readers and a Ruler* carries 20 of the pictures it calls for, most of them Bear's marked pictures and R2W's figures; the full edition notes each picture still missing where it belongs.

THE WALK

A working path through a project — ten stations cutting across all four limbs. Station 0 is for returning sessions. A newcomer starts at Station 1.

Station 0 — What is already banked

Session only. A newcomer skips to Station 1.

Before anything else, find out what has already been tried, what is waiting on something, and what has already failed and been recorded. This station exists because the method has no memory of its own and every session that skips it re-derives work that was finished months ago. **That is the failure this entire document was built to stop.**

Three questions. What is **banked** — results already priced, already named, already closed. What is **held** — items waiting on a specific missing artifact, each of which should name the file that would clear it, so a hold is a one-line question rather than a blockage. And what is a **banked negative** — things that were tried and did not work, recorded with their counts, so that a good idea does not get re-run because nobody wrote down that it lost. Read the register of what landed first, beside the record of what missed: a handover keeps the open items and drops the finished wins (row 821).

The discipline that makes this station work is upstream of it: **an idea merely deleted comes back.** A kill has to be recorded with where it applied, why it lost and what replaced it. And a name with no definition on the record is an orphan wearing a label; if you find one here, that is the first thing to fix.

A fresh session is not a lost one. Before declaring anything lost, check the disk and the live record: the canonical object is the file and the deposit, never the session, and “I do not see it where I first looked” is not “it is gone” (row 89).

A tool carried to another machine should find its companion files by what is in them, not by their names; when one goes missing, let the launcher’s census say what happened before guessing (row 721).

Station 1 — Do your words survive the boundary?

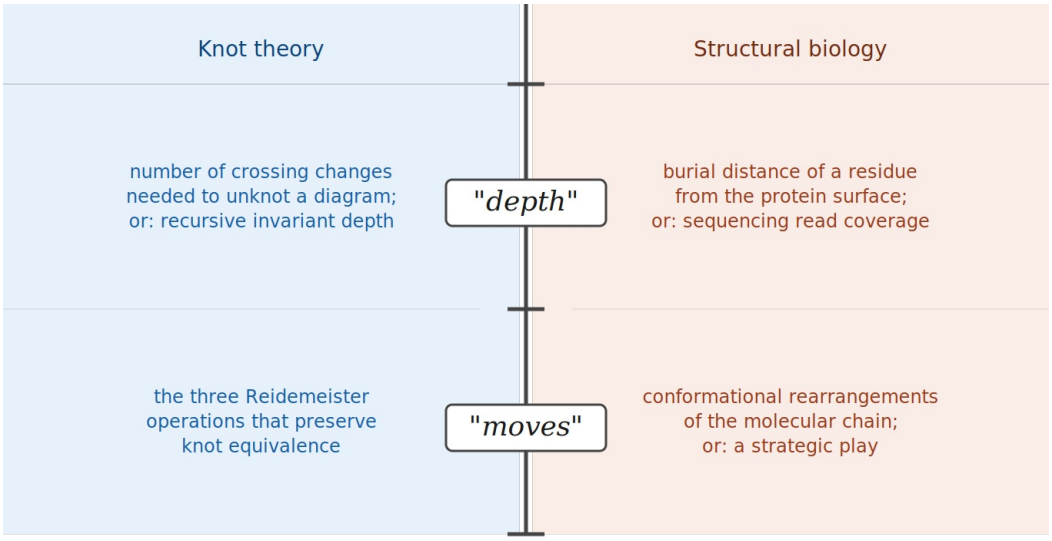


Figure 1. The words fence.

Before you name a single object, check the vocabulary you are about to name it with. **This is the station that comes from the only barrier that ever actually stopped this work.**

Cross-domain words mean different things on either side of a fence. *Depth* and *moves* are the two that cost the most here. And the trap has a second floor that defeats the obvious remedy: **the standards move while you are reading them**, so learning the other field’s term once is not enough. *That is a barrier nobody warns you about and it cost more than any calculation.* The tools were already good. The language was not.

The procedure is one line and it would have caught every instance this corpus has paid for: **before a word carries a claim across a boundary, write down what it denotes on each side and check that the two denotations are the same object.** Do it in writing. The instances that got through were all cases where somebody knew both meanings and did not write them next to each other.

Check the modulus as well as the word. A wrap chosen to compare the directions of lines can fold a quarter turn away, and then a definition crosses the fence inside the arithmetic, where nobody wrote it down (row 554).

And name the pair: two cases that seem to disagree about a phase may be measuring it between different pairs of knobs (row 586).

A word inside a number must survive the boundary too: “travel” was a length in one tool and a signed part along a line in another, and one lap read two ways looked like two strokes (row 612).

When a result is called portable, say in which sense, number for number where the equation is the same or a calibrated ruler everywhere else, and name the machine, since a cousin is not the same machine (row 727).

Station 2 — Can we reach the bundle?

The bundles are always there. The fiber bundle network between all points in your piece of real estate is assumed to exist — it is not absent, it is not optional, it is not something you earn by having the right substrate. What varies is your ability to access it, and that depends on exactly two things: the available sensor data, and the geometry. **Asking whether a bundle exists is the wrong question.** The right question is: do we have the sensors and the geometric exposure to read the bundle that is already there?

This reframes every null at this station. You are not dead — **you are unmapped**, and unmapped is always recoverable. The geometry will offer a vertex, a junction or a seam when the instrumentation can reach it.

Name three things before going further: the **base** you are moving over, the **fibre** you are carrying, and the **connection** that says how the fibre is carried. If you cannot name all three yet, that is a gap in the map, not a gap in the world.

One prerequisite, and it is worth exactly four words: **one pin, no loop.** Holonomy needs a path along which state is transported and around which closure can fail. A single steady source with no differential offers nothing to walk — not because the bundle is absent, but because the sensors have not yet given you a path to walk.

Ask the incumbent bar now, while you can still change the question. Who is already doing something like this, and what bar does this work have to clear? An instrument that claims to be universally superior is a marketing claim, not a scientific instrument.

Begin the expert witness roster here. An expert witness in the courtroom sense: retained in advance, qualified on a specific question before they are needed, independent of the work and therefore able to disagree with it. Name the specialties this project will need. The roster accumulates across projects and starts earning its keep long before anyone is called.

Before asking what a medium remembers, ask whether its whole numbers are latches or labels: held behind a wall, or flying away holding nothing (row 561).

Station 3 — Choosing the loop

The witness roster grows here as the loop geometry becomes concrete.

The loop is not a free parameter and it is not a resolution dial. **Loop shape is a match to the interior’s symmetry, not a ladder toward perfection.** A finer loop reads a smooth interior better and reads a discontinuity *worse*, because it averages across the cut.

Match the aperture to the source’s dimensionality. Getting this wrong produces a clean, honest and entirely misleading null. **The rim sees the mouth and never the neck:** a tight core inside a wide circulation collapses below your station spacing whatever your sensitivity.

Redundancy is the lie detector, and a triangle has none — three vertices are exactly determined, so one bad station corrupts the reading with nothing left over to catch it. **Prefer an odd count**, because an even pointer has a preferred axis and couples to the two-fold disturbance it should reject.

Where the substrate has its own arithmetic, **let the arithmetic choose the face.** A lattice whose smallest closed walk has four steps gives you squares, not triangles.

Station 4 — Declare the blind spots now

Now, before the first reading. Not after.

Start with the ones you can derive for free. A signed quantity is blind to anything that cancels under mirroring — that follows from the definition with one line of algebra and no experiment. A scalar observable is loop-blind by the telescoping identity, permanently. These cost nothing and can tell you a channel is closed before you spend even a minute measuring it. Derive the free blind spots first precisely because they are free.

Then publish it. **The honest instrument ships its own visibility matrix**: which physical modes each channel can see, at what aperture, and — as importantly — which it cannot.

And commit the bar in writing while the data is still missing. Two parts, so that a win on one axis cannot be reported as a win on both. **Then do the step almost nobody does: go back and ask whether the bar could ever have said no.** A target satisfied by construction is not a target.

Before the first reading, ask every toy the four cheap questions: faster, backward, again, and one knob alone (row 584).

Derive the symmetry before choosing where to centre a loop: a loop centred where its quantity is odd under a mirror reads zero to every order, a blind spot you can predict without running anything (row 259).

Station 5 — The floor, before you read anything

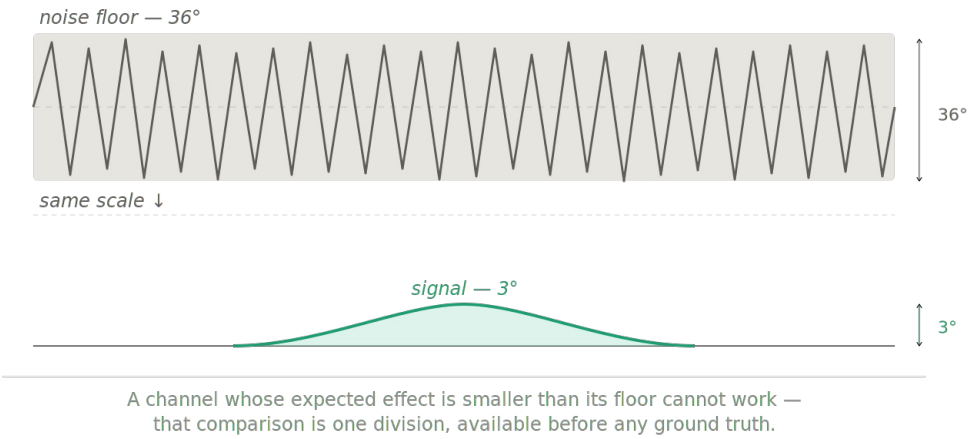


Figure 2. The floor.

This is the cheapest station in the walk and it kills more bad projects than every other station combined. It needs no labels, makes no claim, and takes about a minute.

Measure how far your quantity moves when nothing is wrong. Then compare it to the effect you are hoping to see. **A channel whose expected effect is smaller than its floor cannot work, and no amount of processing downstream will recover it.** That comparison is one division, available before any ground truth at all.

The floor belongs to a substrate *and* a perturbation *and* a corpus. **Using one floor for three substrates is an error**, and it is an easy one because the first floor you measure feels like a property of the method.

The integer can die before the amplitude. Above the matched scale a count is far more robust than a magnitude; below it, white noise does not merely degrade the count, it *manufactures* windings that were never there.

On real data, the rim decides: a count is only as good as the rim it is read from, and the window for a read has two edges and a trap in the middle (rows 566, 568).

Before trusting a statistic, run it on a record of the same length with nothing in it: a median that carries only the record's length is not a clock (row 616).

Say the instrument's own floor before setting a threshold: a bar tighter than the floor fails for the wrong reason (row 337).

Before reading a miss against a bar as noise, simulate how often the bar fails by chance on the null shape (row 723).

Station 6 — The first read

Decide which channel carries your signal, and know that the choice is not between a good number and a bad one.

Some numbers wobble and some numbers tick. Amplitudes and phases are continuous; windings and linking numbers are integers that cannot move by less than one. **A local dent cannot change how many times the phase wraps** — only a perturbation that itself winds all the way around can. That is the whole protection.

Two habits worth more than any single channel. **Read differences of an invariant** — differences kill any uniform background, and an integer cannot drift. And **read the rate, not the level.**

This last one is worth dwelling on. The corpus arrived at it again and again, from substrates that share no derivation: a tornado's core strain, a fault's shear, a clinical array, three commodities' prices, a clinical residual and a walker's ternary rates (rows 105, 294, 460, 492 and 537). It had arrived six times when row 537 was written, and version 7's fold met it twice more (rows 708 and 715). Arrivals that independent are not a coincidence — it is the corpus telling you something about the world. When the same lesson re-derives itself across that much distance, it is not a method preference. It is a structural fact about how information travels.

The integral erases history. A global walk is order-blind, so a true inverse pair vanishes. And **an open arc is a convention; only the closed loop is an invariant.**

Before connecting two numbers that look alike, ask what kind of object each one is. An exponent that tracks a length is a magnitude, however integer-like its digits (row 548).

A count is whole only when it is read as a sum of closed pieces (row 560). And split any sum by the sign of the field before trusting it: the physics fixes which symmetry a quantity must have, and a part with the wrong symmetry is a witness against the instrument (row 575).

After the first read, read the data once more as if no physics applied: what does it look like, and what is it doing? That read is not a result, but it is often where the next test comes from (row 613).

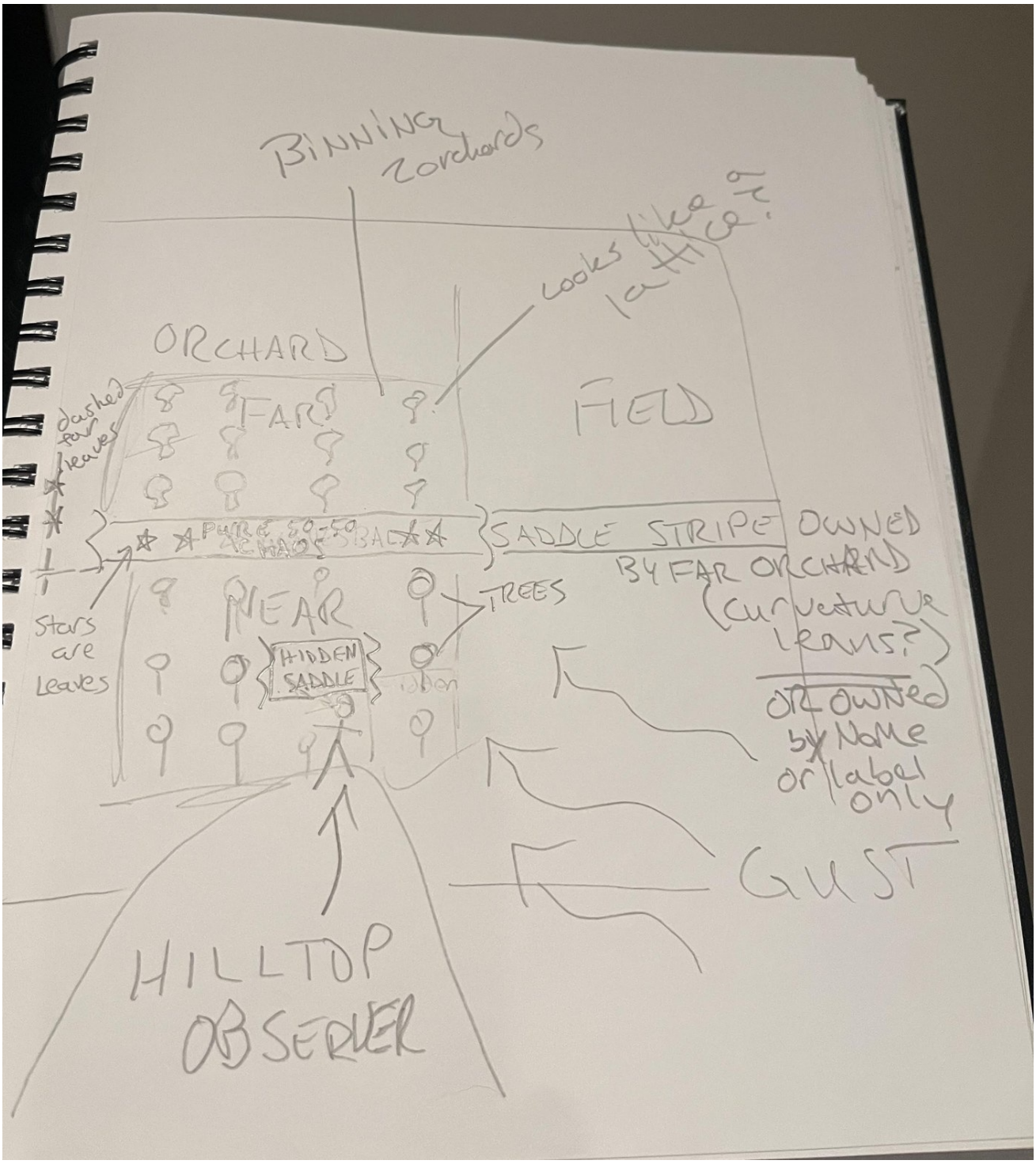


Figure 3. The orchards, drawn. Bear's pencil drawing, 6 October, of R2W's read 33, a read given in words alone and naming no law: two orchards, far and near, a stripe of star-leaves between them, a hidden saddle, a gust and a hilltop observer. In R2W it was the read that showed how much a no-physics read can carry, and R2W's own pattern lessons later set its saddle between two orchards beside the light cone of 28 October as a rhyme.

Count landings, not stumbles: a counter watching a noisy phase at one line logs its jitter as pairs of crossings, and a knee that commits only on landing removes them (row 627).

Name a state only by a test that separates it from its neighbours: its fraction and its period together, read over a whole number of its cycles, and only after it has landed (row 665).

A count of locks is a count made by a reader, so name the reader beside the number (row 722). And judge a transition by a measure that changes smoothly: a yes-or-no line laid across it says only which side of a chosen value a point fell (row 728).

If the data comes as a picture, read it with *Two Readers and a Ruler*, the part after the walk: name the picture, declare what you have already seen, read it blind, and let the ruler decide (rows 713, 369 and 616).

Station 7 — Before you believe it

This is where the expert witnesses take the stand.

An instrument that has only ever agreed with itself has not yet been tested. Internal consistency — nulls, injections, shuffles — is necessary and supplies no external witness. This is where the witnesses you retained at Station 2 take the stand: a specialist qualified in advance who reviews the reading independently, knowing the question but not the preferred answer.

Coherence is the symptom, not the safety. This corpus arrived at that three separate times in three substrates: a control loop defending a corrupted setpoint; a pipeline confident across every seed because one character of an accession was transposed; a game board whose whole map descended from one anchor placed on ground that read clean and was not. **A wrong input makes every internal check honestly wrong together.** A false reality from a single source is more seamless than a true one assembled from independent noisy sensors, because there is no second voice to disagree.

Widen the window until the answer stops moving, then report how wide you had to go — a boundary in your search will masquerade as a structure in your data.

Two checks before the headline is written. Validate its number against a harder version of your own method that changes one thing, and remember that passing proves the solver, not the quantity (row 549). And if you read a magnitude off a staircase across an integer, list everything else that changes with the integer and pin it (row 550).

A second fit of the same data is not a second witness. Two readers of one light, like two solvers of one set of shapes, tell you about the readers (rows 549 and 556); the witness is a second instrument on the same ground.

Some witnesses are already retained: the substrate’s own motion past the instrument, the archive’s own keywords, a second count with the opposite blind spot. Call them, and expect them to testify against you too (rows 567, 576, 577).

Count witnesses at the level of what is independent, the run and not the episode, and re-run what was first seen as unscored controls beside fresh cases from a grid that cannot repeat it (rows 619, 621).

Before judging another room’s number, run that room’s own code here, unchanged, and refine only then (row 650).

When two instruments disagree, do not take the older one as the reference; bring a third that shares code with neither (row 35).

Station 8 — When you hit a wall

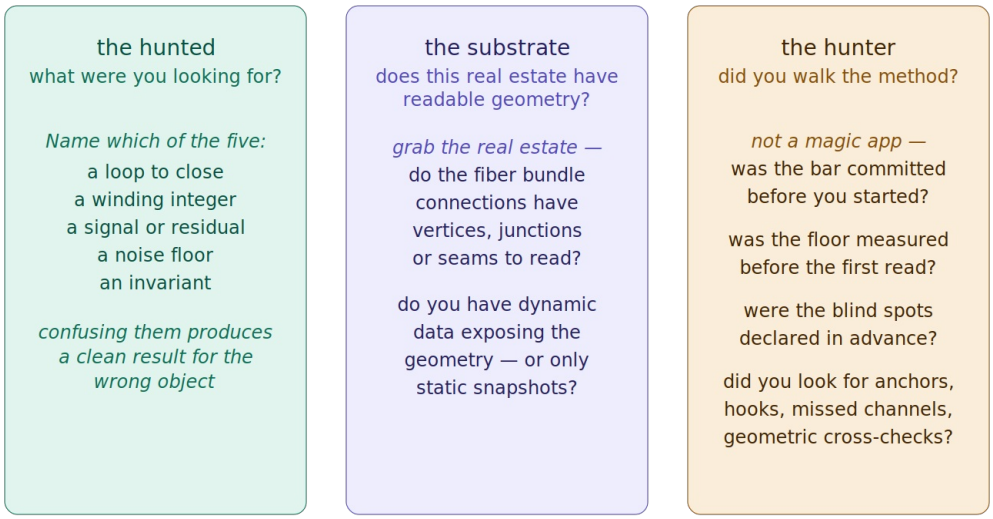


Figure 4. The three-cause autopsy. Flucht nach vorn.

The null is not the exception branch. It is the main road.

The entire guide exists because holonomy delivers failure routinely, and the method demands a new partnership with failure as the teacher. The clause that keeps this from being consolation: **the method does not slack and does not hand out participation trophies.** A lesson is not conferred by having failed — it has to be extracted, deliberately, by looking.

The four places to look before writing any null:

An **anchor** — is there a fixed exterior reference you never identified?

A **hook** — is there a place the structure offers purchase you walked past?

Missed instrumentation — is there a channel already in the data you never read?

A **geometric cross-check** — is there a closure, a parity, a mirror or a symmetry that would test this for free?

Now run the prior-art pass on the route. Not the idea — the route. The incumbent bar was asked at Station 2 and tells you what bar to clear. This is a different question: has someone already mapped this particular wall? Run it here, when you have hit the wall, not when you had the idea, because running it too early means inheriting the field's route including its blind spots.

Do not stop when you find an answer. You will never find the other patches and combs when you quit looking.

When the pass credits a result, read the result — the pages, not the abstract. A citation checked only for existence is a claim you have not verified (row 552).

When a thing gets thin near an edge, compute the edge first and then look; a grid cannot find a sliver (row 631).

When the whole toy turns out to be a textbook object, its map becomes a bench: every landmark is a known length to check the ruler against, and any claim of something new moves from the physics to the reading (row 726).

Station 9 — What you owe

Tell the ending first when the ending is a failure, and tell it in the abstract rather than the discussion. Name your loss before a reader can name it for you. Put the top results there too, one sentence each with their numbers, because a reader, human or machine, may read no further (row 840).

Report the misses in print. A registered prediction that failed belongs in the body with the reason it failed, because the reason is usually the result. **A clean pre-registered null with a caught borrowed parameter is worth more than a positive result**, because only one of the two can be faked by wanting it.

Separate your confidence from your tone. Write down, in a register, what you assumed, which specialty should check it, and the precise question that would settle it — so that the ask can be routed to seven people rather than pleaded to one. Where the trade belongs to somebody else, **state the asymmetry in both directions with numbers and decline to resolve it**, because that judgement is not yours to make on their behalf.

And publish the compliance statement in fixed form, so that **an omission is visible without an audit.**

Read the built artifact back before it ships (row 553), and ask of every check whether it could have failed on the question you are claiming (row 551).

Report an UNDECIDED as written; it is a verdict (row 580).

A withdrawal reaches every document that carries the number, the withdrawing room’s own first, or the correction becomes a second defect (row 588).

The limbs follow: The Method as an Instrument — The Geometry & Substrate You Walk — Holonomy You Write — The Ledger.

Station reference card

Station	Name	Key rows	What they price
0	What is already banked	402, 486, 504, 89, 721, 821	The banked negative; an idea merely deleted comes back; the guide is the solver’s externalised state; a fresh session is not a lost one: check the disk and the record before declaring a loss; find files by content; read the register of what landed first
1	Do your words survive the boundary?	502, 503, 554, 586, 612, 727	Write both denotations before any word carries a claim; the standards move while you are reading them; check the modulus as well as the word; name the pair of knobs; a travel number is a length or a signed part along a named line; say which sense of portable, and which machine
2	Can we reach the bundle?	505, 534, 535, 561	Expert witnesses retained early; name which of the five you are hunting; the substrate is fiber bundle real estate; matter latches and waves label
3	Choosing the loop	355, 374	Prefer an odd count — even pointers couple to even-fold noise; rim sees the mouth and never the neck
4	Declare the blind spots now	2, 21, 337, 584, 259	Declared blindness; the visibility matrix; a bar that cannot fail is not a bar; faster, backward, again, and one knob alone; derive the symmetry before centring a loop
5	The floor, before you read anything	305, 451, 566, 568, 616, 337, 723	The integer can die before the amplitude; rank each channel by effect divided by floor — one division; on real data the rim decides; coarse sampling misses, noise fabricates, likeness erases; run a statistic on an empty record of the same length first; say the instrument’s floor before the bar; simulate the bar
6	The first read	105, 294, 359, 120, 121, 548, 560, 575, 613, 627, 665, 713, 722, 728	The signal is in the rate; read the slope not the value; read differences of an invariant; the integral erases history; an open arc is a convention; ask what kind of object each number is; a count is whole only as a sum of closed pieces; split by polarity before you sum; read once more as if no physics applied; count landings, not stumbles; name a state only by a test that separates it; read a picture with two readers and a ruler; name the reader; judge by a smooth measure
7	Before you believe it	306, 424, 505, 549, 550, 556, 567, 576, 577, 619, 621, 650, 35	An instrument that has only ever agreed with itself has not been tested; coherence is the symptom; expert witnesses take the stand; a harder solver tests the solver; pin what climbs with the integer; a second fit is not a second witness; call the witnesses already retained; count witnesses at the level of what is independent; run the other room’s code before judging its number; bring a third instrument when two disagree
8	When you hit a wall	507, 508, 506, 509, 552, 631, 726	Flucht nach vorn — null is the main road; four places to look before any null; prior art at the wall not the idea; do not stop when you find an answer; credit the theorem that exists, read on its pages; compute the edge before looking for a sliver; a textbook toy’s map is a bench
9	What you owe	439, 475, 495, 551, 553, 580, 588, 840	Tell the ending first when the ending is a failure; publish the compliance statement; state the asymmetry in both directions and decline to resolve it; a check certifies only what it could have failed on; only the built artifact proves an edit landed; UNDECIDED is a verdict; a withdrawal reaches every document that carries the number; the top results go in the abstract, one sentence each

How to use this card. The card leads from any station in *The Walk* to the rows in the register that it calls on most; row numbers are stable, and titles travel with them in every cell. When a station raises a question you cannot answer from the prose, the row numbers here are your entry points into the register. Every row carries its price and its name in the same cell — enough to re-load the lesson with zero context.

Rate-over-level (rows 105, 294, 460, 492, 537) had arrived six times when row 537 was written, and version 7’s fold met it twice more (rows 708 and 715). It is listed at Station 6 because that is where you choose the channel, but it applies everywhere you compare a continuous measure to a categorical one.

The walk cuts across all four limbs. The register organises by limb; this card organises by when. A specialist who arrives at Station 5 via the register (*L1 — The Method as an Instrument*) and a newcomer who arrives at Station 5 via The Walk are reading the same rows — just through different doors.

TWO READERS AND A RULER

How to read a picture of data: a person's eye and a blind second reader propose, and a ruler decides. Taught by Bear to three projects at once in October 2026, and checked against answers known in advance.

A computer reading a chart can miss what a person sees at a glance, and a person can see shapes that are not there. This part teaches a way to use both. The person marks the picture, a second reader marks it without seeing the first, and every mark becomes a test with a number before anyone believes it.

Some of the best readings in this work were made by eye, before any program existed to find them. On 9 and 10 October 2026 Bear taught three projects at once how he reads a picture of data. The **Loop Atlas** book watches a needle, a forced and damped pendulum looked at once per push of its drive. The **R2W** maps a rider on a tilted washboard, passed again and again by a ripple, in the Depth Axis Explorer. The **triple-star sessions** follow a third star that passes an inner pair of stars once per lap. Each project turned his method into code, read pictures blind before it saw his marks, read the other projects' pictures, and put a number on every claim that could be tested. This part is what survived the measuring.

The cast and the dials. A few words from the three toys recur. The needle is pushed round by a drive that repeats; its heaviness decides how much of its start it keeps. Strictly, that dial is its mass over its drag, the time it takes to forget its speed. The Loop Atlas calls it the needle's memory, and with the drag set to 1, as the needle's equation sets it, memory and mass are one number (row 643). R2W's rider drops ledge after ledge down a tilted washboard; its maps are drawn against dials called the drive, the pull and the clock ratio, with the rider's lag, where it stands in turns when the ripple passes, up the side. Its weight is its heaviness and its honey is the drag that slows it. A rider standing still is "the rest", a rider dropping one ledge per tick is "the conveyor", and any other lock on a fraction is a "groove"; "the third face" is a three-step dance found at weight 0.40, and "a coin" is a place where two starts end differently. The triple star's dial is its birth ratio, the number of turns the inner pair makes per lap of the third star (7.200, 7.230 and so on), and its pictures are drawn once per pass. Each project ran in working sessions of its own with an AI partner, Claude; this part sometimes calls a project's sessions its room.

The idea underneath fits in one sentence: **what looks like disorder is often order binned on top of order.** Many signals land in the same holes and lie over one another, and the method takes them apart.

Two readers and a ruler. The first reader is a person's eye. The second is an AI partner that reads the same picture blind and writes its reading down, by the clock, before it sees the person's. The ruler is whatever turns a claim into a number: the data, or the picture's own pixels calibrated on its axes, tested against a shuffle of the same dots and against a made-up case whose answer is known. **The eye proposes; the ruler decides.** What both readers saw is the strongest evidence there is. What only one of them saw becomes a test.

The method in five steps, for an AI working beside a person. First, make pictures for an eye, not for a summary: a strobe picture, the same picture squeezed sideways and tilted to a grazing angle, and colours that do not lump classes together. Second, ask the person to mark it, with strokes and a few words, before either of you sees the other's reading, and write your own blind reading too. Third, turn every mark into a claim a number can judge, and register it before measuring. Fourth, trust each reader for what it does well. In these projects the person's marks were right on place, direction, relation, symmetry and faint, long, bending structure, while the AI's blind reads were good on geometry and weak on classes and depth. Sizes belong to the data, not to either eye. Fifth, score both readers' misses with the same ruler, and keep the readings that open a new test.

This part is an order of operations, like *The Walk*, and it is used in that order. Every lesson in it is also a row in the register, most of them in *L1 — The Method as an Instrument*, and it cites them by number; rows 702 to 719 are its spine. Section 1 says why to read pictures and which ones to make. Sections 2 and 3 give the method and its protocol. Section 4 is the toolbox, section 5 a field key of shapes, section 6 the traps, and section 7 the cases. Section 8 is the mathematical object all three projects turned out to share, and section 9 is the scorecard. Section 10 says how to use all of it, and how not to. The method's most common result is to rediscover known physics quickly, and that result is credited as a check on the toy, never as a discovery.

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1 — Why read pictures, and which pictures to make

A picture keeps things that an average throws away, so a person looking at it finds questions faster than a program does. But a picture also invites shapes that are not there, so which picture you draw matters as much as who reads it.

What the eye catches that a summary drops. It keeps lopsidedness. In one triple-star picture the centre of a band's core held within 0.0004 from start to finish, so its average said nothing happened, while Bear's "short spiky top left, long spiky top right" was a flip in the band's lopsidedness, measured afterwards from the picture's pixels as −0.49 to +0.47 and on the data as −0.18 to +0.29: right in sign, overstated in size. It keeps negative space, and the way to see it is a technique: "I unfocus my eyes and allow the white patches to fill my vision. Each is bordered by little clusters of four or more dots in a line." It may keep walls: where Bear drew walls, the step between one visit and the

next was 0.778 of its usual size, smaller than anything 2,000 random sets gave, but that comparison was chosen after the run, so it is claimed as nothing, and the test registered for the next birth has no recorded result at the triple-star thread's close on 29 October (row 313). And it keeps long reach. On one picture Bear drew six long strings and called it 80 per cent, where a greedy program drew fifty short ones (rows 715 and 707).

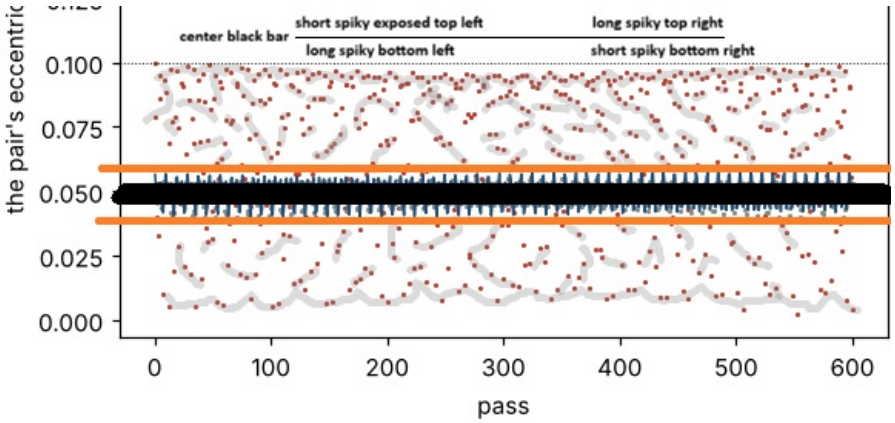


Figure 5. The band, as Bear marked it. The first panel of the triple-star bend figure (birth 7.200, the pair's eccentricity over 600 passes, run 20261009_022616), with his labels: a centre black bar, short spikes at the top left and long at the top right, long at the bottom left and short at the bottom right. Measured afterwards, the band's lopsidedness flipped from −0.49 to +0.47 on the picture's pixels and from −0.18 to +0.29 on the data, while its centre held within 0.0004: right in sign, overstated in size.

What the second reader misses, measured. The AI reader has blind spots of its own, and the projects measured four of them. It names classes badly when a colour scale lumps them: on one map, colouring by denominator painted the rest, the conveyor and two grooves the same orange, and a blind read named a chain above pull 1.0 as the rest, where the data said conveyor, 37 cells at lag 0 (row 717). It reads depth badly: a strobe picture is a shadow of something with more numbers to it, and a join that looks real in two numbers can come apart in a third (row 710). It defaults to a fork, drawing two roads at every split, where Bear draws three, counting the road that goes straight on (row 705). And a tool's sensors see only at their own size: sensors 5 holes long could not see lanes 0.3 turn long (row 715). So the person's reading is not decoration. It catches what the program was built not to see.

The eye is the fastest source of claims there is. Before any fit, Bear read the triple-star bend figures and said that the bend is woven from two phases; adding the second angle took the fit from 0.4302 to 0.9723. He said that two angles make a torus; at 6,000 passes the neighbour ratio fell from 0.1126 to 0.0603. He said that the leftover wave slides, that the corridor has walls that pinch, that the visits pile up at the walls, that the rows arch below the waist, and that the arrows lie along the first angle. Most of it was borne out. The piling at the walls stood beyond anything 2,000 random sets gave, on a comparison chosen after the run and so claimed as nothing, and it did not reach the bar registered for it; the rows' registered test missed by one row (section 9; row 313). In one sitting of R2W the eye's shortcuts were concrete: a second S-shaped curve he drew days before any test looked for it; a word about symmetry that led to a closed form for the pin, until then only "a place where curves cross", which a registered test (run PIVOT-1) had just failed to explain, and so probably saved a day; and a side note that would otherwise have stayed unread. Each saved at least a run of guessing.

And the eye carries its own trap. Look for a shape and you will find it. Bear's warning: "If we are just looking for a fit, we are playing an inspirational game of Texas marksman … frequency illusion will cause it to appear. So I stay blind and tell you what I see." The rest of this part exists to keep that trap shut: a second reader who has not seen the first one's marks, and a ruler that has the last word.

Which pictures to make, in this order. Some pictures teach a reader and some do not, and the difference is a rule.

1. **The thing moving, as frames.** A wing drawn as a strip of snapshots, good timing against bad, taught at once. A graph of push against distance did not, and the same story told hands-first, a rubber band between the fingers and a hand pushed through a sink of water, did. Dot pictures of a strobe, a thing caught once per tick, are the eye's native ground. Drawings side by side teach best, and a person's own hand drawings can teach both readers: a sketch of steep peaks from 2011 became two runs of the Loop Atlas.
2. **The counts.** Put a histogram on each axis, and treat every hole as a counter (row 702).

3. **Only then a derived quantity,** always with its anatomy drawn and with a made-up case beside it. Shown the twin lobes of a third difference, Bear read back-to-back phase-locked vortices, two propellers geared together, four waves through the centre, mirror-math, or a boundary transition. The shape belonged to the ruler: three centred differences turn a corner into a step, a bump and then a pair of lobes (row 711; the made-up corner was run CORNER_PRINT 20261010_061715). Two of his five words, boundary transition and mirror-math, were right. The picture had invited the other three.

The reason, as far as the Loop Atlas could test it: shown a thing moving, or a population landing in bins, the eye reads it truly and often faster than the tools. Shown a derived quantity, such as a force law, a rate or a third difference, the motion that made it is gone, and the eye supplies one. The fix that worked was to draw the derivation as its own picture, step by step: the corner; its slope as a step; the step's change as a bump; the bump's change as the pair.

Draw two angles as a square that wraps. In the triple-star sessions the picture that taught most was the torus of two angles opened flat: one angle across, the other down, both edges wrapping, each dot coloured by one quantity. Nearly all of Bear's best reads were made on it. Colouring by the pass number, his "history", showed which way a weave travels. A picture meant to be marked is drawn plain, with an equal aspect, a light grid and no overlay, and its code is kept, so that it can be redrawn pixel for pixel and a person's strokes become numbers (row 714).

Read at two scales. At one birth of the triple star the dots come in clusters of three, the same place visited at passes k, k + 201 and k + 402 (575 of 599 dots), and the clusters themselves make rows. A lattice whose holes were sized to the dots saw only the clusters. Bear saw the rows. His eye worked at the scale of the return and the scale of the row at once, and a tool needs both (rows 640 and 703).

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2 — The two readers

The first reader is a person, and the second is an AI that reads the same picture without seeing the person's marks. Where they agree, trust grows; where only one sees something, it becomes a test.

The first reader's method: the lattice. On 9 October 2026 Bear gave all three projects the same training text, while he was teaching R2W and the triple-star sessions at once, each with its own pictures, and he pasted it to the Loop Atlas the same night. He began by naming what this guide is about: "Slopes and rates, changes of rates, changes of changes of rates; ternary +1, 0, −1 slope to everything I see. And the geometry is exposed in the data." The lesson is quoted once here, from the triple-star sessions' fuller copy, with its opening taken from his first message of the same lesson, typing slips set right, and a few asides left out where "…" stands:

> "Today, simple. Take every Cartesian coordinate system like these graphs and wrap the graph in a 2D lattice. Each landing point is a hole in the lattice. It gets a count. And establish a histogram on the horizontal and the vertical axis. … Every example you will see is my mind doing curve fitting. I can't do the math, but I can show the curve. Each point or hole has a series of neighbors, and a history. … So here are your rules of the road and your test. This is like Minesweeper: wrap every possible grid point on the coordinate system in a lattice. Add top and bottom axis histograms as counters for binning (depending on the orientation of the axis and the content). Now we loop at each hole where we find a point, and we ask that pixel or point, 'What is your history?' It might be my oscillations, an event-time counter, whatever. It will often show up as geometry in the data plot. Now we start in 2D. We will graduate you to 3D, and then to how to shift perspective around the data in 4D. It's all there once you have the algorithm to expose it. And the important thing isn't that you find every one. I don't. But if you find a few new observations that somehow connect to other observations, then that's a big win. So every hole with a pixel or point needs a history. That history is the geometry, and the interplay of the knobs in the geometry. The scale makes a big difference sometimes, as we saw. Okay, so remember: slope or rate, change of rate, change of change of rate, our core fundamentals since day 1. How does the neighborhood history manifest? Linear, exponential, logarithmic-style curves. So each point has 8 potential neighbors. If a neighbor is immediately present, record the slope. Now we build out, from radius 1 (immediate neighbors) out to a radius of 5. Now we are building the history of the neighborhood. We have a slope for each. Now we are looking for its relatives. The relatives will look similar (slope or rate), or behave similar (change of slope), or share a change of change of slope (family members one step removed, indicating a boundary layer or a big change in the curve). Same slope = the same string, or overlaid strings. A change in slope = now we might be related, maybe not. When they line up tightly it's easy; when they space out, skipping one or more holes to the next pad, they get harder to track, but over time they will always show as steady (the same behavior, the same frequency), or decreasing or increasing at a given rate or slope, or exponentially or logarithmically. So each lattice hole that is unreported is like a little cellular automaton, a Minesweeper game. But each one is also a counter and a sensor for its geometric neighbors. It can answer: what is my slope with my neighbors? Does it continue straight, and does the frequency change? If it curves or bends, does it do so consistently, or at a given rate, exponent or log? So you can imagine you build a little fiber-bundle network of each of these where it makes sense: a network map interconnecting all points, just in case they might be

> related. Each address is a series of qualities, and these qualities may be shared with their neighbors. You should be able to use a radius-5 triangle lattice (or any n-gon you want, a pentagon or whatever) for every pad to each other pad, as a surface where data points are present. I can run it on my laptop, setting up the true structure in the geometry and the data. 3D and 4D are parity checks on outliers, and on shifting your perspective around event time."

That is the first reader's job in one line: **the eye fits the curve, and the ruler does the arithmetic.**

What the lattice is for. Handing the Loop Atlas its lattice test, he said: "Now a single point can share many strings overlaid. … The hole with the data point may have 10 strings overlapping, so that individual data point is actually stacked 10 deep. So what looks like disorder is just binning layered on top of binning, where ten signals are overlapping." And of one such point: "I think of it as a fiber bundle address index. Geometrically, that point is a line or curve through time, at a frequency of appearance. And that is the flow." The triple-star sessions read the words as a design, and built the lattice's next version in their order (section 4). The address is a place on the torus. The fibre over it is the line through time of that place's visits; at one birth there is one visit every 201 passes, which is the frequency of appearance. The flow is the step from one visit to the next. Where he saw "3 dots on top of each other in the z-axis, into and out of the page", the fibre is seen end on.

The same method, as steps. R2W wrote the lesson as a procedure.

1. **Lattice and histograms.** Cut the chart into holes and count the landings in each. Draw a histogram of the counts along each axis.
2. **History.** For each filled hole, a pad, ask where its riders came from and where they go next.
3. **Neighbours.** Look from radius 1 to radius 5 round each pad, and record the slope to each neighbour as +1, 0 or −1. The runs of level holes between steps make a word, and a straight line gives a balanced word, its runs differing by at most one.
4. **Strings.** Link pads column by column into curves. Give each curve its rate, its change of rate and its change of change, and a shape. The largest change of change marks a boundary layer.
5. **Relatives.** Curves with the same slope are one string, or strings laid over each other. The same change of slope suggests relatives; the same change of change marks a family member one step removed.
6. **Negative space.** The empty patches are bordered by short lines of four or more dots, and the patches outline the threads.
7. **Zoom.** Change the magnification. A real shape holds its place; a drawing artefact moves.
8. **Stay blind.** Read before seeing the other reader's reading, and test whatever only one reader saw.

Five habits that went with the steps. His teaching carried habits that every project adopted.

1. **Count on both axes.** A histogram on each axis turns a picture into counts: the bins hold the numerators, and the whole population is the denominator.
2. **Zoom out more often than in.** "Magnification is a big deal on this stuff," and "most of the time we want to demagnify": watch how the shapes grow as the view widens.
3. **Look for mirrors.** Bilateral symmetry is a place to stalk a shape.
4. **Prize lines of dots.** "Straight line clusters and curve line clusters are gold, because then the 3D shape comes out. Toroids. Vortices. Boundary zones."
5. **Count before trusting a crowded hole.** Many strings can overlay one filled hole, and the eye cannot tell whether it is shared by 5 strings or by 50.

And the frame for all of it: looking at data is "a best practice, not a crystal ball. It's art as much as science."

The second reader. In R2W's reads of the pictures Bear sent, and in the Loop Atlas's first two practices, the second reader was Claude, reading blind. It wrote its reading into the record, stamped by the clock, before it opened Bear's marks, and only then were the two compared. What both found on their own held up best: hooks paired across a V by their height, a beat at the centre of a run, a pinch in one angle near 0.42 (read from the picture; the corridor's measured pinch sits near 0.45), and the nodes along a flank. What only one of them saw became a test. Bear's woven strings became the weave test, and his "three phase-shifted waves" set against Claude's "shrinking" wave became a test of thirds; both readings turned out partly right. **Two blind readers who agree are the guard against the Texas sharpshooter.**

How the two readers differ. The two readers are not alike, and their difference is the point. In both projects that compared them, the person's eye was the better reader of negative space, lopsidedness and long reach, and all three have now been measured: lopsidedness and long reach first, then negative space on his dead zones (sections 7 and 9). A fourth was measured on a window of the Loop Atlas's needle: faint, long, bending structure. Of Bear's strokes on a density of chaos, 0.741 (red) and 0.673 (green) lie within 0.004 rad of the computed fold or its first eight images, against 0.36 and 0.37 for the same strokes set at random, and Claude could see the voids and the X but not the faint

curves until they were computed (row 707). Claude's blind reading was better at geometry, and worse on classes (row 717) and on depth (row 710). And the second reader's eye has habits of its own: it defaults to a fork, drawing two roads at every split after reading that Bear draws three (row 705).

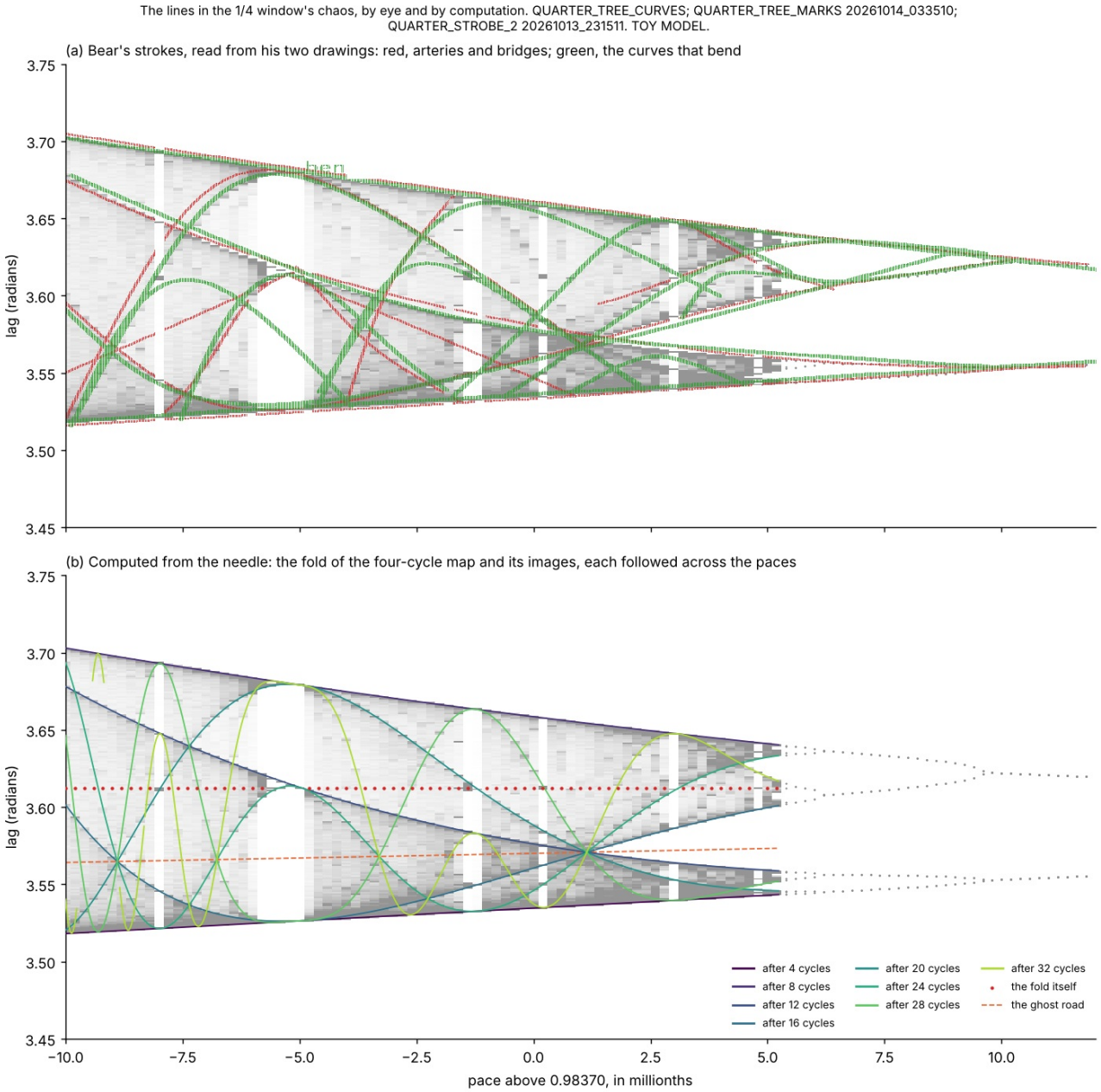


Figure 6. Bear's strokes and the computed curves. Above, his strokes on a window of chaos of the Loop Atlas's needle, read from his two drawings: red for arteries and bridges, green for the curves that bend. Below, the fold of the four-cycle map and its images, computed from the needle. Of his strokes, 0.741 (red) and 0.673 (green) lie within 0.004 rad of the fold or its first eight images, against 0.36 and 0.37 for the same strokes set at random (row 707). TOY MODEL; QUARTER_TREE_CURVES, from QUARTER_TREE_MARKS 20261014_033510 and QUARTER_STROBE_2 20261013_231511.

Hold back the data while the eye is learning. Bear: "Let's finish the examples and then I will give you all the data. I want you to see it, otherwise the learning is shortcut." With the numbers in hand a reader measures instead of looking. Without them it has to look first, and the looking is the skill being taught. The practices that held the data back are what produced blind reads that a scorecard can judge (row 716).

The ruler. Every claim gets a number before it is believed: from the data, or, when the data is held back, from the picture's pixels, calibrated on its own axis ticks and checked on marks whose values are known (row 714). The ruler is checked too, four ways: against a shuffle of the same dots (row 616), against a made-up case (row 711), at a second scale (row 703), and by printing what its own step can and cannot decide (row 686). Bear's rule for his own eye holds for both readers: "Don't let me speak on behalf of the geometry until the data bears witness."

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3 — The protocol

A reading of a picture goes in twelve steps. Say which picture you are reading and what you have already seen, write your reading down before you see anyone else's, turn each mark into a test written before the run, and let a ruler that has itself been checked give the verdict.

The twelve steps are kept in the order a reading goes. R2W ran the core of them on ten pictures in two days. The Loop Atlas and the triple-star sessions each added steps where their own readings had failed without them. One of those steps, the made-up case, was reached by the Loop Atlas on the same day that R2W reached its nearest relative, the simplest separate run.

The steps fall into five stages, in this order: before the reading, the reading, before the run, the ruler, and the score.

Before the reading, name the picture and declare what could colour it.

1. **Name the picture before naming the pattern.** A picture's axes decide which dots sit side by side, so they decide what counts as a neighbour, a weave or a lane. One triple-star run has its dots' nearest relatives 5 passes apart on the page of eccentricity against pass, 14 apart in the reading's own rhythm, 47 apart on the opened torus over 600 passes, and 682 or 663 apart over 6,000 passes at the kit's tolerance, whose arithmetic shaped those two (at the tight tolerance the beat is 722 passes, row 858). Each is right for its picture. R2W read the 5 and the 47 from the pictures' pixels, and the triple-star sessions confirmed both on the numbers: one finding made twice on one dataset, by eye and then by ruler. Say which picture a claim lives in, and draw the same data through a second picture before naming a pattern (row 713).
2. **Declare what you have already seen.** Before a cold reading, write down everything you have read or seen that could colour it. The declaration does not remove the colour. A reader who had read that Bear draws three roads at a split still drew two. Two rules written up as new in one project's practices, the overpass test (whether two strands that touch in one number lie apart in another) and the shuffled null, had been in that project's own notes two days before, in its record of 8 October. But a declaration tells the next reader where to look for the colour (row 369). For the Loop Atlas's practices Bear set one rule, no finished solutions from other sessions: no search of past chats, no sister session's letters or run files on anything a practice touched, and whatever was needed came through him. The rule worked because each practice declared what had already been read.

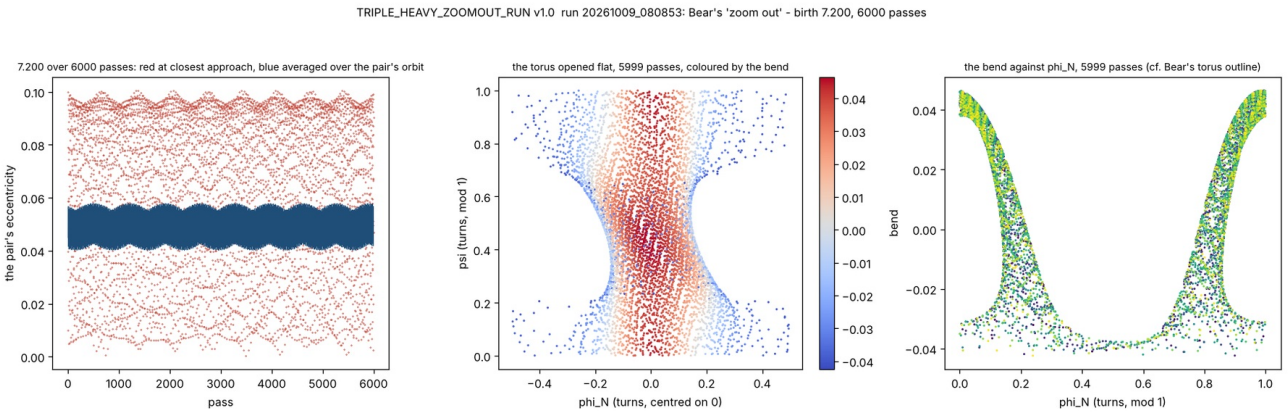


Figure 7. The zoom-out Bear asked for. Birth 7.200 over 6,000 passes: the pair's eccentricity against pass (left), its band scalloped by a slow beat of about 686 passes at the kit's tolerance, whose arithmetic shaped it (at the tight tolerance the beat is 722 passes, row 858); the torus of two angles opened flat (middle), the strands filling a solid hourglass; and the bend against φ_N (right). The same birth over a shorter run is the left panel of Figure 15 (600 passes) and the top left of Figure 8. TRIPLE_HEAVY_ZOOMOUT_RUN v1.0, run 20261009_080853.

The reading: write it down blind, and turn each mark into a test.

3. **Read blind, and write it down by the clock.** Write the reading before you see anyone else's, with three bets, each with a test. Then compare. What both readers saw counts most; what only one saw is turned into a test. Writing the bets first shows exactly which assumption failed when they miss (row 369). A reading made after seeing the data is claimed only after a blind test: chaos on the borders between treads, read at 84 per cent on a map already seen, held at 97 per cent on a fresh grid read blind, while a staircase's turns marked by a change of word failed blind (rows 369 and 655).

4. **Translate each mark into a test with the person who made it, before the run.** The worst miss in the triple-star sessions was a translation: Bear's gold lanes of empty space became, in the test's wording, rows of dots, and the test measured the wrong thing (row 715). Their best result came the other way, from his own words built literally: his change of change of slope found his crossings without seeing his marks (row 573). Better still, let the person draw in the instrument's own numbers. The Explorer's pen keeps each line Bear draws in the chart's own units, numbered, with his word on it, so a drawn line becomes a registered prediction before it is checked (row 573).

   Some marks come with a standing translation. A circle drawn round a group is a hypothesis about a factor: test whether the group's values are a product, one profile times another. Arrows drawn between two maps are a claim about the direction the data moved: test the direction of change. A rhyme named between two pictures, or across rooms, asks for a shared equation before anyone calls it a coincidence, and for the number that tells a shared shape from a shared cause; the cases are in the field key and in section 6. A mark on a proof that asks why something is not being tested is a run request: one such mark in R2W became its fold-direction series. And when the person names a shape, the second reader says plainly whether it saw the shape too, names the shape, and gives that separating number.

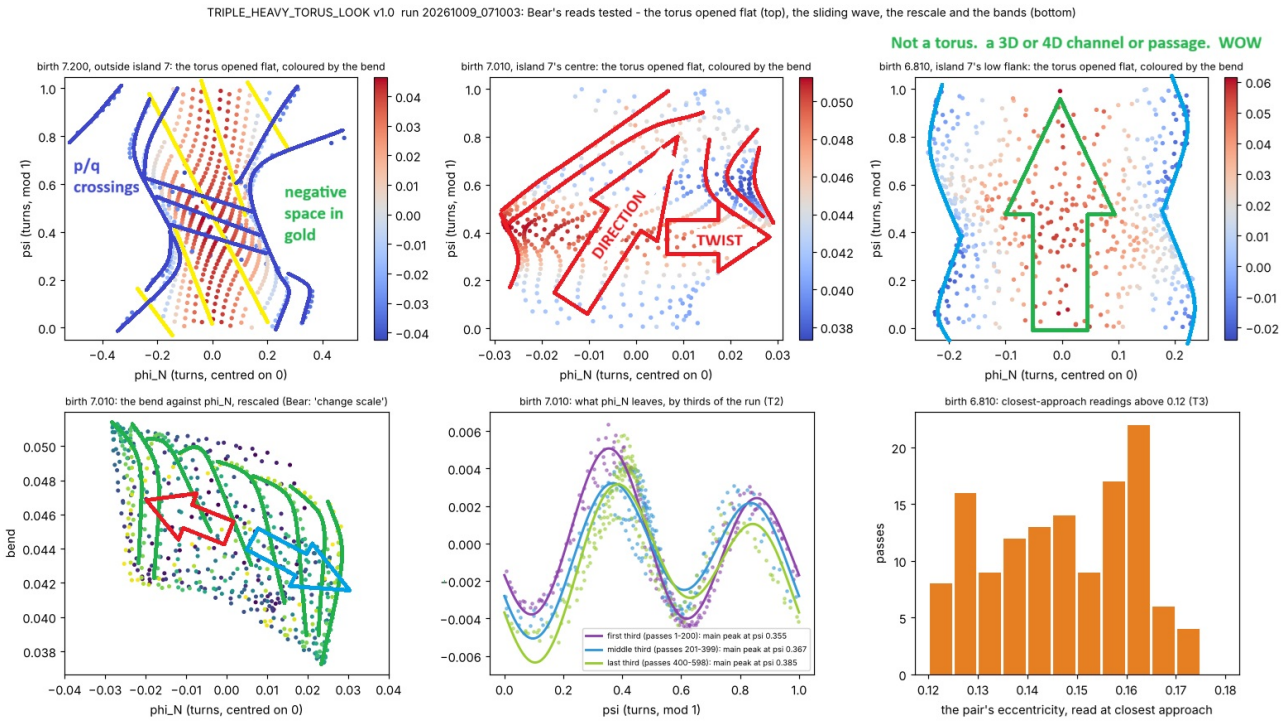


Figure 8. Bear's reads on the torus opened flat. Top, the torus of two angles opened flat at births 7.200, 7.010 and 6.810, with his marks: p/q crossings in blue and lanes of empty space in gold at 7.200, a direction and a twist at 7.010, and at 6.810 a passage with breathing walls, which he called not a torus but a channel. Below, the sliding wave, the rescale he asked for and the bands. The gold lanes were never measured: the lane sensors were too short-sighted. TRIPLE_HEAVY_TORUS_LOOK v1.0, run 20261009_071003.

Before the run, search the record and register with the control built in.

5. **Search the record before registering.** One registration took a birth from a coarse dial's mark, when a run a day earlier had already placed it on a dial five times finer, and half of one test repeated work already passed. Another named the top of a band from a ledger row that the record had corrected two rows later, and missed for that reason alone (row 666).

6. **Register, with the control inside the tool.** Write every prediction down before the tool that judges it exists, with the bar it must clear and the rival reading beside it, and build the featureless control into the same tool before its first run. Three passes in two days were empty because the control came afterwards or never, and the third came after the row that warns of it had been read in full (rows 616 and 620). A rule that can live in the tool should live there, because a reader's memory did not hold this one.

The ruler: every claim gets a number, and the ruler is checked too.

7. **Print what the ruler's step allows.** Under every judged number, print the range the instrument's step permits and say whether it crosses the bar. A ratio that read 7.33 against a registered band of 3 to 7 was written down as failed, while the dial's step allowed anything from 5.25 to 11.5. The failure stands as registered, and it is not decisive. Look for the ruler in every miss (row 686).
8. **Give every claim a number.** Take it from the data. When the data is held back, take it from the picture: calibrate on the axis ticks, check on a mark whose value is known, and say that the number is the picture's (row 714).
9. **Shuffle, with the statistic chosen first.** Ask what the same dots, shuffled, give for the same statistic. A weave test passed as registered, with 72 per cent of its links on two gaps; but nearest neighbours on a page are always close in pass, so the bar was too easy. Against 200 shuffles the real links preferred a gap of 5 passes at +12.4 standard deviations, and only the shuffle showed what was real. Choose the statistic before the shuffle: on the needle's ordered bands, strings of 20 or more pads separated the picture from 100 shuffles by +895 spreads (a spread is the shuffles' own standard deviation), while the count of strings of 4 or more pointed the wrong way, 209.8 in the shuffles against the picture's 17. And a null must keep everything it is not testing: a shuffle of rows that merged 11 per cent of a cloud's pads put the picture +9.4 spreads above chance, and a shuffle of the columns' order, which merged nothing, left +1.8 (rows 616 and 30).
10. **Run the ruler over a made-up case.** Before reading meaning into a shape, give the ruler an input whose answer you know and look at what it draws. The twin lobes Bear read as vortices were drawn the same way by a made-up perfect corner, matching every lane's pair to 0.993 in size at the top doubling and better than 0.9999 at the next two (row 711). R2W's nearest relative is the simplest separate run, a single rider with no ripple, whose answer it could work out by hand: it placed the whole-number grooves (the conveyor within 0.003 in clock ratio), and it showed that a prediction about a memory band had held for the wrong reason (row 683).
11. **Read at a second scale.** Strings found on a lattice of 100 rows were checked at 50; pictures of 600 passes were checked against 6,000. Keep a relation only if it survives the change, and zoom out before naming a shape (rows 703 and 640).

The score: score the sign before the size, and print what could not be seen.

12. **Score, then print what you could not see.** Score an eye's claim on its sign and its place before its size (row 712). On a tie, print "I can't see it", not a peak (row 128). When a guess misses, say aloud whether it can be redesigned, and run the redesign. When a real effect misses its bar, print both facts (row 313). When a bar could never have said no, call the pass empty (row 337). And read every built file back against its source, because a ledger's PDF once lost about 1,600 words to a clipped table without a word of warning (row 553).

When the person beside you reads the picture differently, write both readings down before either of you looks at the other's.

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4 — The toolbox

The three projects wrote different programs for the same jobs. The table puts them side by side, job by job, and says how each one can fool you.

Each project built its own tools, and the jobs came out the same. The table sets them side by side by job, with what each assumes and how it can fool you; most of the warnings in its last column were learned from a miss. Its rows run in four groups, in this order: following the orbit itself, and its birth; Bear's lattice and the other reads of holes and neighbours; measuring a picture, reading a person's marks on it, and redrawing it; and checking the ruler and the reader. The programs are the projects' own and are not yet public.

Job	The tools, by project	What it assumes, and how it can fool you
Follow the orbit itself	The Loop Atlas: STROBE_NEWTON and CASCADE, which find an orbit by Newton's method and read its multiplier. R2W: the Explorer's records. The triple-star sessions: their cascade finder.	Near a fold or a doubling nothing settles in time to be read; follow the orbit and read the sign of its multiplier (row 642).
A birth from both sides	The Loop Atlas: history going down and coming up (LATTICE_READER_2,	A coarse dial makes a birth look gentle: a slope of 26.8 at birth against 160 on a fine

	HISTORY_SETTLE). R2W: CONVEYOR_BIRTH_1 and its reader.	dial. Fit the approach and the departure separately (row 654).
Bear’s lattice in code	The Loop Atlas: LATTICE_READER_2. R2W: LATTICE_NETWORK, and the Explorer’s strings mode, whose strings carry a ternary slope sign. The triple-star sessions: triple_heavy_lattice_look, with pads, counters, rings 1 to 5, families, strings, change of change, empty-pad sensors, a network, a surface and two controls.	The size of the holes decides what a string is. A hole joins one string and the first claim wins. Pads sized by the dots’ spacing see only clusters. A tie can print as a peak. Sizes without signs miss half of the reading (rows 703, 640, 128 and 537).
History at a hole	The Loop Atlas: each rider’s own period (LIFT_3D). R2W: LATTICE_NETWORK’s second step. The triple-star sessions: the pass number as each dot’s history.	A rounded count has a scale of its own and can be wrong by one; count the truth by history (row 709).
Lift into a third number	The Loop Atlas: LIFT_3D, which adds the rider’s speed at each strobe to its angle.	A touch that the third number does not separate is a real meeting, resolved by scale or by history, not by another dimension (row 710).
Find the event in a string	The Loop Atlas: LOBE_TWIN and ZERO_CROSS. R2W: LATTICE_NETWORK’s boundary layer. The triple-star sessions: the lattice look’s change of change.	A centred stencil answers a sudden bend with two lobes of opposite sign; read the crossing between them, and give it room (row 708).
Smoothness without a chosen basis	The triple-star sessions: the torus look, a dot’s five nearest neighbours set against the spread of the quantity.	Noise gives about 1; a wide swing spreads the neighbours. A low-order fit that misses is not a missing variable (row 718).
Nearest returns of a rate	The triple-star sessions: triple_heavy_ladder_check.	One angle’s ladder is not a pair’s (row 719).
Measure the picture	R2W: PICTURE_PIXEL_TOOLS (axis calibration, line envelopes, lopsidedness by parts of a run, crest counting, colour-scale matching) and the weave pixels. The Loop Atlas: a colour digitizer, kept out of its deposit files with the borrowed figure it read. The triple-star sessions: triple_rf_reads_check, which judges another project’s pixel reads on the data.	The plotter draws the last points on top, so colour counts favour late passes; dots merge where crowded, so pixel and blob counts bracket the truth (row 714).
Read a person’s marks	R2W: BEAR_STROKES_MATCH, which finds the pixels Bear added to a screenshot, turns them into the chart’s numbers and scores them against the strings; and the Explorer’s pen. The triple-star sessions: triple_heavy_blind_figure, a plain figure to mark, and triple_heavy_blind_score, which reads the marks back pixel for pixel through the redrawn axes and scores them against random sets.	It needs the clean twin of the marked picture. A figure drawn at a birth on an exact fraction shows a frozen frame (row 648).
Squeeze, tilt and turn a picture	The triple-star sessions: triple_strobe_views, which draws a strobe picture squeezed, tilted as a perspective transform (the tilted picture) and turned, so that Bear’s moves are reproducible. R2W: the squeezed picture counted by its own pixels (row 743).	Squeezing one axis presses columns of dots into one shape whose envelope the eye reads, its symmetry, its waist, its cusps: it is the aspect ratio, which graph-makers have long chosen so that slopes read well (Cleveland’s banking to 45 degrees, from Cleveland, McGill and McGill, 1988; Heer and Agrawala, 2006, bank each of a picture’s scales on its own), used on purpose to bring slow change along one axis into view. Tilting sights along the picture as a carpenter sights down a board: slow trends along the line of sight are exaggerated and detail across it is suppressed. Turned a quarter turn, an up-

		and-down mirror becomes a left-and-right one, which the eye seems to judge at a glance.
The phase-offset view	The triple-star sessions: triple_back_side_look, a delay plot.	A picture of one angle read once a pass shows each rocking edge-on, its two halves on one line; plotting the angle at one pass against the angle a quarter of a rocking later turns the line face-on into a ring, the “back side” Bear felt. The rooms, the parts a long run splits the island into, became hollow rings, the room that keeps its stars a filled disc, and the early centre a needle whose own swing is too slow for that lag (run 20261014_075830). Choosing the lag is choosing the phase offset; the right lag is a quarter of the motion you want to see.
A short view beside a long one	The triple-star sessions: the strobe picture’s first 600 passes beside passes 601 to 6,000, on the same axes.	A moving waist is invisible in either alone.
The shuffled null	The Loop Atlas: SHUFFLE_NULL and SHUFFLE_NULL_2. R2W: the weave pixels and their shuffle. The triple-star sessions: the controls inside the lattice look, and the row-fit control.	A shuffle that changes the counts is a test of its own; choose the statistic before the shuffle (rows 30 and 616).
The made-up case	The Loop Atlas: a made-up perfect corner (CORNER_PRINT). R2W: its nearest relative, the simplest separate run, a single rider with no ripple (CHALKBOARD_W2_FIGURE). The triple-star sessions: a known answer put inside the run.	When the ruler draws the same shape from a known input, only the shape’s place belongs to the data (row 711).
Is the ruler fine enough?	The Loop Atlas: the error a whole column puts on a ratio read from where events sit. R2W: SCALE_FLAG, the off-scale flag.	Print what the step allows under every verdict, and say whether it crosses the bar (row 686).
Keep a reader honest across deliveries	R2W: the seen list, kept by file name and md5.	Without it, a chunk delivered twice votes twice (row 369).

The Depth Axis Explorer is the one tool where both readers and the ruler work on the same picture at once. It is a single page, built for Bear, holding R2W’s maps of the rider, each rider drawn where it stands when the ripple passes. A reader can switch families and grooves off, fit a line through a dragged box, draw a line by hand and keep it in the chart’s own numbers, play the riders passing by passing, and see a rider’s dance on a ring as a star polygon. Since its seventh and eighth versions it lays Bear’s lattice over the picture: holes that count landings, a histogram on each axis, each hole’s slope to its neighbour, its bends, and strings linked hole to hole with their slope, change and change of change. Its ninth version holds R2W’s complete pad atlas, 28 maps, 2,152 chunks and 227,128 riders. The method it carries is cross-domain already: the lattice crossed from R2W’s maps to the triple-star pictures and to the needle. The page becomes a cross-domain tool on the day it loads another project’s table of setting, angle, rider and passing, and that day has its test waiting: read the needle’s doubling staircase, and it should give the splits Newton’s method fixed to ten digits.

Three things to carry from R2W, and one habit. The off-scale flag, which prints what the ruler’s step allows under every verdict; the seen list, which keeps a reader from counting a delivery twice; and the pen, which keeps a person’s drawn line in the instrument’s own numbers. The habit is to ask “what does the ruler decide?” before “did it hold?”.

The lattice’s next version, proposed and not yet built. The triple-star sessions wrote it in the order of Bear’s evening words.

1. Find the frequency of appearance first: the commonest pass gap between a dot and its nearest neighbour (201 at one birth; 47 at another, over 600 passes).
2. Collapse each fibre to its address, and keep the fibre’s own shape. In Bear’s words, “Every three-dot has a curve to it; the arrows lose that detail.”
3. Lay the lattice over the addresses, at the addresses’ spacing, and read slope, change of slope and change of change with their signs, +1, 0 or −1, not only their sizes (row 537). R2W’s strings already carry that sign.
4. Draw the flow: the step from each visit to the next, as a field over the torus.

5. Let empty space vote along whole lines, for the lanes.
6. Put every registered test's featureless control into the same tool, before its first run.

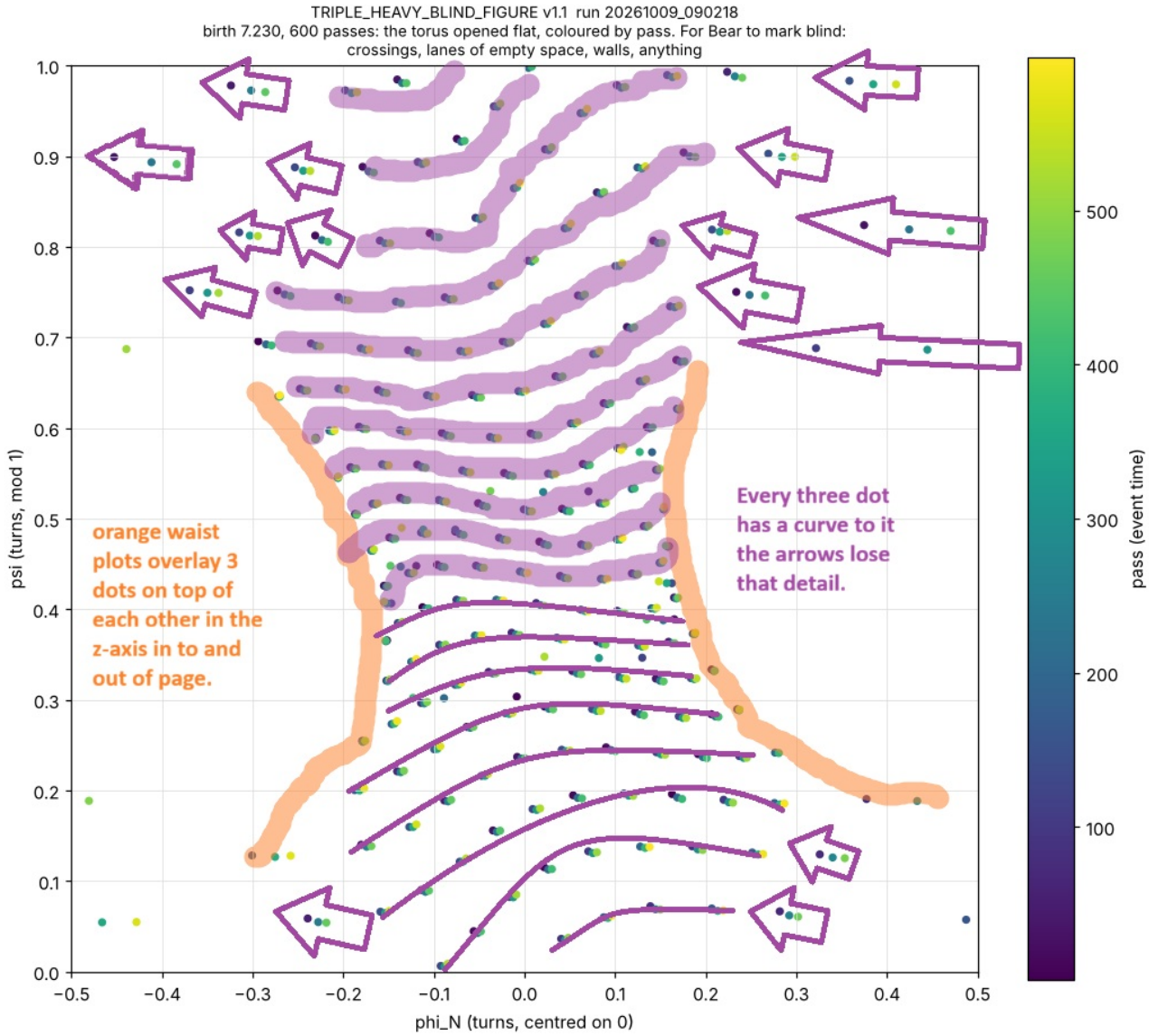
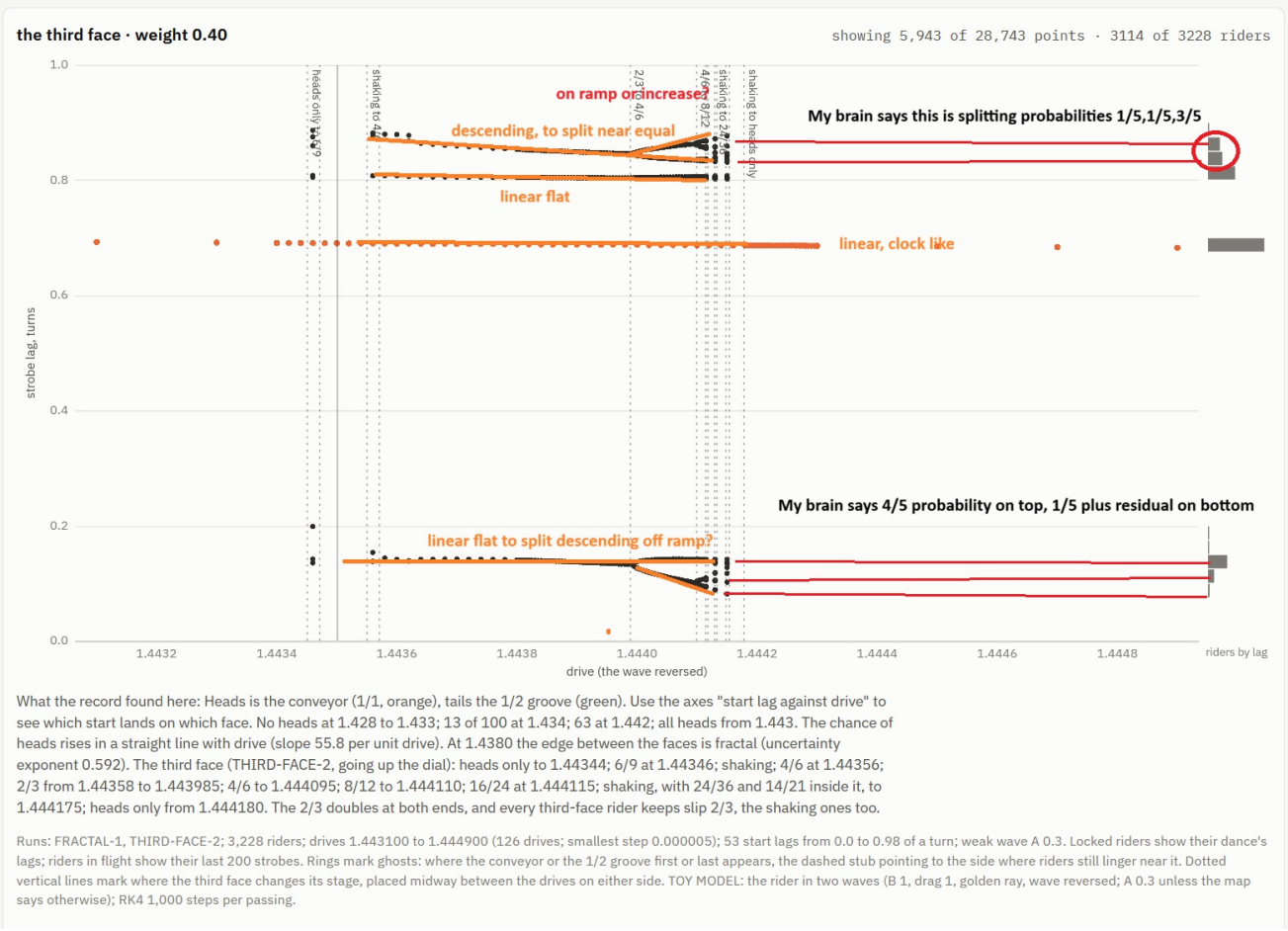


Figure 9. A picture drawn to be marked blind. Birth 7.230, 600 passes, the torus opened flat and coloured by pass. Bear drew rows through the clusters of three, arrows along the first angle and an orange waist, and wrote that every three-dot has a curve to it, a detail the arrows lose. The lattice later found 575 of the 599 dots to be one place visited at passes k, k + 201 and k + 402. TRIPLE_HEAVY_BLIND_FIGURE v1.1, run 20261009_090218.

Open, from the Loop Atlas. Three pieces are still open. The first is event time, the fourth number: read each rider's passings in order and watch a string being visited. The second is Practice 1's own test: mirror and rescale the third face's right-hand tree onto the left-hand tree of the weight-0.50 picture's eye, the place where one road becomes six (the field key), and count the top cluster's shares against 1/4, 1/4 and 1/2 with both histograms (the shares themselves, 0.22, 0.26 and 0.50, were measured on R2W's map and are in the field key; the mirroring and the count with both histograms are what stay open). The third is a lattice of triangular or hexagonal holes, in place of square holes with eight neighbours; it is not built, and the network joins the pads by a triangle mesh instead.



What the record found here: Heads is the conveyor (1/1, orange), tails the 1/2 groove (green). Use the axes "start lag against drive" to see which start lands on which face. No heads at 1.428 to 1.433; 13 of 100 at 1.434; 63 at 1.442; all heads from 1.443. The chance of heads rises in a straight line with drive (slope 55.8 per unit drive). At 1.4380 the edge between the faces is fractal (uncertainty exponent 0.592). The third face (THIRD-FACE-2, going up the dial): heads only to 1.44344; 6/9 at 1.44346; shaking; 4/6 at 1.44356; 2/3 from 1.44358 to 1.443985; 4/6 to 1.444095; 8/12 to 1.444110; 16/24 at 1.444115; shaking, with 24/36 and 14/21 inside it, to 1.444175; heads only from 1.444180. The 2/3 doubles at both ends, and every third-face rider keeps slip 2/3, the shaking ones too.

Runs: FRACTAL-1, THIRD-FACE-2; 3,228 riders; drives 1.443100 to 1.444900 (126 drives; smallest step 0.000005); 53 start lags from 0.0 to 0.98 of a turn; weak wave A 0.3. Locked riders show their dance's lags; riders in flight show their last 200 strobes. Rings mark ghosts: where the conveyor or the 1/2 groove first or last appears, the dashed stub pointing to the side where riders still linger near it. Dotted vertical lines mark where the third face changes its stage, placed midway between the drives on either side. TOY MODEL: the rider in two waves (B 1, drag 1, golden ray, wave reversed; A 0.3 unless the map says otherwise); RK4 1,000 steps per passing.

Figure 10. The third face, read by eye. R2W's rider at weight 0.40, where one road becomes three, with Bear's strokes on the roads and his reading of the split's shares from the histogram, 1/5, 1/5 and 3/5. The tree gives 1/4, 1/4 and 1/2, and R2W's map measured 0.22, 0.26 and 0.50 (the field key); the mirroring onto the weight 0.50 picture's eye stays open. TOY MODEL; runs FRACTAL-1 and THIRD-FACE-2.

Kin, found after. The Explorer was reached without a prior-art search, by Bear's standing order that the run comes first and the prior art after (row 506). The search after found interactive pages for Arnold tongues and for the circle map's devil's staircase. It found none, in one pass, that combines a pen kept in the chart's own numbers, a lattice of strings, and a record of blind reads. That is a statement about one search, not a proof of novelty.

The method's parts have kin of their own. Finding lines and curves in a picture is an old problem in image analysis, with the Hough transform as its classic tool (Duda and Hart, 1972, after Hough's patent of 1962). Keeping the analyst's eye off the answer is blind analysis, which experimental particle physics has practised for decades: "A blind analysis avoids the possibility of experimenters biasing their result toward their own preconceptions by preventing them from knowing the answer until the analysis is complete" (Klein and Roodman, 2005). And the strobe picture itself, one look per tick, is the Poincaré section of the textbooks. What this part adds is the practice of running them together, with a person's eye as one of the instruments.

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5 — A field key of shapes

This is a lookup table of shapes. Find the shape you think you see, and the table says what that shape turned out to be when a reader met it before, in which project, and how far the answer was checked.

How to read the key. Each row gives what the readers saw, what the shape turned out to be, and where it was seen. The status column says how far each was taken: measured on the data; measured from a picture's pixels; a mechanism named but not measured; seen; or a test still waiting. Some shapes belong to the drawing or to the ruler rather than to the data, and the next section gives those as traps. The rows are grouped by the project whose picture showed the shape, in the order the projects were introduced: the Loop Atlas's needle and its other toys, then R2W's rider, then the triple star. Inside each group the rows keep their order.

The Loop Atlas’s needle and its other toys. Shapes from the needle were judged against answers known in advance. The group ends with the Loop Atlas’s tapping rider, its heat with a memory and its tied bells.

What the eye saw	What it turned out to be	Where	Status
Eight strings that read as four	Four tight pairs of an eight-step dance, at holes too coarse to part them	The Loop Atlas’s needle	Measured against the known answer
Twin lobes: vortices, propellers, mirror-math	The ruler’s echo of a corner; only its place, the doubling, belongs to the data	The Loop Atlas’s needle	Measured against a made-up corner
Two strands touching, read as an overpass	A fork’s young daughters, born at the doubling 2 to 7 columns above	The Loop Atlas’s needle	Measured
Each hole holding 128, 64, 32 or 16 landings	The period of the dance, read without following a single rider	The Loop Atlas’s needle	Measured against the known answer
A cloud	Holes that hold one landing each; thickness alone is not disorder	The Loop Atlas’s needle	Measured
A dip ringed in a spread panel, marked with a question about gaps	A window of 10 drive cycles inside a half-turn chaos	The Loop Atlas’s needle, memory 1.7598	Measured
Flickers in a quiet stretch below a chaos	The rungs of a ladder of doubled cycles, 32 down to 2, where needles take long to agree	The Loop Atlas’s needle	Measured
A clean lock (Claude’s reading, from a spread over 200 cycles)	The same ladder, invisible to a ruler that sees only cycles dividing 200	The Loop Atlas’s needle	Measured; Claude’s miss (the Loop Atlas’s misses row 219)
Rungs, Bear’s word for the stretch	A doubling ladder, evenly spaced on a log ruler to within a quarter; ratios 3.43, 4.24 and 4.57 toward 4.669	The Loop Atlas’s needle	Measured

What the eye saw	What it turned out to be	Where	Status
Two songs on two boards	One song: 96 per cent of the gaps exactly the rider’s rest, split by rounding	The Loop Atlas’s tapping rider	Measured, in whole clock steps
A warm brick in ice that Bear heard as quenching	Two cold fronts meeting in the middle knock the brick down at once	The Loop Atlas’s heat with a memory	Measured
Event timing ranked above the water’s memory	The squeeze period sets the sign of the water’s gain; the memory deepens it	The Loop Atlas’s tied bells	Measured
Rings round the bells’ map’s fast columns and its slow columns, with an arrow up the memory axis and a question whether the two are orthogonal	A product of a timing profile and a memory scale: read that way, the thin-cored map’s departure from even is 98.6 per cent that product, and the mended map’s 87.9 per cent, less after the mending	The Loop Atlas’s tied bells, marked by Bear	Arithmetic on saved results, after the marks; not a registered test, and left open at the Loop Atlas’s close
A question whether the map is balanced	A ratio: the cost side runs about twice as deep as the pay side, 0.61 below even against 0.30 above on the mended map, and 0.65 against 0.37 on the thin one	The same map, marked by Bear	Measured
Arrows pressing inward on the thin map and outward on the mended one	The direction the data moved: the mending opened a neutral band in the middle, with gains within 0.03 of even across periods 2.12 to 3 at short memories	The same two maps, marked by Bear	Measured, after the arrows

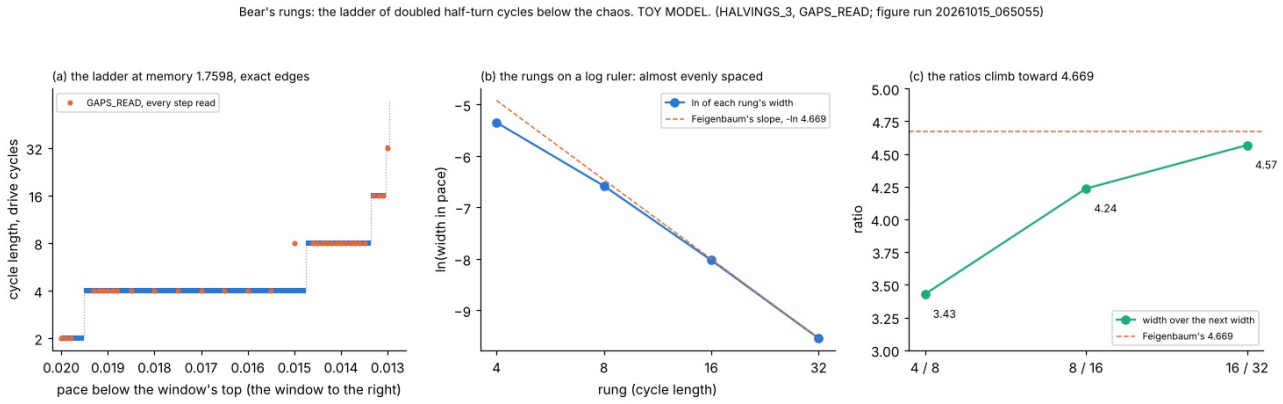


Figure 11. Rungs, Bear’s word for a quiet stretch below a chaos of the Loop Atlas’s needle: (a) the ladder of doubled cycles at memory 1.7598, each rung between its exact edges; (b) the rungs on a log ruler, almost evenly spaced; (c) each rung’s width over the next, 3.43, 4.24 and 4.57, climbing toward 4.669. TOY MODEL; figure run 20261015_065055, from HALVINGS_3 and GAPS_READ.

R2W’s rider. Shapes from its pictures of the rider, at the weights named in the third column.

What the eye saw	What it turned out to be	Where	Status
A “lurch-stroll-lurch” S on a groove’s lower stop	Its slope runs 190 at birth, 2.8 in the middle and 53 at the end; its slowest point, at lag 0.145, is a foothold where level strings cross three other zones	R2W, weight 0.26	Measured
A balanced word of level runs (8, 9, 9)	The digital trace of a straight line	R2W, weight 0.50	Measured
Curves through a staircase converging on a ring	The ghost of a saddle-node: the stops of the dances pile into the birth point, and 1 − slip goes as the distance to the birth to the power 0.507; read from below and from above, the birth agrees to about a millionth	R2W, weight 0.26	Measured
A gentle birth	A square-root birth hidden by a coarse dial: a slope of 160 at birth on a fine dial, against 26.8	R2W	Measured
A “hollow beam”, dots crowding at a column’s edges	A ring seen edge on (the arcsine law)	R2W	Mechanism, not measured
A memory band with a ceiling at 0.31	Hysteresis: below pull 1.0 a rider at rest stays and a runner keeps running; the ceiling is the chart’s own window	R2W’s pad atlas, with the band’s floor read at weight 2	Measured
An orange chain above pull 1.0	Not the rest but the conveyor, which colouring by denominator paints the same orange; the whole-number grooves are Shapiro-like steps	R2W, weight 2	Measured, after the data
One road becoming three at a split, and six at an eye	A probability tree: the parent carries on as a ghost, and each split halves a branch’s share	R2W’s weights 0.40 and 0.50, read by Bear	A reading, made after the measured shares had been seen: 1/4, 1/4 and 1/2 from the tree, 0.22, 0.26 and 0.50 on the map; Bear’s eye read 1/5, 1/5 and 3/5
One flat string, showing through three windows	A staircase of flat strings, at lags 0.144, 0.707, 0.81 and about 0.92	R2W, weight 0.26	Found by R2W’s decoder; the cold reading had seen one
A shape seen twice	A groove’s lower foot found again about 0.010 of drive to the right, its shelf squeezed to about 40 per cent of its length	R2W, weight 0.26, read by Bear	Seen
A string drawn through empty space	Possibly a frequency illusion, a line the sampling makes	R2W, weight 0.26	Test waiting: split each rider’s record in halves, or strobe at another phase
A fan of curves, one per strength, pinching near one place: a waist	A balance of two terms. The drop over the lift crosses the golden value near weight 1.43 for every strength from 0.05 to 0.30; R2W proposed that this is where each direction’s next-order term is the same fraction of its first, a test it wrote down and did not run. The waist slides slowly as the strength grows (1.456 to 1.397), which marks a balance of two terms rather than a symmetry, and it is not at φ, where the ball’s inertia and drag answer the ripple equally	R2W’s fold-direction series (FOLD-DIR-b)	The waist measured (FD-5, FD-6); the balance of next-order terms proposed, its test not run

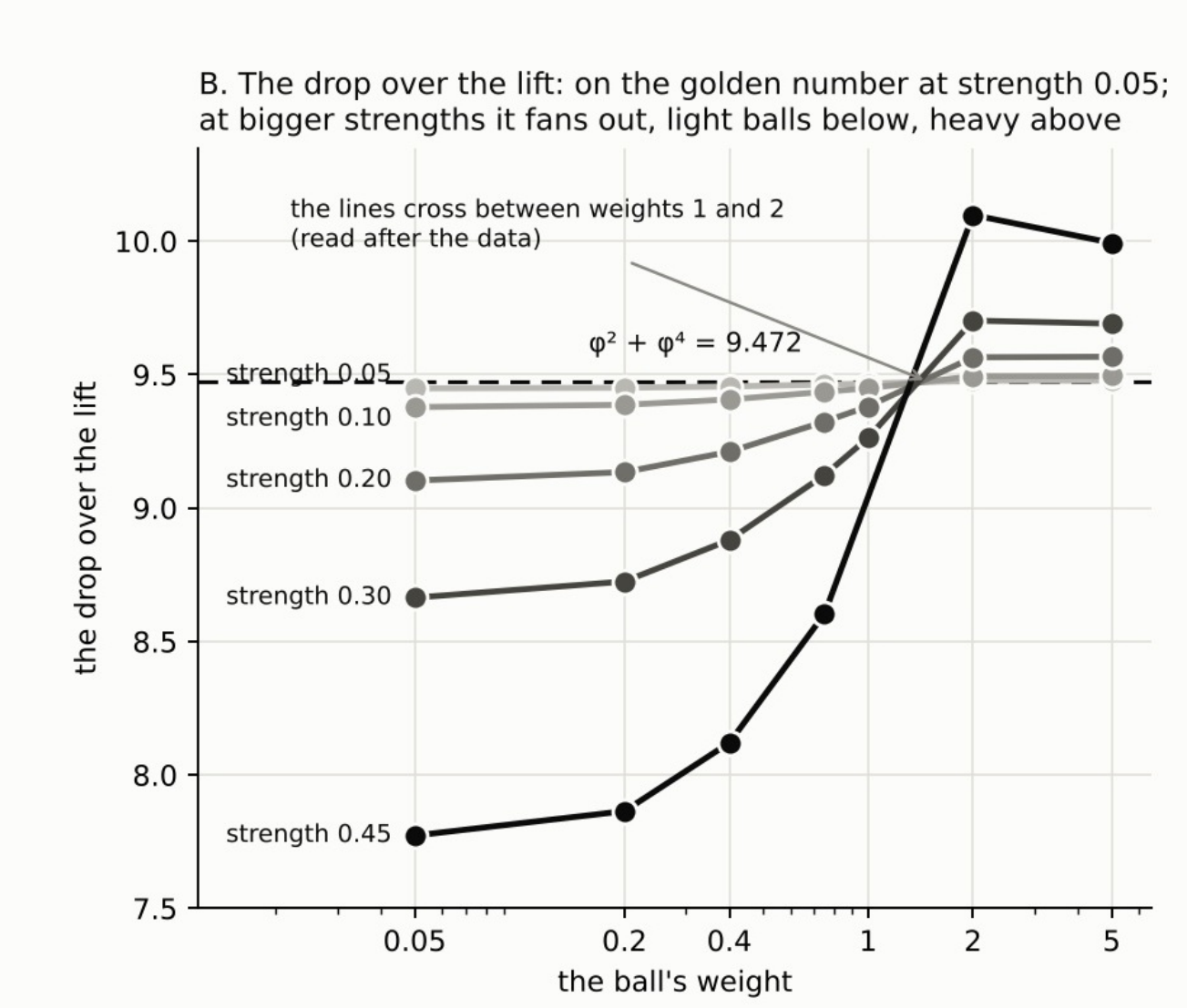


Figure 12. The fan and its waist. Figure 24 of *A Rider in Two Waves*, right panel: the drop over the lift, how far the ripple turned around moves the plain's edge against how far the ripple against the ball moves it, plotted against the ball's weight, one curve per strength from 0.05 to 0.45. At the smallest strength it sits on the golden value $\varphi^2 + \varphi^4 = 9.472$ at every weight; at larger strengths it fans out, light balls below and heavy above, and the lines cross between weights 1 and 2, the waist near 1.43 (read after the data). TOY MODEL; R2W's fold-direction series (FOLD-DIR).

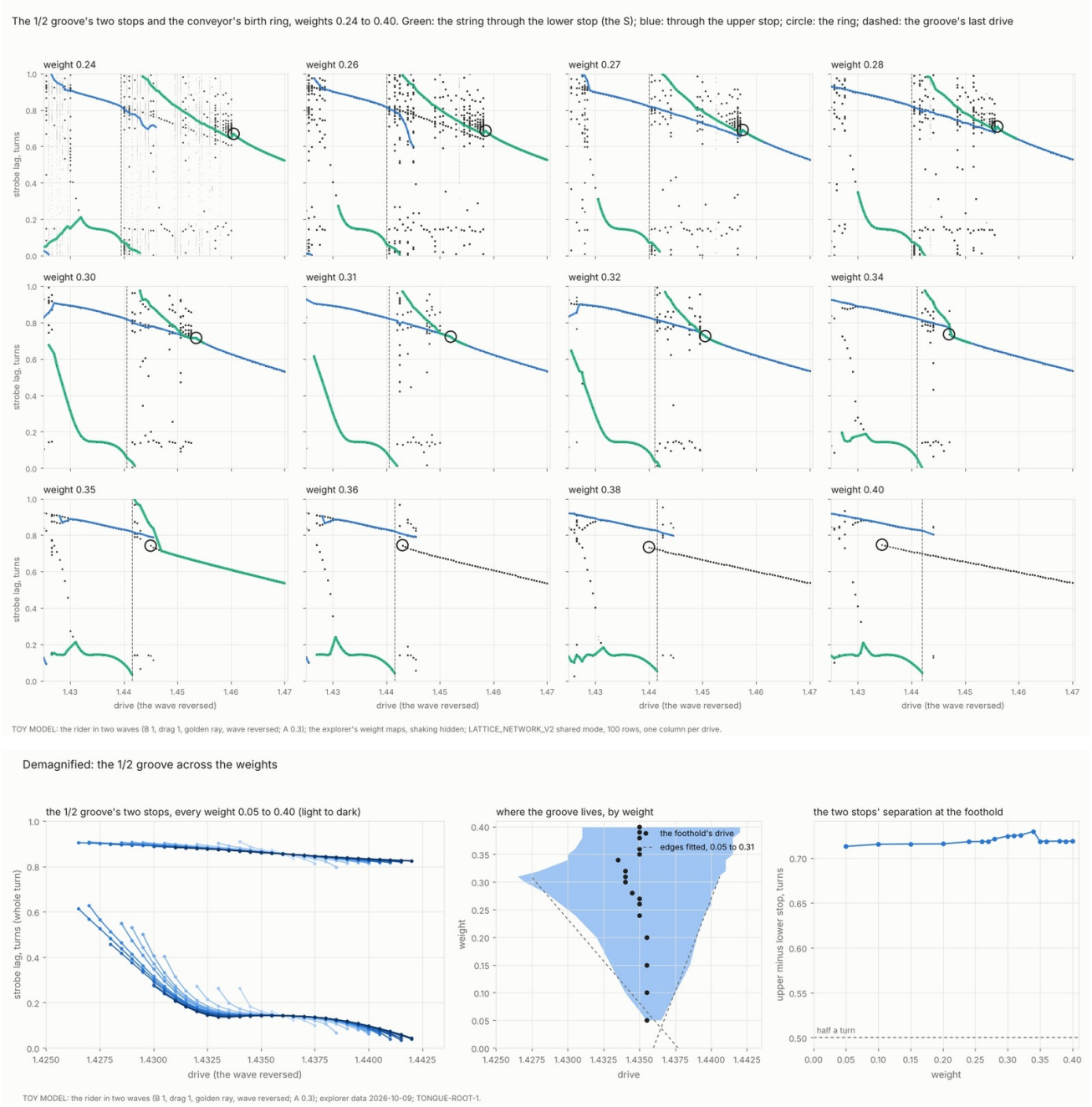


Figure 13. The S, the ring and the foothold. Above, Figure 27 of *A Rider in Two Waves*: the 1/2 groove's two stops and the conveyor's birth ring at weights 0.24 to 0.40, green for the string through the lower stop (the S), blue for the string through the upper stop and a circle at the ring, where the strings converge. Below, Figure 28 of *A Rider in Two Waves*: the groove's two stops at every weight from 0.05 to 0.40, the lower stop flat near lag 0.14 at every weight (left), where the groove lives by weight (middle), and the two stops' separation at the foothold (right). TOY MODEL; LATTICE_NETWORK_V2 and TONGUE-ROOT-1.

The triple star. Shapes from the triple-star pictures: the bend figure, the torus opened flat, the lattice, the blind figure and the strobe pictures.

What the eye saw	What it turned out to be	Where	Status
Woven strands, "thousands of strings"	A rotation seen through a strobe: the dots' rhythm of 0.21257 cycles per pass returns near itself after 5 and 14 passes	The triple-star bend figure: seen by Bear; the mechanism named and measured from the pixels by R2W	Measured, from the pixels and on the data
White patches between the threads	The gaps of a rotation, which come in at most three sizes, in	The triple-star pictures	A test of R2W's, registered and never run on the triple-star

	phase		data
A torus outline; rungs across a V	The bend is a smooth map on the torus of two angles; the hooks pair across the V by height	The triple-star pictures	Measured (the neighbour ratio 0.1126, then 0.0603); the hooks partly
Little curves with little tails at the edges	Folds of a surface seen flat, packed where the swing turns (Whitney’s folds and cusps)	The triple-star opened torus	Seen; R2W’s test registered and never run on the triple-star data
A “jellyfish”; two out-of-phase hills	A landscape on a torus must have saddles (Morse’s count)	The triple-star pictures	A test of R2W’s, registered and never run on the triple-star data
A diamond lattice of crossing lines	The envelope of a standing wave, crossing at its nodes	The triple-star pictures	Seen
A double helix, pulsing	Read two ways: one oscillation seen half a turn apart, a beat (R2W); a bend that depends on two angles, which took its fit from 0.4302 to 0.9723 (the triple-star sessions)	The triple-star bend figure	Seen; measured
A black bar; short spikes up becoming long	The band’s lopsidedness flips sign while its centre holds	The triple-star pictures	Measured: right in sign, overstated in size by the pixels
An orange core line	The readings at the far point of the orbit rise about 0.007 to 0.010 mid-run and come back	The triple-star pictures	Measured from the pixels
Bands on the flank	Two heaps in the readings above 0.12	The triple-star pictures	Weak, open
A passage with breathing walls; “not a torus: a channel”	A corridor whose walls pinch near 0.45 in the second angle; at 6,000 passes the strands fill a solid hourglass with sharp walls, either a section of a torus with more angles or a chaotic layer	The triple-star pictures	The corridor measured; the rest a test waiting
Dense top, lean bottom	The rider lingers where its swing is middle, where the bend is flattest (212 coloured pixels per column against 51); the top is 3.7 to 6 times denser than the bottom	The triple-star pictures, 6,000 passes	Measured from the pixels
Scallops on a band	A slow beat of about 686 passes (666.7 was a bin of the spectrum, row 318, not a reading)	The triple-star zoom-out	Measured
p/q crossings	Three lines across the waist where the strings’ change of change of slope is largest	The triple-star lattice	Seen after the run: a description
Gold lanes of empty space	Not yet measured; the lane sensors were too short-sighted	The triple-star lattice	Open
Clusters of three	The same place visited every 201 passes (575 of 599 dots)	The triple-star blind figure	Measured
Rows that arch below the waist and lift above it	The rows of the clusters’ centres: 6 of 7 arch, 6 of 8 lift	The triple-star row fit	Measured; one row short of its registered bar
A frozen grid of tight clusters	A birth near an exact fraction	The triple-star blind figure	Seen; redrawn a hair away
Bilateral symmetry about a blue axis, on a squeezed strobe picture	The island’s centre at $\varphi_N = 0$, where the runs start: 40 of 40 outlines centred	The triple-star strobe picture, passes 601 to 6,000, read by Bear	Measured, on a bar the wrapped outlines could not fail
A pinch in the middle	The room that keeps its stars, births 7.470 to 7.505, the	The same, read by Bear	Measured

	narrowest outlines of the long view		
A waist on the upper panel	The narrowest swings of the first 600 passes, births 7.525 to 7.535, which did not last	The same picture’s first 600 passes	Measured
Black spacer rungs between pink data bands	The gaps the grid of births draws, one column every 0.005 with its dots spread 0.0018 either side: the drawing’s, not the data’s	Both panels, read by Bear	Measured from the grid
A V: the long runs’ core narrowing in a straight line to a pinch	An outline of the coarse sampling: on a grid five times finer each run’s width follows its own fate	The tilted strobe view, a birth every 0.005	Not borne out at 0.001
One trace turning over, then tracking another like a damped copy of a sharp signal; after the crossings, the two lines steady and parallel	The steady, parallel run held: at 1/8 and 1/9 the slope is the same on both sides of the fraction (−0.239 and −0.240; −0.295 and −0.292); at 1/10 and 1/11 it falls to about a quarter across the fraction (ratios 0.24 and 0.23), and after that the two lines run parallel. The rhyme held in a weaker, measured form: with each trend taken off, the two shapes correlate 0.285 at no shift and 0.521 at the best shift (the tenth moved eight steps), the tenth’s shape 0.75 the size of the eleventh’s, the same kind of mark at about three quarters the size and not the same curve	The triple star’s chart of four crossings, the tenth set against the eleventh, read by Bear (TS1’s ledger, run 20261017_172324)	Measured, reported and not scored; a second reader drew the same crossings within 4.47×10^{-6} and 8.95×10^{-6} (runs 20261017_155035 and 20261017_160624), so the shapes are the orbits’, not the reader’s; the same run also judged Claude’s call of a textbook shape at the tenth, withdrawn when the slope through 1/10 sharpened over a longer record (TS2; section 9 and row 795)

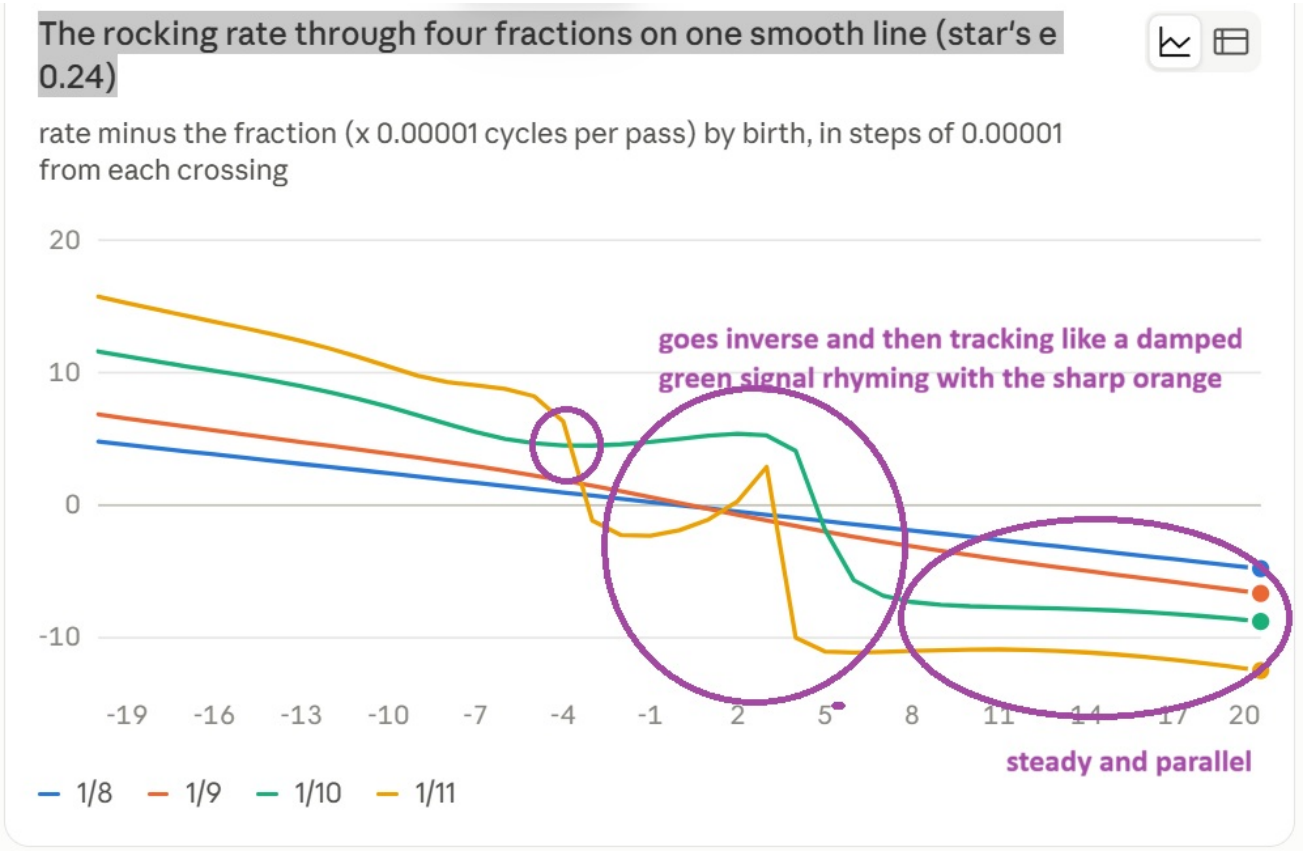


Figure 14. Four crossings, marked. The triple star's rocking rate through 1/8, 1/9, 1/10 and 1/11 at the star's eccentricity 0.24, with Bear's circles and words: one trace goes inverse and then tracks another like a damped copy of a sharp signal, and after the crossings the lines run steady and parallel. The steady, parallel run held; the rhyme held in a weaker, measured form (the field key). TS1's ledger, run 20261017_172324.

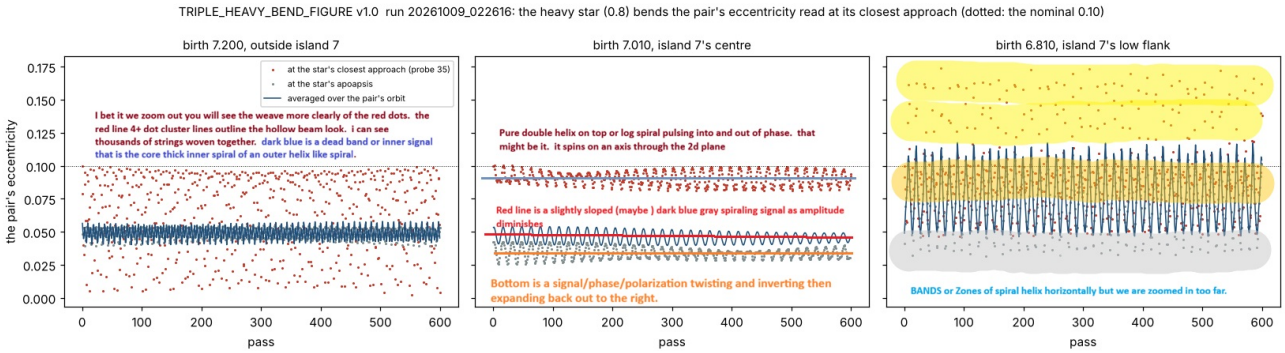


Figure 15. The bend figure, as Bear marked it. The pair's eccentricity read at the star's closest approach over 600 passes, at births 7.200 (outside island 7), 7.010 (its centre) and 6.810 (its low flank). He saw thousands of strings woven together at 7.200, a pure double helix at 7.010 and bands at 6.810, and asked to zoom out. The weave turned out to be a rotation seen through a strobe and the double helix a bend that depends on two angles (the field key); the zoom-out is Figure 7. TRIPLE_HEAVY_BEND_FIGURE v1.0, run 20261009_022616.

Bear's channel, "not a torus", is the top right of Figure 8; the solid hourglass of 6,000 passes is the middle panel of Figure 7.

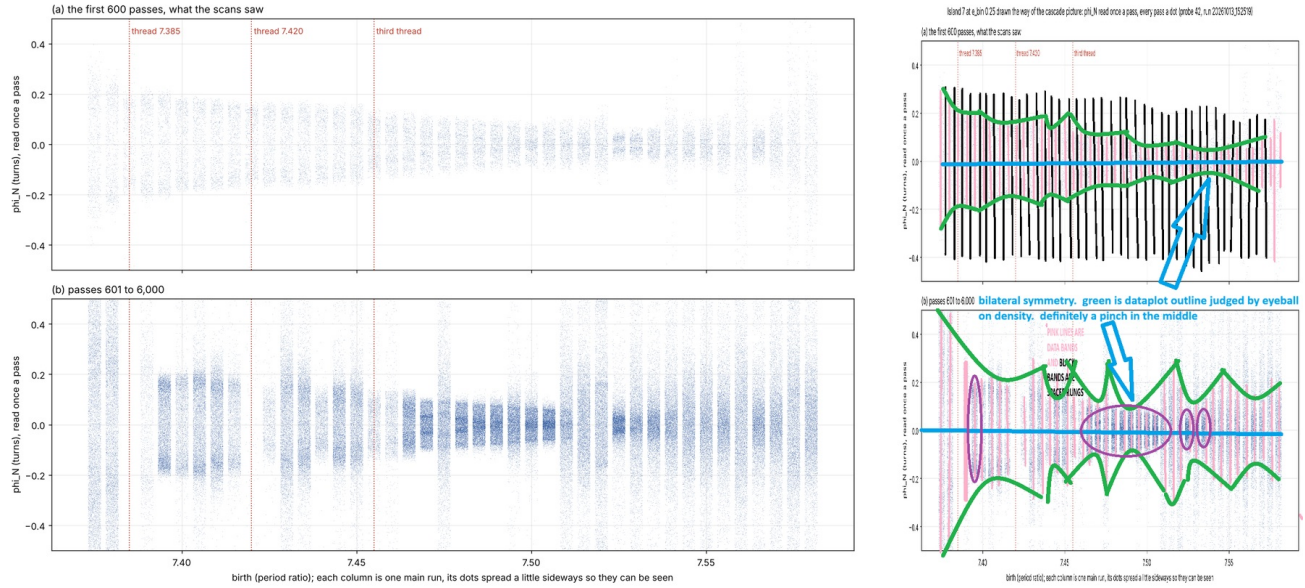


Figure 16. The squeezed strobe. Left, the triple-star paper's Figure 7: island 7 at e_bin 0.25, φ_N read once a pass, over the first 600 passes (a) and passes 601 to 6,000 (b). Right, Bear's working copy of the same picture, squeezed sideways and marked by eye on 14 October before anything was measured: pink data bands, black spacer rungs, a green outline judged by density, a blue axis and a pinch in the middle. triple_paper_figure7_marks v1.0, run 20261021_052614.

6 — Traps, and how each was caught

A picture, a set of bins or a test can make a pattern all by itself, and a reader can bring one along too. This section lists the ones the projects fell into, and how most were caught, so the next reader can walk round them.

Every trap here was walked into at least once, and most were caught by one of the protocol's steps. They come from the drawing, from the bins and grids, from the test itself, and from the readers, and they are grouped in that order, each group opening with its rule.

The first kind comes from the drawing. Its rule: name the picture before naming the pattern.

The ruler's echo. A shape can belong to the measurement. The twin lobes of a third difference were the echo of a corner, drawn the same way by a made-up corner (row 711). Caught by the made-up case.

A line the sampling makes. Riders in flight, strobed once per passing, can line up into a string that is not a dance. In Bear's words: "When we start injecting a 3rd dimension into 2D, you often see what you want to see." Test by splitting each rider's record into halves or strobing at another phase: a dance stays where it is, and an illusion moves or breaks up (row 710).

The plotter's order. A plotter draws the last points on top, so a crowded picture shows mostly the end of the run, and colour counts favour late passes. Where dots merge, pixel counts and blob counts bracket the truth from either side (row 714).

A colour that lumps classes. Colouring by denominator (its figure is in the field key) painted the rest, the conveyor and two grooves the same orange, and a blind read named the wrong one (row 717). Caught by checking the class itself.

Starts laid on top of each other. At one weight, a rider started at lag 0 landed on a groove while one started half a turn round went into a doubling tree, and the picture, which laid both starts together, said nothing of which start made which dot. A coin, a place where two starts end differently, was hidden in it. Give each dot its start (row 664).

A cut through a band. An angle or a lag is a circle, and a picture's edge is a cut. A cut through a band makes one band look like two, and a string that runs off one edge is not thereby the one that enters at the other: a wrap matched by slope alone was withdrawn (row 704). Cut where nothing lands. And a column the dial left empty is not a break in a curve: the Explorer's first lattice read the empty columns between sampled drives as breaks, until it was made to read each slope to the next column that holds any landing (row 704).

A ceiling that belongs to the window. A memory band saturated near 0.31, and the saturation was the chart's own range, not the riders' (row 416). Check whether the chart sets a limit before explaining it.

The second kind comes from the bins and grids. Its rule is row 329's: binning is where the information design happens.

A pattern the grid draws. On a squeezed strobe picture (its figure is in the field key) the pink bands were data, and the black "spacer rungs" between them were the gaps the picture's own grid of births made, one column every 0.005 with its dots spread 0.0018 either side, so they belong to the drawing. It is chaos-and-binning in another coat, a pattern made by the bins, and the rule is the old one: name the picture before naming the pattern. Bear's earlier gaps, the white tops where stars had left, were the data's. Mark the empty space, then ask who drew it.

A shape the coarse picture makes. A V read on a picture with a birth every 0.005 (its figure is in the field key), the long runs' core narrowing in a straight line to a pinch, dissolved on a grid five times finer: at 0.001 the island's fates speckle, and the V and the floor went into the speckle, while the offset between where the island looks calm and where it lasts survived. Before a shape gets a name, look at it on a grid several times finer.

A grid on exact fractions. A birth chosen on an exact fraction drew a frozen frame of tight clusters (row 648). Test a hair either side.

The grid set by a top value. Paces set as shares of each floor's greatest grip mostly landed where the rough floors slip all the time, so the lean could not be judged. Place a grid on each case's own band (row 748).

A hole the size of the dot. Pads sized to the dots' own spacing saw only three-dot clusters and none of the rows they make; at 6,000 passes strings fell to 3 dots (rows 640 and 703). Caught by reading at two scales.

A thick string called a cloud. A crowding rule called a thick string a cloud, and a family's trace ended 3 columns early; the counter, landings per hole, is the cloud test.

A tie printed as a peak. Bins of lane direction gave a three-way tie no taller than evenly scattered dots give. Print "I can't see it" (row 128).

The ruler with holes. A spread over a fixed count of cycles sees only the cycles that divide the count (row 641). Caught by Bear's rings (row 715).

The short window that passes a slow slipper. A lock read in windows of 100 time units passes a rider that slips once every 200 to 400. Caught by the planted band test before the run of record (row 645).

One verdict over a core and a shell. A lock's single verdict can mix two populations. Each long lock's middle, read over 6,000 and 12,000 passes, split into a core that holds on the fraction and a shell that wears away, losing its lock or escaping; the guesses registered for both runs, that the middles would keep closing, were not borne out (LX1′ and LX2). At a fifth, seven births from 7.455 to 7.461 closed on it within 3.5×10^{-9} over 6,000 passes (LX1′, run 20261017_100655), while 7.452 escaped at pass 3,077; taken from that record, the fifth's core held over 12,000 passes, at 3.49×10^{-9} to 4.06×10^{-9} (LX2, run 20261017_200756). At the seventh, twins 10^{-10} apart then showed which members are chaotic: exactly 7.121, 7.122 and 7.123, the births that lost their lock (TW7, run 20261018_192644), while 7.124's twins never parted over 12,000 passes, after TW8 the only core birth read as regular. The reader could see the split only because the tool printed each member's reading beside the group's verdict. Print the members beside the verdict.

The table of bins. The triple-star sessions kept a table of the bins they chose and what each one decided.

Bin or threshold	What it decided	Row
A rest’s tolerance, 2 or 5 per cent of the ratio	At 2 per cent a slow creep was cut into many long rests; at 5 per cent it merged into one	622: two thresholds fixed in advance that disagree are data
A window of 0.1 round a whole number	Perfectly locked rests could reach only 0.44 to 0.53 inside it, so a bar of 0.4 sat under its own ceiling	210 and 416: a bar must sit where the reading can reach it, and a ceiling can belong to the window
A grid of births on exact fractions	One birth drew a frozen frame of tight clusters	648
A grid of births 0.03 apart	It stepped over an island shrunk to a sliver	631
Pads the size of the dots’ spacing	At 6,000 passes strings fell to 3 dots; at another birth the pads were a cluster’s own size and saw only clusters	640 and 703: strings depend on the lattice, curves do not
Bins of lane direction 5 degrees wide	A three-way tie, no taller than randomly scattered dots	128: say “I can’t see it”
Bins of eccentricity 0.005 wide	Bear’s bands passed by a dip of 9 against a bar of 9.6	Thin: the bar should come from a control
A colour scale	The conveyor hid inside one colour with the rest	717
The order the plotter draws in	Later passes cover earlier ones, so colour counts favour the late	714

Three rules travel from the table. **Choose the bin from the structure you are asking about,** not from the data’s spacing alone; at one birth that meant two scales, the return and the row. **Fix two thresholds in advance and read their disagreement as data.** And **run every binned statistic on a featureless record of the same size before believing it**: R2W’s shuffle turned a pixel weave into +12.4 standard deviations, and on the numbers the same linking read +23.4 against the same shuffle.

The third kind comes from the test itself. Its rule: a check needs a check of its own.

The count that points the wrong way. On a clean picture the count of short strings is higher in the shuffles than in the picture; only long reach separates them (row 616). Choose the statistic before the shuffle.

A null that makes its own effect. A shuffle that merged a tenth of a cloud’s pads thinned the null by itself, and the picture’s excess fell from +9.4 spreads to +1.8 once the null kept its counts (rows 616 and 30).

A bar that can hardly say no. A guess that asks only for a sign passes about half the time on noise: a guess about rows passed with shares of positive turns at 0.527 and 0.488, a coin’s half (row 337). And a rule can find its own shape in noise: a rule for S-shaped rows, a reading of Claude’s and not one of Bear’s marks, passed 6 of 7, and its control, built afterwards, called a smooth parabola S 0.718 of the time (row 616).

A rounded count wrong by one. A reference count of distinct points, rounded to a millionth, ran one or two high in six columns, because a value on a rounding boundary rounds two ways, and a prediction judged against it missed for that reason alone (row 709). Caught by counting each rider’s own period instead. The reference needs a check of its own.

A prediction right for the wrong reason. A memory floor was predicted at the pull where a rolling rider is caught again, and it held. A single rider with no ripple showed that a rolling rider keeps rolling down to 0.78 to 0.80, and that the floor, 0.86 to 0.87, is where a rider started half a turn round gets rolling at all. The band had been wide enough to catch the wrong mechanism (row 683).

The finder that reads its own failure. A doubling finder reported four edges, and every one was a place where Newton’s method stopped converging, because the needles’ angles were carried unreduced. Caught by a planted edge with a known answer, not by a check that the edges were found (row 751).

The threshold that reads a draining as a jump. A rule that called the edge where half of sixteen needles had stopped put the leak’s closing between two weights. The count kept beside it showed 9, 7, 2 and then 0 needles falling: a door eased shut, not slammed (row 787).

One edge, three doors. A threshold read from a sweep is a number, and the sweep does not say which way out it marks. In R2W’s toy a ball resting on the plain has at least three ways out: the resting dance stops existing (a fold, its multiplier reaching +1); the dance doubles (a multiplier through −1) and the ball is shaken loose; or the ball sits in a dance whose own fold comes first, the third door, read only after the data (that fold landed at 0.859975 and 0.713061, FX-2c; the reading of two resting dances side by side is withdrawn at weight 2, where the continued dance is unstable,

and unchecked at weight 5). All three appeared in one toy across strengths 0.30 to 0.60 (FOLD-DIR-e, FX-2b), and each gives an edge that looks the same on a sweep of drives. Which door it is decides how the edge moves when a knob turns. Under a gentle ripple a heavier ball's edge moves less. Under a strong turned ripple a heavier ball slips off sooner, weight 5 at drive 0.713 where weight 1 holds to 0.883 (FOLD-DIR-e), because the heavy ball, set down at rest, sits in a dance whose own fold comes first, the third door (FX-2c); R2W first read that edge as the dance doubling, which is how the heavy balls leave at strength 0.45 (FX-2b). A reading of the sweep before it (FOLD-DIR-d), where balls of weight 0.05 to 1.00 all slip off between drives 0.88 and 0.89 at strength 0.60, had the ball resting far down a stiff flank, where a spring shaken slowly answers with its stiffness and not its mass; it predicted that heavy balls would hold on longer, and missed (FX-2). Name the door before trusting the number, and when a knob's effect changes sign between two regimes, look for the mechanism that took over the edge (rows 642, 314, 695 and 739).

The planted print that shows a judged number. A planted case printed whole can show, before the run of record, a value a lean will judge. Say so in the row, and change no judging line (row 716).

A built file that lost its words. A ledger's table ran wider than the page and the builder clipped about 1,600 words without a warning, unseen for days until the PDF was counted back against its source (row 553).

The fourth kind comes from the readers. Its rule: write each reading down before setting it beside another, and translate a mark into a test with the person who made it.

The eye defaults to a fork. The second reader, which had read that Bear draws three roads at every split, the parent's road going straight on, still drew two (row 705). Caught by writing the cold reading down and setting it beside his.

The overpass that was a meeting. Two strands that touched (their figure is in the field key) were registered as crossing at different depths. Lifted into a third number, speed, they were close in every number: a fork's young daughters, and the prediction had carried the very error Bear had warned of shortly before (row 710). Caught by the lift itself, and resolved by smaller holes and by history.

A mark put into the wrong words. Gold lanes of empty space, translated as rows of dots, were tested as the wrong thing, and sensors 5 holes long could not see lanes 0.3 turn long (row 715). Translate a mark into a test together with the person who made it.

A coarse fit read as a missing hand. A two-angle fit of a bend that explained 0.7230 of its variance was read as a third hand at work. A smoothness test that needs no chosen basis found the bend smooth on the torus of the two angles (row 718).

A corridor read by distance. The chaotic stars below the island were told apart cleanly by which way they turned (17 of 18), and not at all by how far from the edge they started. Read a corridor by direction, not by distance.

One birth read as a fate. At fine scale one birth is one sample of the chaos. The patterns that carry are shares (two births in five hold in the room that keeps its stars) and edges (no escape above 7.472).

A rhyme taken at its word. Two figures that look alike may share a cause, or only a shape, and the picture alone does not say which. Bear set R2W's pin figure, every weight's dance crossing at the weightless root at a drive of 1.436, beside the waist of the golden ratio's fan, every strength's ratio crossing the golden value near a weight of 1.43, and saw one picture. It is one shape: a family of curves, one for each setting of a knob, crossing where the knob's first effect vanishes, on the curve the knob-free toy draws, in a narrow region whose width the next effect sets. Chemistry calls it an isosbestic point, correlated-electron physics knows it from specific-heat curves crossing near one temperature (Vollhardt, 1997; Greger, Kollar and Vollhardt, 2013), and an engineer biases a transistor at its zero-temperature-coefficient point for the same reason. It is not one mechanism: the pin's offset grows as the weight to about the 2.5th power (32 times for 4 times the weight), the waist's as the strength squared (4.0 times for twice the strength), and 1.436 and 1.43 belong to different knobs and agree by chance. A rhyme can also be one law. The one he named across the three rooms, a driven, damped pendulum, is the pendulum equation, or Adler's, under three names: the needle lagging a turning field, the rider in a wave, and the angle of an orbital resonance (section 8). Name the shape, use it as a design point, look for the shared equation before calling a rhyme a coincidence, and test any claim of a deeper link by moving a knob that should separate the two (row 146).

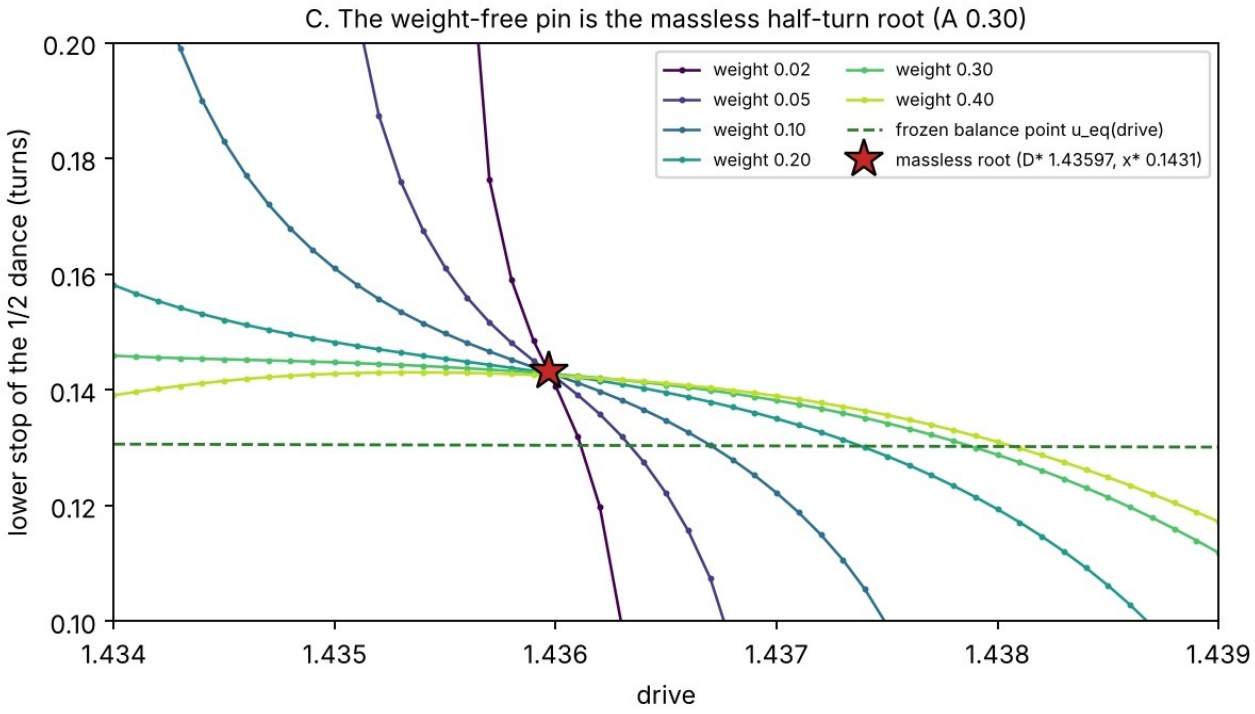


Figure 17. The pin. Figure 34 of *A Rider in Two Waves*: the lower stop of the 1/2 dance against the drive for every weight from 0.02 to 0.40, each curve crossing at the weightless half-turn root, drive 1.43597 and lag 0.1431. The other family of the rhyme, the golden ratio's fan with its waist, is Figure 12. TOY MODEL.

Two traps that are one, and two that are mirror images. A built PDF that loses text and a colour scale that lumps classes are both tools fooling their makers. And "when a sign is free, register the size" and "a prediction can hold for the wrong reason" are mirror images: one held for the wrong reason, the other missed for one.

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7 — Cases

Here the method meets real pictures. Each case tells what a person's eye saw, what the computer missed, the test written down first, and what the numbers then said; the first case is a control, a picture whose every answer was known before anyone read it.

Four cases. The first three come one from each project: the first is the control, the one picture whose every answer was fixed in advance; the second shows three readers on one picture; the third shows blind reads crossing from one project to another and being judged there. The fourth shows Bear's eye working in its own frame on three of the triple star's pictures, each marked first and tested after. Every case keeps one form, in one order: what the eye saw; what the computer missed or could not see; the test written before the data; the result, with its numbers and runs; and the figure the case needs.

The control: the needle's doubling staircase

The Loop Atlas turned Bear's lattice first on a picture whose answers it already knew to ten digits: the light needle's doubling staircase, at mass 0.75 (the dial the Loop Atlas also calls memory), where Newton's method on the orbit had fixed every split. The staircase is a dance of 2 steps becoming 4, then 8, 16, 32 and 64, as the drive is pushed. The bench was 901 drives a millionth apart from 1.68920, two riders each, 256 strobes recorded after 8,000 passings, with the angle and the speed at each. It is the only case in the three projects where every claim of the eye could be scored against an answer fixed in advance. That makes it the control for the rest.

What the eye saw. Here the eye's part was its method and one reading. The tools were Bear's lattice turned into code, and where the method's change of change of slope drew a pair of twin lobes at the splits, Bear gave five readings of the lobes (section 1).

What the computer missed. The lattice reader's first version, registered to see an eight-step dance as eight strings, saw four tight pairs at holes half a degree wide. Three predictions built on eight strings missed, and four could not be read (row 703). The change of change of slope peaked two columns early in every family. And lifted into the rider's

speed, two strands that touched in angle were registered as an overpass, apart in speed, when they were a real meeting.

The test, written before the data. The answers came first, from Newton's method on the orbit. Every prediction was then registered in the data ledger before its tool existed. The tools were LOOP_ATLAS_LATTICE_READER, LATTICE_READER_2 and HISTORY_SETTLE; practices 1 and 2 used figures from R2W's weight sweep (REV-WEIGHT-2), and practice 3 lifted the needle into a third dimension (LOOP_ATLAS_LIFT_3D). The first reader was run 20261009_092420. The second registered 10 predictions, and the lift registered five.

The result. *Its second version held 8 of 10.* It took its families from the side histogram's filled stretches, 18 rows in 4 stretches, and read them on holes of 10^{-4} radian: the same picture and the same method, and only the scale changed. Its counter read the period in every column without following a single rider: with 256 recorded passings, the median hole held 128, 64, 32 and 16 landings in the four bands (row 702). The lanes opened as square roots, with log-log slopes of 0.5115 for the families and 0.4959 for the strands, where the coarse scale had preferred a straight line (rows 703 and 706). Where strands crowd, count pads, not runs: on the fine lattice the 32-step band read 31 runs in columns 135 to 137, so 32 runs showed in only 81.2 per cent of the band, and the 64-step band read 58 to 61, while the counter read 32 pads in every column of the first band and 63 or 64 in the second.

History placed the doublings. The first columns where each lane's period reached 4, 8 and 16 (dances of 16, 32 and 64 steps over the four lanes) fell at 264, 143 and 116, against doublings of record at 262.4, 142.3 and 116.5 (row 709). A dance counts its steps over all four lanes, and the four-step dance visits each lane once, so its doublings fall where each lane's own period doubles; at the doubling of 262.4 each lane's period reaches the 4 that the history read. Settling is slow beside a doubling. After 8,000 passings the history first read the four-step dance's single visit at column 824 (823 in one lane), one column beyond its window, which had ended one column short (the window's size became a follow-on test); settling four times longer moved it to 822, 821, 820 and 822, not to the 819 or 820 predicted from the multiplier (run HISTORY_SETTLE 20261009_093424). An unsettled rider is a reading in itself: at column 111, a fraction of a column above a doubling, neither rider repeats within 128 passings, and at column 117 one reads period 64 and the other 32.

The third number found a real meeting. All five predictions about the lift missed. The two strands touched at columns 135 to 137, 1.58×10^{-4} to 1.84×10^{-4} apart in angle and only 6.41×10^{-5} to 7.51×10^{-5} apart in speed. They were a fork's young daughters, born 2 to 7 columns above, close in every number; holes of 10^{-5} in both numbers parted them, 32 strands in every column from 122 to 140. And the reference count the predictions were judged against was itself one too high in places, because a value on a rounding boundary rounds two ways (rows 709 and 710). That count was the distinct strobe points rounded to 10^{-6}, high in columns 122, 123 and 127. The count of record is each rider's period, the smallest repeat of its passings within 10^{-7} in both angle and speed, and the lift with holes of 10^{-5} agrees with it in every column from 112 to 140 where both riders repeat with one period (run LIFT_3D 20261010_031527). Lifting into a third number helps, though not always: at column 780 the speed separates the widest pair four times better than the angle does, and two of the pairs barely at all.

The twin lobes placed the splits. The peaks taken two columns early fell at columns 815 and 260 in every family, against the four-step dance's doubling at column 817.0 and the next at 262.4. A centred stencil answers a sudden bend with two lobes of opposite sign, and the peak taken had been the lower lobe. Both lobes are present in all four lanes at both doublings, their midpoint within a tenth of a column of the doubling (run LOBE_TWIN 20261010_043453). Read where the lobes cross zero, on columns a millionth of a push apart (run ZERO_CROSS 20261010_043523), the event sat on the doubling, within 0.0031, 0.0014 and 0.0062 of a column, and the ratio of the two stretches between those three doublings read 4.6153 in every lane, against 4.6155 from the orbit. The method needs room: at the fourth doubling, where the next event lay inside the stencil's reach, the crossing landed 0.08 of a column high (row 708).

The lobes themselves were the ruler's. A made-up perfect corner, run through the same three differences (run CORNER_PRINT 20261010_061715), drew the same pair, crossing zero exactly on each doubling and matching every lane's pair to 0.993 in size at the top doubling and better than 0.9999 at the next two. The shape was the ruler's echo of a corner; only its place belonged to the data (row 711). The corner's own size was measured later: each doubling turns the last one's corner over, shrunk by a fixed ratio, -1.966 to -1.968 in all four of the needle's lanes and -1.9644458 on the logistic and sine maps (row 877). Of Bear's five readings of the lobes, "boundary transition" and "mirror-math" were the two the data bore out: the pair marks the boundary between one dance and its doubled dance, and it is mirror-antisymmetric because the ruler is. Of his three other readings, his "phase-locked" has a real cousin in a radio discriminator's S-curve, whose zero is where a receiver locks (from recall).

Record a whole number of the dance's periods. The needle's periods all divide 256. On a logistic-map test bed with period 3 and 256 recorded passings, each lane got 85 or 86 visits, the lanes' centres jittered, and the change of change of slope put its largest peaks in the wrong place (a scratch file, not a run of record).

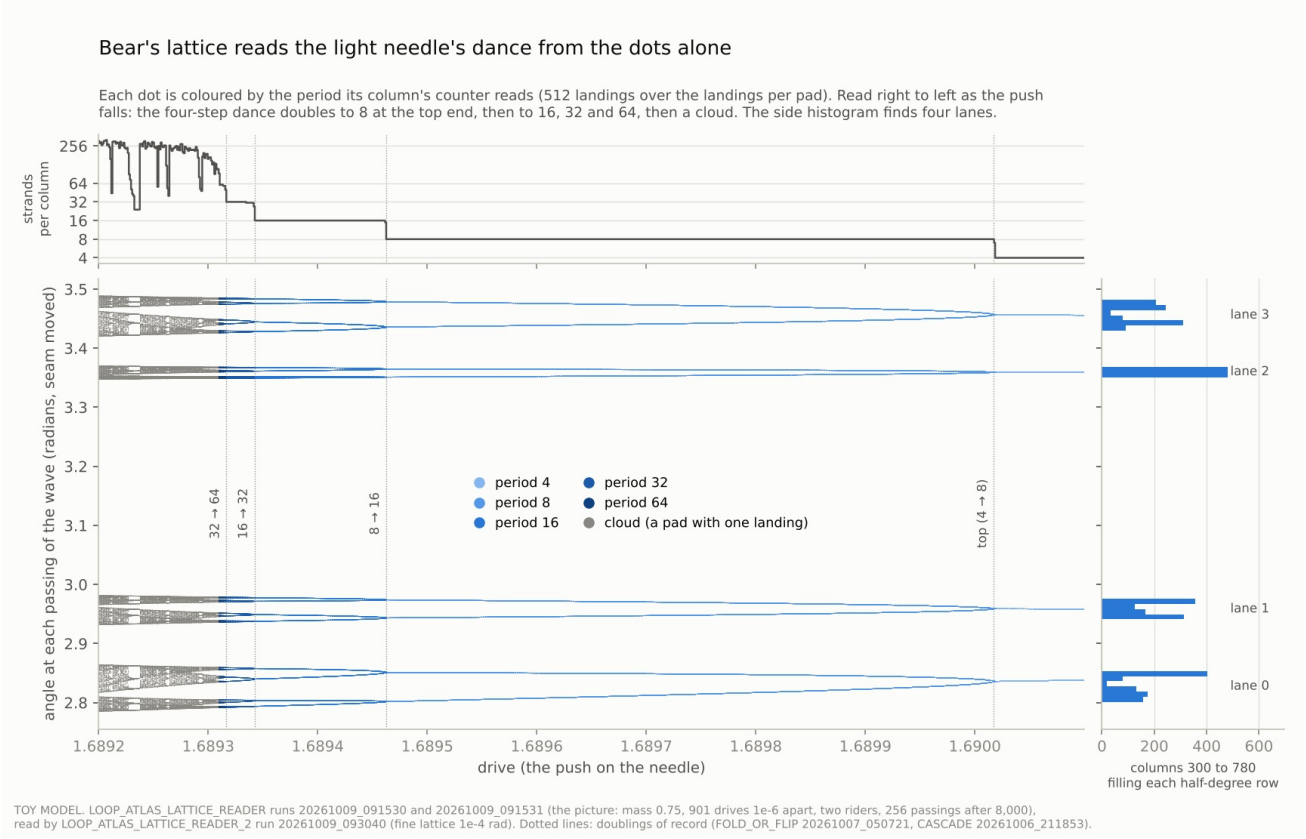


Figure 18. Bear’s lattice on the needle. The light needle at mass 0.75, read from its dots alone: right to left, as the push falls, the four-step dance doubles to 8, 16, 32 and 64 and then becomes a cloud; above, the strands per column climb as a staircase; at the right, the side histogram finds four lanes. TOY MODEL; LOOP_ATLAS_LATTICE_READER runs 20261009_091530 and 20261009_091531, read by LATTICE_READER_2, run 20261009_093040.

Three readers on one picture: R2W’s weight 0.26

Bear handed the Loop Atlas one of R2W’s maps as its lattice test: a strobe picture at weight 0.26, riders’ lags against the drive, with a groove, a conveyor and two clouds. Three readers read it: Bear, Claude reading cold, and R2W’s decoder.

What the eye saw. Bear drew six long strings, circled three populations, and called it 80 per cent, with one string maybe a frequency illusion. One of his strings carried the groove’s upper foot through the middle cloud to the conveyor’s ring. Another was a copy of the groove’s lower foot, its drop, its shelf near lag 0.14 and its final fall, shifted about 0.010 of drive to the right with its shelf squeezed to about 40 per cent (row 706). The seven columns of three dots were three of his strings crossing one window.

What the computer missed. Claude’s cold reading joined pieces the picture did not support: a string leaving the bottom edge to one entering at the top, matched by slope alone, and three strings crossing one window, read as one dance (row 704). R2W’s decoder, greedy by design, drew 50 strings of four or more pads on a lattice of 91 by 100, short chains where the eye drew long ones.

The test, written before the data. The cold reading was on record first, before Bear’s word and the decoder’s reading, so its joins could be judged against both. Its counts came from the picture’s own pixels, read by colour.

The result. *The cold reading’s counting found what neither drawing showed.* The side histogram gives the numerators, how often each lag is visited, and the bottom histogram the denominators, how many riders each column holds; a fraction needs both, and R2W’s own shares were uneven for that reason (8 starts at some drives, 50 at others). Claude read the bottom histogram off the picture by colour: 2 to 21 black shapes per column, exactly 3 in each of the seven columns from 1.4455 to 1.4485, then 12 at 1.4490. It found the lag axis to be a circle with its seam in the wrong place. Only 7 of the 340 black shapes lay between lags 0.27 and 0.52, and counted by tenths of a turn the two black clouds had matching profiles, left 13, 5, 4, 0, 0, 5, 5, 13, 13, 23 and middle 18, 8, 5, 0, 0, 5, 8, 21, 18, 19. That is the misplaced seam: a cut through one band makes it look like two. For comparison, on its own needle the Loop Atlas’s second reader puts its seam in the middle of the side histogram’s widest empty stretch, at 4.9655 radian,

and with it none of the 461,312 dots fell outside a family (run 20261009_093040). The cold reading also found one flat string at lag 0.145 showing through three windows, and the shapes of the groove's lower foot, an exponential settling onto a near-flat stretch and then an exponential fall.

The decoder found the families the eye passes over. It found a staircase of flat strings where the cold reading found one, at lags 0.144, 0.707, 0.81 and about 0.92, each through several windows, and about ten rising strings where the cold reading saw none (row 707). Flat strings come as a family, and rising strings must be looked for on purpose.

Bear's long reach. His six strings explained more of the picture than fifty short chains, and long reach is also what separates a clean picture from its shuffle (row 707). The cold reading's join across the seam by slope alone was withdrawn after Bear's word and the decoder's reading. Asked which string was the illusion, Bear named the cause rather than the string: "When we start injecting a 3rd dimension into 2D, you often see what you want to see" (row 710).

What each reader did best is what the rest of this part asks for: the eye's long reach, the decoder's families, and the cold reading's counts. One reading is not yet tested. There may be a copy across pictures: the flat lags at weight 0.26 (0.81 and 0.144) sit where the third face's flat feet sat at weight 0.40 (0.805 and 0.135).

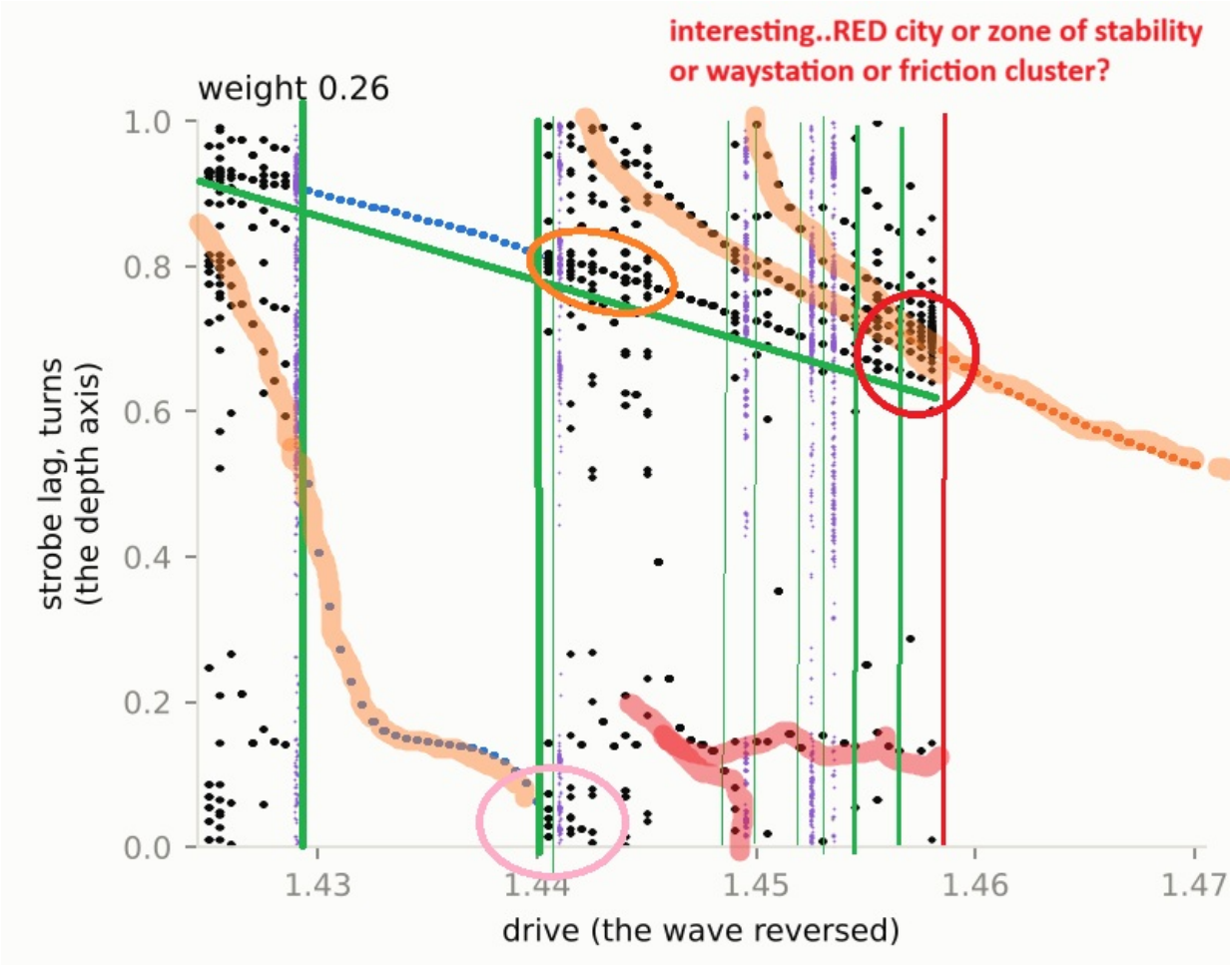


Figure 19. The weight 0.26 picture, as Bear marked it. R2W's depth-axis figure at weight 0.26 (Figure 5 of *A Rider in Two Waves*) with his marks: strings in orange and red, one of them a copy of the groove's lower foot shifted to the right; green lines across the drives; and three circled populations, the red one at the conveyor's ring, with his note: "Interesting… red city or zone of stability or waystation or friction cluster?" TOY MODEL.

Beside the case: the chalkboard. One more lesson came from R2W's pictures, and it is a check rather than a case: explaining it simply is a test (row 667). Partway through the work Bear stopped to ask what the Explorer's picture actually is. R2W drew its weight 2 map for him "like to a smart 12-year-old on a chalkboard": the rider as a heavy ball on a staircase of ledges, the memory band redrawn from the chunks, and the falling ball's beats set against the lines a rider with no ripple would follow. Every label had to name what the data says, so each was checked as it was drawn. The plain account named the whole-number grooves as Shapiro steps (Shapiro-like, as R2W later tightened the name, since a junction's microwave adds a push while the ripple moves the bumps), sitting where the rider's own pace with no ripple is 1, 2 or 3 bumps per passing (the conveyor within 0.003 in clock ratio from pull 0.90 to 1.20). And it found

mistakes while it was being drawn, among them a blind read that had named the conveyor as the rest (row 717), and a memory prediction that had held for the wrong reason (row 683). Bear's two questions at the chalkboard, who sets the clock and whether the honey only steals speed, are the ones section 8 answers.

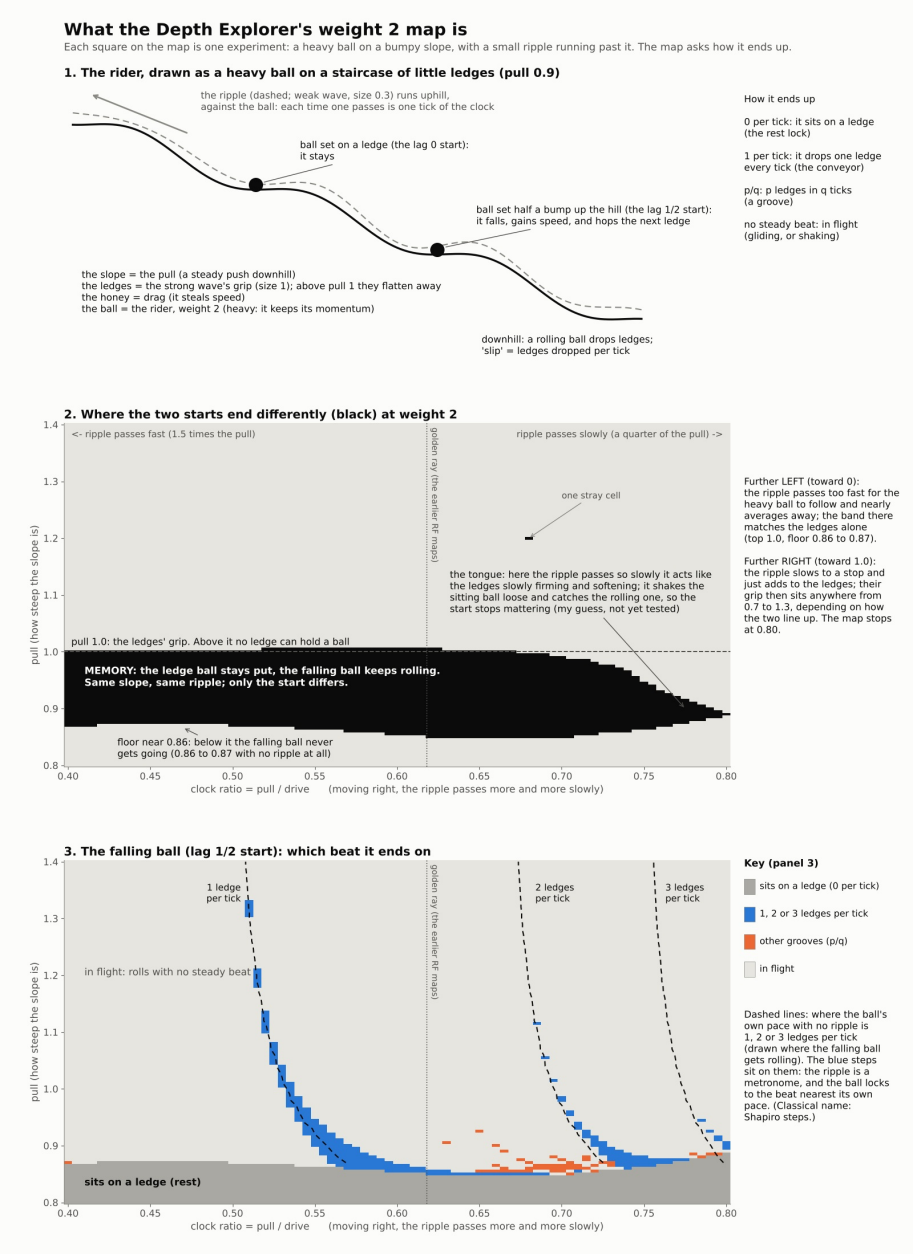


Figure 20. The chalkboard. R2W's drawing of its weight 2 map for Bear, 10 October (Figure 2 of *A Rider in Two Waves*): the rider as a heavy ball on a staircase of ledges, with the ripple running uphill against it; where the two starts end differently, the memory band, with its floor near 0.86; and where the falling ball ends up, its beats of 1, 2 and 3 ledges a tick sitting on the dashed lines a ball with no ripple would follow. TOY MODEL; CHALKBOARD_W2_FIGURE.

Blind reads across projects: the triple star

Two readers read the triple-star sessions' pictures, and the sessions judged both on the numbers behind them: R2W, from pixels alone, and Bear, whose words built the sessions' lattice and whose eye read birth 7.230.

What the eyes saw. R2W read the triple-star sessions' figures blind, from their pixels alone, because the data was held back from it. Three of its reads were registered in its own words: the dots on the page of eccentricity against pass link 5 passes apart and shun gaps of 1 and 3; the band's lopsidedness flips sign across the run while its centre holds; and at 6,000 passes the top edge of a band has 8 crests about 681 passes apart. Bear had written that strings can skip holes ("when they skip 1 or more holes to the next pad"), and the sessions built his lattice literally from his words. At birth 7.230 he drew no crossings, since he saw no obvious p/q, but he added all the lines, and he saw the

dots stack: “3 dots on top of each other in the z-axis, into and out of the page”. His eye collapsed each cluster of three into one place before following the rows, followed a row whole, saw lanes of empty space, and saw the walls as places where the visits pile up.

What the computer missed. R2W could not see the numbers, only the page. The sessions’ lattice tool had the numbers and did none of those four things: its links broke one row in half at exactly the stretch he had drawn flat, and its sensors, looking 5 pads each way, could not see the lanes of empty space. Each of the four is now a line in the lattice’s next version (section 4).

The test, written before the data. The triple-star sessions judged R2W’s three reads on the numbers behind the pictures, with each guess written into the tool before it computed anything. The ladder’s reading, a gear between two angles found at birth 7.200, was then run blind at two more births, 7.190 and 7.220: read off both angles’ median net turns, the gear had to name the lattice’s three commonest gaps in their order, and it was run again with the angle shuffled.

The result. All three of R2W’s reads were borne out, and the sizes of two were off in opposite directions: the weave was stronger on the data than on the page, probably because merged dots hide links, and the lopsidedness was weaker, probably because a drawn line’s width and joins exaggerate its spikes. Section 9 keeps the score. At birth 7.200 Bear’s lattice found the dot families climbing a ladder in which each pass gap is the sum of the two before, 19, 28, 47, 75 and 122, and the multiples of gap 47 pointing the same way within 1.5 degrees: one string skipping holes, as he had written. The ladder belongs to a pair of angles together, not to either angle alone; a check of their data made while this guide was assembled found the pair geared one to three (row 719). Run blind, the gear named the lattice’s three commonest gaps in their order at 7.190 and 7.220, and failed with the angle shuffled. The gear and the lattice read one record, so their agreement checks the two readers, not the stars (row 556). At birth 7.230 the lattice found the stacking he saw: 575 of 599 dots are the same place visited at passes k, k + 201 and k + 402.

Bear’s eye in its own frame: the maypole, the dead zones and the squeezed strobe

Three more of the triple-star sessions’ pictures, of 12 and 14 October, each marked by Bear first and tested after. Each is told in the same form.

The maypole, marked. The maypole, Bear’s word, became a disc picture of the triple star: at each look the angle is φ_N and the radius the pair’s eccentricity, with the pole at the centre.

What the eye saw. On panel A he drew a pink band on the slide lane. On panel B he drew star-polygon paths, “24-gon? Skipping 4 vertices for each line like clockwork.”, with circled 2s and 3s at the corners, “Order?” inside, and “3+” and “Chaos?” on the slide side. On panel C he drew a green mirror line from the pole along $\varphi_N = 0$, “Symmetry and Asymmetry”, with thin pink rays and a faint violet chord. He confirmed Claude’s reading of his marks the same day, adding that each 2 or 3 can be traced to its sister points and that the vertex skipping is solid. His eye read the picture in the circle’s own frame, a clock about its centre.

What the computer missed. Claude’s sealed read did not see the clockwork star polygon. The tools read the picture in the pole’s frame.

The test, written before the data. Claude wrote its translation of his marks and fingerprinted it before opening its own sealed read (6687e26ce2c620bd). The sealed read carried a test of its own, M1.

The result. The seal and his eye both put the figure’s axis on $\varphi_N = 0$. The seal held a circle wrapping the pole by 0.002 at birth 7.200, read as a fixed arrow plus a turning one, which he did not mark. His clockwork held, and it is the inscribed-angle theorem (row 726). The seal’s own test, M1, read nothing by its control; its redesigns found the count switching on at 7.130 while the fitted circle still sits beside the pole, and the narrow window’s single miss is a tie at 7.130 (row 201). Two readers in two frames saw two true things, and a theorem 2,000 years old joined them (runs 20261012_033532 to 20261012_062723).

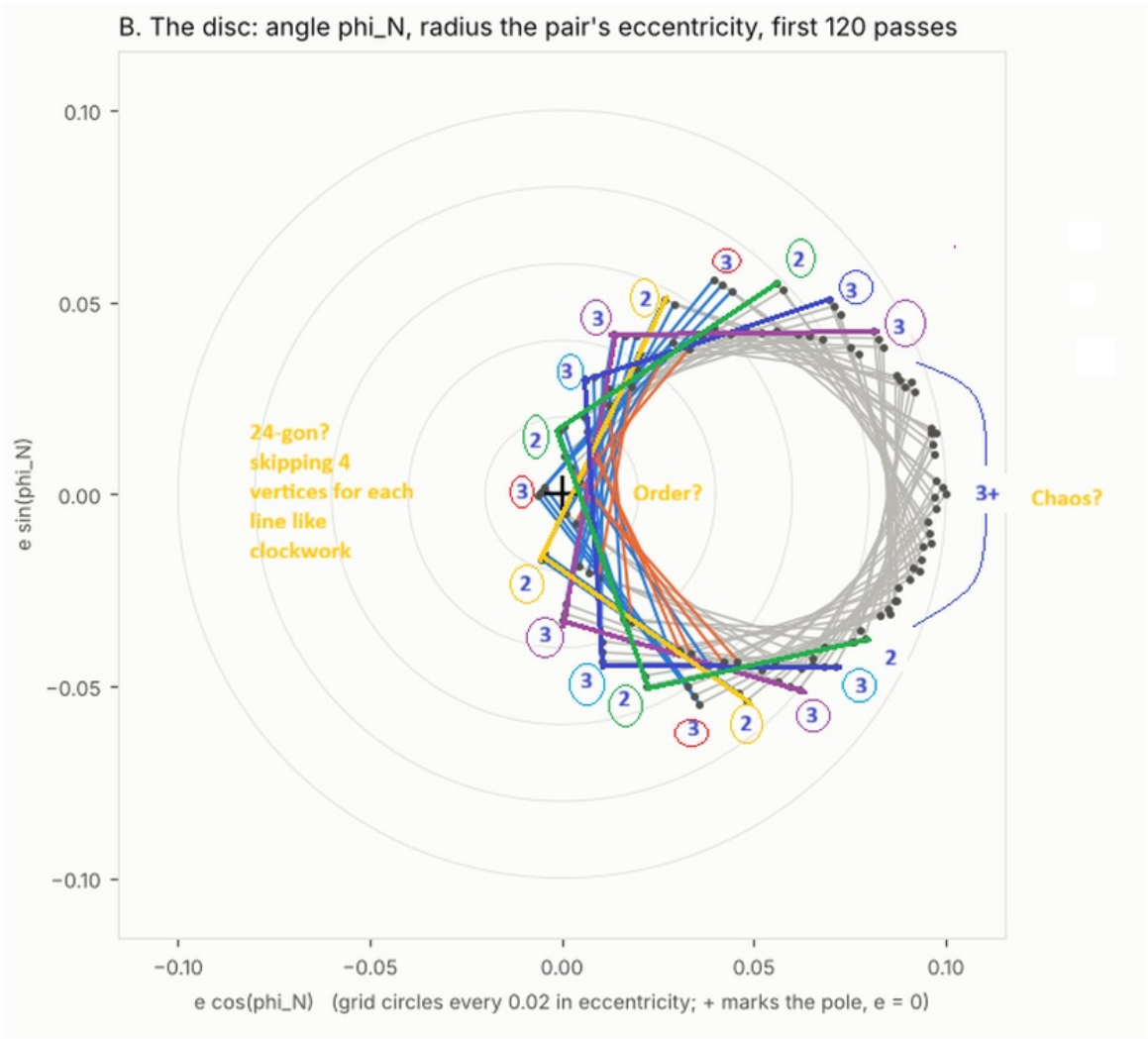


Figure 21. The maypole, marked. The disc of φ_N with the pair's eccentricity as its radius, birth 7.200, first 120 passes, and Bear's note: a 24-gon, skipping 4 vertices for each line, like clockwork. His clockwork held: the median step about the circle's own centre was 0.2154 turn against 5/24 = 0.2083, the inscribed-angle theorem at work. From the triple-star paper's figure of the eye, run 20261025_033626.

The dead zones, on an old figure. He shared a figure of 6 October, marked: the island map, a log scale of φ_N's band against birth.

What the eye saw. He drew four lines at about 1.2, 4.5, 135 and 600 turns and wrote "Dead Zone Skips" between each pair. He traced each phase's runs as a curve, with arrows where the curves jump the lower zone, and on the third panel he traced the pair's least and most eccentricity. His eye found a gap in a log-scale axis, negative space again as with his gold lanes, but this time measurable, and it joined two panels by tracing both.

What the computer missed. The figure, of 6 October, already showed the start sitting at the top or the bottom of the pair's swing, the reason the thread moved to the mean-ellipse start after the checks of 8 October. His trace shows it on the older figure.

The test, written before the data. His marks came first. The zones were then tested on data he never drew on.

The result. On data he never drew on, both zones held (the scorecard, section 9; row 715). On that figure's own runs the zones are a description: its held runs reach 0.880 and its circulating runs start at 5.897, and the three runs above 130 are exactly the three whose pair was knocked to an eccentricity of 0.61 to 0.64. The lower zone's run at 7.130 is the maypole's tie, the one birth where the narrow window's swing toward the pole and the circle's edge agree to 0.0001; a test called E1 was registered to try the same at the zone's other run, 6.785, and no result of it was recorded before the triple-star thread closed on 29 October. His traced third panel held too: of the 41 runs started on the beat, 33 never let the pair's eccentricity rise above 0.101, while all 41 started half a beat late rise, to 0.189 to 0.636, and 40 never fall below 0.092 (run 20261012_073126).

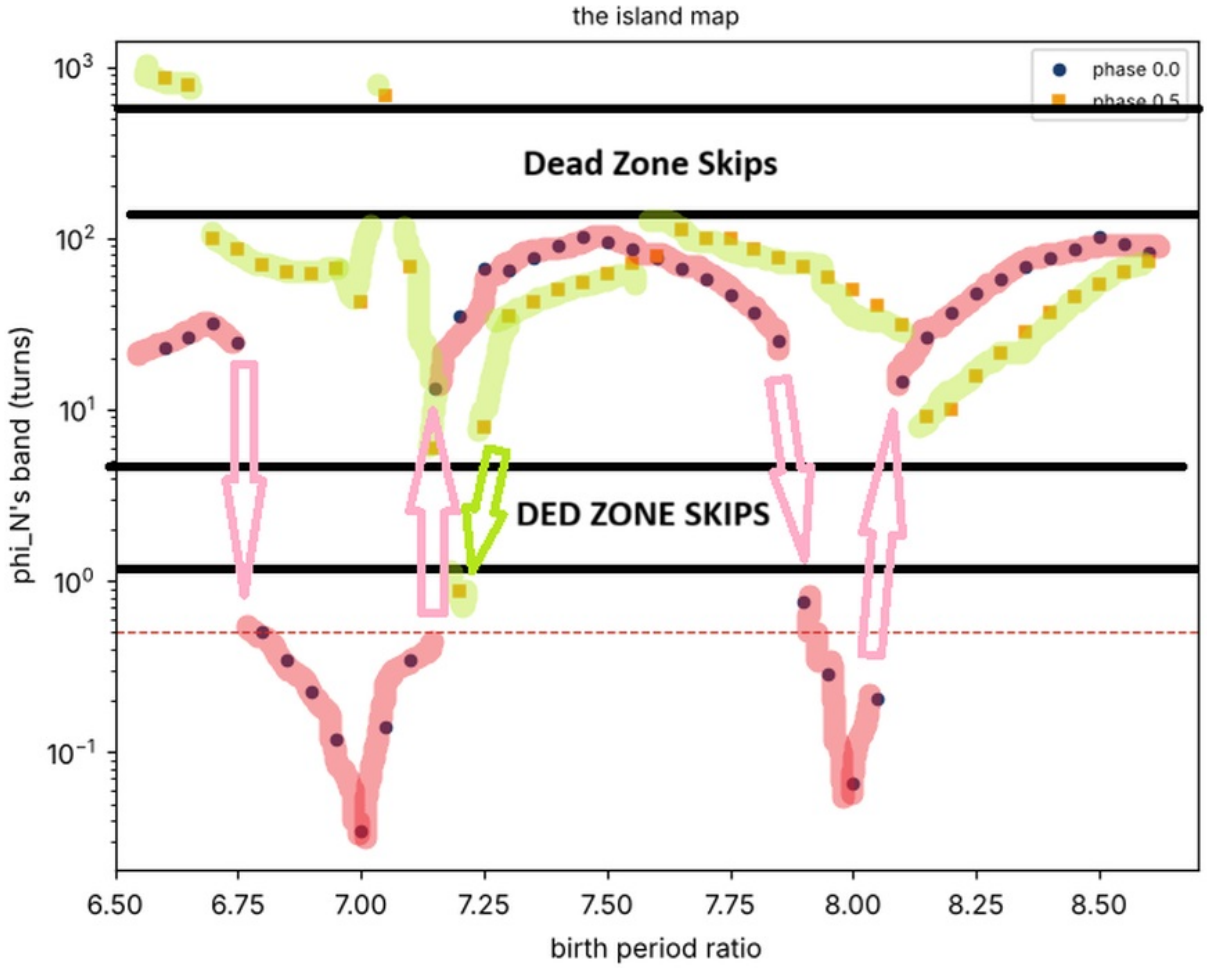


Figure 22. The dead zones, marked. The island map of 6 October, φ_N's band against birth on a log scale, with Bear's marks of 11 October: four lines with "Dead Zone Skips" between each pair, each phase's runs traced as a curve, and arrows where the curves jump the lower zone. His lower zone held on 600-pass runs he never drew on: 3 of 484 runs fell inside it, against 64 just below and 11 just above. From the triple-star paper's figure of the eye, run 20261025_033626.

The squeezed strobe. The strobe picture (island 7 at the pair's eccentricity 0.25, every pass of every main run a dot, birth across) went to him unmarked on 14 October.

What the eye saw. He marked pink "data bands" on the columns of dots and black "spacer rungs" between them. Then he squeezed the picture sideways to about half its width, drew by eye a green outline of where the dots are dense and a blue axis along $\varphi_N = 0$, and wrote: "Bilateral symmetry. Green is dataplot outline judged by eyeball on density. Definitely a pinch in the middle." Last, he turned it a quarter turn on his screen and photographed the screen from a grazing angle; the file's name says it was marked before the squeeze, with the eye's phase offset pretended at 75 degrees "to make it severe". He named the two strategies himself, to sit beside zooming out, zooming in and looking at a dot's neighbours: "I cheat the phase offset view by angling my screen a little or by compressing event time."

What the computer missed. Where the island survives: the centre that a scan of the first 600 passes reads is not where the island lasts.

The test, written before the data. Three reads drawn from his marks were registered, marked first and tested after: his symmetry, his pinch, and a moving waist that Claude read from his two arrows.

The result. Two of the three held as registered, his pinch and the moving waist, and his symmetry passed only by construction (the scorecard, section 9). And the squeezing found what no one had asked: the births with the smallest swings over the first 600 passes, 7.525 to 7.535, are not the ones that last. Their outlines over passes 601 to 6,000

widen to 0.46 to 0.75 turn, while the room that keeps its stars sits 0.02 to 0.06 lower. Only the short view and the long view side by side, each squeezed into one outline, showed it.

Figure 16 is the strobe picture of island 7 and his squeezed copy, with its green outline and blue axis.

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8 — The common object

The three projects turned out to be watching the same kind of thing: an angle going round a circle, looked at once each time an outside clock ticks. This section names that shared object, the question that splits it into two families (does it keep area, or lose it?), what the eye reads on it, and why finding it already named in the textbooks is the usual result, not a discovery.

The section goes in four steps: the object in its three costumes, with the clocks that drive it; the fork that splits it, area kept or lost; the three readings the eye makes on it, the count, the corner and the leftover, each with its rule; and where the shared method stops.

One object under three costumes. All three projects watch an angle carried round a circle by a flow, and look at it once per tick of an outside clock. That is a circle map, strobed: the Poincaré section of the textbooks. Its currency is the winding number p/q, p turns in q ticks, the same holonomy currency that runs through the rest of this guide. The register's own transfers agree: on 16 October it recorded 20 transfers between the three projects, 14 held, 3 held weaker and 3 failed, and none of the six breaks was put down to the substrate or a wall. R2W sent 14 of them (8 to the Loop Atlas, 6 to the stars), the Loop Atlas 4, and the triple star 2, both ways of reading pictures (rows 734 and 743).

In each toy	The Loop Atlas's needle	R2W's rider	The triple star
The outside tick	One push of the drive	One passing of the ripple	The third star's pass, at closest approach
What turns	The needle's angle	The rider's place on the washboard	The inner pair's phase, and the resonance angle φ_N (Mardling's [N:1](2) angle)
A lock	A gear: whole turns per push held over a band (the one gear; the 1:2 gear)	A groove: the slip, ledges per tick, sits on p/q; the whole-number grooves are Shapiro-like steps	An island: φ_N rocks about a fixed value
Outside the lock	—	The rider is in flight	φ_N slides round
Friction	Damping: area is lost	The honey: area is lost	None: area is kept
The rider's own pace	Set by its slope and its damping	Set by its slope and its honey	The pair's own turns, and a slow beat of about 686 passes at one birth
Memory of the start	Where two locks overlap (row 670)	The memory band	Total: every start keeps its own orbit, and the start chooses which locks can be seen
A birth by a fold	The glide's extra turns grow as the square root (slopes 0.46 to 0.48)	The lurch-stroll-lurch S; below a conveyor's birth, the bottleneck exponent 0.507 (row 654)	—
A birth by a flip	Ratios closing on 4.669 (4.6153 read from the dots)	The third face's first ratio	An island doubles once and its stars escape; an area-keeping test map closes on 8.72

R2W's rider is the needle. R2W's chalkboard equation is m u'' + u' = A cos(u − Wt) + B cos u − F. With B = 1 and u = π/2 − L it becomes the needle's m L'' + L' + sin L = F − A sin(L + Wt): the same ball on the same washboard, with the angle measured from a different mark. So the chalkboard's weight 2 is the needle's memory 2. Three numbers that answer three different questions can then sit side by side without contradiction. The table sets them out, with the exact answer to the second question, and the textbook's estimate for heavy balls, beside them.

Question	Number	Where it comes from
Where a ball already rolling is caught again, read on a slow ramp down	0.773	The Loop Atlas's run
Where a ball already rolling keeps rolling, with no ripple at all	0.78 to 0.80	R2W's single-ball run
The same, computed exactly	0.7899	This guide's check (row 711)
The heavy-swinger estimate, 4/(π√m), good only for heavy balls	0.900	The textbook
Where a ball set still half a turn round gets going	0.86 to 0.87	The chalkboard's floor

In a junction's terms the weight is the Stewart–McCumber number, and with the ripple off a memory band first opens at weight 0.7026, computed for this guide (row 711).

The triple star's angle, as a picture. Marking the passage of the triple-star thread's paper on the pair's eccentricity and the reading that bends it, Bear asked whether the bend is the pair lining up with the third star, its tug growing stronger when the star passes in their direction, twice in each turn, since either star of the pair can be the one in front. It is. The star pulls harder on the nearer star of the pair, and the pair looks the same after half a turn, so the stretch comes round twice a turn, and where the star finds the pair in that rhythm is φ_N itself. The paper now says so in a plain sentence after the passage. One question was left open when the thread closed on 29 October: whether his picture is the same bend that a test called GK1 measured once a pass at the star's closest approach, at 1.116, 1.117 and 1.115 of Georgakarakos's prediction as the triple-star thread's paper gives it (run 20261018_193951).

Three timings, and one word kept narrow. The **outside tick** is set from outside and never feels the rider: a passing of the ripple, a star's closest approach, a drive's period, and in the world a lab source, a transmitter or an orbit. The **rider's own pace** is set by its slope and its drag. A lock is the rider's own pace giving in to the outside tick where the two are already close. The Loop Atlas met a third timing, the **event**, which opens a gate when something arrives. A bead with a blade, in an even soup of waves, drifts 0.009 when the blade is switched by an outside tick, no more than its own scatter of 0.008. Switched by listening, on when the push points the way it wants to go, it drifts 0.124, though a later toy that billed the charge showed that drift to be a free lunch the first toy had not billed for (row 590). Bear's words for the contrast, given before it was measured: "event timing like the jellyfish instead of master clock timing." Two clocks that each keep time by their own events and are tied to each other are **coupled clocks**: they come home together after the denominators of the continued fraction of their ratio, and a lock is where they agree (row 719). "Two clocks" keeps only this narrow sense in this guide's vocabulary.

The fork that decides most of what follows: does the system keep area, or lose it? The triple star keeps area. Nothing settles, every start keeps its own orbit for good, and islands sit in a chaotic sea. At a lock's clean edge the time to slip grows like a pendulum's logarithm, over the ladder as a whole: 7.57 and 7.37 passes per decade of distance at two clean rims, against 7.29 and 7.83 forecast from the island's own width. Step by step it departs: 5 to 11% short far out, a quarter steep nearest the edge, and 1.76 times beside a chaotic layer (row 780). The doubling constant of its class is 8.72. R2W's rider loses energy to its honey. It settles onto attractors, and the start matters only where two attractors coexist (the memory band, 0.2112 of the cells at weight 2). The time to slip at a lock's birth grows like an inverse square root, and the doubling constant of its class is 4.669. A lightly leaky map reads the frictionless constant for its first doublings and then crosses over, once one trip round its orbit keeps less than about half its area (0.440 to 0.524 across five leaks) (row 680); read later across fifteen dampings, the departure falls on one curve of the area kept, its 10 per cent crossing at a median of 0.4905 (KN1 and KN2, run 20261016_201627; section 9). So the first question about any edge is which kind it has: a chaotic layer, or a band where two starts end differently.

A rhyme, not a test. The three-body problem without friction is reversible in time, as R2W's toy is at zero weight. The triple star's rope check has two doors that are mirror images in φ_N (-0.140 and $+0.141$ to $+0.155$), and its bands mirror each other. Whether that is the same kind of mirror as the toy's is a question for the triple-star sessions' own pictures, under their own rules, and it was not taken up before the thread closed on 29 October.

Three readings, and they are what the eye is good at. The **count** is turns per tick, the winding number p/q, which the eye sees as bins. The **corner** is where a count changes, which the eye sees as a bend. The **leftover** is what a loop keeps when it comes home, which the eye sees as an offset. The Loop Atlas's spine is the third: a loop comes home to where it started, so what did it leave behind? Each reading has its rule, and the rules follow in that order.

The count: nearest returns come in a ladder, and its neighbour is a theorem. A steady rotation seen once per tick returns nearest to where it started after the denominators of its continued fraction. This is the best-approximation property of the convergents, and when the partial quotients are 1, each return time is the sum of the two before. That is the ladder the triple star's families climb: 19, 28, 47, 75 and 122 passes. Its neighbour is the three-distance

theorem, conjectured by Steinhaus and first proved by Sós (1958), Surányi (1958) and Świerczkowski (1959): the first N points of a rotation leave at most three gap lengths between them, the largest the sum of the other two. That is what the white patches between the threads would test. Say which of the two a picture shows.

The count as a rate: a rate says how it was read. The triple star's φ_N slides -0.1254461 of a turn per pass at one birth, read end to end. Read by a fitted line, or by the median step, it is -0.11175 or -0.10572. All three are right. The angle's single steps fall in three bands, the slide itself and the tide's flips of about half a turn either way, and each way of reading weighs the bands differently. The flips come mostly where the pair's orbit is nearly round as the star passes: 87 per cent of the steps off the slide come where its eccentricity reads below 0.03, against 38 per cent of all steps. A rate quoted for an angle that does not slide steadily says how it was read, and where its steps come from (row 558).

The corner: births come two ways. By a fold, a lock opens as a square root, and a slow ghost is visible before it is born: riders in flight crowd toward the place where it will appear (row 654). By a flip, a dance doubles, its multiplier passing -1, and the ratios of the doubling gaps close on a constant set by the shape of the top: 4.669 for a rounded top, 7.2847 for a quartic, and 8.721 for a map that keeps area (row 680). Near either, nothing settles in time to be read, so follow the orbit and read the sign of its multiplier (row 642). A window can open and close by the same road, a cascade at each end (row 697).

Two ladders, one finish. When two systems end at one number, compare how they come to it. In R2W the 1/2 groove's doubling ladder reaches Feigenbaum's ratio in three steps; the 3/7 groove's starts far lower (2.41) and takes six. Near the end the two close in the same way, each gap to 4.669 about a quarter of the last (0.210, 0.054, 0.012 and 0.218, 0.050, 0.012). Universality is in the approach as well as in the limit (row 680), and one toy's road can forecast another's: R2W's rule, registered as a forecast of the needle's fourth ratio, gave 4.6585 and 4.6591, and the needle gave 4.6599 and 4.6604, then a fifth, forecast at 4.6672 and 4.6673, of 4.6673 from above and 4.6672 from below (row 872).

The leftover: a loop comes home turned, never moved. That holds for a body with nothing to push on. The Loop Atlas book's falling cat and its chair come home turned, never moved, while a swimmer pushes on its water and does come home moved (rows 242 and 630). The projects meet this one wall from two sides: the book's spine, what a loop leaves behind, and its "a ship cannot tow itself"; R2W put the same momentum wall on page one.

Where the shared method stops: it rediscovers known physics quickly. The triple-star sessions rebuilt Mardling's harmonic from the inside before reading her paper (R. A. Mardling, 2013), found the ladder of nearest returns, and met Arnold's resonance of order three, which predicted before the run that an island's centre would rock across a third as the star's eccentricity grew; it did, between the star's eccentricities 0.325 and 0.330 (probe 31, run 20261007_151231). R2W met Shapiro-like steps, saddle-node bottlenecks and Stewart and McCumber's hysteresis. It then found its whole toy in print: a particle in two waves, one standing and one moving, which without the honey and the slope had been built with an electron beam in a travelling-wave tube, where a devil's staircase appeared as the second wave was turned up (F. Doveil, A. Macor and Y. Elskens, 2006). The Loop Atlas met Feigenbaum's ratio and square-root births. Each match felt like a discovery for an hour, and each was a check that the toys work. Finding it already named is the normal outcome, and it is credited as such. Not every reading ended at a name: the ratio by which each doubling turns the last one's corner over, -1.9644458 for a rounded top (row 877), and the cause of the odd-order errors in a published table of doubling constants, a sign slip, after McCartney (2020) had found the errors (row 874), were found named nowhere in a limited pass, each a pointer for a specialist. Once the whole toy is named, its map becomes a bench: every landmark is a known length to check a ruler against, and any claim of something new moves from the physics to the reading (row 726).

A part can be a textbook object too. The maypole's clockwork was the inscribed-angle theorem (section 7). In R2W, five groups of three at 12 and 36 degrees swing twice the golden ratio and twice its inverse of one group (run FIFTEEN-1): five equal arrows, each turned 36 or 108 degrees from the last, which is the textbook pentagon array factor. Look at a measured ratio for a known shape before calling it new.

The method reads what a picture shows; it does not decide what physics allows. One project hoped a rider's motion might move a craft that pushes on nothing. No picture could rescue it: momentum is conserved, a wall no reading could have shown, and the method was the part of that work that still stood.

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9 — The scorecard

This is the score sheet. Each claim made by eye was written down before its numbers existed and then judged on the numbers; the tables show which claims held, which missed, and whether a miss lay in where a thing sits and which way it points, or in how big it is.

The scorecard holds eye claims registered before the numbers existed, judged on the numbers, against shuffles. It is kept by the kind of claim, because the readers are good at different kinds. It is scored in separate columns for sign and place, for size, and for which members of a set stand out, because a claim can be right in one and wrong in

another (row 712).

The tables come in three groups, by how the read was made: the second reader's blind reads; the Loop Atlas's predictions, judged against answers fixed in advance; and marks made on figures, most of them Bear's, registered in his words before they were tested. What the columns add up to comes last.

Blind reads from pixels. R2W's reads of the triple-star pictures, judged by the triple-star sessions on their data.

Claim	Kind	Sign and place	Size	Verdict
The weave: dots link 5 passes apart, and shun gaps of 1 and 3	Geometry	Right: +23.4 standard deviations on the data; gaps 1 and 3 at −3.9 and −4.3	Stronger on the data than on the page (+12.4 from the pixels)	Borne out. Gaps 2 and 6 shun as hard (−4.6 and −4.3) and gap 14 stands tall (+9.6): the read named the tallest gap rightly and two of several short ones
The ripple: the band's lopsidedness flips sign while its centre holds	Shape	Right: −0.18 to +0.29; the centre held, 0.04893 to 0.04911	Overstated by the pixels (−0.49 to +0.47)	Borne out on the sign
The zoom-out: 8 crests about 681 passes apart	Count	Right: 8 crests	Median gap 686 (666.7 was a bin of the spectrum, row 318)	Borne out

The first row is also R2W's registered test of the weave against a shuffle. Four more of its tests were registered for the triple-star data, never reached that room in their exact wording, and were not run before the thread closed on 29 October: Morse's count of critical points on each orbit's own torus; the wave's shrink and its slide; three gap sizes, in phase; and folds and cusps.

R2W's blind reads in one sitting, scored. The reading order for this sitting, set at Bear's suggestion that the unmarked pictures be read first as a self-test, was: check provenance (every picture must match a file of record, and anything already seen is declared; three fragments found by a text search had been seen); read blind, writing a blind read of each unmarked picture before opening the next; check the pattern; read his marks, and the earlier sitting's reads of the same pictures; score.

On the weight 0.26 map the blind read hit the upper stop, the S on the lower stop and the conveyor. It was partial on his two curves into the ring, stopping where the dots crowd while he followed both into the ring, and where he circled three populations it saw two bands. It missed his "maybe" stroke, which was a second S, and his steep curve from 1.452. It was wrong on a "period-3 dance" and on "in flight". It was right, where he made no mark, on the empty middle and the ghost at the ring. A grey stripe at 1.4535 stays open.

On the bend figure the crest counts by eye were wrong (23 against 37, 38 against 43), while the order and the decay were right; the woven strings in the outside readings were missed. On the triple star's rope check, read blind by eye and then graded by pixels, one reading was right, one mostly right with a knot misplaced, one refined into "cells" like sausage links, and one wrong; the read missed the mirror and the straight middle lines that his marks showed.

In sum, the blind reader missed his second S, his steep curve, the woven strings, the rope check's mirror and the straight middle lines. It saw, where he made no mark, the empty middle and the ghost at the ring. And it saw the symmetry only after he pointed, when it took minutes to measure and became the night's result.

Bear's marks on the weight 0.26 map are Figure 19.

Claude's own reads by eye, registered the same way. In the triple-star sessions Claude's calls made by eye were registered as Bear's marks were, before anything was computed, and withdrawn as plainly when they failed.

Read	Kind	Result
From the bench’s cascade ledger (TRIPLE_CASCADE_CROSSOVER_LEDGER_20261007_051639): the damped cascade’s departure from the frictionless ratio is a function of one quantity alone, the area an orbit keeps over one trip, with its 10 per cent crossing near 0.49	Shape and place	Borne out: one curve across fifteen dampings (KN1), and a median crossing of 0.4905 (KN2; triple_friction_knob v1.1, run 20261016_201627). It became the bench’s dial in the triple-star thread’s paper; R2W’s note reads it as γT near 0.71, a derivation not yet tested on its own map
A shape at the tenth, called textbook	Shape	Withdrawn: its cause turned out to be the record’s length
A rule that each closest-return record is built from earlier ones (LD1b)	A rule for records	Not borne out: it held for 1 of 3 records, and 21 of 200 random pairs of rates obeyed it (row 616)
A stall called unusually slow (RW1)	Rarity	Not borne out: 79 of 200 random pairs matched or beat it, so the stall was ordinary (row 616)

Predictions against known answers. The Loop Atlas’s registrations about its own pictures, each judged against an answer fixed in advance.

Registration	Kind	Result
The lattice reader’s first version, registered to see an eight-step dance as eight strings	Count	1 of 9: the fidelity check held; at holes half a degree wide the eight strands came as four tight pairs, so three predictions built on eight strings missed and four could not be read, and one more, on the fold’s ghost, missed (run 20261009_092420; row 703)
The second version, families from the side histogram’s filled stretches, read on holes of 10^{-4} radian	Count, place and shape	8 of 10: the counter, the rate, the square-root opening, the change of change of slope, the cloud test and three Newton checks held; the history missed by one column, and one band read 32 runs in only 81.2 per cent of its columns (run 20261009_093040; row 703)
Lifted into speed, the strands per column equal the distinct points, and the touching strands are apart in speed (three predictions and their two finer follow-ons)	Depth	0 of 5: the touch was a fork’s young daughters, and the reference count was wrong by one in places
Two lobes of opposite sign lie either side of each doubling, their midpoint within a column of it	Shape and place	2 of 2
The crossing between the lobes sits on the doubling registered for it, the sixteen-to-thirty-two (Z2)	Place	1 of 1: within 0.0062 of a column, against a bar of 0.05; printed beside it and not judged, the doublings at columns 817 and 262 within 0.0031 and 0.0014, and the one at column 116, inside the stencil’s reach, 0.08 high
The lobes are the echo of a corner	Shape	1 of 2: the made-up corner crosses zero on each doubling and matches to 0.993 and better; a signed correlation, registered where the corner’s sign was free, missed as written

Bear’s marks on the triple-star figures, registered in his words and tested. These are his marks on the triple-star figures, from the bend figures and the blind figure to the maypole, the dead zones, the squeezed strobe and the pictures of 15 and 16 October.

Mark	Kind	Result
The bend is woven from two phases	Shape	Borne out: the fit went from 0.4302 to

		0.9723 with the second angle
Two angles make a torus	Geometry	Missed by a hair on its first run (the neighbour ratio 0.1126); borne out over 6,000 passes (0.0603)
Walls where the visits pile up	Geometry	Under its registered bar. Beyond 2,000 random sets (0.778 against never below 0.856), but that comparison was chosen after the run and is claimed as nothing; the test registered for the next birth has no recorded result at the thread's close (row 313)
Rows that arch below the waist and lift above it	Shape	Dot by dot, a coin's shares (0.527 and 0.488), so an empty pass; on cluster centres, missed by 0.015; fitted whole, missed by one row (6 of 7 arch, 6 of 8 lift)
Arrows along the first angle	Direction	Borne out: a median of 8.1 degrees from the axis, his direction "ambivalent" by his own word
Gold lanes of empty space	Negative space	Not borne out, because the sensors could not see them: the test measured the wrong thing
p/q crossings at a blind birth	Count	No reading: he drew none
His clockwork on the maypole: "24-gon? Skipping 4 vertices for each line like clockwork."	Count	Borne out: the median step about the circle's own centre 0.2154 turn against 5/24 = 0.2083, and 0.2579 in shuffled order. It is the inscribed-angle theorem: the slides step about the centre at twice their step about the pole, −0.2112 against −0.2114
The jumps, read on his clock	Direction	Borne out on one test and missed on the other: the jumps run a little off the clock, up-jumps landing late and down-jumps early, and C24d held at 7.190 and 7.195; T3 missed as registered, holding at 13 of its 15 births and missing only where down-jumps were three and twelve (rows 544 and 644)
His lower dead zone, on the 600-pass runs that followed his figure, bands as drawn	Negative space	Borne out: 3 of 484 runs inside, against 64 just below and 11 just above
The same, as a rate (Claude's rescaling)	Negative space	Not borne out: 10 inside, 2 below, 28 above; the rescaling moved the held runs too
His upper dead zone separates broken pairs, on the later 600-pass runs from the mean-ellipse start	Negative space	Borne out: 1,462 of 1,475 rows agree; all 12 broken pairs above it; 13 healthy runs also above it, four at exactly two thirds of a turn per pass
His bilateral symmetry on the squeezed strobe (SQ1)	Geometry	Passed at both scales, but empty by construction (row 337): over passes 601 to 6,000, 40 of 40 outlines centred on zero within a fifth of their width, against a bar of at least 80% (three outlines wrap the whole turn and are symmetric by construction, and the island's centre is $\varphi_N = 0$ because the runs start there)
His pinch (SQ2)	Place	Borne out: registered at 7.465 to 7.505 for the long view and 7.520 to 7.550 for the short; the narrowest long outline at 7.505 (0.147 turn; 7.470 to 7.490 at 0.173 to 0.188), the narrowest short one at 7.525 (0.103; 7.530 and 7.535 next). The long view's narrowest is one run, a single

		sample of the chaos
The waist moves (SQ3, Claude’s, from his two arrows)	Place	Borne out exactly at the bar: the long view’s narrowest birth 0.020 below the short view’s, against at least 0.02
A very regular pattern inside the centre, traced as a swell	Shape	Borne out: his red line has 37 highs 53.5 passes apart, the beat (BC0, registered before it was measured; run 20261015_133950; row 581)
“The ratio stays fixed, the rates don’t”	Relation	Not borne out over 600 passes, holding at 3 of 8 slices, the misses showing where the beat passes through zero (RF1); borne out as registered over 2,400 passes (SM1; runs 20261015_140219 and 20261015_151629; row 581)
Fractals within fractals, read as islands round islands	Shape	Partly: the chains are there, 121 locked runs within 10^{-4} of a simple fraction against 34.5 at decoys, but the falling law is barely seen, and a parity appeared instead (E4 and E6, registered before anything ran; runs 20261015_225444 and 20261015_230449; row 581)
His marks on a sheet of plain figures	Place	Measured, not scored: his circle sits on a lock at a seventh, found without its label (run 20261016_054242; row 581)
“5 twists around 4” on the egg figure	Count	Confirmed and tested; his other readings of that figure were translated into test forms and not confirmed (row 782)

How the dead zones and the squeeze reads were scored. The dead zones were scored on data he never drew on (triple_dead_zone_check, run 20261012_062504). The three squeeze reads drawn from his marks (SQ1 to SQ3, the third Claude’s, from his two arrows) were registered before anything was measured, and scored by triple_strobe_views (run 20261014_072942). A run’s outline is the 2nd and 98th percentiles of its φ_N, with the 5th and 95th as a second scale, and a verdict holds only if both scales agree. Of the three reads made by squeezing and tilting and registered before the numbers, two held as registered at both scales, the pinch and the moving waist, with a caution: the pinch’s narrowest long outline is one run, a single sample of the chaos, and the waist moved exactly as far as its bar, measured from that same run. The symmetry was centred by construction, because the runs start at the island’s centre, so the outlines could not fail its bar and its pass is empty (row 337). The one read made on the tilted view, the V, did not survive a grid five times finer (the next table). Here the eye proposed and the ruler decided.

The maypole disc with his marks is Figure 21; his dead zones, on the figure he drew them on, Figure 22; the squeezed strobe, short view beside long, with his pinch and his two arrows, Figure 16.

Reads on a finer grid and over longer runs. A run of the 0.25 island five times finer than before (a birth every 0.001) for 6,000 passes, and two follow-on runs after it, tested reads made by eye on coarser or shorter pictures, Bear’s and Claude’s alike, each registered before its numbers were computed.

Read	Where it was made	What the finer or longer data said	Verdict
The V: the long runs' core narrowing in a straight line to a pinch	The tilted strobe view, a birth every 0.005	At 0.001 each run's width follows its own fate, narrow where it held and wide where it did not; neither a V nor a floor fits better than a smooth curve	Not borne out: an outline of the coarse sampling
The offset: the room that keeps its stars lies below the early centre	The squeezed view, short beside long	The narrowest early swings at 7.524 to 7.531; 22 of the 24 births that hold lie at 7.505 or below	Borne out again
Neighbours alternate at the sealing edge	The fine run's window as it ran	Fates speckle between neighbours across the whole grid; but no 0.005 of birth holds more than half its runs, and the escapes stop sharply at 7.472	The alternation seen everywhere, as speckle; the registered climb in the held share not borne out: the edge to holding is no edge
The walls sit where the rocking crosses a ninth	The flanks of the coarse walls	The edge lies 0.001 from a ninth, but the rocking sits near a ninth almost everywhere	Borne out, weakly
A chain of small islands locked at a ninth in the wall	R2W's rhyme	No birth in the band reads at a ninth	Not borne out; a finer look is registered
The door is a corridor: stars cling at its inner end, roam at its outer end, and run the wrong way	A figure of the triple-star thread's paper, at one eccentricity (Bear: "Agreed on edge or ledge")	The wrong way: 17 of 18 chaotic stars below the island that circulate turn the far side's way, at three eccentricities, and all 13 that escaped from below were turning that way. The ledge: the clinging stars are scattered through the layer	The direction borne out; the ledge not
The shark's fin: the apse turn peaks in a point at the island's upper edge	Bear's green trace on another figure of that paper	Proposed tests at first, then tested on nine lines	The next row
The shark's fin, tested: the tip at the hold's edge, and the fin's finer form	The same trace, read on one line of that figure; tested on nine (runs 20261017_030927 and 20261017_032258)	The tip's place at the hold's edge held on all nine lines (FN1). The finer form did not: the return to free within 0.050, gentler than the climb, held on four of nine (FN2), the two-sided logarithmic point on one (FN3), the wake on six (FN4). On a grid five times finer the two-sided logarithm went away, the split seen at 0.005 in birth being the spacing's (FR1), and a ragged corner showed instead: 7.358 held between unheld 7.357 and 7.359, the fastest turn 0.017237 at 7.356	The place borne out; the outline mostly not

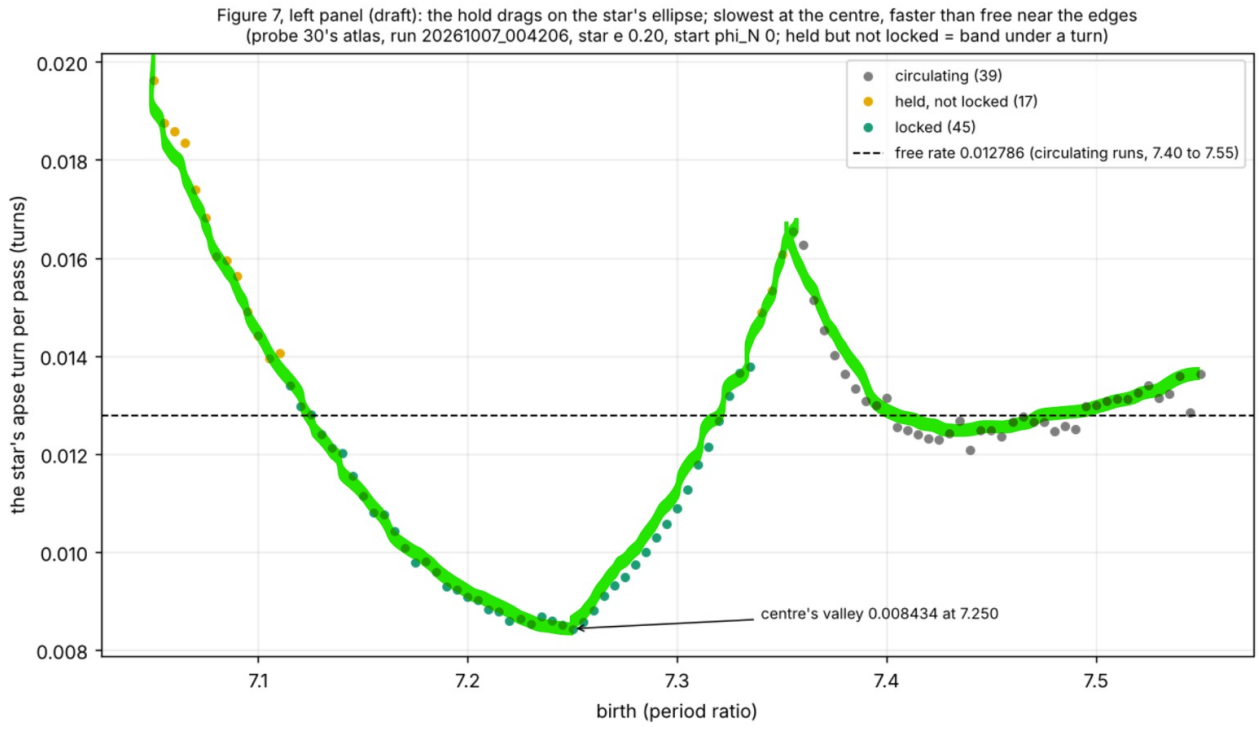


Figure 23. The shark's fin, as Bear traced it. His green trace on a draft of Figure 7 of *Held, Rocking, Locked*, left panel: the star's apse turn per pass against birth (probe 30's atlas, run 20261007_004206, the star's eccentricity 0.20), slowest at the centre's valley, 0.008434 at 7.250, and peaking in a point near 7.355, at the hold's upper edge.

Relations held better than shapes. Reads that named where one thing sits against another, which way something turns, or which side something is on held here more often than reads that named a shape (the V, the floor, the ledge). These are few cases: a guess, not a rule, and a question to carry to the next picture.

Relation beats shape, tested blind. The guess, Bear's own sense and the triple-star thread's, was then registered as RC1 before anything was counted: reads naming a relation should be borne out more often than reads naming a shape, by at least 0.2 in share, with at least eight reads in each class. It was blind by construction (run 20261017_040300). One agent listed the 74 registered picture-reads, most of them Bear's, with their verdicts, and a second, given only the reads' wording with every result word removed (21 quotes had carried one and were reworded), sorted them into relation, shape or neither. Relation reads were borne out 19 of 30 (0.633) and shape reads 11 of 28 (0.393): +0.240 against a bar of +0.2, a narrow pass. The sorter's note that about seven shapes might move to neither on a stricter reading was reported, and the count was left as registered. The shark's fin on nine lines (the table above) shows the same lesson from the other side: where the fin sits was a relation and held, and its outline was a shape and mostly did not. For reading with Bear, trust where his eye puts things before the outline it draws. Claude's own reads by eye showed the same from the other side: they held on mechanisms and failed on rarity and on rules for records, better at where and how things move than at how rare they are (row 616).

Bear's marks in one Loop Atlas session, registered in his words and tested. His marks on figures, his names and his predictions, each registered before its run and judged on the numbers, in the columns of the table of his triple-star marks above.

Mark or prediction	Kind	Result
His rings on a dip and on four flickers in the Loop Atlas’s spread panel, with a question about gaps	Place	Borne out: both were real, a window and a ladder’s rungs; the ruler of record saw the window and not the ladder
Rungs, his name for the Loop Atlas’s ladder	Shape	Borne out: evenly spaced on a log ruler to within a quarter (registered as B1)
Event timing more than the water’s memory, for the Loop Atlas’s tied bells	Size, as a ranking	Borne out: a swing of 0.63 across the period against 0.25 across the memory in the toy as first built, and 0.6381 against 0.2671 with the rings’ cores given their own size (run BELL_SOFT_MAP 20261021_000736; row 807)
The mass-1 passing lane fills the road (a sentence of Claude’s, marked by Bear)	Extent	Held to 1.09; missed from 1.10, past the map it was read from
A power gauge: the worn rider works harder	Sign	Borne out: the effort rises 3.05 times before the fall
A true rate sensor in the skater’s ear (his trade’s knob)	Size	Borne out: it recovers 88 per cent of the drift lost to looking every quarter cycle, against 45 per cent for a slope from the ear’s own two looks (run SLOPE_EAR 20261015_190935)
A master oscillator keeps the count	Sign and place	Partly: only if loud, and only at or below the bead’s working pace, three quarters of its tune (run BEAT_SWING 20261015_204207)
His pause ear, left, pause, right, pause, with a dead zone	Shape	Borne out as a knob: the ear latches, its push per switch rises at every widening, and its price never turns over (runs PAUSE_BEST 20261015_220655 and PAUSE_WIDE 20261015_223744)
The dive dream, its four claims	Several	Three borne out in the toy (gravity’s clock; no air needed for the turn; the mat for the shape part); one waiting (count or event, the Loop Atlas’s card 199)

What the column says, for the Loop Atlas. Bear’s marks on figures were right on place both times in one Loop Atlas session. On 14 October, on a window of the needle, his marks were right on place three more times: his red and green strokes on a density of chaos lie within 0.004 rad of the fold or its first eight images (0.741 and 0.673 of their pixels, against 0.36 and 0.37 set at random; read with a tool, its expectation printed before the run), his black arc is the fold’s third image, the curve that ends the window (0.604 of its pixels within 3×10^{-4} rad, against 0.076 at random; registered before its run), and his rule for the windows’ cycles, read off a drawing, gave $12 = 4 \times 3$ and $20 = 4 \times 5$ in one piece and $24 = 4 \times 2 \times 3$ in two, as measured (rows 707 and 705). His predictions and his names held on sign, and the one that held only in part, the master oscillator, held wherever the bead’s working pace allowed it. Where a marked sentence missed, the miss was in its extent, past the map it was read from, and that miss belonged to the sentence’s writer, Claude. Claude’s one reading of a picture in the same days, the clean lock, missed because its ruler could not see the thing that was there.

His remarks that became exact. In one thread of R2W Bear mostly looked at one figure, the self-mirrored dances of each fraction at its weightless root, and said what he saw. Each remark was turned into a registered test or a counted rule, and most became something exact: one failed as worded and became exact in another form, and two are still open.

His words	What it became	How it was checked
“This is order, not chaos … why do you think it behaves so precisely?”	The order in layers: forced by the Möbius map and the mirror, measured, argued, open	The thread’s seed document, and the headline section of R2W’s paper
“it has a repulsion quality to it”	Red and purple strictly alternate around the circle; two dances of a rotation in disguise cannot pass each other	READ-E1, nineteen rows
“counting left to right and if we round up, the red dots always match the denominator”	q stops per dance; his half dot is the seam at ±0.5; the numerator is how many stops the ball skips back each passing	READ-E, all 18 dances of the first nine rows
“the biggest jumps happen when nobody is dancing in the cent[re]”	Failed as worded at 3/5; became the window count q − p − 1 (fifteen of fifteen after the data, four of four predicted)	READ-E2, PIN7-1g, PIN68-1w
“stretched out grippiness at bottom”	Read as the bottom rows: the ball slips most of a turn each passing, and its only grip is the pair, 21 to 28 times slower than any other stop outside	READ-E’s strobe speeds; which reading he meant is open
“Does the dot coming around the circle change its frame?”	The loop’s multiplier is its holonomy on a neighbouring step: exactly 1 without weight, about 0.95 for red and 1.02 for purple at weight 0.01; the passing matrix to the q-th power is (−1)^p	Algebra, and Newton’s multipliers
“that record player is still a read head in my brain”	The drive as the radius, each groove a fraction, played in the order of the fraction tree	TONGUE-MAP, in an earlier sitting

His marks on R2W’s other figures were measured too. On a coin map at weight 0.40 his long lower fin, an echo of the main arc’s lower edge, was registered and run at 35,000 riders and ran parallel at 31 of 31 drives (COIN-3; row 706); his dipole-and-mirror picture held in kind and his staging ground in every arrangement measured (row 128); and of his two pencil lines on figure AA, read after the data and not registered, one sits exactly on the gear break and one is unexplained (row 573).

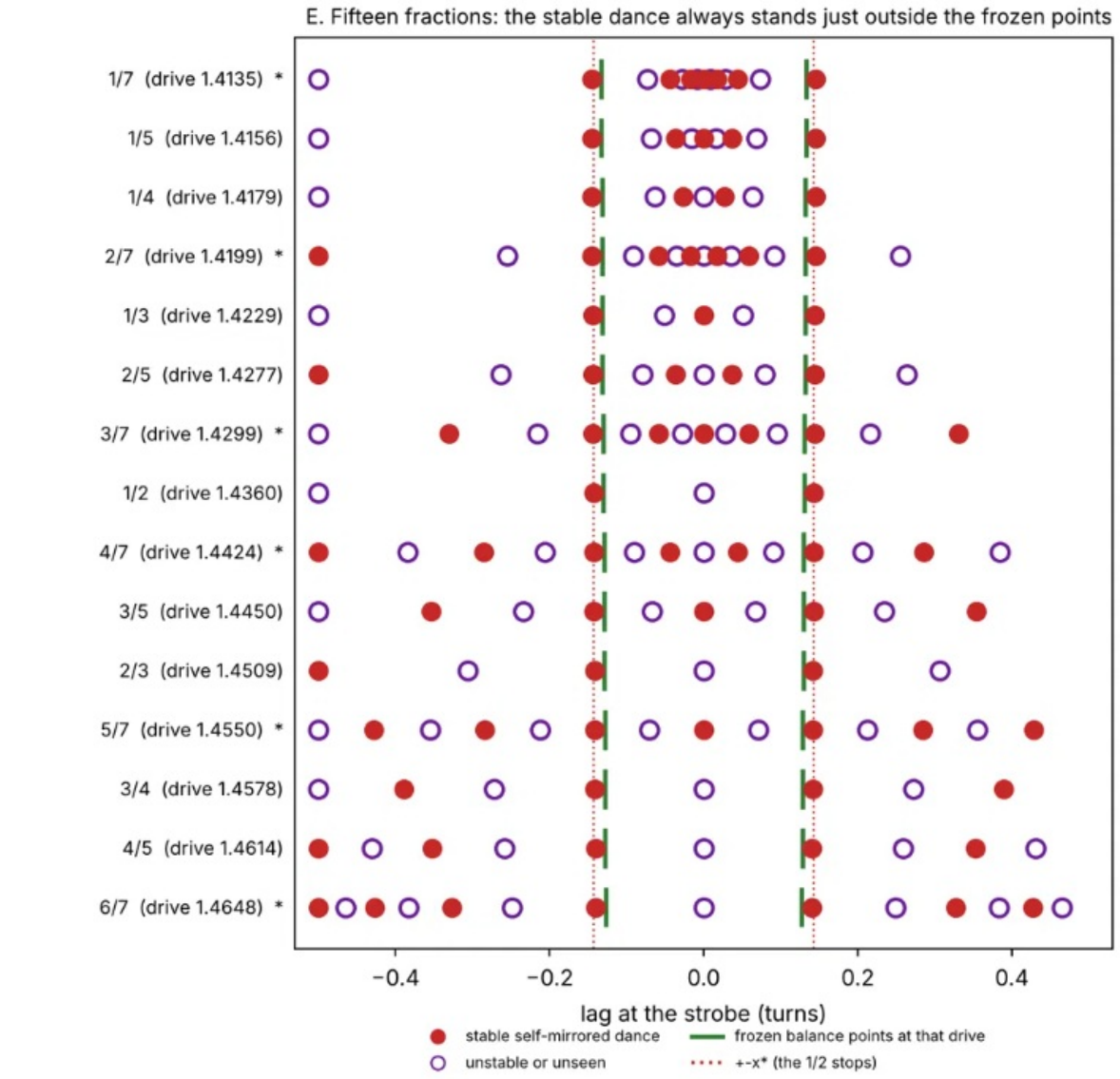


Figure 24. Fifteen fractions at their weightless roots. R2W's figure of its toy's self-mirrored dances, from 1/7 to 6/7, each at the drive of its weightless root (1/2 at 1.4360): red the stable dance, purple the other, green ticks the frozen balance points. Bear mostly looked at this figure in one thread of R2W, and the table above gives what each of his remarks became.

What the columns say. An eye, either reader's, proposes sign and place well, and the size belongs to the data. Claude's blind reads were good on geometry and weak on classes and on depth: they named the conveyor as the rest, and bet on an overpass that was a meeting. Bear's misses here were in the translation and in the bar, not in the seeing: lanes that no sensor could see, walls that beat 2,000 random sets on a comparison chosen after the run and fell under a strict bar, rows that missed by 0.015 and by one row. And the column of predictions against known answers is the one to trust most, because every row in it has a known answer.

Why sign and size are scored apart. The two sides of the sign-and-size lesson were reached on the same day from opposite directions. One project found the eye right on sign and wrong on size; the other registered a sign that was free and missed on it. That is why sign and size are scored apart.

10 — How to use it, and how not to

Reading pictures is a fast way to find claims and a poor way to prove them. Here are the rules for using it, the ways it goes wrong, and how to tell whether it worked.

The rules come in two lists, how to use the method and how not to, and end with its measure of success.

Use it as a generator of claims, never as the test. Every claim from either reader gets a number, from the data or from the picture's calibrated pixels, before it is believed.

Use it in order, the order of section 3.

Use it first on a picture whose answers are known. A method that has never been scored is a story.

Use the two cheapest rulers every time. Does the pattern survive the same dots shuffled? Does the ruler draw the same shape from a made-up input? The cheapest test is often the strongest (row 584).

Use it as a conversation between two kinds of eye. What both readers saw counts most. What only one saw becomes a test. When the person beside you reads a picture differently, write both readings down before either of you looks at the other's.

Use it at two scales and through two pictures. One scale and one picture are a single draw.

Use the marks themselves, not a description of them. A paper that describes the eye's way of guessing in words, without one mark shown, hides the method it describes. The triple-star thread's paper now sets its Figure 7 beside Bear's own working copy of the same strobe picture, squeezed sideways, tilted to a grazing angle and marked by eye on 14 October before anything was measured (triple_paper_figure7_marks v1.0, run 20261021_052614). His picture was taken by its fingerprint, 9c1034e2c9af4676, and left unchanged but for a working note lifted from its foot. Its readings had been registered as SQ1 to SQ3, scored in section 9, and VC1, the V, which was not borne out and was redesigned as the pinch grid. A picture he pasted later was matched pixel for pixel to his marks of 15 October on the ratio atlas (BEAR_RATIO_ATLAS_MARKED_2026-10-15.png, on run 20261015_140219) before it was read.

The don'ts are of three kinds, in this order: taking what the method finds as proof, reading a picture past what it can show, and bending the procedure.

Do not use it as evidence on its own. The eye proposes and the ruler decides. Claude's own eye invented a third hand in one bend and S-shaped rows in another picture, and the rulers took both away (rows 718 and 616).

Do not use it as evidence of new physics. When a shape matches a named result, the literature is the first stop, and the match is a check on the toy, not a discovery.

Do not let the eye speak for the model. Bear's rule for his own eye holds for every reader: "Don't let me speak on behalf of the geometry until the data bears witness."

Do not read a mechanism off a derived picture without its anatomy. Bear looked at a picture of twin lobes and saw vortices, propellers and mirrors. They were a corner. Taking the picture apart into a corner, a step and a bump showed something truer than either reading alone, and two of his five words had been right from the start (section 1).

Do not read someone else's picture without saying so. With a borrowed figure and no data, measure the picture, say that the numbers are the picture's, and keep the figure out of any public deposit.

Do not read across a wall no reading can cross. Read a picture's frame before its shapes.

Do not use it as a rulebook. The lessons are questions to ask quietly. In Bear's words, "Nothing about holonomy is square or harsh". A lesson that blocks a run is being misused.

Do not re-aim until something hits. Redesigning after a miss is right, and it needs registrations stamped before each run and the ruler check beside every verdict, or a loop of redesigns becomes a search for a hit.

Do not use a bar that cannot say no, and do not name a peak on a tie.

The measure of success. The shuffle and the made-up case are the reason this method can be trusted. Its measure of success is the one the lattice lesson set (section 2): not to find every shape, but to find a few new observations that connect to other observations.

Sources cited in this part

This section lists the books and papers this part leans on, for a reader who wants the originals.

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Reference card for this part

This card maps each section to the rows in the register it calls on most, and says what those rows price.

Section	Key rows	What they price
1 Why read pictures	715, 707, 702, 711, 714, 640	The eye keeps lopsidedness, negative space, walls and long reach; counts before derived pictures; a derived picture with its anatomy and a made-up case; draw plain for marking; two scales
2 The two readers	716, 705, 710, 717, 686	Hold back the data while the eye learns; the eye defaults to a fork; blind reads are weakest on depth and class; the ruler’s own step
3 The protocol	713, 369, 573, 666, 616, 620, 686, 714, 30, 711, 703, 712, 128, 313, 337, 553	Name the picture; declare and read blind; translate marks with their maker; search the record; the control inside the tool; what the step allows; shuffle with the statistic chosen first; the made-up case; a second scale; sign before size; read the build back
4 The toolbox	642, 709, 710, 708, 654, 648, 718, 719, 537, 743	What each tool assumes and how it fooled its makers
5 The field key	702, 703, 705, 706, 707, 710, 711	What each shape turned out to be
6 Traps	329, 622, 210, 631, 416, 664, 704, 683	Binning is where the information design happens; two thresholds that disagree are data; a ceiling can belong to the window
7 Cases	702, 703, 706, 707, 708, 709, 710, 711, 667, 719, 556, 726, 201, 715	The control with known answers; three readers on one picture; blind reads judged across projects; the eye in its own frame
8 The common object	680, 642, 654, 670, 697, 719, 558, 726	Area kept or lost; the count, the corner and the leftover; the outside tick, the rider’s own pace and the event; births by fold and flip; a named toy’s map is a bench
9 The scorecard	712, 313, 337, 544, 644	Sign and place apart from size; both facts printed; a bar that cannot fail
10 How to use it	584, 506, 616, 711	The cheapest test; prior art after the run; the two cheapest rulers

The Register

Every ornament earned on every project. Use the station reference card to find the rows a given station calls on most. Use this section when a station raises a question the Walk cannot answer.

The headline register: what was achieved goes on the front page

The guide is good at catching a result that is wrong. Until version 7 it had no habit for keeping a result that is right, and the work behind this version showed what that costs. Bear counts it at about thirty-seven papers whose results never reached the abstract, and his ruling is to fix it from here on: a result that is not on the front page has, for any reader, not happened.

What went wrong

Wins had no channel. Among the markdown files the three projects sent the guide from 15 to 17 October were 18 lessons files, 17 files of no-physics reads, 8 of pattern-recognition lessons, 5 reviews, 3 ledgers of guide items and 2 records of misses. Not one was a record of results. A miss gets a numbered row in a register the guide reads line by line; a prediction that held gets a clause inside a row. In the build of 18 October the word “held” appeared 472 times, spread across hundreds of rows, and nowhere were those results gathered.

The register teaches the brake and not the accelerator. In the same build, 33 of its row titles carried the words of limiting or checking a claim, among them “Claim the connection, never the gap” (row 130), “The novelty ranking, corrected downward” (row 177), “Size the new claim yourself, before anyone else can” (row 280) and “Name the stricter claim a reader would want, and decline it” (row 347). Two titles stated a headline (rows 215 and 542), and none told a session to collect its results; rows 821, 840 and 882, added since, do. Each honesty row is right on its own; together, with nothing on the other side, they train every new session to shrink what it found.

Size checks net results away. A size check that weighs one result against a whole field (“medium for engineers”, “small, but important to a builder”) is honest about that result and silent about the sum. Twenty small results reported one at a time read as twenty small results; listed together they can be a capability no one else has. Bear’s picture for it is a disability rating: many separate awards that add to well over a hundred per cent, combined by a rule that reports far less. Handovers make it worse. A session’s handover carries forward its lessons, its misses and its open work, and its results only if someone wrote them down, so a result that lived in one exchange’s reply is gone by the next session.

A known road met mid-project became the road. The triple-star thread began from scratch on 29 September. By its lessons file of 10 October a published stability paper (Mardling, 2013) is cited six times, and by its treasure chest of 15 October the thread asks whether its own escape door “is just the known boundary toward the pair, or Mardling’s overlap”. Bear’s reading of the finished draft is that later sessions measured the toy in that paper’s terms instead of measuring that paper against the toy’s own results, so the paper came to read as a companion to work it had started independently of. The toymaking rule (row 506) already says to meet prior art at the wall, not at the idea; nothing told a session what to do once the wall was met.

The rule

Keep every honesty row. Add the other half: **every result that held is a headline candidate, written in the register the exchange it lands, sized against the right comparison, and counted, never netted.**

- **The right comparison is the nearest known method or result in the same niche, with numbers**, not the field at large. “Ours against known” is a pair: the known value and its source, our value and our method, and the margin between them.
- **A benchmark matched is a result.** Reproducing a known number with a new instrument validates the instrument, and it is said that way, with the number.
- **Count, do not net.** Small results stay listed one by one, and the register’s count is reported beside its largest entries.
- **When a known road appears, measure it against ours at the wall, and say where our road went that the known one never tried.** Do not switch roads.
- **The top of the register goes on the front page:** the abstract, the record’s description and the opening of the paper. A result on page 53 might as well be invisible.

Size both ways. A front page can be wrong in either direction, and the triple-star thread’s islands paper met both when its front was rewritten to its register. Plain sentences are where overclaiming sneaks in: on 21 October three of its new section openings were changed when read against their own subsections (Section 6’s ‘for many thousands’ was not measured there, ‘the clearest one we found’ was a ranking the section does not make, and Section 7’s opening named two rhythms before its first subsection introduced them), and cold reviews found 26 items to mend on the headline page, among them a call that the locks are exact and a fit called a prediction. A tool now lists every sentence that has a claim word and no number (claim_words v1.0, 9a02ea5631941e79: 135 of 1,354 sentences in the

test of 22 October), and a message in the conversation is a claim too: a reply's 'to six decimals' was really about four, and the paper says 8.721 and 4.669. The read cuts both ways: set against their sections on 21 October, the front's sentences came out eight trimmed and two strengthened. Underclaiming is an error too. The headline audit of 18 October found that the register had logged verdicts and never weights (each landed result's size check, how big and what is ours; row 868), collected 21 section headlines no one had gathered, and found the known road above at work: crediting Mardling had turned into framing through three of Claude's sentences, under no decision of Bear's, though the core was the thread's own by 8 October. The register's Part D now keeps weights both ways: what is the thread's own as far as the literature passes went, what is known and confirmed here, and the corrections to the size of headlines. And a count can be worded against its own numbers: Appendix D's 'Most guesses miss' stood against 208 landed and 194 missed, and became 'Many'. Read each new plain sentence against the section it opens, keep the weights beside the verdicts, and word every count by its numbers (the triple-star sessions)

The register a session keeps

One file per project, opened on the first day, carried in every handover and in the project's final crate to the guide. Its columns:

No.	What held, in one line	Run and date	Compared with (known method or result, and its source)	Ours against known (numbers)	Kind	Who it matters to	Front page?
example	Four tests tell a true nothing from two hands cancelling when a knob has no sign	ARRAY-1, NINE-1, LOBE-1, RATCHET-63, RATCHET-64, FLAT-75, FLAT-50	the reversal technique, which needs a knob with a sign	a symmetry's nothing at 6.994×10^{-15} against one rider's 2.599775×10^{-2}; a balance placed blind within 0.04% and 0.006%	method, extends a known one	anyone reading a null on a knob that cannot be flipped	the record's description

Kinds: first measurement; benchmark matched; benchmark beaten; known result reached by a new route; new connection between fields; engineering knob with its setting; tool. A row is written when a registered prediction lands or a result is read from data. It is never removed for being small, only marked superseded, with the run that superseded it.

At the close of a paper the abstract is drafted from the register's top rows before anything else is written, and the record's description lists them. The guide carries each finished project's register in a table of its own, with the project's concept DOI, so a reader of the guide can see what the method achieved and not only how it is done. All three projects had published when this version was built, and their tables follow the counts below; the two-voices papers, published after them, follow the three.

The three projects' registers at their close

Each project counted its register at its move to final publishing on 22 October and went on registering and counting after it, so each count here carries its date. At its move the triple-star thread had registered 444 predictions before their runs: 208 landed, 194 missed and 42 could not be read, and 31 of its 80 redesigns of earlier misses landed (counted by script over the whole thread, runs 20261019_062115 and 20261022_202112); at its paper's publication on 28 October it had registered 490: 236 landed, 211 missed and 43 could not be read, and 35 of its 92 redesigns landed (run 20261028_210006). At its move the Loop Atlas had judged 1,064 registered predictions, 712 held, 323 missed and 29 undecided, among them 990 distinct statements of which 668 held, its session G alone judging 109 of them, the count going from 955 to 1,064; at the close of its session I on 31 October it had judged 1,106, 736 held, 341 missed and 29 undecided, among them 1,031 distinct statements of which 691 held; and at its move to session K on 6 November, 1,143, 756 held, 358 missed and 29 undecided, among them 1,068 distinct statements of which 711 held. At its book's publication on 9 November it had judged 1,104, 706 held, 369 missed and 29 undecided: in its session K the register sorted every verdict line by its own tool's word for it, and 75 lines the tools call fidelity or rig checks moved from the predictions to the planted checks, which number 572, 555 held. Its 584 runs of record, 386 of which judged a prediction, are printed whole in its record, *The Loop Atlas: the record* (concept DOI 10.5281/zenodo.24100980), with its 656 misses, each with its reason. R2W's whole-record count of 19 October graded 402 tests against predictions written first: 331 landed and 71 missed outright; at its move its headline register held 202 landings, 62 misses or voids, and 39 misses redesigned into a landing. Its whole-record recount of 1 and 2 November first reproduced its count by eye of 24 October, which stops at 22 October (425 graded tests, 352 landed, 73 missed outright) before reading a later line, then found 515 graded tests to 2 November, 436 with at least one

prediction landed (214 with every prediction, 193 with some and 29 only through the named rival) and 79 missed outright, and its book states that count; at its move of 2 November its headline register holds 365 landings, 275 of them predictions written first, and 161 misses or voids, 87 of them redesigned into a landing; at its book's publication on 5 November it had graded 527 tests, 448 with at least one prediction landed (223 with every prediction, 195 with some and 30 only through the named rival) and 79 missed outright, by its counting kit's run on the published extract, and its register held 386 landings, 296 of them predictions written first, and 173 misses or voids, 95 of them redesigned into a landing. The three count different things, a prediction, a judged statement and a graded test, so they are not added; each is its project's own.

The table below is the triple-star thread's register at its close, for *Held, Rocking, Locked*, published on 28 October (concept DOI 10.5281/zenodo.23755473), which follows the islands of a body a millionth of either star's mass round a coplanar pair of equal stars at eccentricity 0.10 (scanned from 0.05 to 0.25), near the 7:1 and 8:1 resonances, a circumbinary planet's setting. It is drawn from the thread's headline register of 18 and 19 October and its closeout of 29 October; its count is given above.

What held	Ours against known	Kind	Run; rows
Mardling's resonance pendulum, tested piece by piece	her relation between the rocking rate and the island's width within 1.1 per cent at the pair's eccentricity 0.10, failing from 0.15; her widths 20 to 31 per cent too wide; the centre's rocking quickening 1.435 times from the pair's eccentricity 0.05 to 0.20, where her pendulum gives 1.98 (Mardling, 2013)	test of a published prediction	20261008_133322; rows 506, 678
A one-sided door past the stability limit	with the star's eccentricity at 0.20, 163 of 166 escapes leave the island on its side toward the pair, where the three published stability rules call the whole resonance unstable (one) or stable (two); started circular, none of 82 runs escapes	new in its details, as far as the passes went	20261016_182117, CS1 20261018_201130; rows 599, 863
Slow chaos under short islands	at the pair's eccentricity 0.25, 22 of the 28 births held over 600 passes lose their hold within 6,000, and every pair of twins parts exponentially; stickiness predicts such look-alikes	a known kind, measured here; method	20261013_152519, 20261014_090328; row 732
The edge's clock	the time to slip grows 7.57 and 7.37 passes per decade of distance at two clean rims, against 7.29 and 7.83 forecast from the island's own width with nothing fitted; a damped lock's square-root law would have grown the slip period 4.5 times over the span, and it grew 2.4	test; knob	triple_door2_edge_law 20261006_201311; row 675
A frictionless lock rests centred	every locked run rests within 1.1 degrees of centre over swings of 0.06 to 0.46 turns, where a damped lock's resting angle slides to a quarter turn at its edges	test; knob	row 675
The drag turns the ellipse back	the resonance holds back the turning of the star's orbit in a straight line with its grip (R^2 0.958); the centre's ellipse stands still at the star's eccentricity 0.2811, registered	test and new	rows 450, 661

	at 0.2817, and turns backward beyond it		
One doubling, then escape	island 7’s centre doubles once and every star round it escapes within 1,600 passes; the bench gives the area-keeping cascade’s 8.721082 on a known map (Greene, MacKay, Vivaldi and Feigenbaum, 1981) and the damped one’s 4.669197	new and test; benchmark matched	probe 32 20261007_224303, stage A 20261007_051341; row 680
What the arithmetic fakes	at a relative tolerance of 10^{-9} every unlocked rocking rate falls about 2.8×10^{-8} cycles a pass in every pass, faking a lock’s slide and part of its wear; 100 times tighter leaves 0.0093 of it, and the energy’s drift falls by the same powers per tenfold tightening as the creep left in the rates, 1.46 and 1.92 against 1.41 and about 2.0	method; a flag for a numerical analyst	FA0 20261018_031414, DR1 20261018_031349, EN1 20261028_140844; rows 741, 858
Verdicts on a second machine	the long runs made again from nothing on a laptop: of 103 lines, 83 the same, 20 the same verdict with other digits, none different	method	XC1 20261028_132326; row 810
Reads by eye	reads naming a relation held 19 times in 30 (0.633), reads naming a shape 11 in 28 (0.393): +0.240 against a bar of +0.2, a narrow pass	method	RC1 20261017_040300; rows 100, 714

The table below is R2W’s register at its close, for *A Rider in Two Waves* (*Different music, the same moves*), published on 5 November (concept DOI 10.5281/zenodo.23999853), which reads a damped pendulum pushed by a wave, and the standard model equation of a Josephson junction run on the same bench, with a stroboscope that counts. It is drawn from the book’s own list of what it achieved and its abstract; its count is given above. A dash in the last column marks a result no row of this guide yet tells.

What held	Ours against known	Kind	Run; rows
A law with nothing fitted for how a locked pair of dances answers an extra tone	the move $a \cdot \sin(2\pi x)/(2\pi\, n^2\, \omega^2)$, the tone’s size times a sine of the pair’s place, divided by 2π, by the harmonic number squared and by the drive’s frequency squared; the first-order theory, worked out after the moves were measured, reproduced all 56 of them (harmonics 1 to 8 at seven dances) within 0.00004 of a turn, and the tenth and twelfth harmonics, written down first, landed 14 of 14 within 0.000015	law; test	KERNEL-1, EAR-2, EAR-3; rows 699, 737
One gearbox in two media	each extra step of a dance multiplies its extra grip by the same factor, in the toy and in the junction: blind at 8 of 8 rows in the toy and 4 of 4 in the junction; one curve fitted on 28 rows predicted the stretch no run had visited in both within 0.017, and the toy’s bend within 0.0073 where the plain power	new connection between media	TOY-SIDE-3g, 4g, JJ-SMALL-1c, GAP-1, TOY-BEND-1, TOY-WALK-1; rows 79, 649, 742, 840

	law missed by 0.10 to 0.18		
A selection rule and its break, each predicted	a touch of weight keeps the dance that leaps across the far side (19 of 19 toy fractions, 44 junction roots, 48 junction settings); counting the radio wave’s quanta says one tone can never flip that choice and two can; the grip’s power read blind as 1.06 to 1.07, 1.95 to 1.98 and 2.78 to 2.89, closing on 1, 2 and 3; the flip moves as the strength squared within 3 to 6%, came where predicted, and held under real weight to four figures at three weights	test; knob	FAR-1, PINCH-1, PINCH-2, WALK-1; rows 106, 308, 666, 686, 736, 739
Exact counts from the toy to the junction	a lattice count built on the toy and registered for the junction agreed with 205 of 205 junction rows; the window count, found after the data, then landed exactly at four new fractions	count; test	LATTICE-JJ-1, PIN68-1w; rows 514, 821
The junction’s grip has one knob	the grip follows the wave’s strength over its frequency squared: predicted blind where the rival disagreed, held at ten fresh settings within 0.063; in the strong, slow corner, where the straight line fails, one curve predicted eight new settings within 4.5%	engineering knob with its setting	JJ-4a, JJ-GRID-1, JJ-CORNER-2; row 541
The grip forgets and the slip remembers	written down before the read: at weight 0.05, 99.1% of 110 locked runs had a stretch rate below −0.005 and all 101 slipping runs sat within 0.005 of zero; at weight 0.25, 100.0% and 99.0%	first measurement	MEM; rows 686, 813
A weight-proof setting for a two-tone rectifier	at one drive ratio the window moves 60 times less than on the usual setting at weight 0.04; one number, the weight times the frequency, sets how far that holds: predicted before each run, the band ended at 0.2446 against 0.2445 (frequency 0.40), 0.1395 against 0.138 (0.70), 0.11475 against 0.11508 (0.85) and 0.09742 against 0.09743 (1.00), at one value of the weight times the frequency within 0.44%	engineering knob with its setting	RATCHET-26, 30, 65, EDGE-85, EDGE-100; rows 731, 801
A dark point placed by a symmetry	the mirror pins the dark point to one timing, and runs placed it within seven parts in a million of the prediction; its sibling, the flat point, placed blind from four runs of a small dial within 0.16%, at fast drives within 0.14%, the dark points within 0.012%	test	RATCHET-53, 63, 64; rows 400, 581, 672, 786, 794, 799, 805
A weight meter that drift cannot fool	the drive’s drift predicted to move the push by 1.8607×10^{-6} and measured at 1.862829×10^{-6}, the weight’s channel	instrument; test	RATCHET-71, WITNESS-1, RATCHET-72; rows 75, 106

	exactly blank; a second engine written from scratch agreed to 5 parts in 10 million of the move’s size		
A whole number inside the rectifier	the timing that gives no drift makes exactly one half-turn along the weight knob at all nine settings run; its heavy end, 181 degrees divided by the weight times the frequency, predicted from one frequency, came out 183.2, 180.4 and 179.3 at three others	test	RATCHET-8, 11 to 13; —
A fade predicted to the fourth decimal	the rectifier’s height fades nearly as the weight to the minus fourth; the next rung’s power, written down as −3.977, came −3.9771 at weight 24.5, and at all seven heavy weights from 8.2 to 24.5 doubling the second tone doubles the height to one part in ten thousand	test	RATCHET-80; row 821
Bear’s coin	a rider lands heads or tails by where it is set down and nothing else, along an edge between the faces that stays ragged however closely it is examined, with an uncertainty exponent of 0.592; a second engine written from scratch, sharing only the equation, flips every one of the coin’s 1,800 riders of record the same way	first measurement; a second engine’s check	the book’s §4.17, WITNESS-COIN-1; —

The table below is the Loop Atlas’s register at its close, for *The Loop Atlas* (*A loop comes home to where it started. What did it leave behind?*), published on 9 November (concept DOI 10.5281/zenodo.24100672) with its record, *The Loop Atlas: the record* (concept DOI 10.5281/zenodo.24100980), and its note on the odd-order constants (concept DOI 10.5281/zenodo.24101154), which follows one question across toy machines and bodies: when a loop comes home to where it started, what did it leave behind? It is drawn from the book’s own list of what it found, in its record’s description, each result read against the row of this guide that holds it; its count is 1,104 registered predictions judged, 706 held, 369 missed and 29 undecided, with 572 planted checks beside them, 555 held. One result that list gives, a listening blade’s drift of 0.124 in an even soup of waves against a clock-switched blade’s 0.009, is left out: the book bills it as a toy that did not pay for its charge, and billed honestly the pull comes only from a difference in push across the blade’s length (row 590).

What held	Ours against known	Kind	Run; rows
Fifty-one meetings with the published record	from Feigenbaum’s ratio, met at 4.66920159 for a rounded top, to Kane and Scher’s falling cat and Tong and Dullin’s diver, the diver’s model rebuilt from its own table within 0.02 to 0.15 per cent once one entry their table most likely drops is restored (theirs to confirm)	benchmarks matched	TONG_DULLIN_SCAN 20261017_181953; rows 771, 814, 818
One cat under two rules	a cat that twists at its waist and a cat that turns twist-free turn alike, at every shape, through a fixed half turn of the rates, and on every path: within 2.842×10^{-13} degrees on nine swelling paths	new, claimed on a limited, noncommercial review of the citations and references	rows 820, 836
A widely copied table’s odd entries	the published δ for maps of odd order is off at every odd order and right at every even one and in α: 5.9679687038 for the maps’ own 6.0846890743 at order 3, 8.3494991320 for 8.3443847487 at order 5; one sign may explain the difference	correction of a published table, as its own record	rows 874, 875
A rider who listens works on any swing	timed by the swing’s own phase, every rider on swings of 0.5 to 12.5 m passed 90 degrees within 12 of its own swings; reading by the wrong rate has a price that can be written down	not found in print in the pages reached; knob	row 681
Smoke rings that leapfrog	given exactly a leapfrogging pair’s budget, one ring travels at 0.84 to 0.88 of the pair’s speed, so on one budget the pair is 14 to 19 per cent faster	first measurement on one budget	RING_BUDGET 20261004_080107; row 788
A maintenance alarm read where nothing hides the wear	at equal catch, one fixed level of a balancing rider’s own lag threw away 0.031 to 0.051 of a life against the calendar’s 0.250 to 0.333	engineering knob with its setting	RESTED_LAG 20261016_205504; rows 636, 746
A diver cued by the body	a kick-out cued by the body’s angle missed by 0.420 of a clock-timed one’s miss, a number predicted before the run	test of a prediction	DIVE_CUE 20261017_082454; row 581
A latch that holds until the push turns	above an edge of 1.5 of the push’s typical size a latch and a memoryless crest-firer switch exactly as often, and the latch pushes 1.36, 1.51, 1.75 and 1.82 times as much at edges of 1.5, 2, 2.5 and 3	engineering knob	LATCH_EVENT 20261015_232643; row 669
An ear that stops where its own ring dies	a spring bead’s listening ear stalls at critical damping, sharply: 0.054 to 0.175 of the soup’s push at 0.95 of critical damping, and exactly nothing at and above it	first measurement	row 753

The two-voices papers' registers

Two papers of the author's two-voices series, on what a learning mind keeps when what checks its work cannot be trusted, were published after the three projects closed. *Question the Substrate: Why a Learning Mind Can't Grade Its Own Homework* (published on 9 November; concept DOI 10.5281/zenodo.24094163) ran 41 experiments on a small network that counts how many times a loop winds about a point, a task with an exact checker, so every checker under test could be scored against the truth; it registered 212 predictions before their runs: 122 held, seven of them with a qualifier, 89 missed and 1 held in part. *Two Arrows: Why a Learning Mind Heals Its Answers Long Before Its Self* (published on 10 November; concept DOI 10.5281/zenodo.24120415) registered 170, counting one independent run registered and carried out in the first paper's sessions: 87 held and 83 missed, and 4 of its 11 redesign re-checks held; the independent run held 6 of 6. Their lessons are rows 883 to 891 and additions to 20 earlier rows. The first table is drawn from the first paper's close-out, its lessons each with its run and its after-action review; the second from the second paper's headline register. A dash in the last column marks a result no row of this guide yet tells.

The table below is *Question the Substrate*'s.

What held	Ours against known	Kind	Run; rows
A mind's own certainty is a rich-get-richer filter	keeping the surest half of its own labels lost 25.3 points in world 1 and 12.9 in world 2, and an answer holding 11 per cent of the data went extinct, 10.6 per cent to 0.0, while overall accuracy barely moved; in world 3, tidy clusters, the rule came second best (+3.6)	known at the scale of large language models; measured here against an exact checker	E1e, E1w2, E1w3, E14; rows 280, 883
Every failure grades itself, and every cure is an outside reference	the same half-size cut, ranked by the frozen first network instead of the chain, lost 2.1 points against 25.3 and kept an 11 per cent answer alive (survival 0.83 against 0.02); a twin taught once from the truth and then held fixed kept the chain level (−2.5)	test of the paper's spine	E41, E35; row 77
Who ranks matters more than how many are kept	the same keep-half rule lost 25.3 points when the chain ranked its own labels and gained 2.7 when a strong outsider ranked them	test	E26; row 77
Buy any real check	a checker that catches a quarter of the errors earns 40 per cent of the benefit, one that catches half 68 per cent; even a half-blind, wasteful checker beats the mind's own certainty, by 28.6 points in world 1 and 10.9 in world 2	engineering knob with its setting; the knobs are ratios	E7; row 884
Every listener is deaf to its own echo	the student's batch of its surest labels pulled 0.004, about 50 times quieter than any other; a strong outsider ranked batches by their true share of wrong labels at +0.829, against +0.213 for the student	first measurement	E8, E8b; row 884
One knob makes it causal	tidy clusters gave +3.6 with equal answers and −14.7 with unequal ones, the two smallest answers falling from 12 to 0.7 per cent and from 8 to 1.1	test	E15; row 584
The field's known fix, gauged both ways	selecting within each answer removed the survival cliff, then over-predicted the small answer, 1.17 times its true share at 12 per cent and 2.51	test of a known fix	E22; row 883

	times at 3 per cent, while accuracy rewarded the extinction (+2.1) over the rescue (−0.4)		
A tipping zone	ten starts at a 1-point step found a band between 14 and 12 per cent of the world where identical setups live or die by their start; the end-state switch shows the same between size ratios of 0.61 and 0.64	first measurement	E31, E37r; row 886
A noise floor’s fingerprint	the aimed fix’s deaths matched $(1 - p)^{30}$ start by start; the survival cliff moved about 0.8 point of share per doubling of the data, so it is mostly not a noise floor	test; method	E38z, E39, E39b; row 218
A copying chain’s restoring force	a chain that held level for six generations slid 7.9 points at a leak of 5 per cent of its labels a generation, and absorbed a leak of 0.5 per cent entirely	first measurement	E25, E34; row 887
A contradiction settled by running it	a published rule whose abstract seemed to contradict the paper, confidence filtering that prevents collapse (Fu and colleagues, 2025), held when run in the paper’s chain: its confidence belongs to a frozen base model, a fixed outside reference	known result reached by a new route	E41; row 77

The table below is *Two Arrows*’.

What held	Ours against known	Kind	Run; rows
An area law in a learner	after a closed loop of tasks, the part of the leftover that flips with direction doubles when the enclosed area doubles: log-log slopes 0.9545 to 1.0092 against the loop’s radius squared, five of five seeds, and 0.9503 to 1.0067 in the hidden layer	new in this setting; the closed-loop form of the lesson comma	s1; rows 419, 644
Answers come home; the inside does not	after 30,000 practice steps the answers keep 0.0167 of their change and the hidden layer 0.3049, so the answers come home 18.49 times better (the mean of five starts’ ratios)	first measurement	r3; row 888
Only a kept record of the self restores it	a pull toward a record of the hidden layer from before the loop leaves 0.0003 of the inner mark; the true answers 0.3049, its own answers recorded on return 1.7621, its own current answers 4.6840, a stranger’s 5.6064	first measurement	b2; row 889
Hold on after the trip, not during it	holding the inner record throughout the journey follows the moving task 1.382 to 2.156 times worse; switched on only after the trip it leaves 0.00019 to 0.00085, at no cost to the	the stability-plasticity trade-off, known in kind, with a number	b16, b17; row 889

	journey		
A forecast replaces 30,000 steps	linear relaxation, with no practice run, within 25 per cent in 37 of 45 cases, median ratio 1.003; re-linearised in segments, the two hardest cases within 25 per cent in 9 of 10	instrument	b7, b18; row 259
Correct from the first map	a lineage corrected against the first generation’s map stays level for twelve generations (largest over smallest 1.02 to 1.06); one corrected against its parent ends 4.97 to 7.83 times higher	first measurement	b11; row 308
The yardstick decides whether drift is heard	against a kept copy of home, the healed mark is 8.888 to 9.876 times its home gap on inputs three times wider, 6 of 6 predictions held (registered and run in the first paper’s sessions); a later run resized it: a soft random change of the same home gap is about as loud	test; resized	bridge_b1, b12; row 884
A relabelling, not a drift	half a turn on a direction-blind task exchanged two hidden units; renaming them removes 68 to 85 per cent of the distance from home, and the exchange is one event, just past a quarter turn	first measurement	b4, b5; row 890
A real outside anchor bounds identity drift	after 12 laps: no anchor 147.070 degrees, a real anchor 3.106, a yes-man and a mirror identical to none, an internal copy 146.884	test	q3; —
Flat worlds heal, curved worlds keep the area	a rigid partner on a circle drifts at most 2.226×10^{-14} degrees; on a globe the drift equals the enclosed solid angle within 1.4023 degrees, 0.036779 with the exact rotation rule	known geometry, reproduced	q2, q2_redesign; —
Right after a crisis, subtract the slow arrow	one thinking round: the best mix, at −42 degrees, scores 0.7775 against 0.4426 for the outer arrow alone	a lead compensator, known engineering	q5_redesign; row 633
The one command, rerun in a fresh copy	after one runner fix, 148 files compared: data and figures byte for byte, printed output apart from timing lines	benchmark of the paper’s own crate	rerun of 10 November; row 862

A specimen from the guide’s own session

On 17 October the guide’s publishing session called R2W’s lesson 178 “a solid addition to the toolbox, not a big moment for the guide”, sizing it against the guide as a whole. Sized against its niche it reads differently: four tests that separate a true null from a cancellation for knobs that cannot be reversed, where the classical reversal technique needs a knob with a sign, each test carried by numbers from named runs (row 805). That is a method result, and it belongs in a register.

What kept arriving

Three projects, the triple-star sessions, R2W and the Loop Atlas, ran side by side, each on its own machine, and each sent this guide its lessons. Some lessons arrived again and again, from projects that never shared a run. Those repeats are this guide's own finding: a lesson that keeps arriving on its own is a lesson about the work, not about one machine. So from 6 October the guide counted them. Each family of lessons was defined in one sentence before it was counted, so that the count cannot drift with the definition. An arrival is a project meeting the idea in its own run, on its own system; a lesson relayed through two letters is one arrival, and a duplicate file is none. And each arrival says whether its session had read the guide first, because a lesson found before reading weighs more than one found after.

Rank	Family	What it says	At version 6	Counted by 10 October	Counted after	In version 7's fold	Found blind	Projects and papers
1	**A check that cannot fail**	A test whose outcome was guaranteed, or whose bar the data could not miss, says nothing by passing.	2	21	110	131	2	5
2	**The built artifact is the evidence**	An edit or a build is a claim until the finished artifact is read back.	0	15	75	90	1	6
3	**A word that changes meaning across a boundary**	One word or symbol carries two denotations across a boundary, and the claim rides on whichever was assumed.	10	14	54	68	0	7
4	**Chaos and binning**	A system near a boundary between bins, locks or levels flickers across it, and the apparent signal belongs to the boundary, not to the physics.	12	12	51	63	3	3
5	**A second reader**	An independent re-reading of the same files finds what the author's own checks were blind to.	not counted	15	35	50	0	4
6	**Coupled clocks and event time**	A loop keeps time by its own events, and when two such clocks are coupled, what is measured depends on which clock reads it and on how nearly the two agree.	one vocabulary entry	17	20	37	2	3
—	**The eye proposes, the ruler decides (proposed)**	A reading of a picture or a pattern by eye, a person's or a model's, is a proposal until a measurement that could have failed settles it.	not counted	—	29	29	0	3
7	**The mirror audit**	Run the mirror on the pipeline, and read what flips and what stays; a symmetry operation tests handedness more sharply than randomising.	6	12	16	28	1	4
8	**Count the right unit**	A count's unit must be the independent thing: runs not episodes, time not episodes, landings not stumbles, net crossings not raw ones.	not counted	11	14	25	1	3
—	**A law carried past its range (proposed)**	A fit or a law measured over one range is used outside it, and the claim rests on a regime the measurement never saw.	not counted	—	22	22	0	3
9	**The integer diagnostic**	A count must be able to come out only whole; a near-integer is a measure until it is read from closed pieces.	3	11	8	19	1	4
10	**The knee**	A switch or counter that commits only past a margin holds its choice, and a round trip costs the area between the two thresholds.	0	7	11	18	0	3
11	**Rate over level**	The signal is in the rate of change, not the value.	5	1	1	2	0	2

The families, ranked by their arrivals in version 7's fold, from 6 to 23 October; families with equal counts share a rank. Version 6's counts are only a baseline: four of them are a judgment or a sample of the register. Found blind: met before the session had read the guide. Projects and papers: how many separate sources the fold's arrivals came from. The two families marked proposed were defined on 23 October and are counted from that day's sort, not ranked; coupled clocks, proposed on 7 October, was accepted on 22 October and is ranked from 23 October. Every arrival is listed, family by family, in the appendix after the ledger.

What the ranking shows. The lesson that arrived most often is not about any machine's physics. A check that cannot fail arrived 131 times, from five projects and papers: bars set below the instrument's own floor, controls that passed as readily as the guesses they were meant to test, and tests that a theorem or the run's own setup had already decided. Second comes the built artifact is the evidence, 90 arrivals. The two commonest lessons are both about the tools of checking, the test and the build, before they are about the system under test. Third, since 23 October, is a word that changes meaning across a boundary, 68 arrivals, which also came from more projects and papers than any other, seven. Chaos and binning, 63, is fourth, and it is the family that arrived most from the physics itself: one session of the Loop Atlas met it nine times in three days, in its needle, its long snake, its rings, its spring bead and its counters of crossings, each time a count or a threshold that sat on a boundary and read the boundary instead of the toy; and where the family's own question was put to a test, the chaos on the borders between R2W's treads came out the same at three chaos thresholds, structure and not binning (row 655). It also carries the strongest sign that these lessons are found rather than taught: 3 of its arrivals came from sessions that had not yet read the guide. None of the 446 arrivals counted after 10 October was found blind: 445 came from sessions that had read the guide, and for one, the Loop Atlas's arrival on the jellyfish paper's swimmer (row 769), it is not known, so from here the count measures how often a lesson is met again, not whether it is discovered. Rate over level, which version 6 had counted five times, arrived twice in the fold: Bear's lattice taught it to three sessions at once, so it was counted once (row 708).

Two families defined on 23 October. The ranking of 17 October promised a sentence for each lesson the readers kept meeting outside the families before the next count, and the two they had named most often got theirs on 23 October, before that day's sort. They are counted as proposed families, as coupled clocks was. A law carried past its range (a fit or a law measured over one range is used outside it, and the claim rests on a regime the measurement never saw) arrived 22 times, meeting a forecast registered on 17 October of at least 20 arrivals from at least three projects, and the eye proposes, the ruler decides (a reading of a picture or a pattern by eye, a person's or a model's, is a proposal until a measurement that could have failed settles it) 29 times, each from all three projects. Both were counted in the lessons placed since 17 October and in the 37 earlier units whose readers had flagged them, so each count is a floor: an earlier unit no reader flagged was not read again.

How it was counted. The count of 10 October was made by the session that gathered version 7, arrival by arrival. Everything placed after it, 1,527 lesson units, R2W's and the triple-star thread's lessons of 16 October, R2W's and the Loop Atlas's of 17 October and the three projects' sends at their move to final publishing on 22 October among them, was sorted by fourteen fresh readers against the families' own sentences, each arrival backed by a quotation from its source that a script then found word for word; 46 of their entries were arrivals already counted. Where one event reached the guide through several records, a lesson, a record of misses, a letter, it counts once (a further reader found 37 groups of entries that were one event, and on 23 October 4 more pairs of entries of one family naming the same run were joined): 446 new arrivals from 607 entries, and 582 in the fold. One arrival counted on 17 October was withdrawn on 23 October because its own project had withdrawn the reading behind it: the Loop Atlas's tied bells read as choosing between two answers, which its session G found was the toy's thin cores (row 807). One more reader sorted 60 of the first 801 units, drawn at random, without seeing the first readers' answers. The two agreed on 85 per cent of them (Cohen's kappa, which discounts the agreement chance alone would give, was 0.67), and wherever both saw an arrival they named the same family, 16 times in 16. The second reader was stricter about what counts as an arrival, keeping 16 of the 24 sampled units the first readers had marked; with the later arrivals scaled down by that rate, the ranking keeps its order.

What fits no family yet. The readers also met lessons that recurred and fit none of the families. The commonest named on 23 October were testing every scale a toy carries (6 of the Loop Atlas's units), a later finding reaching the claims already written (5 of the triple-star thread's), and keeping a register of what landed (4, from the Loop Atlas and the Field Guide sessions); earlier rounds had named a session's files as its only memory, a finish line and what waiting costs, and a floor or a control measured before the law. They were noticed while counting, not defined before it, so they are not counted here; each gets its one sentence before the next count. From here a second example of a lesson adds a count to its family and a line to the appendix, not a new row (row 798).

How the lessons travelled, and where they live

Where each lesson lives. Every row of the register and every arrival in the ranking carries one to three domains from a fixed list of sixteen, written before any tagging so that a tag cannot drift with its reader: three body, driven oscillators, maps and chaos, kinematics, waves, quadrature, geodesy, solar, biology, number theory, materials, control and circuits, topology, reading pictures, records and tools, and measurement. The appendix “The register by domain” lists every row under each domain, so that a reader, or a reader’s AI, can start from the domain it knows and walk out from there. A lesson that keeps arriving in new domains is a lesson about the method, not about one system, and the families differ most in how far they reach: a word that changes meaning across a boundary arrived in 15 domains and a check that cannot fail in 14, while the built artifact is the evidence, though it arrived 90 times, stayed almost entirely inside the projects’ own records.

Family	Domains its arrivals reached	The commonest three
A check that cannot fail	14	measurement (126), driven oscillators (42), records and tools (32)
The built artifact is the evidence	8	records and tools (89), measurement (12), driven oscillators (3)
A word that changes meaning across a boundary	15	records and tools (38), measurement (25), driven oscillators (24)
Chaos and binning	13	measurement (57), driven oscillators (29), maps and chaos (18)
A second reader	9	records and tools (29), measurement (24), three body (13)
Coupled clocks and event time	10	quadrature (19), driven oscillators (19), three body (14)
The eye proposes, the ruler decides (proposed)	7	reading pictures (23), three body (13), driven oscillators (11)
The mirror audit	11	driven oscillators (17), measurement (14), kinematics (5)
Count the right unit	10	measurement (25), driven oscillators (12), three body (5)
A law carried past its range (proposed)	9	driven oscillators (17), measurement (14), maps and chaos (5)
The integer diagnostic	9	measurement (12), driven oscillators (9), topology (7)
The knee	11	driven oscillators (16), measurement (6), maps and chaos (5)
Rate over level	4	reading pictures (2), maps and chaos (1), measurement (1)

How lessons travelled. Bear’s question of every carried lesson is “it worked in X but not as well in Y? Is it a wall, a seam, a substrate?” Read row by row through row 847, the register records, on 322 of its rows, 519 places where a lesson, a tool, a law or a reading was carried from one system or setting to another and the text says how it fared. It held 222 times, held weaker 79 times and failed 218 times. When it did not hold, the break had an owner:

Owner	What broke	Weaker or failed
ruler	the instrument, grid, threshold or reading was too coarse or wrong there	62
not stated	the text does not say	60
substrate	the new system lacked the geometry or ingredient the method reads	58
seam	a boundary between regimes lay between them	48
knob	another setting had moved between the two places	41
wall	a symmetry or structural limit of the method itself	28

Of the breaks whose owner the text names, two lead and are close: the ruler, the instrument, the grid or the threshold that read it in the new place (62), and the substrate, a new system without the geometry the method reads (58), against 60 whose text names no owner. The lead is not settled. When the 96 rows that changed on 17 October were read again by fresh readers, the ruler’s count among them fell from 20 to 13 and the seam’s rose from 10 to 16, so which of the two comes first depends partly on the reader. Every reading agrees on the other end: a wall of the

method itself owns the fewest breaks, 28. The ruler's share is the ranking's own finding met from the other side, where a check that cannot fail leads every family, and the substrate is the owner that the guide's three causes of a null call by the same name. Every transfer is listed in the appendix "Transfers, row by row".

How it was read. Six fresh readers tagged the register's first 792 rows, and three more its first 326 arrivals; one more tagged the 24 rows changed or added on 16 October and one more the 18 arrivals counted that day, on 17 October two more tagged the 96 rows changed or added and one more the 60 arrivals counted, and on 23 October four more tagged the 141 rows new or changed from 18 to 23 October and two more the 179 arrivals counted that day, against the fixed list and definitions, each transfer backed by a quotation from its row that a script then found word for word. On 1 November one more reader gave domains alone to the 34 rows new from 848 to 881, matching the sets of like rows already tagged, on 2 November the publishing session tagged row 882 the same way, and on 10 November rows 883 to 891, from the two-voices papers; their transfers are not yet read, so the transfer counts here are of rows 1 to 847. The four readers of 23 October said where they drew the line, and drew it in slightly different places: one counted a lesson named a second specimen in another project as carried, another counted only what the text marks as carried and says how it fared, so the newest rows' transfers are the least evenly read. A second reader, blind to the first, tagged 40 rows drawn at random: every one of the 40 shared at least one domain between the two readers, 32 shared the first domain, and the two sets of domains overlapped by 0.87 on average. The two readers agreed on whether a row records any transfer at all in 37 of the 40; where both found transfers, the commonest outcome matched in 7 of 8. The second reader found 15 transfers where the first readers had found 12 in the same rows, so the count of 519 is more likely low than high.

The data. The register's domains, the arrivals and the transfers also travel with this guide as three tables of data, FieldGuide_v7_ROWS.csv, FieldGuide_v7_ARRIVALS.csv and FieldGuide_v7_TRANSFERS.csv, so that a later reader can sort them by domain, family or owner and keep adding to them; each transfer there carries the words of its row that record it.

L1 — The Method as an Instrument

What you do, and the instrument you do it with.

Declared blindness (2) — Before any claim to see, write down the combinations the instrument cannot read, so that a silence is never taken for an absence. The cochlear instrument did this explicitly: 8 measurement modes across 6 signal channels, not every combination readable, and writing that down before the read is what makes the instrument honest.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 2.

Which loops exist and which have opened (5) — A system can already hold a complete feedback loop in its hardware that nobody uses, so look for idle loops before concluding there are none to walk. Most modern cochlear implants carry ECAP telemetry, which measures how the auditory nerve responds to electrical stimulation and is a complete closed-loop control capability; it is used once at fitting, and then the loop is left open and never asked again.

Holds: not yet carried elsewhere. *Met in:* control and circuits, biology.
Shown: version 6; full guide row 5.

Acceleration-centroid tracker (10) — A slow drift riding on a fast wave is a reading of its own. The centroid of the seismic wavefield moves through the crust at 1.9 and 4.8 km/day, 10^{5}–$10^{6}\times$ slower than any physical wave, at plate-tectonic rates while the wave it reads travels at seismic speed, and that slow motion is what this instrument tracks.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 10.

Frequency-footprint address (11) — Knowing where a substrate's signal lives in the spectrum, and where its background noise lives, tells you where to look and what to expect there. The cochlear frequency ladder runs down from electrical mains hum at 50 or 60 Hz through Schumann resonances at 7.8 Hz, Earth's electromagnetic background, microseismic noise at 0.1 Hz and tidal periodicities at 10^{-5}–10^{-6} Hz to corridor-travelling one-off events at 10^{-7} Hz.

Holds: not yet carried elsewhere. *Met in:* quadrature, geodesy.
Shown: version 6; full guide row 11.

Mod-N aliasing, discrete-Stokes redemption (12) — A modular winding count folds values above N/2 into false negatives, so a cell that gathered 120 half-turns can read back as 0. The discrete-Stokes construction lets each cell add at most one count to the boundary sum, and 120 cells returned 120 distinct values, none aliased. The same wall bounds a writer: a ring of N writers writes a twist only up to N/2.

Holds: carried to one, held there: a ring of writers (Latch Atlas). *Met in:* topology, materials, geodesy.
Shown: Latch Atlas; full guide row 12.

Shared null space (13) — A differential read is immune to a background that arrives at every sensor at once, because subtracting sensors from each other removes what they share. The block-vs-block winding matrix organised 229 individual station readings into 16 common-mode series and read only what the groups disagreed on: inter-block rotation reached 3.587 cycles, and station P062's 6.395 cycles exceeds the largest common-mode pair.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 13.

Byzantine rogue census (14) — When sensors cannot be trusted, map the pattern of their failures, because the pattern is itself a signal. A seismic station census sorted 26 stations as convict, consistently wrong, 37 as suspect and 148 as reading clean; station CLCV-PSPT-P736 showed +1 versus −128 against its neighbours, and all 35 standing edge-pairs fired in opposition, with an opp-frac of 1.000.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 14.

The matched-filter law (16) — Know the noise level at which a filter stops working before trusting what it outputs. A matched filter made 25 detections out of 25 trials at noise level $\sigma = 0.3$ and had only an 8% success rate at $\sigma = 0.6$; running two smoothing passes before filtering restored the full 25 out of 25 even at high noise.

Holds: carried to one: failed in the same filter at noise $\sigma = 0.6$ (ruler). *Met in:* measurement, quadrature.
Shown: version 6; full guide row 16.

The projection law (17) — Counting signs throws the angle away, and under phase noise the angle is what survives. Ternary classification (−1, 0, +1) reaches only 52% accuracy at noise $\sigma = 0.6$ radians, while a continuous-phase Euler representation, using 1/cos(pitch) rather than a sign, is accurate to 2×10^{-14} at the same noise; when phase noise is high, track the full continuous angle using Baker–Campbell–Hausdorff.

Holds: not yet carried elsewhere. *Met in:* kinematics, quadrature.
Shown: version 6; full guide row 17.

Detection floor is not measurement floor (18) — A detector's silence can mean an effect sat below its own trigger rather than that it was absent. The measurement floor is how much a signal moves in a quiet period; the detection floor is how high an effect must be before a specific detector notices it. With an external gate supplying a second measurement, a seismic event detector that found 3 of 40 events on its own found 40 of 40.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 18.

The dipole liar (19) — A real physical relationship weakens with distance and a spurious one does not, so test any strong correlation against separation. A seismic station pair correlated at r = 0.98–0.997 at near-zero p-value, and exactly as strongly at 20 km separation as at 0.4 km: both stations were responding to the same large-scale background, not to each other.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 19.

Event time (20) — When the signal is carried in the relative timing of events, dividing by a time average erases it; here time normalisation drowned the seismic signal by two orders of magnitude. Driving by events is a separate question: in the memory bell, firing when the flow at the bell peaked only tied a clock at the same pace, so a trigger has to earn its place against a clock at matched pace (the Loop Atlas sessions).

Holds: carried to one: failed in an event trigger driving the memory bell (seam). *Met in:* driven oscillators, quadrature, geodesy.
Counted: Coupled clocks and event time 1. *Shown:* Loop Atlas; full guide row 20.

The visibility matrix (21) — An honest instrument ships a table of its measurement modes and signal channels with the inaccessible cells declared, and that table travels with every report it produces. The cochlear instrument's matrix is 8 modes × 6 channels; its shear row was closed after the channel was built, when the instrument was declared structurally unable to read it.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 21.

The residual does two jobs (24) — Treating a large residual as noise to be averaged away throws out the very signal that says something has broken. The Kalman innovation, the gap between what the model predicted and what arrived, is at once the correction that updates the state estimate and the anomaly detector that grows when the model has drifted; any quantity that accumulates that mismatch does both jobs, so read it as a correction and also as a probe.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 24.

Opponent as sensor (25) — Track two numbers per opponent, the estimate and its reliability, range-centre and trust-or-variance, and never the internals: this anti-bloat rule stops the model sprawling into a decision tree. The opponent is a source emitting a signal you cannot fully trust, and von Foerster's non-trivial machine, whose internal state cannot be determined from input-output pairs alone, is the theoretical home.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 25.

The wraith (26) — A consistency check can price a rigid liar, whose fixed false value geometry can triangulate as a defect, but it can never detect a fully adaptive, opponent-dependent one. The wraith changes its report per observer, so no sum-check or signed-ledger test can catch it, and when it reads as unreadable the instrument is working as specified; the honest job is then cost-imposition rather than detection (row 117).

Holds: carried to one: failed in a fully adaptive, opponent-dependent liar (wall). *Met in:* measurement.
Shown: version 6; full guide row 26.

The blackjack diagnostic (27) — Ignore a surface layer only when it carries no information about what lies beneath it. The test is whether the layer can be permuted freely without changing the substrate's statistics: in blackjack the dealer's manner can, so count the deck; in poker the other minds are the substrate, so the theater is the read.

Holds: carried to one: failed in poker, where other minds are the substrate (substrate). *Met in:* measurement.
Shown: version 6; full guide row 27.

Two codes, opposite enemies (50) — Each code is strong against one kind of damage and helpless against the other. At equal mean power with the decoder clamped, the winding code errs 0.063 against 0.643 at $\sigma = 0.6$ under local damage, while under smooth global deformation winding sits at chance (0.899) and magnitude is perfect (0.000). The first codec gave winding an energy advantage and an unclamped decoder, both fixed: the law survives, the margin does not.

Holds: carried to two: failed in winding code under smooth global deformation (wall); failed in magnitude code under local damage (wall). *Met in:* topology, waves.
Shown: version 6; full guide row 50.

The medium reports its own damage in the units it stores (56) — Crack propagation connecting voids is the most likely real failure mode for a topological memory, and it is readable in the memory's own count: the zeroth Betti number b_0, the connected components, drops as cracks join voids into clusters, while b_1, the hole count, may rise as new enclosed spaces form. A medium failing from within looks different from one degraded uniformly from the outside.

Holds: not yet carried elsewhere. *Met in:* topology, materials.
Shown: version 6; full guide row 56.

Coarse reading as a classifier — and it is the same knob as over-binning (58) — Fine detail is where records differ and coarse structure is where categories live, so a deliberately lossy read can sort categories cheaply. A coarse-binned read classified category as well as or better than a full high-dimensional read at roughly 60× compression; over-binning is that same coarse-graining run past the edge, so one mechanism is suffered as a limit or exploited as triage.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 58.

Net capacity peaks at four dimensions — and the fence that makes it honest (59) — A capacity peak has to say which property of the axes produced it, or it reads as a universal cap. A Monte Carlo encode, noisy read and decode peaks at four axes and declines beyond, because added axes share the same physics and arrive at SNR near unity; that is a property of CORRELATED axes, and a separate simulation confirms dimension grows unbounded when axes do not interfere.

Holds: carried to one: failed in axes that do not interfere (substrate). *Met in:* measurement, quadrature.
Shown: version 6; full guide row 59.

Storing an invariant does not make the read a holonomy read (64) — The trap is sharpest where the substrate could hardly look more holonomy-shaped. The chiral memory's bit is a topological winding, but its readout is not a loop walk: sign by circular dichroism, magnitude by bag count, position by strain-induced frequency shift, all local and direct, a false friend of the sharpest kind.

Holds: not yet carried elsewhere. *Met in:* topology, materials.
Shown: version 6; full guide row 64.

Healing and remembering are in tension unless the bit is topological (66) — Ordinary annealing repairs a crystal by erasing defects, which erases the memory; the architecture can heal and remember only because its bits are non-local, so local repair of the bulk leaves the invariant untouched. It is the same tension as two-codes-opposite-enemies in another domain: gel heals the crack and invites the global deformation.

Holds: carried to one, held there: self-healing topological memory gel. *Met in:* topology, materials.
Shown: version 6; full guide row 66.

Correlated noise shaped like the invariant beats the topological code first (72) — No decoder can correct a genuine logical-operator-shaped error. Under streaks aligned with the logical operator, a calibrated greedy syndrome-matching decoder fails 20–120× more than under i.i.d. noise at matched sub-threshold density, while majority vote, blind to phase and winding, holds; the size depends on the model and aspect ratio, but the direction is robust and decoder-validated.

Holds: carried to one: failed in streaks aligned with the logical operator (wall). *Met in:* measurement, topology.
Shown: version 6; full guide row 72.

A self-confirming instrument reads its own output back as evidence (77) — When the loop closes through the thing being measured, the discriminator has to be an exterior referent rather than anything internal. Within the event the two signatures overlap (1.72 against 1.17) and they diverge only afterwards (0.01 against 1.18). The order of operations falls out of it: stabilise the regulator, restore the capacity, then re-anchor, and the paper's safety rail travels with the rule.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: Question the Substrate; full guide row 77.

Closure depth — superseded by row 134 (83) — This row is kept as the trail rather than deleted, because the path from an unverified summary claim to a sourced result is itself part of the record; the priced result is row 134.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 83.

The balanced dipole and the matched offsetting lie are the same blindness (84) — Proven by hand for all N: a signed sum of ternary values is zero exactly when the plus and minus counts match, so a hidden configuration in either world is k matched pairs, and a sum-preserving bijection maps pentagon to plus-one, heptagon to minus-one, hexagon to zero. It holds only on the intersection of ternary and ordered, and its residual is the citable core: the adaptive liar has no static-geometric twin, because geometry's defects sit still.

Holds: carried to one: failed in the adaptive liar (wall). *Met in:* measurement, topology, materials.
Shown: version 6; full guide row 84.

Match your suspicion to the substrate's freedom to lie (94) — The adversary model is a design input, not a disclaimer. Nature rarely lies, so on sky, genetics and weather the adversary is bad data and confident misreading, and suspicion is best spent on your own anchors; prions were chosen as a target because they hold still against a dynamic noise background, so a cheap passive multi-pass wins. Over-suspicion of derived physics is as much an error as under-suspicion of free sensors.

Holds: carried to one: failed in sky, genetics and weather data (substrate). *Met in:* measurement, biology.
Shown: version 6; full guide row 94.

A finer loop is better only if the interior is smooth (97) — Loop shape is a match to the interior's symmetry, not a ladder toward perfection: at a discontinuity a smoother high-vertex loop is worse, because it averages across the cut. Many vertices suit smooth curvature and few-with-an-axis suit a seam; the price was paid at Parkfield, where the pentagon went null and the geometry predicts a rectangle.

Holds: carried to one: failed in a discontinuity (the Parkfield seam) (seam). *Met in:* topology, geodesy.
Shown: version 6; full guide row 97.

Redundancy defends magnitude, never consistency (98) — More vertices catch the outlier, the bad sensor or the static liar, but are structurally blind to the coordinated, frame-variant lie, because every station added is also inside the network: two axes, and adding sides only ever buys one of them. This is the sharp operational half of row 96 and the counterweight to row 14's census.

Holds: carried to one: failed in the coordinated, frame-variant lie (wall). *Met in:* measurement, geodesy.
Shown: version 6; full guide row 98.

Integer-ness, not magnitude — and nested loops are a family you check, not an average (103) — A real winding is a topological count, forced near a whole number across a family of enclosing loops (0.98, 1.02, 0.99), while noise has no closure constraint and scatters (1.6, 4.3, −2.1). Gating on agreement-near-an-integer across nested radii took false alarms from 2.75 to 0 at the sharp cell and from 90 to 3 at the cell that had killed the previous build: threshold on integer-ness, never on size.

Holds: carried to four: held in a rider driven by two waves; held in the same riders in the container; weaker in the same riders in the container (break not stated); held in the same readings in the container. *Met in:* measurement, driven oscillators, topology.
Counted: The integer diagnostic 4, Chaos and binning 1. *Shown:* Loop Atlas, R2W; full guide row 103.

Modulus-1 deletes the axis the noise lives on (104) — Forcing the radius to one keeps the angle, which carries the winding, and discards the amplitude, noise spray and common-mode level, cheaper than decomposing the field. It can delete the axis a count lives on too: on a ring of angle alone a jump up and a jump down draw the same near-diameter, and which side of the centre a jump passed is carried by the discarded radius, where the middle look of every two-look down-jump was nearest the centre (1.000 at six births) (the triple-star sessions).

Holds: carried to one: failed in triple-star jump counts on a ring (wall). *Met in:* topology, three body, quadrature.
Shown: triple-star; full guide row 104.

The signal is in the rate (105) — The corpus's most expensive recurring correction was paid three times in one sitting: shear rather than area, rate-of-holonomy rather than holonomy, slope rather than level. Sorting building, steady and dissipating by the slope of core strain scored 96% against 33% chance, where the same quantity read as a time-averaged level scored 39%, because averaging smears an early-caught riser and a late-caught faller onto the same mean.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 105.

Correlated agreement is one vote, not three (112) — Subtracting the shared level is the wrong repair, because a real multi-channel event raises the shared level too and the subtraction eats the signal; down-weight correlated channels instead. Independence, not count, is the discriminator: a real vortex sings, glows and drops pressure on genuinely independent substrates, while a turbine emits on the same three but correlated.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 112.

Detection fails; taxation works (117) — A suspicious stance converts evasion into a cost, because the moves that dodge a detector fight the clean fast operation the deception needs. The adaptive liar cannot be caught, so the instrument's honest job is to raise the price of undetectable deception, not to achieve the detection that was proved impossible.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 117.

The integral erases history — keep the catalogue, never collapse the trace (120) — A global loop walk is the integral of local curvature, and an integral is order-blind and cancellation-prone: it keeps the net total, not the path. Path-ordered matrix products partly dodge this because they keep order, but any scalar summary re-integrates and a true inverse pair vanishes again under a trace, so the dodge holds only while the matrices stay uncollapsed.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 120.

Calibration fidelity and trust fidelity are different jobs (124) — Checking a reading against a known reference is always worth doing and needs no adversary; asking whether a reporting node is honest is worth doing only where something has the freedom to lie. Blurring the two wastes the suspicion budget and does real harm: treating a value forced by a conservation law as an unverified claim is how a trust-root gets invented where none was needed.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 124.

A known-answer border is a free instrument check (126) — A flat cell's seam has a derived value of exactly zero, so a walk along it that fails to read zero shows the instrument drifting, not the cell lying, and each such border is a calibration at no extra cost. Where the drag fixes the multipliers' product exactly, that product gauges a numerical Jacobian: in R2W's FX-2d, one-sided differences ran 25% low at drive 0.90, while central differences met e^(−T/m) within 0.008% (FX2d-J).

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: R2W; full guide row 126.

Suspect the sensor, not the universe — priced (127) — A first real-data run returned a two-ledger correlation of 0.74 where the crude proxy had given 0.97, and the drop was not physics: once the storm weakened, the tracker's centre teleported to other pressure minima in the box. Leashing the search to within 3° of the previous centre took it to 0.89 and corrected the peak deficit from 15 to 18.8 hPa; a too-clean first number would have hidden the seam.

Holds: carried to one: weaker in the first real-data storm run (ruler). *Met in:* measurement, waves.
Shown: version 6; full guide row 127.

Say “I can’t see it” rather than reporting a number (128) — The repaired tracker flagged 236 of 312 hours as not reliably trackable and greyed them on the plot instead of fabricating a circulation. The hypothesis the flag suggested, that trackability itself might discriminate, was then killed cheaply, a real hurricane tracking 65% and a weak system 62%: the refusal to lie was right, and the idea it suggested was wrong. In the triple-star sessions, a tool that printed the first of a three-way tie as its peak now refuses to name a peak on a tie.

Holds: carried to two, held in both: R2W’s rider (row 712); R2W’s rider, before the runs. *Met in:* measurement, reading pictures.
Counted: Coupled clocks and event time 2, Chaos and binning 1, A second reader 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 128.

Know your substrate’s normal, and know that the normal drifts (129) — A fixed threshold is a fixed anchor in a world where the anchor moves, so re-derive the baseline from the quiet background with the hunted anomalies removed. Replacing a whole-box median with a local drifting ambient, the mean in a 300–600 km annulus around the candidate centre, took one storm’s two-ledger correlation from 0.83 to 0.95 and honestly moved another from 0.89 to 0.72.

Holds: carried to three: failed in near the pole, one pair tight (ruler); failed in later runs of the same sweep (break not stated); failed in the same laptop through the night (break not stated). *Met in:* measurement, records and tools, three body.
Shown: Loop Atlas, triple-star; full guide row 129.

Closure depth beat the magnitude observable, and here are the numbers (134) — Over the full 2005 Atlantic season (n = 12), closure depth, how deep a low digs against its own 300–600 km surroundings, separates five hurricanes (36, 39, 48, 54, 70 hPa) from seven sub-threshold systems with a 16.8 hPa gap containing zero storms, where the standard scalar separates them by 2.6 m/s. Its fences travel with it: peak-versus-peak classification, not forecasting, and one season is a strong pilot, not a publication sample.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 134.

Non-redundancy is what lets a head-to-head have a winner (136) — Circulation peaks after peak wind, by 12 hours in one storm and 4 in another, and if the two channels were redundant they would peak together. That cheap measurement, which most comparisons skip, is the precondition for comparing two measures at all.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 136.

Redundancy is the lie detector, and a triangle has none (140) — One bad station corrupts a three-vertex loop outright, the Byzantine hole. Five vertices are overdetermined, so the residual itself becomes a lie detector; and the five-vertex cell is also the +1 disclination tile, so on this substrate the model and the instrument turned out to be the same object.

Holds: not yet carried elsewhere. *Met in:* measurement, topology, geodesy.
Shown: version 6; full guide row 140.

One atom, two integer reads, and they count different things (145) — Treating two integer reads of one atom as one object would be Station 1’s trap, one word for two objects, with the most famous possible example. Shell structure reads integers off a phase-closure condition plus exclusion and counts states; Bader’s quantum theory of atoms in molecules counts critical points of the electron density, maxima at nuclei, saddles at bonds and rings, minima at cages, with the counts of each type constrained by Poincaré–Hopf.

Holds: carried to one, held there: a single atom (shell structure, QTAIM). *Met in:* topology, materials.
Shown: version 6; full guide row 145.

A round number that appears far too often is a placeholder, not a measurement (153) — Histogram your inputs and look at the spikes before you look at the science. 41.7% of a global earthquake catalogue carried depths of exactly 10.0 and 35.0 km, defaults where the depth could not be resolved; and two cold reads each returned exactly 100 findings, where a reader told there was no target and no limit logged 149.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: Loop Atlas, R2W; full guide row 153.

The aperture has to match the source’s dimensionality (160) — Using the wrong reader produces a clean, honest, entirely misleading null: a point, a line and an areal source each demand their own. In the Tripwire the twenty-year transient hunt was empty above floor and the sub-annual Parkfield hunt returned an honest empty at the loop aperture, an areal loop pointed at a line source; the next instrument named itself, a cross-fault station pair, and CARH-HUNT measured real fault slip.

Holds: carried to one: failed in a line source (fault strain transients) (substrate). *Met in:* measurement, three body, geodesy.
Shown: triple-star; full guide row 160.

The analog residual is a reading too (164) — Reaching only for the integer throws away the channel that carries magnitude. The holonomy residual is valuable even when it is not quantised: continuous holonomy is curvature and field content, the analog channel, the gravitometer reading, and only the loop integer is quantised. Bear's own correction puts both on the page as real.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 164.

The non-result that is the correctness receipt (170) — A character on the tower graph is pure gauge, so re-phasing the nodes must leave the spectrum unmoved, and this is the one measurement whose failure would mean the apparatus is wrong. Sweeping characters moved the spectrum by at most 2.04×10^{-14} and random node phases by 2.00×10^{-14}, against a declared tolerance of 10^{-10}, while flux through the faces moved it by 0.725.

Holds: not yet carried elsewhere. *Met in:* topology, number theory.
Shown: version 6; full guide row 170.

Temperature is apodization (172) — Written in the aperture's own coordinates, a Boltzmann weight is an exponential taper across the barrel, so a thermodynamic reading becomes a resolution question: cooling the gas shortens the effective aperture. The splitting point climbs 1.2110, 1.2183, 1.2287, 1.2525, 1.2661 as the damping rises, so a constant stated without naming its temperature has no address.

Holds: not yet carried elsewhere. *Met in:* number theory, quadrature.
Shown: version 6; full guide row 172.

Solve it rather than sample it (174) — Reading the constant off a sampled grid was good to about seven figures and drifted between working precisions, while recasting it as a curvature zero-crossing and solving it gave twenty digits with a residual of 2.4×10^{-46} and no drift. In the Loop Atlas sessions, five solves gave the motor snake's exact best where 486 grid solves had missed it by 0.16 to 0.54.

Holds: carried to one, held there: 1/2 groove at ripple strength 0.25. *Met in:* driven oscillators, three body, kinematics.
Counted: Chaos and binning 5. *Shown:* Loop Atlas, R2W, triple-star; full guide row 174.

Strong apodization does not blur the pair — it deletes one of them (179) — A window chosen for cleanliness can remove the very thing being looked for. Hann, Hamming and Blackman windows never split the pair at any barrel length, carrying exactly one maximum that creeps toward the far tooth (0.415, 0.949, 0.988, 0.998), because one tooth is a comb's main lobe and the other a sidelobe-regime creature, crushed along with the sidelobes.

Holds: not yet carried elsewhere. *Met in:* number theory, quadrature.
Shown: version 6; full guide row 179.

Know which currency your limit charges in (180) — Every array engineer's instinct is to thin a redundant aperture, but here that instinct spends the wrong resource. The filled aperture carries 19,881 ordered pairs across only 737 distinct baselines, and thinning it saves 4.3×10^5 in elements while paying 9.6× the barrel; the horizon charges in barrel, and the closed form had already made elements free, so thinning pays in the scarce currency to save in the abundant one.

Holds: carried to one: failed in the filled aperture, charged in barrel (substrate). *Met in:* number theory, quadrature.
Shown: version 6; full guide row 180.

A null is a place ruled out (188) — Vary loop size and placement and the turns triangulate what cannot be seen. Four loop walks across a cross-section of sand with a chest buried in it return a turned arrow only where they enclose the chest, and the rest read zero. The same holds for one candidate among many: in the Loop Atlas sessions, a map pairing two falling cats with unequal halves missed by up to 0.178, which rules out that map, not every map.

Holds: carried to two: held in test of one candidate map; failed in falling cats with unequal halves (break not stated).
Met in: measurement, topology, kinematics.
Shown: Loop Atlas; full guide row 188.

Fidelity is the margin, not the presence of the invariant (189) — Which integer you are in is defined at every ring of the bullseye, the rim included; only the margin you can still spend shrinks outward, to zero at the critical edge. The row carries its own prior-art tag, which is why it is trustworthy: this is the radial form of the eye diagram and the noise margin, the right picture, not a new one.

Holds: not yet carried elsewhere. *Met in:* topology, quadrature.
Shown: version 6; full guide row 189.

Mirror the pipeline, not the label (190) — Naive loop reversal is vacuous; run the estimator itself forward and through a mirror and split the result, the odd part being the signal channel and the even part a free artifact meter. With ε as even over odd-plus-even, green under 0.1, amber under 0.5, red at 0.5 or any integer flip, the Loop Atlas's mirror audit read 54 rows, 53 green and 1 amber, and no card moved.

Holds: carried to one: failed in a locked rotor, conveyor and clock blade (seam). *Met in:* measurement, driven oscillators, kinematics.
Counted: The mirror audit 3. *Shown:* Loop Atlas, R2W; full guide row 190.

Two different quantities were both called "the hypotenuse" (193) — The flat chord is the distance home ignoring height, and the full reach is the distance home in three dimensions. The closure deficit uses the chord, which shares a plane with the walk so the triangle inequality forces it non-negative, while against the reach the same deficit goes negative for almost every integer: the failure mode of one definition per quantity (Station 1), caught by the author in his own paper.

Holds: not yet carried elsewhere. *Met in:* measurement, number theory.
Shown: version 6; full guide row 193.

The walls are curvature-free by measurement, so the defect has to live in the loop (198) — The golden-angle offsets all share the vertical axis and so commute, which makes transport along a cylinder wall abelian, with no holonomy of its own; the construction's only closure defect therefore lives in the loop as a whole, measured, not assumed. Measured afterwards, transport across the caps and bases carries holonomy in the rotational channel (row 207) and not in the vector-area channel, and row 215 prints both side by side.

Holds: not yet carried elsewhere. *Met in:* topology, number theory.
Shown: version 6; full guide row 198.

The blur was the renderer, not the structure (199) — Before reading a smear as the structure dissolving, check the display. An early renderer turned a million integers into a formless smear, but on a logarithmic vertical axis with point density mapped honestly the cylinders sharpen from scattered dots into coherent walls as the count climbs through four decades; the instrument had been reporting its own display limit as a property of the object.

Holds: not yet carried elsewhere. *Met in:* number theory, reading pictures.
Shown: R2W; full guide row 199.

A pre-registered diffusive reading, falsified and printed (200) — Repeated prime factors contribute perfectly collinear steps, so the construction is stiffer than a random walk. Mean horizontal spread grows with the prime-factor count as a power law of exponent 0.977, nearly ballistic and decisively not the 0.5 a diffusive random walk would give; the square-root reading had been pre-registered, and its falsification is reported at equal weight.

Holds: carried to one: failed in the prime-factor construction's horizontal spread (substrate). *Met in:* measurement, number theory.
Shown: version 6; full guide row 200.

The one exception, measured and named rather than excused (201) — Explain a single miss from the mechanism, and leave a coincidence named rather than claimed. The octave law has exactly one exception among primes tested to 17,389: p = 5 sits at rung 3, whose heading is the most strongly northward of any low rung (cosine +0.608), and that boost lifts its reach just past a rung. That five is also the determinant of the cinquefoil torus knot is noticed and deliberately not built on.

Holds: not yet carried elsewhere. *Met in:* measurement, three body, number theory.
Shown: triple-star; full guide row 201.

A taller climb makes the loop thinner, not more twisted (209) — A prediction that fails is reported at the same weight as one that survives. The holonomy was predicted to rise with the size of the rung jump, but across fourteen climbs only 5 of 13 steps rise and the envelope falls, 0.2247 at a climb of one against 0.0625 at a climb of twenty-eight, as the tangent indicatrix encloses less area.

Holds: not yet carried elsewhere. *Met in:* topology, number theory.
Shown: version 6; full guide row 209.

One question, two channels, opposite headlines — and both are right (215) — Tower v3.4 measured the vector area, an abelian Stokes object to which a purely vertical climb contributes nothing, and answered no; a second session measured the ordered product of transport rotations, which the climbs break, and answered yes. Printed in one table the two agree completely: boundary loops read the same $S_z = -0.6755$ as wall loops, but rotation 0.224748 against 0.000000.

Holds: not yet carried elsewhere. *Met in:* topology, number theory.
Shown: version 6; full guide row 215.

The area is a rotation angle, so it lives mod 2π — and here you read the complement (224) — The raw signed area of these quadrilaterals comes out just under 2π every time, −6.058, +6.221, +5.863, −5.969, so the holonomy is the complement, 2π minus that. Computed without the fold it reads 6.06 where the instrument says 0.22: the modulus is not optional.

Holds: not yet carried elsewhere. *Met in:* topology, number theory.
Shown: version 6; full guide row 224.

Antiparallel, not parallel — the mechanism corrected without changing the conclusion (225) — The climbing legs were said to go nearly parallel, but their dot product runs −0.9487, −0.9899, −0.9989, −0.9999, −1.0000: they go to opposite poles. The patch narrows because it degenerates toward a lune between two meridians whose width is the heading difference alone, which is also why the value wanders with (b−a)·α instead of decaying smoothly; same conclusion, different mechanism, and the difference is testable.

Holds: not yet carried elsewhere. *Met in:* topology, number theory.
Shown: version 6; full guide row 225.

A convention that names an angle is not a convention that names a rotor (238) — A loop multiplies rotations, so it also multiplies every choice a convention leaves open. Reducing a misorientation by the crystal's point group returns a unique minimum angle and a non-unique rotation, twelve equally valid rotors for hexagonal and twenty-four for cubic; the convention in everyday use specifies the angle and leaves the rotor unchosen, harmless for anything built from angles and not for anything composed around a loop.

Holds: carried to one: failed in rotors composed around a loop (ruler). *Met in:* topology, materials.
Shown: version 6; full guide row 238.

The cost of an unnamed gauge, measured on identical bytes by identical code (239) — Two machines ran one public file through one script, and of the 22,485 cells compared, 18,013 were bitwise identical, but 453 differed by more than a degree and 201 changed defect classification. One cell read 0.0000° on one platform and 146.6018° on the other, because a one-ulp difference in the library sine and cosine flipped which tied representative the selection returned, while everything angle-based reproduced bit-exact.

Holds: carried to two: held in the same reads on another platform; weaker in the same file and script elsewhere (knob). *Met in:* measurement, records and tools, materials.
Shown: version 6; full guide row 239.

Five defensible tie-break rules, a factor of 2.3 in the answer (240) — Boundary-seam counts under five equally valid rules came out 951, 975, 2205, 2197 and 2207; none is more correct than another. The paper's closing line is the rule worth carrying: a number that moves ten percentage points when you reorder a list of twelve matrices was never data.

Holds: not yet carried elsewhere. *Met in:* measurement, materials.
Shown: version 6; full guide row 240.

Build a case whose true answer is exactly zero, and read the spread (242) — Any spread that appears on such a case is pure representative freedom and nothing else. Triple junctions built with zero closure, swept over all 13,824 combinations of cubic rotors, gave median closure spreads of 28.74° at low misorientation, 73.94° in the middle band and 120.73° at high; in the Loop Atlas, the swimming snake read 0.002 at a grip ratio of 1, where the answer is zero, and the fault was the stepping rule's.

Holds: not yet carried elsewhere. *Met in:* measurement, materials, kinematics.
Counted: A check that cannot fail 1. *Shown:* Loop Atlas, R2W; full guide row 242.

The detector floor is a threshold, not a slope (243) — Knowing whether a floor is a ramp or a wall changes what a margin means. On a synthetic map with charge identically zero, false charge read 0.0000% at noise widths of 0.006°, 0.057°, 0.5° and 2.0°, then 28.63% at 10° and 85.74% at 30°: exactly zero false alarms across a three-hundredfold increase in noise, and then a cliff.

Holds: carried to one: failed in noise widths of 10° and 30° (seam). *Met in:* measurement, materials.
Shown: version 6; full guide row 243.

Agreement and certainty move apart (247) — Replicate agreement is the signal nearly every field trusts, so report it beside internal purity. Pushing one structure from 42 to 604 closures took agreement from 50% to 100% while purity fell from 0.834 to 0.759, so the wrong answer got more reproducible; on a second structure agreement went from 50% to 0% while purity rose from 0.691 to 0.792, recovering the honest hedge.

Holds: carried to one: failed in molecular-scale closure replicates (break not stated). *Met in:* measurement, biology.
Shown: version 6; full guide row 247.

A checksum proves the bytes arrived, not that you asked for the right thing (251) — Assert identity out of band on every accession, checked against the deposited file before any computation runs: print the header, the title and a per-chain census. Nothing in this corpus starts at residue 1, one structure beginning at 82 and others at 4, at 10 and at 19 — so a domain cut written as residues 1-240 silently became 20-240, and a coverage assertion costs one line.

Holds: not yet carried elsewhere. *Met in:* records and tools, biology.
Counted: The built artifact is the evidence 1. *Shown:* triple-star; full guide row 251.

A molecule is not a shape, it is a cloud of shapes — and the width of the cloud decides what you may claim (254) — Across 1,172 molecules and 24,864 conformers the frame defect angle moves 35.81° when nothing is wrong, while knotted and unknotted proteins differ on that channel by 3.3°, one eleventh of the cloud, SNR 0.09. Four experiments already run on that channel were predictable failures, without labels, from a single unattended measurement taking about a minute.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 254.

The registered expectation was backwards, and the direction of the surprise is the lesson (256) — An assumption about which substrate is noisiest is not safe in either direction, and the measurement is cheap. RNA was predicted noisier than protein because it is more flexible, but its floor came out 2.4-3× tighter, 0.0640 against 0.1886 on writhe and 14.25° against 35.81° on defect angle, the likely cause named as corpus composition rather than physics: short heavily base-paired stems against flexible loops and disordered tails.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 256.

One floor for three substrates is an error, and they span a factor of three (257) — A floor belongs to a substrate and a perturbation, so one floor cannot serve three. The floors are 0.0640 for RNA, 0.1886 for protein and ~0.18 for DNA duplex axes, and the author’s own work broke the guide’s rule by using one floor for all three, then repaired it in public.

Holds: carried to one: failed in RNA, protein and DNA duplex substrates (substrate). *Met in:* measurement, biology.
Shown: version 6; full guide row 257.

Agreement to three decimals because everyone used the same arbitrary seed (258) — On a closed curve the frame defect is a genuine holonomy, seed-independent to machine precision with a spread of 0.000 across twelve random starting normals; on an open arc it is not a rotation about anything. The same unchanged structure returns 78.635°, 97.325° and 110.220° from three arbitrary starting directions, so six angles that reproduce to ≤0.03° agree because the same arbitrary seed was used, not because the quantity is that precise.

Holds: carried to one: failed in open arcs of the same structure (substrate). *Met in:* measurement, topology, biology.
Shown: version 6; full guide row 258.

A blind spot you can predict without running anything (259) — Signed writhe is antisymmetric under mirroring, so an amphichiral knot cancels to zero, one line of algebra and no experiment required. The figure-eight reads a signed writhe of 0.08 against 9.09 unsigned, and AUC is 0.9808 on chiral knots against 0.5128 on amphichiral ones; restricted to chiral knots the filter gives a 3.62× work reduction, which must be quoted alongside the amphichiral fraction of the target corpus, never separately.

Holds: not yet carried elsewhere. *Met in:* topology, biology, kinematics.
Counted: The mirror audit 2. *Shown:* Loop Atlas, Question the Substrate, Two Arrows; full guide row 259.

The largest signal in the work is measuring a modelling artefact (262) — Where gaps are routine, as in real RNA, declaring them rather than bridging them stops being hygiene and becomes load-bearing. A 2,740-nucleotide rRNA reads Wr = −91.57 as parsed against −8.67 on its longest gapless run, so twelve bridged gaps inject roughly 83 units of fabricated writhe, a 460× effect against a crystal-replicate floor of ~0.18 that is measuring a straight line drawn through missing residues.

Holds: carried to one: failed in real rRNA with bridged gaps (substrate). *Met in:* measurement, biology.
Shown: version 6; full guide row 262.

The restoring force we apologised for is the entire definition of control (269) — A property is not good or bad; it is good or bad for a question, and changing the question is free. Each of the five fatal findings is turned over in one sentence: a device that fails to return to its setpoint is not thrilling, it is broken, and silence-until-read becomes a sensor that needs no battery and does not run itself down while it waits.

Holds: not yet carried elsewhere. *Met in:* control and circuits, biology.
Shown: version 6; full guide row 269.

A local dent cannot change how many times the phase wraps (284) — As local phase noise rises in a communications setting, the amplitude reader crowds and fails early, while the winding reader, taking the sign by a loop integral around the mode, holds. Only a perturbation that itself winds all the way around can change the count, which states the protection as a requirement on the adversary rather than as a property of the signal.

Holds: carried to one, held there: a communications winding reader. *Met in:* topology, waves.
Shown: version 6; full guide row 284.

Let the medium do the searching (289) — Checking a candidate is cheap and finding one from nothing is dear, so the ocean's currents or the body's circulation do the hunting while each pass answers only whether something is anomalous. A target seen once is a maybe, and one confirmed on many independent passes is exponentially unlikely to be a false alarm, so the specificity comes from voting an easy check many times, not from any unit being a good detective.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 289.

The target is never detected — it is accumulated (290) — A single tripwire on a single pass is noise, a raindrop that might be anything. Lay thousands of passive tripwires and walk them again and again, and the thing that does not belong becomes a shape, the one signal that keeps appearing in the same place while everything else washes away: identification by what refuses to change against a moving background, which is the definition of an invariant.

Holds: not yet carried elsewhere. *Met in:* topology, biology.
Shown: version 6; full guide row 290.

A map that decays cannot lie to itself for long (292) — The humility of the map is the safety feature. Each unit reports confidently only what it can physically stand on and touch, its own cell and its immediate neighbours, and every confident report decays with time and distance, so a fact nobody has re-touched fades rather than hardening into a false certainty; that second half is the one almost nobody builds.

Holds: not yet carried elsewhere. *Met in:* control and circuits, biology.
Shown: version 6; full guide row 292.

Read the slope, not the value — third independent arrival (294) — What a loop-walking array does best is read change: it needs only whether the local topology shifted since the last pass, so swept passively it maps rates, change of state, change of rate and the rate of that rate. This is the corpus's third independent arrival at rate-over-level, after the tornado work priced it at 96 percent against 39 and the Cascadia arc reached it through shear-not-area.

Holds: carried to one, held there: a clinical loop-walking array design. *Met in:* measurement, biology.
Shown: version 6; full guide row 294.

Some numbers wobble and some numbers tick (301) — A detector built on integers is deaf to noise by construction, within their binning scale, and that qualifier is the honest part. Amplitudes, phases, energies and positions are continuous, so noise smears them and an adversary nudges them; windings, wrapping degrees, sign-change counts, framing twists and linking numbers cannot move by less than one, so noise, which can only wiggle, cannot move them.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 301.

A topological channel without a matched pre-filter is not partial — it is broken (304) — Every loop channel is operationally a derivative operator, differencing the field around a circuit. Derivatives annihilate smooth systematics exactly, since the curl of a gradient is zero, and amplify white noise mercilessly, so the property that makes the channel immune to drift makes it defenceless against white noise; the pre-filter is not an optimisation, and this was confirmed in three unrelated instruments in a single session.

Holds: carried to one, held there: three unrelated instruments in one session. *Met in:* measurement, topology, quadrature.
Shown: version 6; full guide row 304.

The integer can die BEFORE the amplitude (305) — Integer channels are more robust only above their matched scale. On a degree-one hedgehog with per-vertex noise, the integer reads 25 of 25 correct at low noise while continuous fidelity is already down 9 percent, but at higher noise the raw degree collapses to 8 percent correct while the continuous channel stays readable at 0.70, because white noise manufactures spurious wrappings; pre-smoothing to the texture scale restores 100 percent.

Holds: carried to one: failed in the same hedgehog at high noise (ruler). *Met in:* measurement, topology, materials.
Shown: version 6; full guide row 305.

Our own statistic, confessing that it measures trend-sharing rather than slip-sharing (307) — Test a correlation against distance before claiming it. Raw cumulative correlations between any two smooth decays run 0.98 to 0.997 with permutation p about zero and carry zero co-location information, so every raw-shape correlation was rejected on that control first; with slow components removed, a sub-seasonal band gave r = 0.585 at 0.4 km, surviving Bonferroni over thirty tests, peaking on the co-located meter and decaying with distance.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 307.

Correlated maintenance is not malice, and single-culprit masking breaks on it (309) — A maintenance campaign steps many witnesses at once, so masking has to be correlated-by-date rather than per-station, and any station with an equipment change near an event is dropped from that event's stack. Five stations were re-antennaed in two days inside a months-long rolling wave of receiver and elevation-cutoff changes, and three direct step-on-event collisions were caught this way; the benign correlated failure is more dangerous than the adversarial one, because nobody is looking for it.

Holds: carried to one: failed in a correlated maintenance campaign (ruler). *Met in:* measurement, geodesy.
Shown: version 6; full guide row 309.

The counter-example gets its own line instead of being averaged in (315) — Migration ran dominantly one way, eleven of sixteen events, while one event ran cleanly the other way at rank correlation −0.90 and about 7 km per day, and it is reported on its own because a clean reversal at full strength is information about the mechanism. On the Sun, Dancing on the Sun describes NOAA 12297 of March 2015 as negative where every other region is positive or near zero, without averaging it away.

Holds: not yet carried elsewhere. *Met in:* three body, geodesy, solar.
Shown: Loop Atlas, triple-star, Dancing on the Sun; full guide row 315.

Zero-padding manufactures fake smoothness — use native resolution to measure (318) — Interpolation between real samples is not measurement, and a number read off a padded spectrum can be wrong by a factor of three while looking smoother. Padded splittings of 0.10 to 0.16 collapsed to 0.04 to 0.06 at native resolution; in the triple-star sessions, a reader padded sixteen-fold returned 7,500 passes at 7.1985, the longest period it was allowed, which says only "longer than this".

Holds: carried to one, held there: splittings read off a spectrum. *Met in:* three body, quadrature, solar.
Counted: Chaos and binning 1. *Shown:* triple-star; full guide row 318.

Recovered by the right tool, with three controls and a named boundary (319) — A first-light result should ship with its controls and a stated ceiling. Recovered by autocorrelation across the whole structure, the quantity came with three controls: the known large-scale spacing re-recovered in the same transform, the value shown to differ from the confusable spacing, and window power at the claimed frequency measured at 0.0003, ruling out a sampling-gap alias. The depth-resolved version needs a further tier of machinery and is not claimed.

Holds: not yet carried elsewhere. *Met in:* measurement, quadrature, solar.
Shown: version 6; full guide row 319.

Binning is where almost all the information design happens (329) — An animal does not store a photograph of a place; it stores forage patch, depleted, a coarse category with a few operational tags. Binning throws away within-category detail and keeps between-category signal, and both edges cost: the over-binning cliff, where the categories collapse into one useless class, against paying to remember pebbles. The window between is the design.

Holds: carried to one, held there: animal place memory (forage patch, depleted). *Met in:* measurement, biology.
Shown: version 6; full guide row 329.

The channel's cost drives the code — and the test is whether it inflates when the cost drops (331) — A pair of just-so stories becomes an experiment once it has a designed discriminator. A caller on an expensive channel compresses to the operational core and a speaker on a cheap one inflates toward status and nuance; if signal complexity scales up freely when channel cost drops, the compression is a constraint and not an optimum, the same shape as the minimal-memory rung's test.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 331.

A coherent wrong answer is a systematic, not noise (336) — First light read a coherent glide of 6.45 plus or minus 1.03 microarcseconds per year pointing 128 degrees away from the published apex, coherent at 3.5 sigma above its own shuffle null, deterministic, and wrong. It was localised three separate ways: a rostering defect had drawn reference sources known to carry photocentre jitter into every ring, a magnitude-split mirror put the attenuation at the bright end, and the verdict grammar itself was defective.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: version 6; full guide row 336.

Two axes, one monotone and one broken, both reported at the registered strictness (338) — Recovery was pre-registered to improve monotonically from bright to faint. The direction axis is strictly monotone, separation from the published apex running 62, 27, 24 and 6 degrees, which confirms the bright-jitter diagnosis independently, while the amplitude axis broke monotonicity at the final step at only 1.0 sigma; the locked verdict is recorded as non-monotone, with the break read as tranche noise rather than promoted to an effect.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: version 6; full guide row 338.

Derive your own instrument's transfer function from its geometry, then confirm it blind (340) — Knowing which part of a deficit the instrument's own geometry owes lets the remainder be a finding instead of a mystery. A deterministic dilution of about 0.80 was predicted from the construction itself, since a mesh vertex sits roughly 37 degrees from its face centre, so a ring reads a dipole shrunk by the cosine of that angle; measured blind, the constant was confirmed in both arms.

Holds: not yet carried elsewhere. *Met in:* measurement, topology, three body.
Shown: version 6; full guide row 340.

The robustness check that doubled as a sensitivity measurement (348) — A pass-or-fail robustness check throws away how strongly the instrument responds to the systematic it probes. Perturbing the pinned constants within their published errors left all 85 detected patches standing, and injecting three kilometres per second of fake solar motion spawned only five to seven marginal rings while the 85 real ones sat unmoved, so one test certified robustness and measured sensitivity together.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: version 6; full guide row 348.

THE WORK LIVES IN THE PHASE GAP, AND IT PEAKS AT A QUARTER TURN (352) — Two things in phase carry nothing between them, and a perfectly locked system does no work because there is no gap to flow across. Power between two coupled resonators goes as the sine of their phase difference, zero in phase, zero net anti-phase, maximum at quadrature. The best gap depends on what is driven: a quarter cycle for two knobs, half a wave along the body for a chain, and in the Loop Atlas's worm and hinge toys each hinge past about 5 to 7 adds little.

Holds: carried to one: weaker in a hinged chain body (worm toys) (substrate). *Met in:* driven oscillators, waves, kinematics.
Shown: Loop Atlas; full guide row 352.

A harmonic requires a nonlinearity, so the octave is work climbing through strain (354) — No strain, no octave: a purely linear resonator makes only a sterile pure tone, and generating a harmonic takes a strained, out-of-phase region, so the harmonic is energy doing work to climb from the fundamental into a higher mode. The tension between a resonance and its harmonic is not a metaphor; it is literally the strain that does the work of making the harmonic.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, waves.
Shown: version 6; full guide row 354.

Odd-fold balancers are structurally immune to even-fold noise (355) — An even pointer has a preferred axis, so it couples to and is fooled by any dipole disturbance, the very thing it is supposed to reject. An odd pointer is axis-less and gives a two-fold dipole nowhere to grab, which is why this corpus keeps reaching for pentagons and triangles, stated as a symmetry theorem rather than a preference: three legs, three-phase power, odd spokes.

Holds: not yet carried elsewhere. *Met in:* measurement, topology, control and circuits.
Shown: version 6; full guide row 355.

Odd modes cannot close cleanly, so they always leave something to read (356) — In Bear's own derivation, an alternating ternary, each cell the opposite of its neighbour, closes cleanly only for even modes, where plus-minus-plus-minus comes home balanced and silent. Odd modes frustrate: the loop cannot alternate all the way round without returning a plus beside a plus, and that leftover mismatch is the residual, the signal, so the instrument is built on odd because odd cannot close.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 356.

Keep the two spaces labelled — fusing them is the trap (360) — A sensor's physical anchor in real space and the orientation state it carries, a point in pose space, are two different spaces: attach a pose to a vertex, but never equate the vertex's position with the pose's axis. This is the specific conflation that tangles solo attempts at this material, the same class of error as the two-quantities-one-name trap, arriving at the level of spaces.

Holds: carried to one, held there: sensor anchor versus pose space. *Met in:* measurement, kinematics.
Shown: version 6; full guide row 360.

Zero-strain register at the coincidence sites, accumulating pose strain between them (362) — A mismatch between two symmetries can itself be a read mechanism. Bring an axis-locked square grid against an axis-less round rim and they lock at one fiducial, agreeing elsewhere only at coincidence points; between those points the orientation disagreement is pose strain, so peak strain means the read has drifted off a coincidence site, and the grid reads the loop by where it locks and where it strains.

Holds: not yet carried elsewhere. *Met in:* materials, kinematics.
Shown: version 6; full guide row 362.

Free in magnitude, paid in phase (364) — Knowing which ledger a move is charged to keeps a free lunch from being claimed. Riding the unit circle costs nothing in magnitude, since there is no radius to pay for, but every quarter-loop deposits a quarter-turn of phase, and that phase is the currency: if it were free it could not carry a bit.

Holds: not yet carried elsewhere. *Met in:* topology, quadrature.
Shown: version 6; full guide row 364.

Median minus mean is a signed tilt, and its sign is the handedness (365) — The mean is symmetric and throws away which side a distribution leans, while the median knows order and position. Their difference is a ternary whose sign is the direction of the lean, robust against outliers that would fool a magnitude read entirely: a handedness detector built from two statistics everyone already computes.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 365.

Three points at 120 degrees is the minimum that locks a disk without over-constraining it (366) — Two slide freedoms and one rotation have to die, and three points is the least that kills all three; fewer and the disk can still rock, more and it fights itself, like the wobbling four-legged table. That is why tripods, telescope-mirror mounts and lug patterns are three, and why the equilateral triangle is the unique stable balanced mount for a disk under load.

Holds: not yet carried elsewhere. *Met in:* control and circuits, kinematics.
Shown: version 6; full guide row 366.

Information capacity costs strain (371) — Start at the quiet mode and add ripples on purpose, because capacity you did not choose to buy is strain you did not budget for. A pure uniform swell gives one ternary, load shared evenly across three pins with no shear, while an added odd ripple gives a ternary at every lobe, many bits, but concentrates the load at the peaks so the pins see shear; this prices row 329’s binning window mechanically.

Holds: carried to one, held there: a three-pin swell with odd ripples. *Met in:* control and circuits, materials.
Shown: version 6; full guide row 371.

A controller needs a setpoint, and air has none — so build it from agreement (376) — A vane reads position, the loud part, while the discriminator lives in accumulation, which rock gives free and air will not, and in rate; the session’s miss was running on position. The repair is architectural: a co-breathing outer ring acts as the live setpoint and rejects common mode, so subtract the ring next to you, not a quiet day.

Holds: carried to one: failed in air, read by a vane (substrate). *Met in:* waves, control and circuits, geodesy.
Shown: version 6; full guide row 376.

Use a principled projection, not a max-over-search (377) — The metric the permutation null killed was a maximum over bearings, a free parameter searched for its best value, which random data supplies too easily. The replacement used the same five stations and took a single principled mode projection: same data, same sensors, and only the number that came from a fixed question has a null you can trust.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 377.

Lock for the index, slip for the load (380) — Perfect phase-lock kills in two ways, a resonator going sterile when the coupling floods and a density wave starving at corotation, so the engineering target is a hair off perfect closure. A best setting sits where a ratio is near 1, met twice in the Loop Atlas’s toys: a scallop’s dial gave its best at 0.89, and a spring bead made its best ear when its loss was about as large as its tune.

Holds: carried to one, held there: the Loop Atlas scallop and spring bead. *Met in:* driven oscillators, kinematics.
Shown: Loop Atlas; full guide row 380.

A phase-conjugate return lights up only the defect (392) — Send a wave back time-reversed and it retraces its own path; interfered with the forward wave it cancels everywhere except at the defect, so the seam is the only thing that lights. It is a real deployed technique in imaging and non-destructive testing, and the read-side twin of row 367’s reciprocity: the same geometry run backwards to find the thing that broke the symmetry.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 392.

Mask the degenerate case rather than trusting floating point to produce it (400) — A zero the mathematics forces and a zero the arithmetic happened to produce look alike, and the second survives every test until the day it does not, so adjacent segments are masked out explicitly. In R2W, RATCHET-53's first dry run put the lean's dark point exactly on a grid point, and a sign-change test with a strict inequality counted that exact zero as no crossing; an exact zero is the root, counted once.

Holds: carried to one, held there: RATCHET-58 reader. *Met in:* measurement, driven oscillators, topology.
Counted: A check that cannot fail 2, The built artifact is the evidence 1. *Shown:* R2W; full guide row 400.

The same defect changed from rare to routine when the domain changed (401) — A hazard's frequency is a property of the corpus, not of the method. Backbone gaps were rare in the small structures the instrument was built on and routine in the larger ones it was pointed at next, one chain carrying twelve, and a bridged gap joins two points with one long artificial segment that was never in the molecule, so the code reports both numbers and names which to trust.

Holds: carried to one: failed in larger structures, one chain with twelve gaps (substrate). *Met in:* records and tools, biology.
Shown: version 6; full guide row 401.

The break test: remove the constraint and watch the protection vanish (404) — The mechanical tier loosens the belt and the rotor drifts freely, with no lock at all, which is the only way to show that a property came from the constraint rather than from the apparatus. In the Loop Atlas sessions the same move read what a surrounding adds: the tied bells' clock was mostly their plain stroke, the water giving only 1.013 of BELL_CLOCK's 1.427.

Holds: carried to one, held there: the tied bells in a remembering water. *Met in:* measurement, control and circuits, kinematics.
Shown: Loop Atlas; full guide row 404.

Two tests, one for the claim and one for the instrument (406) — A rig that can only run the test of the claim has no way to be wrong, so beside the protection test, which perturbs one element's phase, sits a substrate test that mis-calibrates one channel on purpose. Test the reader on a known case too: in the triple-star sessions a lock angle went round 5 to 8 turns on a run known to be locked, while the angle its dominant term acts on held within 0.06 to 0.08 turns over 300 passes.

Holds: carried to one: failed in pair at 0.25 (ruler). *Met in:* measurement, three body, waves.
Counted: A check that cannot fail 1. *Shown:* triple-star; full guide row 406.

Precision and meaning run opposite, by construction (415) — The first question about an impressively precise match is how long a walk bought it. Climbing the ladder improves the central value, the miss falling from 0.0700 to 0.0035, but that rung is a forty-nine-step walk carrying a 48 percent band, because the rungs that agree best with observation agree with everything else too; only the short extreme is punished, so the best member is the shortest qualifying walk.

Holds: carried to one: weaker in the ladder's walk-length rungs (substrate). *Met in:* measurement, number theory.
Shown: version 6; full guide row 415.

The comma is the zero-memory limit of the mirror test (420) — Naming which half of an analogy does not carry turns it from a picture into an instrument. The comma and a swimmer's mirror residual are both loop-closure deficits, but the comma has no time in it, pure arithmetic and pace-independent, while the residual moves with pace and with Reynolds number; so the comma is what the residual would be if the medium forgot instantly, and everything above it is the memory term.

Holds: carried to one: weaker in a swimmer's mirror residual (seam). *Met in:* number theory, kinematics.
Shown: version 6; full guide row 420.

Above threshold, even your "I don't know" has been captured (425) — In the honest setting a corrupted sensor says so, and ignorance is visible and a calibrated win is always reachable. Past a corruption threshold the sensors return a confident, plausible, wrong value identical to a true read, the lie is stable so re-checking confirms the fabrication, and two agents supporting each other manufacture a false consensus that reads as more trustworthy: the lie is not blindness but certainty.

Holds: carried to one: failed in past the corruption threshold (seam). *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 425.

The corruption with no agenda is harder to catch than the one with an agenda (426) — A liar's errors correlate toward something, which is what lets you triangulate it; a drifter corrupts the substrate with no agenda, silently re-authoring ground already confirmed from outside the frame the honest sensors were built to model. The

counters are structural rather than detective, anchors that resist rewriting and a bounded right to force contacts to identify themselves, the third arrival on this axis with row 309's correlated maintenance and row 291's mis-anchored friend.

Holds: carried to one, held there: a drifter corrupting with no agenda. *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 426.

A measured crossover with a control that passes (428) — The two arms differ only in the communication graph, and no agent is hand-set to ignore information. With no corruption the centralised arm wins, 0.10 against 0.53, because pooling cancels noise; under a targeted adversary at ten percent corruption the result crosses, centralised 29.98 against distributed 9.59, swept 0 to 50 percent over 200 seeds per point against two adversary types.

Holds: carried to one: failed in a targeted adversary at ten percent corruption (seam). *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 428.

A smarter aggregator does not save the centre (429) — Giving the hub a robust median rule changed nothing under the targeted attack, because when the corrupted node is the hub the aggregator is the source of the lie. Every median, trimmed mean and consensus rule is computed downstream of the thing that was captured, so no internal cleverness substitutes for an exterior reference, and here that is measured rather than argued.

Holds: carried to one: failed in a captured hub under targeted attack (wall). *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 429.

Blur is a classifier, and the window is the design (442) — Discarding high spatial frequency removes what separates members of a class and keeps what separates one class from another, so a read need not always be as sharp as possible. A deliberately lossy coarse-binned read classified category as well as or better than the full read at roughly sixtyfold compression, by shedding within-class noise; blur too little and you pay for detail the question does not use, too much and the categories smear.

Holds: not yet carried elsewhere. *Met in:* measurement, reading pictures.
Shown: version 6; full guide row 442.

We built a system optimised to undo change, then expressed surprise that it would not change (444) — One property can be prized in one paper and fatal in the next, depending entirely on the question brought. Every piece of the sense-compare-correct loop, the reference-correction and the proofreading, applies a restoring force toward a fixed setpoint, and a controller with a fixed setpoint was confirmed stable but trapped, exploration suppressed by construction; it is row 269's exact mirror image, where the same force is the definition of control.

Holds: carried to one: failed in an architecture meant to evolve (knob). *Met in:* control and circuits, biology.
Shown: version 6; full guide row 444.

Rank each channel by effect divided by floor before running anything (451) — A channel whose expected effect is smaller than its own floor cannot work, and no processing downstream will recover it; the check costs one division, before anything is built. A ruler too coarse for the effect reads the grid, not the world: in R2W, a law predicting lifts of 0.0003 to 0.0034 on a pull grid 0.01 apart turned "exact" into a question of which cell the edge fell in, so the edge was registered for a grid 0.001 apart (runs BAL-H and FINE-H).

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, three body.
Counted: A check that cannot fail 1, Chaos and binning 1, A second reader 1. *Shown:* R2W, triple-star; full guide row 451.

Three channels chosen because they go blind under different conditions (452) — That is not a coincidence of the design; it is the design, since convergence buys three to ten times against independent channels and nothing at all against a common driver. Wind, background flash rate and thermal drift are three different blindnesses, so a turbine lifts one channel while a thunderstorm lifts all three from one cause; score a mitigation by the between-channel correlation it removes, not by the false alarms it rejects.

Holds: carried to one: failed in a common driver such as a thunderstorm (wall). *Met in:* measurement, waves, solar.
Shown: Dancing on the Sun; full guide row 452.

The divergence between channels is itself the diagnostic (460) — Three commodities under one shock diverged in step with the mechanism: one eased as its input normalised, one stayed elevated because its binding constraint was never restored, and one never moved because its supply was never short. The spread drives the allocation directly: the one showing least relief is the one the largest section addresses, and its replacement can be drawn from stock already in the ground.

Holds: carried to one, held there: commodity prices under one shock. *Met in:* measurement, biology.
Shown: version 6; full guide row 460.

The binding constraint is not the thing that is scarce (461) — The exposure most easily missed sits one step upstream of the thing being priced. Processed phosphate prices moved while phosphate rock did not, because the constraint is the sulfur needed to process it, with roughly two-thirds of tradable supply out of the market; read at the wrong node a shortage is a mystery, and read one step up it is arithmetic.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 461.

A tenfold spread in partition — and the ratios survive the assumption (464) — Across four soils the fraction of added phosphorus that stays in solution runs 0.054% to 0.538%, and because one product greatly exceeds the solution volume in every soil, the ratios between soils depend only on measured parameters, the free parameter divided out. R2W's ripple gave a second specimen: the ratio of the edge's two shifts stays at −0.10551 to −0.10584 over a hundredfold range of weight (FOLD-DIR).

Holds: carried to one, held there: R2W toy machine's edge shifts. *Met in:* measurement, driven oscillators, biology.
Counted: The mirror audit 1. *Shown:* R2W; full guide row 464.

Resonance is the objective for the absorber and the failure mode for the structure (468) — A rule written without scope will either forbid the machine from working or permit the structure to destroy itself. An energy absorber is often designed to resonate, because a small force applied repeatedly at the right moment produces a large motion, while a structural member that resonates accumulates motion it has no way to spend, and fails; where the two meet, the mount between them becomes a compliance item, demonstrated rather than assumed.

Holds: carried to two: failed in a structural member, where resonance is failure (knob); held in absorber and structure rules. *Met in:* waves, control and circuits.
Shown: version 6; full guide row 468.

No periodic driver at all (470) — Forced resonance needs an external periodic driver matched to a structural frequency; a self-excited instability needs none, because the structure's own motion changes the forces on it so as to feed energy back into that motion, and the amplitude grows under steady flow. Frequency separation is therefore no defence, and the defence is damping and section shape; ice accretion can create the cross-section galloping requires, on a member that was stable when it was built.

Holds: carried to one: failed in self-excited galloping under steady flow (substrate). *Met in:* driven oscillators, control and circuits.
Shown: version 6; full guide row 470.

Do not escalate correlated quantities separately and then multiply them (474) — The joint probability of a correlated set is not the product of the marginals, and wave height, surge and wind are physically correlated within the same storm. Where a project lacks the data for a joint analysis, the limitation is recorded in the design basis rather than concealed by conservative single-variable factors, because a margin applied to hide an unknown is indistinguishable, downstream, from a margin applied to a known.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 474.

A factor-of-two error in the direction that flatters the project (479) — The regular-wave power expression and the irregular-sea one differ by exactly half, and mixing them is the commonest error in the field, so the paper names which form every figure uses and what it yields. The direction of the bias is the part that matters, because an error that flatters is one nobody goes looking for.

Holds: carried to one: weaker in irregular seas (break not stated). *Met in:* measurement, waves.
Shown: version 6; full guide row 479.

Two numbers that are different numbers (480) — Label a hard ceiling and an assumed output separately, so the ceiling cannot be quoted as an output. Incident flux is the ceiling nothing can exceed, and deliverable at an assumed capture fraction is the capacity of record; the falsification section adds that the capture fraction has never been independently estimated for any family, and every capacity figure in the document rests on it.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 480.

The veto is seasonal, not annual — and an annual test errs in both directions (481) — An annual mean can hide the case for building as easily as it can manufacture one, so collect a monthly curve with the complement's monthly curve beside it. A flat coast passes the annual test with no season in which it beats its complement, while a monsoon coast fails it though it delivers 65 percent of its energy in exactly the four months when the complement is weakest.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 481.

A factor of 2.4, from one pipe (483) — Leverage taken from a boundary condition rather than from a machine is the cheapest kind, and a device-centred search never finds it. Moving condenser cooling from tropical air at a realistic coefficient of 3.0 to deep cold water at 7.1, with the same compressor and no new physics, drops the electricity per tonne from 30.8 kWh to 13.1, and the pipe route is shared with another subsystem so its cost is paid once.

Holds: not yet carried elsewhere. *Met in:* control and circuits.
Shown: version 6; full guide row 483.

Holonomy hides in plain sight wherever a field has learned to distrust its own amplitudes (489) — What a system accumulates around a closed path is magnitude-blind by construction, since no uniform rescaling can change a winding rate, and that is exactly the right invariance when the dominant nuisance acts approximately as a uniform gain. The fence comes in the same paragraph: holonomy is not being imported as a framework; it is already the operative principle of the established method.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 489.

Every safety-relevant control system in forty years is closed-loop because plants drift; this one is not, because this plant does not (491) — A design choice can be right for the population it was made for and wrong for the neighbouring one. For a stable deaf ear the plant does not change and measurement costs power, so leaving the loop open is correct; for a nerve under slowly increasing compression the plant drifts and the device is blind to it, though threshold, response latency and fibre recruitment are all measurable with hardware already inside the device.

Holds: carried to two: failed in a nerve under slowly increasing compression (seam); held in the cochlear device's open-loop decision. *Met in:* control and circuits, biology.
Shown: version 6; full guide row 491.

Watch the residual and its derivatives (492) — The residual is the gap between the response commanded and the response achieved: stable means a stable pathway, and steady drift means steady change. A residual whose rate of change is itself increasing says the driving process has accelerated, and that second derivative is the decision variable, rate-over-level's fifth independent arrival and the first where the decision rides on the second derivative rather than the first.

Holds: carried to one, held there: a clinical residual read from serial imaging. *Met in:* measurement, biology.
Shown: version 6; full guide row 492.

The binding constraint is learning, not bandwidth (494) — The reported limitations are perceptual rather than technical, so encoding choice dominates hardware. People keep a device that clearly works and abandon one that fails: non-use ran at 2.8% for one device class against 27.3% for another, where failure and complication were cited. Put first effort into encoding studies on cheap reversible equipment, because if an encoding does not work through a headset, implanting it will not rescue it.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 494.

The design lever is the opposite of the intuition (500) — A wavelength-set resonator has no implantable version at audio frequencies, because the pipe is the wavelength; a mechanical resonator escapes that wall, since its frequency comes from stiffness and mass. Frequency scales with the square root of the stiffness-to-density ratio, so the design lever is a low ratio, the opposite of the reflex that reaches for silicon.

Holds: carried to one, held there: an implantable audio-frequency resonator. *Met in:* waves, biology.
Shown: version 6; full guide row 500.

A project is dead only because we chose not to look for four things (508) — Before any null is written down, look in four places, each with its own question: an anchor, a fixed exterior reference never identified; a hook, where the structure gives purchase; missed instrumentation, a channel already in the data never read; and a geometric cross-check, a closure, parity, mirror or symmetry that would test it for free. It is the three-cause autopsy's positive twin, asking what was never looked at.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 508.

The mechanism was wrong and the instinct was right — move the anchor, keep the phrase (514) — A correction that preserves the reading is rarer and more useful than one that discards it. The phrase "anchors of probability" became true once the anchor moved from friction to fitting, a standing wave that must close on a whole number of wavelengths; in R2W, "nobody in the centre" failed at one row (3/5) and became the window count $q - p - 1$ (fifteen of fifteen after the data, four of four predicted).

Holds: not yet carried elsewhere. *Met in:* driven oscillators, waves, maps and chaos.
Shown: Loop Atlas, R2W; full guide row 514.

Good rule, sharp domain (517) — Keep a claim with its domain named rather than its scope quietly widened. That the ordinary gap exceeds the log gap is exactly true whenever both numbers sit above one: the doubling ladder gives gaps of 1, 2 and 4 against log gaps of 0.693 each, while below one it reverses, 0.5 against 0.693. It holds for every prime, since all are at least two.

Holds: carried to one: failed in values below one (seam). *Met in:* number theory.
Shown: version 6; full guide row 517.

Every tooth divides the period — and only some divisors are teeth (520) — Read a containment in the direction it was measured. Across eight periods there were 21 teeth and zero false positives, yet only 21 of the 29 divisors in range stood up, so dividing the period is necessary and not sufficient. The arithmetic decides what is possible, a prime period of 13 producing no teeth at all, but not what appears.

Holds: not yet carried elsewhere. *Met in:* number theory, quadrature.
Shown: version 6; full guide row 520.

Apodization kills by APERIODICITY, not by depth (522) — Same unevenness, opposite outcome: four weight patterns rescaled to one max-min ratio split by shape, the aperiodic ones reading 0.76 and 0.84 for the taper and 0.84 and 0.91 for random rows, against a flat reference of 1.85 and 2.57, while the periodic weights read 1.57 and 1.92, and 1.84 and 1.51. A taper of the same depth that repeats is invisible to the comb.

Holds: not yet carried elsewhere. *Met in:* number theory, quadrature.
Shown: version 6; full guide row 522.

The teeth MOVE — which is what makes it the lattice and not the integers (528) — A prediction that something already measured will vanish is worth more than one that something new will appear. Holding the lattice and changing only the period, q = 5 stands at 1.55 on a ten-row lattice and lies flat at 0.52 on a twelve-row one, and q = 7 stands at 1.52 on fourteen rows and flat on the others, so the comb is commensuration.

Holds: carried to one: failed in the same tooth on a twelve-row lattice (knob). *Met in:* topology, number theory.
Shown: version 6; full guide row 528.

One line of bonds turns the comb on, exactly (531) — What separates a comb from its absence is whether the chain closes. Against a fixed ring size, six ring counts on an open chain gave five teeth that divide nothing; closed by one more line of bonds, the same counts gave fifteen teeth, every one a divisor, with zero false positives. In the Loop Atlas sessions a ring conveyor agrees: teeth stand at the divisors of the ring's wave count, none on the open line.

Holds: carried to two: held in a ring conveyor rider (Loop Atlas); failed in the same chain left open (substrate). *Met in:* driven oscillators, topology, number theory.
Shown: Loop Atlas; full guide row 531.

Holonomy hunts five things — name which one before you start (534) — Each target decides what floor to measure, which channel to read and what counts as a find: a loop to close, a quantised integer or winding, a signal or residual left after the integer is extracted, a noise floor, and an invariant that survives the full loop. The noise floor comes first, before any of the other four can be trusted, and confusing the hunts gives a clean result for the wrong object.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 534.

Geometry carries the load at vertices, junctions and seams — not in every cubic centimetre (536) — Spacetime holds more data than can be imagined, but that granularity is not required, because geometry stores its information at the interesting places: a vertex, where paths meet and curvature concentrates; a junction, where topological type can change; and a seam, where something does not close. Everything in between is roughly flat and roughly boring, so multi-channel holonomy-friendly instrumentation reads where the geometry acts rather than sampling everything.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 536.

The rate of change, and the rate of change of that — ternary all the way down (537) — Decompose the change a walker needs into rates, and rates of those rates, and read each level with the ternary: +1 for a slope climbing, zero at its tangent kiss between up and down, −1 for a slope falling. This is the architectural reason rate-over-level keeps arriving: not only that rates are more robust, but that the geometry is rates, down to the sign, whether in a sine wave or in a curve's curvature.

Holds: carried to one, held there: the walker's ternary rate decomposition. *Met in:* measurement, topology, quadrature.
Shown: version 6; full guide row 537.

The enclosed sum is not bounded by the walk's ceiling — and that is the headline (542) — A failed assertion can be the result itself. One asserting that no enclosed sum can exceed its own boundary's n/2 was violated in 4,420 of 4,800 loops in the smoke run: at k=120 the enclosed sum held 120 while the 34-edge boundary could only ever have read ±17. Walking the boundary wraps the number away and summing the interior triangles recovers it, so it is reported as a headline.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 542.

Degenerate bins print, but they do not vote (544) — A bin whose value is set by discreteness rather than physics cannot be allowed to drive the answer, though it still renders in full for the audit trail. A median of three is one: in the triple-star sessions a registration with no floor on its count missed at the birth with 3 down-jumps and at one with 12, and held at the 13 others, so every median in a registration carries a floor on its count.

Holds: carried to two, held in both: small-area contrast bins; triple-star registration medians of three. *Met in:* measurement, three body, geodesy.
Shown: R2W, triple-star; full guide row 544.

Writhe is geometric, not topological — and the probe's conservation is slow drift, not conservation (546) — Do not call a quantity conserved because it is slow. Run 5× longer, the probe's writhe drifted from −3.357 to −3.411 while the filament stretched from 28.82 to 29.15; the knot's topology is preserved by construction, but writhe changes under smooth deformation, with twist taking up the difference. Read the topology from the Alexander polynomial and the conserved dynamics from helicity.

Holds: not yet carried elsewhere. *Met in:* topology, waves.
Shown: version 6; full guide row 546.

An exponent that tracks a length is a magnitude wearing an invariant's clothes (548) — Before connecting two numbers that look alike, ask what kind of object each one is. A comb line bouncing between about 1.5 and 0.75 and a filament energy exponent near three-halves looked like a rhyme, but both were magnitudes: the same tooth read 0.76 at full aperture and 0.93 at a tapered one, and the latch exponent, 1.554 after relaxation and 1.565 before, belonged to the shapes the construction built.

Holds: carried to one: failed in the latch filament energy exponent (ruler). *Met in:* measurement, materials, number theory.
Counted: The integer diagnostic 1. *Shown:* version 6; full guide row 548.

A quarter turn hid inside a wrap of ±45, and the fence was the arithmetic (554) — Fix every sign convention against the area's orientation before reading a sign. Dancing on the Sun compared HMI's azimuth with half the arctangent of Stokes U over Q wrapped into −45..+45 and built two forward models on it; wrapped into −90..+90 instead, a full refit sat 88.2 to 90.3 degrees from HMI's at every band, because HMI counts Q with the opposite sign to the textbook Milne-Eddington model. The wrap folded the convention away, and both models inherited it.

Holds: carried to three: held in the Sun's azimuth comparison; failed in HMI's reported azimuth (knob); held in the Loop Atlas polarisation toy. *Met in:* measurement, waves, solar.
Counted: A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, Dancing on the Sun; full guide row 554.

A fitted nuisance absorbs part of a planted mismatch, so a crossover moves less than the arithmetic says (557) — Every free parameter a fitter has pays part of a planted mismatch, so the pre-written expectation is kept as a bracket, not a miss. In Dancing on the Sun the Faraday crossover sat at 943 G at 34 mÅ; the expectation written before the run was 690 G at 25 mÅ and 1,390 at 50; it measured 780 to 821 and 1,162 to 1,240 and moved the same way, about half as far.

Holds: carried to one: weaker in planted widths of 25 and 50 mÅ (ruler). *Met in:* measurement, solar.
Shown: Dancing on the Sun; full guide row 557.

A count is whole only when it is read as a sum of closed pieces (560) — The question is not whether a read comes out near a whole number but whether it is built from closed pieces, so that it must. In the Latch Atlas a skyrmion's count, added up triangle by triangle, came out −1.000000000000 for whirls of every size, while the smooth textbook formula on the same grids gave −0.669, −0.906 and −0.976: right in the limit and wrong on every real grid.

Holds: carried to one: weaker in real grids of skyrmion whirls (ruler). *Met in:* measurement, topology, materials.
Counted: The integer diagnostic 1. *Shown:* Latch Atlas; full guide row 560.

On real data, the rim decides (566) — A count read from a rim is only as good as the rim: through two X-class flares, 1,589 and 1,352 loops whose rims stayed readable kept their counts, every change falling where a rim went weak. An angle likewise needs a gate on its amplitude: in R2W, RATCHET-22's reader graded an offset whose amplitude, 3.48×10^{-5}, sat below its coefficients' errors, so RATCHET-23's reader asks for at least four times the larger error.

Holds: carried to one: failed in flare-site loops whose rims went weak (ruler). *Met in:* driven oscillators, topology, solar.
Counted: A check that cannot fail 1. *Shown:* R2W, Dancing on the Sun, Latch Atlas; full guide row 566.

A measure takes the instrument's turning in full; a count reads through it (567) — Counts and measures have different blind spots, and the difference between them is where the instrument shows. A field rotation put in on purpose at 5.5 degrees across 38 sunspot maps moved every current in full but only 31 of 154,267 readable loop counts, and at 20 degrees only 97; the offset in every sunspot's current behaves like such a turning, and survived blind tests on 24 of 26 and 45 of 50 fresh regions.

Holds: not yet carried elsewhere. *Met in:* measurement, topology, solar.
Shown: Dancing on the Sun, Latch Atlas; full guide row 567.

The window for a read has two edges and a trap in the middle (568) — Coarse sampling misses a count, noise fabricates one, and smoothing by likeness erases it. In the Latch Atlas, scattered clean stations under-count a whirl and never invent one; noise past about 0.6 radians makes false whole numbers, not fractions; and averaging arrows that look alike, wherever they are, outvotes the rare core that carries the count.

Holds: not yet carried elsewhere. *Met in:* measurement, topology, materials.
Shown: Latch Atlas; full guide row 568.

Split by polarity before you sum (575) — The physics fixes which symmetry a quantity must have, so a part with the wrong symmetry is a witness against the instrument. A current the Sun makes is polarity-odd, so Dancing on the Sun assigned the polarity-even part to the instrument or the pipeline, predicted it and found it blind twice, on 24 of 26 and 45 of 50 regions. In R2W the same split turned a steering knob into a pure size knob (RAMP-1).

Holds: carried to two, held in both: third star's apsidal motion; R2W silent pair's trim. *Met in:* driven oscillators, three body, solar.
Counted: The mirror audit 2. *Shown:* R2W, triple-star, Dancing on the Sun; full guide row 575.

Three questions on every toy, and one knob alone (584) — Four cheap moves caught real structure every time across the Loop Atlas's toys: faster asks whether an effect follows the rate or only the shape, backward whether it changes sign, again whether it is a property or an accident, and one knob alone which input owns it. In R2W's BOARD-2 the damping ratio was held fixed, so softness could not be told from damping; BOARD-3 moved the softness alone.

Holds: carried to one, held there: a ratio test and a stiffness test. *Met in:* measurement, driven oscillators, kinematics.
Shown: Loop Atlas, R2W, Question the Substrate; full guide row 584.

Name the pair of knobs before comparing two cases (586) — Two cases can only agree or disagree about the same pair of quantities. In the Loop Atlas sessions a skater moving "in step" and a wing moving "a quarter apart" looked like a contradiction until it was seen that each phase was measured between a different pair of knobs. Before reading one phase against another, write down which two quantities each is measured between.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, kinematics.
Shown: Loop Atlas; full guide row 586.

A zero and a small number both need a second look, for opposite reasons (587) — A zero is suspected of being the code and a small number trusted as physics, and in one Loop Atlas sitting each was wrong once: the stalled blade was real, and the snake's 0.002 was a bug in the stepping rule. An exact zero may also mean a test cannot fail: an arrow started down a ridged surface read exactly 0.000e+00 by symmetry; rebuilt with a slanted arrow, the test could fail, and passed.

Holds: not yet carried elsewhere. *Met in:* measurement, kinematics.
Counted: A check that cannot fail 1. *Shown:* Loop Atlas; full guide row 587.

A toy that harvests needs a supply ledger (590) — A simulation that draws work from a medium without a ledger of what the medium can supply is a free-energy machine by omission. In the Loop Atlas sessions a listening blade got a steady pull from an even soup, its charge appearing from nowhere; with the opposite charge left on the ship, the pull is exactly zero at zero tether, and the old listener's 0.124 was a free lunch the toy did not bill for.

Holds: carried to three: held in the listener's ear; held in the listening blade (counter-charge toy); failed in a slope ear (SLOPE_EAR) (break not stated). *Met in:* waves, control and circuits.
Shown: Loop Atlas; full guide row 590.

A reflected starting nudge fakes a flip (601) — Parity cannot tell a reflected starting condition from a ride, so every blade or rider toy prints its starting nudge's arithmetic beside its result. In the Loop Atlas sessions a draft horse with no soup to pull through read +0.00048 and flipped sign under the rider mirror, which looked like a ride; it was the starting nudge: 0.01 of momentum over an outfit of 21 is 0.000476.

Holds: not yet carried elsewhere. *Met in:* measurement, kinematics.
Counted: The mirror audit 1. *Shown:* Loop Atlas; full guide row 601.

A census without a slope test carves fractions out of a smooth curve, and the control the tool names must be run (603) — A plateau is accepted only with flat shoulders and only after the null case has run. A survey listed 26 to 30 plateaus per case and only two were flat, and the control, run with no second rhythm at all, found 11/5, 9/4 and five more fractions at the same paces. Read four times longer, a 4/3 “lock” moved away, its mean departure 10.856 times larger.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, three body.
Counted: Chaos and binning 1, Coupled clocks and event time 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 603.

The price of listening is strength over rhythm (604) — A listening rider’s harvest grows with the medium’s strength and its bill with the medium’s rhythm, so place every listening rider on the price map before building it. Across 25 cells of the Loop Atlas’s price card the measured price ran 0.941 to 1.109, the flip rate sat at 0.97 of Rice’s zero-crossing rate, and the card broke only where beads were caught, at strengths of 0.4 to 0.5.

Holds: carried to two: failed in strengths 0.4 to 0.5, where beads were caught (seam); held in the swing equation (SWING_GRID). *Met in:* driven oscillators, control and circuits.
Counted: The knee 1. *Shown:* Loop Atlas; full guide row 604.

Engagement is depth and speed together (606) — Which of two rhythms captures a rider is not decided by closeness of speed alone, so sweep depth whenever a handover is reported. In the Loop Atlas sessions the handover from the strong rhythm to the weak came at paces 0.97, 0.80 and 0.59 as the weak rhythm’s depth rose through 0.1, 0.3 and 0.6, correcting a prediction of a synchromesh handover; three points cannot fix the depth law, so it is the follow-on test.

Holds: carried to one: weaker in a rider between two rhythms (knob). *Met in:* driven oscillators, kinematics.
Shown: Loop Atlas; full guide row 606.

A small-stroke rule is the summit of a landscape (607) — When a small-stroke rule fails at large stroke, map the curvature before blaming the tool: a large loop integrates a grip that varies across it and can change sign. On the Loop Atlas’s big loop, Stokes on the mapped curvature came within 1.9 per cent at 60 degrees while the square rule ran 25 per cent high, and the loop’s room did not shrink as predicted; the grip varies.

Holds: carried to three: failed in the big loop at 60 degrees (seam); held in the same loop walked by wave plates; held in ridges as steep as 76 degrees. *Met in:* topology, waves, kinematics.
Shown: Loop Atlas; full guide row 607.

Counts are gears and measures are friction wheels (608) — Tag every new loop as a count or a measure first, because the tag decides which test applies, and two counts add only when they share a base. Of the Loop Atlas’s 43 loops tagged on one base, 33 transmit only a measure, 9 carry a lock and transmit a count, and 1 needs quadrature; a first tally of 12 beside 37 (49 of 43) had counted rows that are not loops.

Holds: not yet carried elsewhere. *Met in:* measurement, topology, kinematics.
Counted: Count the right unit 1, The integer diagnostic 1. *Shown:* Loop Atlas; full guide row 608.

Resonance can reach a fence that strength alone does not (609) — Check any closed medium for resonant gain against its catching fence before trusting its numbers. The Loop Atlas’s ring conveyor, on its first run at a gain of 0.25, pumped the grip to 22 through resonance and the bead caught the carrier, so the run was retired and repeated at a gain of 0.01.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, kinematics.
Shown: Loop Atlas; full guide row 609.

A travel number says whether it is a length or a signed part along a named line (612) — Opposite signs, which a comparison of sizes drops, can be the clue that one lap was read two ways. The Loop Atlas’s big loop at a swing of 60 degrees travelled +0.054869 in one tool and −0.05277 in another: one returned the length of the shift, the other its signed part along its own x line, and the 3.83 per cent gap is the cosine of a 15.9-degree tilt, most of it the frame’s.

Holds: not yet carried elsewhere. *Met in:* measurement, kinematics.
Counted: A word that changes meaning across a boundary 1. *Shown:* Loop Atlas; full guide row 612.

Read the data once more, as if no physics applied (613) — After the numbers and the surprises, say in plain words what the data looks like and what it is doing, naming no law and writing no formula. A read is never a result, but it is where the next test comes from: “memory lives in the slip, not the grip” went to another session as a sentence and came back as a registered prediction that held, and seven of R2W’s pictures became registered tests.

Holds: carried to two: held in another session’s registered prediction; failed in one session’s own reads (break not stated). *Met in:* records and tools, reading pictures.
Shown: Loop Atlas, R2W, triple-star, Field Guide sessions; full guide row 613.

An event is read from the conserved quantity, not from one coordinate (614) — A threshold on a position is a guess about an event; the conserved quantity says whether it happened. A triple-star tool stopped a run when a star passed 40 units from the centre and called it an escape, but checked against the star's energy, 6 of the 15 runs it called escapes were still bound, and those stars came back. An escape is far and unbound.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: triple-star; full guide row 614.

Count where the time goes, not how many episodes there are (615) — Counting by episode lets many short events outvote the long ones that hold almost all the time. A triple-star test asked whether a gentle triple's rests lasted at least three passes by median, read 1.0 and gave no reading; weighted by time, the same table gives a median rest of 207 passes, and 14 of 18 runs never left the rest they were born in. It is the inspection or waiting-time paradox of probability.

Holds: carried to one, held there: a switch's mistakes (relay, timer, ear). *Met in:* measurement, three body, control and circuits.
Counted: Count the right unit 2, Coupled clocks and event time 1. *Shown:* Loop Atlas, triple-star; full guide row 615.

Before trusting a statistic, ask what a featureless record of the same length gives it (616) — Run the statistic on a record with nothing in it first, and build the control into the tool before its first run. In the triple-star sessions the Kozai triple's recurrence lines had a median lag of 218 and 219 strobes, first read as one eccentricity cycle, but lines spread evenly over its 748 strobes give 221: the median carried only the record's length, and the reading was retracted.

Holds: not yet carried elsewhere. *Met in:* measurement, three body, maps and chaos.
Counted: A check that cannot fail 5. *Shown:* Loop Atlas, R2W, triple-star; full guide row 616.

A crowding statistic cannot tell a comb from a pile (617) — A share counts crowding of any kind, teeth or heap, so measure spikes against the spectrum's own smooth trend instead. In the triple-star sessions the Kozai triple's lag spectrum is a plain comb to the eye, yet the share of points on its most crowded lags came to only 1.69 and 1.92 times that of clean chaotic runs, short of the registered 2.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: R2W, triple-star; full guide row 617.

A selection rule must require the property the test depends on (618) — Write down what the test needs of a case, and make the picking rule demand it. A triple-star rule for choosing which run to strobe picked one whose inner pair changed partners 22 times, so its kicks landed on close approaches 0.082 of the time against 0.100 by chance; on the two runs whose binary never changed partners, the same test gave 0.900 to 1.000.

Holds: carried to two: failed in run whose inner pair swapped 22 times (substrate); failed in a grid five times finer (ruler). *Met in:* measurement, three body.
Shown: triple-star; full guide row 618.

Shuffle at the level of what is independent: the run, not the episode (619) — Episodes from one run are not independent replicates; counting them so is pseudoreplication (Hurlbert, 1984). In six triple-star control runs, one carried every bit of an apparent effect at the 2 per cent threshold, 10 of its 13 long rests within 0.1 of a period ratio of 6; on 48 fresh runs the reading was 0.209, matched or beaten by 0.425 of null draws that treat each run as one witness.

Holds: not yet carried elsewhere. *Met in:* measurement, three body, maps and chaos.
Counted: Count the right unit 2. *Shown:* Loop Atlas, triple-star; full guide row 619.

Build the fair control into the starting point (620) — Choose a start that cannot fake the result, so that its opposite becomes a fair control. The triple-star boundary-layer runs start at a period ratio of 6.25, as far from 6.5 as from 6, so half-integers fairly control for whole numbers; at the 5 per cent threshold rests sat near a whole number 0.603 of the time and near a half-integer never, a level that row 456 shows was luck, the replication reading 0.302.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: triple-star; full guide row 620.

Re-run what was seen as unscored controls, and draw fresh cases from a grid that cannot repeat it (621) — Test a sighting on cases that could not have produced it, and keep code trials on the controls. In the triple-star sessions the six phases where a rest at ratio 6 was first seen were re-run beside 48 fresh phases offset by half a grid step: all 1,338 of their excursions reproduced exactly, unscored, and on the fresh phases the rest at 6 dissolved into a slow creep around the starting ratio.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: triple-star; full guide row 621.

When two thresholds fixed in advance disagree, ask what each one merges and what each one splits (622) — A disagreement between two rulers fixed before reading is data about what each can see. In the triple-star sessions a 2 per cent threshold cut a slow creep near the starting ratio into many long rests, while at 5 per cent the creep merged into the first rest and only rests after a real departure remained, 14 in 10 runs, near 7 and 8: the finer ruler chops a rocking star, and the coarser keeps it whole.

Holds: carried to one, held there: triple-star rests between departures. *Met in:* measurement, three body.
Counted: Chaos and binning 1. *Shown:* Loop Atlas, triple-star; full guide row 622.

Count landings, not stumbles (627) — A counter watching a noisy phase at one line logs the jitter across it as pairs of crossings, so commit a slip only when the needle lands in the next well. In the Loop Atlas's turning needle, backward slips came out only 7 to 15 times rarer than forward ones, though a true backward slip essentially never happens; given a knee, no row showed one, and the net count moved by 3 at most.

Holds: carried to two: held in a counter of the turning needle's slips; failed in a heavy third star (ruler). *Met in:* driven oscillators, three body, quadrature.
Counted: Count the right unit 4, Chaos and binning 1, The integer diagnostic 1, The knee 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 627.

An event that lands on a sample point is counted by rounding; count in events, never in seconds (628) — Where the record is made of steps, count in whole steps, or rounding decides the result. In the Loop Atlas sessions 96 per cent of a tapping rider's gaps are exactly its rest, 200 clock steps; read in fall times, rounding split that pile differently on two boards (Kolmogorov-Smirnov distance 0.4224), while counted in whole clock steps the boards agree to 0.0065. A bar that is a whole number of grid steps lands on the grid, so set it at a half-step.

Holds: carried to four, held in all four: tapping rider's rest timed in seconds; tapping rider's gaps on two boards; spring bead's beat count; R2W DOOR-3 bar on a 0.001 sweep. *Met in:* measurement, driven oscillators, quadrature.
Counted: Chaos and binning 1, The integer diagnostic 1, The knee 1. *Shown:* Loop Atlas, R2W; full guide row 628.

Two honest counters differ only at the ends of the watch; find the identity that relates them and check it exactly (629) — A disagreement between two counters is settled by the identity that relates them, not by choosing one. In the Loop Atlas sessions a ratchet logging a slip on first topping the barrier and a knee logging it on landing agree on the rate, and ratchet minus knee equals the needles past the top at the end less those mid-slip at the start, to the count ($25 = 31 - 6$, $20 = 37 - 17$).

Holds: carried to three: weaker in a knee ear watched now and then (seam); held in a watcher's looks (BRIDGE_COUNT); held in the never-blinking reference watchers. *Met in:* measurement, quadrature.
Shown: Loop Atlas; full guide row 629.

Find a toy's clock by doubling its rhythm (630) — Whatever stays fixed when the rhythm doubles is counted in the toy's own events; human seconds enter a loop only through a force that lasts for a time or a reaction delay. In the Loop Atlas sessions the wing went 1.873166480074 m per flap at 1.5, 3 and 6 flaps per second, and only the skater counted seconds, because its legs push with a force for a time.

Holds: carried to two: failed in the skater (substrate); weaker in two tied bells in a remembering water (knob). *Met in:* kinematics, quadrature.
Counted: Coupled clocks and event time 2. *Shown:* Loop Atlas; full guide row 630.

A grid cannot find a sliver; aim it with the exact boundary (631) — When a thing gets thin near an edge, compute the edge first. In the Loop Atlas sessions, near the square root of two, the safe island of gains for balancing an upright board with a delayed push is 1.222×10^{-4} tall at a delay of 1.40 and gone at 1.415; grids of 0.05 and 0.01 missed it twice, and a grid placed by the boundary curve found it.

Holds: carried to one: failed in two higher eccentricities, births 0.03 apart (ruler). *Met in:* measurement, driven oscillators, control and circuits.
Counted: Chaos and binning 4. *Shown:* Loop Atlas, R2W, triple-star; full guide row 631.

A switch amplifies the last digit, so event-driven runs are compared by their statistics (632) — Compare event-driven runs by their statistics and gap histograms, never tap for tap. In the Loop Atlas sessions two tapping riders whose starting tilts differed by 10^{-12} rad tapped identically for 28 taps and then parted, while their statistics agreed within their spread. Past the predictability horizon, report fractions and patterns, not fates: which triple-star births hold is a sample of the chaos, and the fraction that holds is what carries over.

Holds: carried to one: weaker in fold refiner run on two machines (seam). *Met in:* measurement, driven oscillators, maps and chaos.
Counted: Chaos and binning 1. *Shown:* Loop Atlas, triple-star; full guide row 632.

Grade a rhyme before building on it, then test the chapter's object, not its metaphor (633) — A rhyme across projects can be a coincidence of wording while the real object sits one aisle over. Bear heard the Floer market of his Crossover Atlas in the Loop Atlas's push price, and the Atlas's own verdict tags graded it a coincidence; but the

turning needle's lag fit the chapter's actual object, Novikov's circle-valued setting, its locks always coming as stable and unstable pairs (stable minus unstable 0 in all 39 rows).

Holds: carried to four: failed in Loop Atlas push price (break not stated); held in Loop Atlas turning needle's lag; held in Loop Atlas rhyme test; failed in soup toy's local tilt (ruler). *Met in:* driven oscillators, topology, solar.
Shown: Loop Atlas, Dancing on the Sun, Question the Substrate, Two Arrows; full guide row 633.

Durations are the one form the ruler can't move (638) — A result written as ratios of durations survives any change of the board's size; one written in seconds does not. On a log scale a board four times longer slides the whole song of taps by exactly log 2 and leaves every ratio of gaps alone, 2.000000000 across 373 gaps; and a rider counting in the board's own fall time sings the same song on every board, while its taps per second rise from 2.99 to 11.98.

Holds: carried to one, held there: Loop Atlas book's tapping rider. *Met in:* measurement, driven oscillators, quadrature.
Shown: Loop Atlas, R2W; full guide row 638.

Read slipping itself, never distance from where it started (639) — Count net turns over a window, and widen the window until a bounded motion cannot reach the reader's threshold. A climb-back reader in R2W, calling a slip at 0.02 turn per passing over 20 passings, was tripped by a band that wanders inside the gear and reported scissors of 0.0047 and 0.012 of drive; on 100-passing windows, which a band narrower than a turn cannot trip, the openings read zero within about 0.002.

Holds: carried to one, held there: triple-star tread threshold. *Met in:* measurement, driven oscillators, three body.
Counted: Count the right unit 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 639.

A local law has a neighbourhood; register it (640) — A law read near an edge says where it holds before the run, and its neighbourhood is registered from the machine under test. In the triple-star sessions a drag law that explained 97% to 99% pooled over six nearby start lines was registered to explain 90% farther out and explained 44% to 71% (E7c missed); read line by line, it fell from 99.4% to 99.7% on its own line to 12.2% to 33.7% at 0.12.

Holds: carried to six: failed in three decades of distance (seam); held in the Loop Atlas needle's gear 2 edge; held in R2W's stops at 0.40; failed in fresh start lines farther out (E7c) (seam); weaker in the next two stretches (RATCHET-61) (ruler); weaker in the road read on to 1.80 (seam). *Met in:* measurement, three body, maps and chaos.
Counted: Count the right unit 2. *Shown:* Loop Atlas, R2W, triple-star; full guide row 640.

Read at a fixed moment of the cycle, or over one whole cycle, and say which (641) — A spread read at an arbitrary moment measures the moment, not the state: within one passing of a strong pattern a cloud of noisy copies is squeezed about a million-fold in variance, and read at the most-squeezed moment the wobble law held to 1 per cent. Compare riders at the same phase, never the same clock time: two walkers that disagreed by up to 2.56 radians agreed within 0.009 at the next passing.

Holds: carried to one, held there: a clutch's frame within one lap. *Met in:* driven oscillators, three body, quadrature.
Counted: Coupled clocks and event time 2. *Shown:* Loop Atlas, R2W, triple-star; full guide row 641.

Near a fold or a doubling, never classify by waiting for things to settle; follow the orbit and read its multiplier's sign (642) — States near a transition settle at a rate that goes to zero there, so follow the orbit itself by Newton's method and read its multipliers: −1 is a doubling, and +1 a fold or a doubled orbit's birth. Two riders needed 1,500 passings to sit within 8.5×10^{-8} of their period-8 orbit, and the Loop Atlas's period-8 end at 1.690017021, first called a fold, is the period-4 orbit's doubling.

Holds: carried to one: failed in a doubling (wall). *Met in:* records and tools, driven oscillators, maps and chaos.
Counted: A check that cannot fail 1, A second reader 1. *Shown:* Loop Atlas, R2W; full guide row 642.

Check whether two dials are one before sweeping both, and compare equations before comparing results (643) — A number from one room is a witness for another only once the dials are matched. The Loop Atlas's needle, R2W's rider and a grid generator's swing are one pendulum equation, and raising a toy's drag with the pull held steady is exactly the same as lightening the rider: one dial, mass over drag. Matched, the needle's switch 0.70256 sits inside R2W's measured 0.70 to 0.71.

Holds: carried to three: failed in R2W rider at dial 1.0 (knob); weaker in R2W's rider pendulum (knob); held in toy with cores given own size. *Met in:* driven oscillators, waves, kinematics.
Counted: A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, R2W, Field Guide sessions; full guide row 643.

Keep the numbers small, and cancel a background in the code, not only in the reading (644) — The physics cancels a uniform background; the floating point does not, unless the code does. A triangle of stars carried 400 units aside kept its loop law to the same residual as at rest, yet its history parted from the unmoved run's after its first two loops, because a faraway number is rounded more coarsely; and a clock that added 0.02 to itself 1,300,000 times read 26000.00000047, so count time in whole steps.

Holds: carried to one: weaker in same triangle carried 400 units aside (ruler). *Met in:* records and tools, driven oscillators, three body.
Shown: Loop Atlas, R2W, triple-star, Two Arrows; full guide row 644.

A window must hear three beats of its slowest beat, and a rule that can be computed belongs in the tool (645) — A remembered rule can lapse; a guard in the tool that counts the beats in each window and refuses a run below three cannot. In the Loop Atlas sessions a notch appeared in runs of 2,000 time units where the kicks and the teeth beat once every 5,776 units, a third of one beat, and at 10,000 units it was gone; the rule was broken hours after it was written.

Holds: carried to two, held in both: a beat of the beat, a level down; the triple star's 600-pass windows. *Met in:* measurement, three body, quadrature.
Counted: Coupled clocks and event time 4, A check that cannot fail 1, A word that changes meaning across a boundary 1, The integer diagnostic 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 645.

Labels are not measurements: compare positions, and measure a dial before naming it (646) — Name a quantity after it has been measured, and say in plain words which way a law runs. A gear's locked words matched the film's words at only 3 of 9 paces, because a rolling lap does not close, while compared by where the dots land the frames sat within 0.001 to 0.019 of the cut at 8 of 9; and a dial called "kick size" shrank the kick by only 11 to 14 per cent.

Holds: carried to three: failed in traders' constant 0.5826 for sampled switch (substrate); held in traders' constant 0.5826 for sampled switch; weaker in sampled switch dead band at coarser looks (knob). *Met in:* measurement, driven oscillators, number theory.
Counted: A word that changes meaning across a boundary 2. *Shown:* Loop Atlas, R2W; full guide row 646.

Repeat a known point inside every new sweep (647) — A known point inside a new sweep is the cheapest witness a sweep can carry. In the Loop Atlas sessions 483 runs were first scored against a wrong target, the product of three factors with one of them squared, and the slip was caught because rows that repeated an earlier test's points missed at every filter where they had hit before; the table was retired, and the speeds reproduced row for row.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: Loop Atlas, R2W, triple-star, Question the Substrate; full guide row 647.

A grid that lands on exact fractions samples frozen frames, so test a hair either side (648) — At an exact fraction the phase between two rhythms is frozen at whatever the run starts with, so the result is a setting, not a property of the ratio. Extra grip appeared at ratios of exactly 2.000 and 1.500 and at no neighbouring grid point (1.998, 2.002, 1.498, 1.502); in the triple-star sessions a blind figure placed by rule at 7.250 landed on 29/4, drew a frozen frame, and was redrawn a hair away.

Holds: carried to one: failed in same line at 0.001 in birth (ruler). *Met in:* measurement, driven oscillators, three body.
Counted: The mirror audit 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 648.

Register in the regime where the rival readings disagree, and name what carries an effect before naming its null (649) — A test cannot tell two readings apart where both predict the same: a first test of a slope chose such a range, and after two redesigns it held blind at −0.573. In R2W's FAREY-1, two laws given bands of 15 per cent differed by only 0.6 to 5.2 per cent, both held, and the run could not say which law the junction follows.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators.
Counted: A check that cannot fail 3. *Shown:* Loop Atlas, R2W; full guide row 649.

Rerun the other room's own code before judging its numbers (650) — The strongest check of another room's number is that room's own code, run here unchanged. A three-body law reported to a mean residual of 0.0014 radians was checked by loading the other room's tool, which printed its energy line exactly and found the same 58 loops; only then was the reading refined with nothing else changed, and the law held to 0.00003.

Holds: carried to one, held there: this room, run unchanged. *Met in:* records and tools, three body.
Counted: A second reader 2. *Shown:* Loop Atlas, triple-star, Field Guide sessions; full guide row 650.

A correlation reads pattern, not size (651) — Quote the residual's size beside any correlation of it. A memory test in the Loop Atlas book read +0.87 for the correlation of a residual with loop duration, which looked like memory; with 24 outlying loops set aside it read −0.22, and it read +0.87 again after finer reading had shrunk their offset fiftyfold, to 0.00007 radians.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: Loop Atlas, triple-star; full guide row 651.

When a knob changes mid-run, carry the right state into the new frame (652) — Write down which quantity is continuous across the turn of a knob. A sweep in the Loop Atlas book switched a needle's pace mid-run and kept the needle's own speed, when what had to carry across the switch was the speed of its lag behind the field; it read false

climb-back lines, 0.8105 and 0.5518, against the true 0.7899 and 0.5413 found by shooting from the saddle, and was retired.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators.
Shown: Loop Atlas; full guide row 652.

A ramp reads an edge; a hold asks what lives there (662) — A ramp cannot tell shut scissors from a lag, so test the picture behind a bound by holding riders at one drive. One room's ramps read an opening that halved as the ramp rate halved, the mark of a lag, while another room's ramp, sixteen times slower, read none; held at mass 0.75, every rider started running fell into gear 1 below the fold, and the bound was withdrawn by its authors.

Holds: carried to two: failed in another room's ramp sixteen times slower (ruler); failed in the 11/7 where the speed swung (ruler). *Met in:* measurement, driven oscillators, maps and chaos.
Counted: Chaos and binning 1, The knee 1. *Shown:* Loop Atlas, R2W; full guide row 662.

A reading window reports its middle, and the window that holds a change belongs to neither state (663) — Write a prediction about a change against the first whole window after it, with a tolerance wider than one window plus the lag. A 200-passing window spans about 0.0024 of the drive, so a departure can be read up to about 0.0012 early; a prediction that a rider would land in a second gear "within one window" of leaving the first missed for that reason alone, and held when read from the first window after.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, quadrature.
Counted: Count the right unit 1. *Shown:* Loop Atlas, R2W; full guide row 663.

A chart drawn from one start is blind to memory (664) — Run several starts per setting, and read where they part. The textbook tongue map, the sine circle map, run with 64 starts per drive, agreed at all 241 drives below its critical line (strength 0.9), but at strength 1.2 the starts disagreed at 42 drives, and a single start stepped backwards at 6; a chart of locks drawn from one start cannot show two states where two exist.

Holds: carried to one: failed in sine circle map at strength 1.2 (seam). *Met in:* measurement, driven oscillators, maps and chaos.
Counted: The knee 2. *Shown:* Loop Atlas, R2W; full guide row 664.

Name a state only by a test that separates it from its neighbours (665) — A reader with fewer bins than states gives the extra states a neighbour's name, so key a state by its fraction and its period together. A four-bin reader called "running" what a strobe showed to be mostly tumbling, and at one drive 12 riders dance a cycle of 6 passings and 52 a cycle of 12, both losing two turns in three, a memory no reader keyed by fraction alone could see.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, three body, maps and chaos.
Shown: Loop Atlas, R2W, triple-star; full guide row 665.

Look in the record, yours and the next room's, before registering (666) — Before registering, search your own record and the latest letters for the rule that sets each side. Two predictions modelled a relay's second stage by its steady-drive fold when the session's own record, measured on 3 October, already said that edge holds only when the filter is slower than one over the room, and both failed as the record had predicted; a rung sent out as an open question had already been measured (1.6924902823).

Holds: carried to four: failed in the relay's second stage under a faster filter (seam); failed in this room's registrations (break not stated); held in the Loop Atlas book's frame; held in the old law's tables (DELTA-WEAK-1, FAR-1). *Met in:* measurement, records and tools, driven oscillators.
Counted: A check that cannot fail 1, A second reader 1. *Shown:* Loop Atlas, R2W, Field Guide sessions; full guide row 666.

Change the pace to separate dynamics from geometry (668) — Run a loop at several paces and read the intercept: geometry decides whether, dynamics decides how much. Walking the same loop of drive and mass at three speeds gave drifts of 797, 390 and 185 turns, and fitted as a constant plus a slope times the walking time, the constant was −18 turns, under 5 per cent of the largest drift, so the drift is almost all dynamics.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, topology, kinematics.
Shown: Loop Atlas, R2W; full guide row 668.

A prediction can hold for the wrong reason: grade it held as worded, and say what stays open (683) — Test the mechanism behind a held prediction with the simplest separate run, and search the record first. A memory floor predicted at the retrapping pull held (floor 0.86, band 0.85 to 0.95), but a one-ball run with no ripple showed a rolling ball keeps rolling down to 0.78 to 0.80, so the band had caught the wrong mechanism; such a result is graded "held as worded" and given a follow-on.

Holds: carried to one, held there: R2W memory floor at retrapping pull. *Met in:* measurement, driven oscillators, three body.
Counted: A check that cannot fail 2, A word that changes meaning across a boundary 1, Chaos and binning 1, A

second reader 1, The knee 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 683.

A verdict is only as fine as the ruler under it: print what the step allows, say what the ruler does decide, and look for the ruler in every miss (686) — Under every judged number, print what the instrument's step allows and whether that range crosses an edge of the bar. A ratio of 7.33 against a band of 3 to 7 was written down as failed, but the dial's step allowed anything from 5.25 to 11.5, so the failure decides nothing; and a "none" means none larger than the threshold.

Holds: carried to two: held in SEAM_SCAN's turning-point threshold; failed in lines high above the edge (ruler). *Met in:* measurement, driven oscillators.
Counted: A check that cannot fail 1, A word that changes meaning across a boundary 1, Chaos and binning 1, A second reader 1. *Shown:* Loop Atlas, R2W; full guide row 686.

A cut makes its own maximum (687) — A line drawn through a two-dimensional object reports the line's geometry as readily as the object's, so a "fastest line" read along one line is not yet a twistless torus. The test is cheap: at equal swing, do the two flanks rock alike? In the triple star's islands they never did, at any eccentricity of the inner pair.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: triple-star; full guide row 687.

Compare two sides at matched swing (689) — Before comparing two sides of anything, match them on the quantity that changes along the way. In the triple-star sessions a comparison of an island's two sides made at its edges compared runs of different swing, and at equal swing the answer reversed in 16 of 16 comparisons.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: triple-star; full guide row 689.

A starting label is not the quantity kept (690) — A width or a position read in the starting label carries an offset that moves with the other knobs, so read it in the quantity the system keeps. In the triple-star sessions the birth landing on an island's centre moved from 7.155 to 7.445 as the inner pair's eccentricity grew, while the period the star keeps there stayed between 7.010 and 7.022.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Counted: A word that changes meaning across a boundary 1. *Shown:* triple-star; full guide row 690.

Tell a fixed function from a force with two lines of arithmetic (691) — A per-pass change that tracks the sine of an angle, with a coefficient growing in step with the angle's own rate, is a fixed function of the angle, not a force through it; a force gives a fixed coefficient, and a swing that shrinks as the rate grows. In the triple-star sessions this told a reading artefact from a dynamical kick before any run was repeated.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: triple-star; full guide row 691.

A reader can count a sideband instead of the signal (693) — When a frequency misses, look for the strong line beside the one wanted, and read a rhythm on the quantity that owns it. A triple-star rocking reader read 0.07058 cycles per pass against an expected 0.09045, because it was counting the lower beat sideband of the star's semi-major axis (0.0704); read on the resonance angle's own spectral peak, the rocking came to 0.09044.

Holds: carried to three: failed in the same test at birth 7.220 (substrate); failed in nearly round orbit, rocking below the floor (ruler); failed in HM1 reader's first 2,400 passes (knob). *Met in:* measurement, three body, quadrature.
Counted: Chaos and binning 1. *Shown:* triple-star; full guide row 693.

Measure about a set that holds the objects (694) — A symmetry offers more than one set to measure about; use the one the objects live on. In the triple-star sessions the split of a cascade's newborn orbits, measured about the line $y = -x$, on which none of those orbits lies, had no reading; measured along the map's other symmetry set, a parabola, it shrank by about 4.018 per level and flipped side at each.

Holds: not yet carried elsewhere. *Met in:* measurement, three body, maps and chaos.
Counted: The mirror audit 1. *Shown:* triple-star; full guide row 694.

A reduced problem is a feedback loop: ask on the attractor, and compute the loop's gain before believing its zeros (699) — Before counting the zeros of a reduced problem, compute its loop gain, because a sign change through a pole of the gain is not a root. In the Loop Atlas book a test found orbits outside a lock's edges, but bisected sixty times the residual stayed at 4.91×10^{-10}, a jump; on the 11/7 lock the reading fell to 0.43 of the truth in one place and rose to 1,176 times it in another.

Holds: carried to one, held there: R2W reduced closing condition. *Met in:* driven oscillators, maps and chaos.
Shown: Loop Atlas, R2W; full guide row 699.

A loop reading is a remainder; the whole count lives in its history, watched faster than half a turn per look (700) — A frame carried round a loop turns by the enclosed curvature only modulo a whole turn, so keep the path, not only the endpoint, and look often enough. In the Loop Atlas book, 1,801 circles each 0.002 wider than the last counted 11.966913598514; watched at every 10th circle, where the largest change between looks was 3.0183 radian, the count held, and at every 11th, at 3.3638, it slipped a whole turn.

Holds: carried to one, held there: frame carried round widening loops. *Met in:* topology, kinematics, quadrature.
Counted: The integer diagnostic 1. *Shown:* Loop Atlas; full guide row 700.

A map's peaks are not a count of sources: check what stands under a peak's centre with a tiny loop (701) — A local maximum of a loop reading may belong to two sources at once, so a peak counts as a source only if a tiny loop at its centre turns. In the Loop Atlas sessions three hidden mountains read back from frames gave a fourth peak midway between two, turning the frame 1.3786 radian with plain under its centre; a circle of radius 0.02 there turned it by exactly 0.

Holds: not yet carried elsewhere. *Met in:* topology, kinematics, reading pictures.
Shown: Loop Atlas; full guide row 701.

Every hole is a counter: a histogram on each axis gives the numerators and the denominators, and a hole's landings give the period (702) — A fraction needs both histograms, since the side one alone mixes a dance's shares with how densely the dial was sampled, and each hole is also a counter of its own. With 256 recorded passings a hole on a dance of period p collects 256/p landings, so on Bear's lattice the median hole read 128, 64, 32 and 16 in every column of four bands, giving the period without following a single rider.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, number theory, reading pictures.
Counted: Count the right unit 1. *Shown:* Loop Atlas, R2W; full guide row 702.

The size of the holes decides what a string is: choose the scale from the tightest pair to be separated, count before naming, and keep a relation only if it survives a change of scale (703) — The wrong scale gives a confident wrong shape, so choose the scale from the tightest pair to be separated and say it with the claim. A lattice reader registered to see an eight-step dance as eight strings saw four tight pairs at half a degree, and three predictions missed; read on holes of 10^{-4} radian, the same picture held 8 of 10.

Holds: carried to one: weaker in hill whose slope changes along a leg (ruler). *Met in:* driven oscillators, kinematics, reading pictures.
Shown: Loop Atlas, R2W; full guide row 703.

Frame the picture before reading it: cut a circle where nothing lands, and record a whole number of the dance's periods (704) — An angle or a lag is a circle and a picture's edge is a cut, so a cut through a band makes one band look like two. In one Loop Atlas picture only 7 of 340 shapes lay between lags 0.27 and 0.52, and a reader that put its seam in the middle of the side histogram's widest empty stretch left none of 461,312 dots outside a family.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, reading pictures.
Counted: Chaos and binning 1. *Shown:* Loop Atlas, R2W; full guide row 704.

A split is a tree with three roads, and the tree sets its own denominators (705) — Count a split's offshoots including the road that goes straight on, the parent carrying on as an unstable ghost, because each split is a coin that halves a branch's share. Of one cluster's two strings in the Loop Atlas sessions, one never splits and the other halves, which predicts shares of 1/4, 1/4 and 1/2; the eye read 1/5, 1/5 and 3/5, and the map measured 0.22, 0.26 and 0.50.

Holds: carried to one, held there: Loop Atlas quarter tree, tassel from 1/4. *Met in:* driven oscillators, maps and chaos, reading pictures.
Counted: A second reader 1. *Shown:* Loop Atlas, R2W; full guide row 705.

Name each curve in the shape alphabet, fit the shape you named, and look for its copies (706) — Shapes repeat, and a shared shape hides a shared mechanism: name a curve in Bear's alphabet (flat; linear; ramp on or off; hill or bridge; split near equal or lopsided), fit the shape named, and look for its copy. On an R2W coin map at weight 0.40 Bear outlined a long lower fin; registered and run at 35,000 riders, the echo ran parallel at 31 of 31 drives (COIN-3).

Holds: not yet carried elsewhere. *Met in:* driven oscillators, reading pictures.
Counted: A word that changes meaning across a boundary 1, The mirror audit 1. *Shown:* Loop Atlas, R2W; full guide row 706.

Long reach beats many short chains; look on purpose for the strings the eye does not follow (707) — A string earns its place by how far it carries one smooth shape, and the eye follows the dominant slope, so the strings it skips must be looked for on purpose. On one picture Bear drew six long strings and called it 80 per cent, while a greedy decoder drew 50 strings of four or more holes, among them pieces of his long ones, and the six long strings explained more of the picture.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, maps and chaos, reading pictures.
Counted: A second reader 2. *Shown:* Loop Atlas, R2W; full guide row 707.

The change of change of slope marks an event: read it at the zero crossing between its two lobes, and read rates from where the events sit (708) — A centred stencil answers a sudden bend with two lobes of opposite sign, so read the event where the lobes cross zero, not at the larger peak. In the Loop Atlas book a needle's two largest peaks fell two columns early in every family; read at the crossings, three doublings were placed within 0.0031, 0.0014 and 0.0062 of a column, and the stretch ratio read 4.6153 against 4.6155 from the orbit.

Holds: carried to one: weaker in fourth doubling (ruler). *Met in:* driven oscillators, maps and chaos, reading pictures.
Counted: Rate over level 1. *Shown:* Loop Atlas; full guide row 708.

Geometry links pieces; history says which are relatives and what the true count is (709) — Only a rider's history, which rider and which passing, says which pieces are feet of one dance. In the Loop Atlas book, at half a degree, where the geometry saw one thick string per lane, history counted the doublings, the first columns with period at least 4, 8 and 16 falling at 264, 143 and 116 against doublings of record at 262.4, 142.3 and 116.5. Count the truth by history, not by rounding.

Holds: carried to one: failed in inside the hand-off window (seam). *Met in:* measurement, driven oscillators, reading pictures.
Counted: Chaos and binning 1, Count the right unit 1. *Shown:* Loop Atlas, R2W; full guide row 709.

Before joining two pieces, lift them into a third number; a touch the third number does not separate is a real meeting (710) — A strobe picture is a shadow of something with more numbers: a third number that runs on smoothly across a join suggests one piece, and one that jumps marks an overpass. Not every touch is one: two strands touching in angle, registered as an overpass more than 10^{-4} apart in speed, were 6.41×10^{-5} to 7.51×10^{-5} apart, a fork's young daughters, and holes of 10^{-5} in both numbers separated them.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, reading pictures.
Shown: Loop Atlas; full guide row 710.

Test a borrowed method on known answers, and run the ruler over a made-up case before reading meaning into a shape (711) — Score a borrowed reader first on answers already fixed, and check that the known answer is exact for the case run. Bear's lattice first read a needle's doubling staircase, whose splits Newton's method had fixed to ten digits: its first version missed and its second held 8 of 10 predictions. In R2W a textbook estimate used as the known answer was 14 per cent high at weight 2 (0.900 against 0.7899).

Holds: carried to three: weaker in needle's doubling staircase with known splits (break not stated); weaker in R2W stopping pulls at weights 10, 5, 2 (break not stated); failed in RD1a′ planted test at 1/10, 1/11 (ruler). *Met in:* measurement, driven oscillators, reading pictures.
Counted: A check that cannot fail 1, Count the right unit 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 711.

Name the picture before naming the pattern: each picture has its own neighbours (713) — A picture's axes decide which dots sit side by side, so every claim about a weave, a lane or a neighbour says which picture it lives in. One run of the triple star, at birth 7.200, has its dots' nearest relatives 5 passes apart on the eccentricity-against-pass page, 14 in the reading's own rhythm and 47 on the opened torus of two angles over 600 passes, each right for its picture.

Holds: carried to one, held there: triple-star sessions' numbers. *Met in:* driven oscillators, three body, reading pictures.
Counted: A word that changes meaning across a boundary 3. *Shown:* Loop Atlas, R2W, triple-star; full guide row 713.

When the data is held back, measure the picture, and know the plotter's traps (714) — Calibrated on a picture's own axis ticks and checked on marks of known value, pixel counts turn eye claims into numbers that reproduce, such as a top 3.7 to 6 times denser than its bottom. A count by eye is a guess: in R2W an "about four" came out at about every 80 passes once the pixels were counted, and a plotter that draws the last points on top makes colour counts favour late passes.

Holds: carried to two: failed in next panel, paler dots (ruler); failed in Bear's hand-marked saved file (ruler). *Met in:* driven oscillators, reading pictures.
Counted: The built artifact is the evidence 2, A check that cannot fail 1, Coupled clocks and event time 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 714.

A colour scale that lumps classes fools the eye; check the class before naming what a colour shows (717) — On one map, colouring by denominator painted the rest (0/1), the conveyor (1/1) and the 2/1 and 3/1 grooves the same orange. A blind read named a chain above pull 1.0 as the rest, and the data said conveyor, 37 cells at lag 0; a colour that stands for several classes cannot name any one of them.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, reading pictures.
Shown: Loop Atlas, R2W; full guide row 717.

Test smoothness before adding a dimension (718) — A low-order fit that misses is not evidence of a missing variable. In the triple-star sessions a coarse two-angle fit that explained 0.7230 of a bend's variance was read as a third hand at work, but a nearest-neighbour test on the opened torus of the two angles, where noise gives about 1, found the bend smooth at 0.1126, and smoother over 6,000 passes, at 0.0603.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Counted: Coupled clocks and event time 1. *Shown:* triple-star; full guide row 718.

A lock depends on its reader: name the reader with the count (722) — A count of locks is made by a reader, and the reader goes beside the number. In the Loop Atlas sessions the remembering needle at pace 0.95 was "locked" 1,051 times by a knee on its speed and caught 69 times by where its lag sits; matched needle by needle, every one of the 69 catches had its lock, and the speed reader's extras were hesitations past a shallow top.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators.
Counted: A word that changes meaning across a boundary 3, Count the right unit 2, Chaos and binning 1. *Shown:* Loop Atlas, R2W; full guide row 722.

Price a shape bar by simulation before blaming noise (723) — Simulate a bar's own false-alarm rate on the null shape before reading a miss as noise. In the Loop Atlas sessions a bar asking that the median of 256 lifetimes reach 0.6 of their mean fails by chance only 2.2 per cent of the time when the lifetimes are truly exponential, so three paces of six falling under it was a shape, two drains, not bad luck.

Holds: carried to one, held there: smoke-ring trains' lean, TRAIN_EDGE. *Met in:* measurement, waves, maps and chaos.
Counted: A check that cannot fail 1. *Shown:* Loop Atlas; full guide row 723.

Judge a transition by a measure that changes smoothly, and define a category by its mechanism, not by a symptom (728) — A stumbling ball and a crawling one can both show a small slip, and a yes-or-no line cuts a gradual change in two. In R2W's TRICKLE-3 and TRICKLE-4 a chaos line at +0.05 put weight 0.68 on the tumbling side by 0.0017, while across nine weights from 0.40 to 0.74 the chaos measure rose at every one and the crawl constant bounced, so the smooth measure is the ruler for the transition.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators.
Counted: A check that cannot fail 1, Chaos and binning 1. *Shown:* R2W; full guide row 728.

Read a control's number as data, and register the bar against a control (729) — A guessed outside target can call a large effect a miss. In the triple-star sessions a bar asking a slow beat's low half to run at an outside rate missed, 0.1608 turn per pass against 0.2069, yet against a control the low half ran 1.785 times as fast as the high half, against 1.01 for random cuts; the control was the right bar.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: R2W, triple-star, Question the Substrate; full guide row 729.

Print a quantity's range across the cases before building a rule, a rank or a ratio on it (730) — A measure that barely moves cannot switch a count that climbs. Across births 7.125 to 7.200 in the triple-star sessions a count climbed from nothing to 78 turns while three measures of the looks barely moved, one from 0.0104 to 0.0113, so every rule of the form "the count switches on where the measure is under X" scored exactly what one constant scored, 22 of 23.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, three body.
Counted: A check that cannot fail 2, A word that changes meaning across a boundary 1, Coupled clocks and event time 1. *Shown:* Loop Atlas, triple-star; full guide row 730.

A fake run checks the plumbing, and it must borrow nothing real (731) — A fake run can find in the container the crash that a real run would have hit overnight. In the triple-star sessions a fake worker whose runs carried zero seconds stopped a tool with a division by zero in its pace line; but a fake drive that gave its new cases the saved records of four real ones printed data the test had registered as unread.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Counted: A check that cannot fail 6. *Shown:* Loop Atlas, R2W, triple-star; full guide row 731.

The twin test has two floors: how long it watches, and the arithmetic it shares (732) — State a run's length and the e-folding it implies beside every twin verdict. In the triple-star sessions twins started 10^{-10} apart and called chaotic at 10^{-3} turn could see in 600 passes only chaos that e-folds in fewer than about 37 passes, and an island at 0.25 that was regular by twins over 600 passes dissolved over 6,000, every twin pair parting exponentially.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Counted: A check that cannot fail 1. *Shown:* triple-star; full guide row 732.

Register the exact symmetries first: a symmetry seen is a hint, the transformation that makes it is a test, and it is the cheapest bug detector (734) — An exact symmetry costs one line to register and catches the errors no physical guess would. Before an R2W run on a second tone with a relative phase, two exact symmetries were written down from the equation; the first reading broke the mirror at every phase except 0 and π, which nothing in the physics can do, so the instrument was wrong (RATCHET-1).

Holds: carried to three: failed in second tone with a relative phase (substrate); held in R2W's 1/2 dance stops; held in new path through the engine's code. *Met in:* measurement, records and tools, driven oscillators.
Counted: The mirror audit 4, A check that cannot fail 1. *Shown:* R2W; full guide row 734.

Solve the exact limit first, then the first order, then check the full problem at a small parameter (737) — Register the limit, then measure how fast the world leaves it. In R2W first-order bars on a pin and its stops, tested at weights 0.10 to 0.40, failed at two of three ripple strengths, while the limit itself, tested at weights 0.02 and 0.05, held at all three, each mirror drive equalling the weightless root to five decimals.

Holds: carried to three: weaker in first-order bars at weights 0.10 to 0.40 (seam); failed in drift's zero as weight grows (RATCHET-23) (knob); weaker in the book's drive (BEAT-2) (seam). *Met in:* measurement, driven oscillators.
Counted: Count the right unit 1. *Shown:* Loop Atlas, R2W; full guide row 737.

A number's rounding is part of its ruler: count the digits before blaming the steps or the physics (738) — A difference near a file's last kept digit may belong to the file, so check what the file keeps before reading it as physics or numerics. In R2W, NINTH-1 put a gap of 7.701×10^{-13} between two engines down to the engines, and STEP-1 found that 7.521×10^{-13} of it was the record's engine writing each edge to 12 decimals; with 15 the engines agreed to 2.598×10^{-14}.

Holds: carried to one, held there: every file of record (STEP-1). *Met in:* measurement, driven oscillators, maps and chaos.
Counted: A check that cannot fail 3, Chaos and binning 2, A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, R2W; full guide row 738.

Follow a branch by continuation, from where it was (739) — Anything that picks by distance from a mirror's centre mistakes one twin for the other the moment both exist, and that moment is the interesting one. In R2W a tracker that took the stable stop nearest lag 0 jumped between mirror partners at -0.016 and $+0.014$, so a registered one-way walk failed on the tracker; followed from its last position, the walk is one way, -0.0068 to -0.0251.

Holds: carried to one: failed in far cells out to -0.22, heavy weight (ruler). *Met in:* driven oscillators, maps and chaos.
Counted: A word that changes meaning across a boundary 1, Chaos and binning 1. *Shown:* Loop Atlas, R2W; full guide row 739.

A small, steady effect that is the same everywhere is the ruler's fingerprint: replay the reading on a known answer before explaining it (741) — Before explaining a small effect, replay the reading on a case whose answer is known. A groove's width came out 3 to 6 per cent short at every weight; replayed on a case with no table, no weight and no gear, the reading reproduced its own shortfall within 0.4 per cent, so the shortfall was the reading's.

Holds: carried to two: held in little cluster near memory 1.72; failed in pulsed needle with memory (break not stated). *Met in:* measurement, driven oscillators, three body.
Counted: A check that cannot fail 3. *Shown:* Loop Atlas, R2W, triple-star; full guide row 741.

Watch what stays put while the knob moves: a fan through one point, a pinned edge, a band that looks narrow (742) — Something holds a pinned point fixed, so ask what leaves it unmoved, and try one setting either side before calling it a constant. In R2W a pair sat 0.027 to 0.028 inside its frozen points at every root at strength 0.30, yet was ten times wider at 0.20 and ran the other way at 0.40, because 0.30 sat near where the drift with drive changes sign (TOY-SIDE-3d and TOY-SIDE-4f).

Holds: carried to one: failed in strengths 0.20 and 0.40 (knob). *Met in:* measurement, driven oscillators.
Counted: The mirror audit 1. *Shown:* Loop Atlas, R2W; full guide row 742.

Squeeze a picture to see its outline, mark its layout first, and check the outline by numbers (743) — Squeezed sideways, a comb of columns becomes one silhouette whose waist, flare or spike can be read, while a grazing look finds and never measures. On a triple-star strobe picture Bear coloured the data columns and spacers first, squeezed it from 1,690 pixels to 820 and drew its outline by eye; counted by pixels, every part survived, the waist's spread averaging 0.155 turn against 0.306 elsewhere (STROBE_PIXEL_CHECK).

Holds: carried to three: held in triple-star sessions' registered reads; failed in the paler panel (ruler); held in Loop Atlas quarter tree. *Met in:* three body, maps and chaos, reading pictures.
Counted: Coupled clocks and event time 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 743.

A person's physical words point at theorems: name the theorem in his words, then check it (745) — The instinct names the mechanism and the run checks it. Bear's "it has a repulsion quality to it" was interleaving: two dances of a rotation in disguise cannot pass each other, so their stops strictly alternate round the circle, in all

nineteen rows read. In R2W his question about an odd number of blades pointed at the right symmetry, and ODD-1 confirmed it: three timings read what six read to 3.376×10^{-15}.

Holds: carried to three: held in a dance's fraction across its tongue; weaker in junction's tone and second harmonic (break not stated); held in the meter's timing counts (ODD-1). *Met in:* driven oscillators, topology, control and circuits. *Counted:* The mirror audit 1. *Shown:* Loop Atlas, R2W, Latch Atlas; full guide row 745.

To read a memory in a sequence, split it, ask whether it chains, and find the state on a section; then take that state out along its own curve (747) — Partial out the state on a section of the motion, not a duration that summarises it. In the Loop Atlas sessions the remembering needle lands at a fixed lag, so its speed there is everything it carries forward, and holding that speed fixed took the fall-to-fall correlation from +0.058 to −0.001 on 372,123 triples, where holding the climb's length fixed took only three fifths of it (FALL_SHARE).

Holds: carried to one: weaker in the barrier-top section (ruler). *Met in:* measurement, driven oscillators.
Shown: Loop Atlas, R2W; full guide row 747.

Before naming a tail's shape, size the sample, set the bar from its own scatter, and cut long enough to keep out the merely slow (748) — A small sample's own scatter can make one tail look like two. In the Loop Atlas sessions two of six paces failed an exponential test with 256 needles each, and with 1,024 both passed, the original 256 inside the larger samples with their lifetimes unchanged; and for 64 draws from one exponential, 13 per cent of true traps reach a median over mean of 0.8, so a bar there tests luck as much as shape.

Holds: carried to one, held there: Loop Atlas's FALL_CHAIN test. *Met in:* measurement, driven oscillators.
Counted: A check that cannot fail 1, A second reader 1. *Shown:* Loop Atlas, R2W; full guide row 748.

Ask whether an orbit repeats by the gap between its repeat and everything else, not by a sieve (749) — A sieve can catch the arithmetic's scatter at quiet moments and call it an orbit, while the gap between a true repeat and everything else stays wide. In the Loop Atlas sessions the needles at pace 0.99375 repeat after 19 drive periods at 1.04×10^{-12} in median and at no other gap below 0.72; the longer repeats of 95 and of 399 and 437 had been the sieves' doing (ORBIT_19).

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, maps and chaos.
Counted: A check that cannot fail 1, Chaos and binning 1, The integer diagnostic 1. *Shown:* Loop Atlas; full guide row 749.

A fold fools a trajectory: shoot at it, or find it by its multiplier reaching 1, not by asking whether a run stayed (750) — Just past a fold the vanished orbit's ghost holds the state for a passage that grows like one over the square root of the distance, so a finite trial reads the ghost as an orbit. In the Loop Atlas sessions an edge bisected with 400-drive-period trials sat 1.6×10^{-8} past the fold Newton's method found; one minus the multiplier, falling with slope +0.5015 against the distance, finds the fold itself.

Holds: carried to three: held in Loop Atlas memory switch; held in weightless passing map at frequency 0.40; failed in turned ripple at strength 0.45 (seam). *Met in:* driven oscillators, maps and chaos.
Counted: A check that cannot fail 3, A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, R2W; full guide row 750.

Look for an orbit by Newton's method before calling it absent, and follow it through: it may be there and unstable (751) — When a lock's least repeat is never q but always 2q, the q-orbit may be there and flip-unstable. In the Loop Atlas sessions the needle turns once in two cycles across 0.9951 to 1.01 yet never repeats after two, and Newton's method found its 2-cycle orbit of one turn, with multiplier −2.14 to −2.04, while the needle lives on a 4-cycle orbit (HALF_BUBBLE).

Holds: not yet carried elsewhere. *Met in:* maps and chaos.
Shown: Loop Atlas; full guide row 751.

Past an inner fold the chaos keeps refilling the ghost: count the returns, not just the first passage (752) — A trial of fixed length near a fold can catch the needle on a return into the ghost instead of its first passage. In the Loop Atlas sessions, at seven distances past the 1/4 window's fold, the needle came back into the vanished orbit's ghost 231 to 730 times in 200,000 drive periods, and near the fold a trial of fixed length was a lottery, 34 of 200 answering on at random.

Holds: not yet carried elsewhere. *Met in:* measurement, maps and chaos.
Counted: Chaos and binning 1. *Shown:* Loop Atlas; full guide row 752.

Turn the noise off, down and up before giving an edge to the controller or to the noise (753) — An edge past a controller's textbook limit may be set by the noise or by the controller, and only a run with the noise off tells which. In the Loop Atlas sessions the never-blind sized tap's edge moved with the jostling, 1.70, 1.55 and 1.25 fall times at jostling 0.01, 0.02 and 0.04; with the jostling off it held to 1.75 and 2.00 fall times, so the noise had set those edges.

Holds: carried to one, held there: a memory-5 needle at pace 0.75. *Met in:* driven oscillators, control and circuits.
Counted: Chaos and binning 2. *Shown:* Loop Atlas; full guide row 753.

Read a stretch with the map's own algebra: eigenvalues beside the singular value, and the frozen rates added up (755) — A shear stretches with small eigenvalues, while growth has an eigenvalue near the stretch. In the Loop Atlas sessions the needle's largest one-period stretch, 20.9, has an eigenvalue of 20.5, so it is growth; and the frozen growth rates added up through each drive period ranked with its true stretch at 0.97 to 0.99 on five fresh paces.

Holds: carried to two: held in flagging round-off in a product; failed in the needle settled at rest, only ringing (break not stated). *Met in:* driven oscillators, maps and chaos.
Counted: A check that cannot fail 1. *Shown:* Loop Atlas; full guide row 755.

A window's end in banded chaos is a crisis: find it by the orbit born with the window, and below it read the stays (756) — A periodic window in banded chaos ends where the chaos's band edge lands on the unstable partner born with the window's orbit at its fold. In the Loop Atlas sessions the 1/4 window runs from about 0.983688, below which the chaos fills the whole circle, to its fold at 0.98375283; below that end the needle's mean stay in the window's old strips grows as the inverse square root of the distance, slope −0.4982, the plain one-dimensional 1/2.

Holds: carried to one, held there: the needle's 1/4 window crisis. *Met in:* maps and chaos.
Shown: Loop Atlas; full guide row 756.

A window's shape outlives its size and its company: measure its ratio, not its width, and start the needles close inside the motion (757) — When every number changes, ask what stays. In the Loop Atlas sessions, as the needle's squeeze was made seventy times lighter, the 1/4 window shrank fivefold, top to first doubling 28.8 to 5.9 millionths, yet its end stayed 2.253 to 2.258 times as far below its top as its first doubling, the one-hump law of 9/4 (HEAVY_SHAPE).

Holds: carried to six: held in heavy needle's 1/4 window; held in Hénon map windows, periods 8 to 18; weaker in Hénon map windows, periods 6 and 7 (break not stated); failed in edges at frequencies 0.40, 0.55, 0.70 (knob); held in frequencies 0.40, 0.55 and 0.70; held in lighter needle (LADDER_LIGHT). *Met in:* driven oscillators, maps and chaos.
Shown: Loop Atlas, R2W, Field Guide sessions; full guide row 757.

Look for the machine's own stop near any wall you chase, and read a closing gap in two units, up to its last point (758) — A gap can close in one unit and recede in another, so read it in its own units and in the size of the thing it closes on. In the Loop Atlas sessions, from memory 1 to 1.6, the stop closed on the heavy needle's window in pace, 0.00323 to 0.00044, while in the window's own width it receded, 112 to 306 widths, and a straight line through four points that had closed it at 1.48 missed.

Holds: carried to one, held there: the heavy needle's window at memory 2. *Met in:* measurement, maps and chaos.
Counted: The knee 1. *Shown:* Loop Atlas; full guide row 758.

A saddle has two clocks, and each keeps its log law only near the saddle (759) — A log law near a saddle is an asymptote, so check the speeds where it is meant to hold before betting on its slope. In the Loop Atlas sessions, launched without noise, the needle's fall kept the saddle's log law to a thousandth at crossing speeds of 10^{-4} to 10^{-3} (slope −0.6941 against −0.6949), but only 0.79 of its slope at the replays' speeds.

Holds: carried to two: held in the needle's climbs into the barrier top; weaker in the needle's noisy falls off the top (seam). *Met in:* driven oscillators, maps and chaos.
Shown: Loop Atlas; full guide row 759.

A window can be narrower than the step used to look for it: set the step from the narrowest window you mean to find, and find the edge from the window's own orbit (760) — A sweep finds only windows wider than its step, and a sweep that finds nothing says nothing about narrower ones. In the Loop Atlas sessions the remembering pulsed needle's 1/4 window is 4.8×10^{-5} wide from its fold to its doubling pile-up, under half a sweep step of 10^{-4}, so a sweep of 401 paces cannot land in it (SPIN_START).

Holds: carried to one, held there: the rider's 3/2 tongue at mass 3.0. *Met in:* measurement, driven oscillators, maps and chaos.
Counted: A check that cannot fail 2. *Shown:* Loop Atlas, R2W; full guide row 760.

Follow a narrow window by where its stable orbit exists, and call a cycle stable by every multiplier (761) — A stability test must hear from every multiplier, since the one it ignores can be large. R2W's FX-2c engine kept only the larger multiplier by sign, so a dance reported as strongly stable had its other multiplier at −7.8 to −69, and a claim of two resting dances side by side was withdrawn; print every multiplier a verdict depends on.

Holds: carried to one: failed in the strongly squeezed map at memory 1.6 (ruler). *Met in:* maps and chaos.
Counted: A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, R2W; full guide row 761.

A chain of followers is sensitive along its length and steady at any one place: read its growth as an average over links, and close it into a ring to see it jam (763) — Each follower can settle while the chain amplifies, so read growth along the chain, not at one place. In the Loop Atlas sessions a stumble grew down a 30-walker column by 15,904 to 140 million times while every walker's swing died to under 10^{-10} of its own largest; closed into a ring, thirty walkers turned a 10 cm nudge into a growing stop-and-go wave wherever the open column's largest gain exceeded 1.

Holds: carried to one, held there: followers closed into a ring. *Met in:* waves, control and circuits.
Shown: Loop Atlas; full guide row 763.

A floor has an owner: find it before reading a law under it, test a suspect by how it scales, and halve the step before giving it a mechanism (764) — A floor that does not shrink with the force but does shrink with the step belongs to the step. In the Loop Atlas sessions the waddle's floor of 10^{-5} per top-up read 1.01×10^{-5}, 6.40×10^{-6} and 4.52×10^{-6} at steps of 10^{-5}, 5×10^{-6} and 2.5×10^{-6}, and the three quarters of it that shrank in proportion to the step were the arithmetic (WADDLE_BOOKS).

Holds: carried to one: failed in couple 1 (ruler). *Met in:* measurement, driven oscillators, kinematics.
Counted: Chaos and binning 2, A check that cannot fail 1. *Shown:* Loop Atlas, R2W; full guide row 764.

Probe with the tool itself, run blind, before registering: a check must touch the very number the test will use, and say whether the test can decide (766) — A probe run on a neighbour of the test's input lets faults through. In one night of the Loop Atlas sessions two instrument misses, SEAM and SEAM_2, got past probes that checked the thing beside the thing used, a redrawn figure matching the file pixel for pixel while the axes were read at the wrong resolution; run blind with the tool itself, the probe then caught a fault before a registration in every exchange after it was adopted.

Holds: carried to one: failed in a planted case's value (knob). *Met in:* measurement, driven oscillators, three body.
Counted: A check that cannot fail 2, The built artifact is the evidence 1. *Shown:* Loop Atlas, triple-star; full guide row 766.

Read a doubling ladder's ratios with care: bracket each rung finely enough for the number wanted, keep each refinement inside its bracket, and expect a ratio to swing round its limit (771) — A ratio creeping toward a universal number needs brackets fine enough to see its last step. In the Loop Atlas sessions a doubling cascade brought Feigenbaum's ratio in from one side, 4.2365, 4.5513, 4.6458, 4.6642; bracketed to 10^{-13}, the fifth ratio sat at 4.66801, 0.0012 short of 4.669202, and with brackets of 10^{-11} the fourth had read 4.6642 instead of 4.6640.

Holds: carried to five: held in needle's 1/4 window cascade; failed in quartic-topped map (substrate); failed in area-preserving map (break not stated); weaker in first rungs of needle's ladder below chaos (break not stated); held in needle's doubling ladder rungs. *Met in:* measurement, maps and chaos.
Shown: Loop Atlas; full guide row 771.

Trace a number to its run, and ask its owner, before calling it carried or wrong (772) — A number that cannot be traced is unverified, not wrong, so ask the session that owns it for the trace before writing it off. The session reading row 619's 0.909 could not trace it and guessed it was carried from elsewhere; the session that owned it traced it to its run file, and the rerun reproduced it.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: A word that changes meaning across a boundary 1. *Shown:* triple-star, Field Guide sessions; full guide row 772.

A backward pass at a pause: a reader who did not do the work looks for orphaned threads, its claims are checked before acting, and each item says whether it was verified (780) — A separate reader goes back over everything it can be handed, and its main claims are checked against the files before any is acted on. In the triple-star sessions one such pass corrected a run count to 486, not 729, because the census tool had counted a table's second block, and narrowed "exactly the slope" to "within 6% at the two clean edges, over the ladder as a whole".

Holds: carried to one, held there: the triple-star edge-slope reading. *Met in:* records and tools, three body.
Counted: A second reader 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 780.

A block rule is cut by a single thread, and a width is read in the quantity the system keeps (781) — A reading rule that takes the unbroken held block lets one thin thread cut an island short. In the triple-star sessions the island at 0.25 read 0.135 and the paper said it "halves", but births 0.001 apart showed 7.420 alone chaotic, a thread narrower than 0.002, and the island is 0.170 wide (7.390 to 7.555), narrower by about two fifths, not half.

Holds: carried to one: failed in big swings of the sling (ruler). *Met in:* measurement, three body, kinematics.
Shown: Loop Atlas, triple-star; full guide row 781.

Set a bar from the record's own numbers: at the method's floor, with room above the largest difference seen, and on the median of a heavy-tailed scatter (784) — A bar finer than the record's own floor cannot be met, so read the floor off the record before the run. In the Loop Atlas sessions a bet that the laptop would match the

container's record to the 12 printed decimals at eight folds missed on three drives, by up to 2.8×10^{-11}, though the record already showed the fold refiner's final residuals up to 9.7×10^{-10}.

Holds: carried to two: weaker in Bear's laptop (ruler); weaker in KD0 reproduction on another machine (ruler). *Met in:* measurement, driven oscillators, three body.
Counted: A check that cannot fail 3. *Shown:* Loop Atlas, triple-star; full guide row 784.

What the integrator's step owns: a thin lock's place, a gauge that moves with it, a foot's order, and a switch that looks every step (785) — What moves when the step is halved belongs to the integrator, not the rider. In the Loop Atlas sessions halving the step moved the 8/5 groove's edges by 6.9×10^{-9} of drive, some seventy times a guessed 10^{-10}, while its width changed by only 1.6×10^{-7} of itself, so the width is the rider's and the step moves where the thin lock sits (GROOVES_10).

Holds: carried to one, held there: spring bead's blade switch counts. *Met in:* measurement, driven oscillators.
Counted: A check that cannot fail 1, Count the right unit 1, The knee 1. *Shown:* Loop Atlas, R2W; full guide row 785.

A reader's window decides what it can see: a majority rule, a low barrier, a window copied from a shorter run, a gap rule of fixed size, a yardstick that holds what it measures (787) — A majority rule hides whatever the minority is doing, so keep and read the count under every threshold verdict. In the Loop Atlas sessions a climb-back reader that counted a pace as spinning only when 8 of 16 needles spun missed pockets of 6 and 3 that kept spinning below the edge, and the pockets were a 1:2 gear beside the 1:1 lock.

Holds: carried to two: weaker in a 600-unit run (ruler); failed in ride of 1,000 time units (knob). *Met in:* measurement, driven oscillators, maps and chaos.
Counted: A check that cannot fail 1, Chaos and binning 1, The knee 1. *Shown:* Loop Atlas; full guide row 787.

Compare on one budget, keep a law inside its range and its stage, count what wears, eliminate a suspect by its share and its scaling, and read the step ratio first (788) — A winner named on unequal budgets may be the budget's winner. In the Loop Atlas sessions two rings merged into one first seemed faster than two rings leapfrogging, but the merged ring had carried 1.41 to 1.73 times the pair's energy for the same push; given exactly the pair's budget it travels at 0.84 to 0.88 of the pair's speed, so the leapfrogging pair is 14 to 19 per cent faster (RING_BUDGET).

Holds: carried to five: failed in needle at memory 1 (seam); failed in Loop Atlas's needle relay (seam); held in toy fleet's calendar in running turns; failed in frequency 4 (FLAT-75) (seam); failed in a third swing (FLAT-50) (knob). *Met in:* measurement, driven oscillators, control and circuits.
Counted: Count the right unit 3. *Shown:* Loop Atlas, R2W; full guide row 788.

A run statistic rests on a thin tail, a ramp reads where it crosses, and a transient can look like a second state (789) — A run ends only at a long gap, so a run statistic is decided by the few breaks in its tail. In the Loop Atlas sessions breaks were about 200 to 250 of 15,000 gaps, the remembering needle's run statistics moved with where a slip was timed, and five registered predictions about them missed; time a slip at the bottom of the turn or the landing, and print the break line and the tail beside the statistic.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators.
Counted: Chaos and binning 1, Count the right unit 1, The knee 1. *Shown:* Loop Atlas; full guide row 789.

Follow the orbit itself, and read the doubling's square root and critical point from it (791) — A tool that finds periodic orbits by Newton's method and follows them sees locks a grid misses. In the Loop Atlas sessions LOOP_ATLAS_STROBE_NEWTON sized locks 4×10^{-5} to 7×10^{-5} of drive wide that a 0.0005 grid could not see, and just below the light needle's doubling the period-16 strands are 1.662 times the square root of the distance apart (slope 0.5002).

Holds: carried to two: failed in period-8 orbit at mass 0.75, downward (ruler); weaker in the needle's doubling (a bent road) (substrate). *Met in:* driven oscillators, maps and chaos.
Shown: Loop Atlas; full guide row 791.

Three habits that misread: a reading made after the data, a number read from printed digits, and a run named from its header (793) — A number drawn from printed digits is computed by script before it is said, and a run that is not yours is named from its own files. In the Loop Atlas sessions "the same 8/5 edge to 3×10^{-30}" was said of edges 2.7×10^{-30} to 2.77×10^{-29} apart, and a laptop slowdown of 5.0 to 6.6 times, first read as the laptop's speed, measured the crowd of three sessions' runs.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Counted: The built artifact is the evidence 1, A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, R2W; full guide row 793.

Set a bar with a number in hand, before counting, only where the run can feed it, and at a named crossover (794) — A bar guessed without a number in hand can fail while the picture holds. In the Loop Atlas sessions climbs over two periods were guessed at 1 per cent or more of all climbs and came to 0.46 per cent (69 of 14,969) while the

picture of a fast pass held; the catch rate, about one climb in 217, is the number to carry.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, maps and chaos.
Counted: A check that cannot fail 2. *Shown:* Loop Atlas, R2W; full guide row 794.

A queue for the paper gives each item why it matters, its cheapest test and whether that was checked (797) — A run that runs out of time is not an escape, so every long run's fate is reported by class. In the triple-star sessions one backward pass's queue counted that at 600 passes only 8 of the 12 births below escaped, two ran out of time, one circulated and one held; the paper's runs ten times longer, 6,000 passes, saw eleven of the twelve leave, and the twelfth was never held again.

Holds: carried to two: weaker in the triple's slip near the edge (break not stated); weaker in the Loop Atlas needle under noise (break not stated). *Met in:* records and tools, three body.
Shown: Loop Atlas, triple-star, Question the Substrate; full guide row 797.

Straight lines lost all day: between two known ends a straight line is a guess, not a law; register the bend as the lean and the straight line as its rival (799) — When a guess with no law behind it misses, let the miss choose the law, and register two or three candidate laws side by side before the run. In R2W, R24-b put the second harmonic's edge at drive ratio 1.3956 on a straight line through 0.99 and 1.875, and it lay near 1.3078; RATCHET-25's parabola through five ratios put it at 1.3065, and it measured 1.3056.

Holds: carried to four: failed in the drift's zero along another road (break not stated); held in ladder rungs beyond its range; failed in another drive (FLAT-75) (seam); failed in a third swing (FLAT-50) (knob). *Met in:* measurement, driven oscillators.
Shown: R2W; full guide row 799.

Flip the sign to find the two hands: a result read at one sign may be two parts cancelling (801) — A null, or an effect that seems swallowed, read at one sign only may be two parts cancelling at the setting that happened to be run. On R2W's rectifier, RATCHET-41 had read "the mix keeps the octave's c" as the octave swallowing a third tone; running each size again with its sign flipped (RATCHET-44) split it into an odd part, +15.06 at size 0.38, and an even part, −14.98, that nearly cancel.

Holds: carried to two: held in Loop Atlas's tied bells; failed in BEAT-4, a quarter turn off mirror (substrate). *Met in:* measurement, driven oscillators.
Counted: Count the right unit 1. *Shown:* Loop Atlas, R2W; full guide row 801.

When the knob has no sign, tell a true nothing from two hands cancelling: sweep the size, move another knob, turn the timing half a turn, weigh each hand alone (805) — A symmetry's nothing sits at the arithmetic floor at every size, while two hands cancel at one setting only. In R2W eighteen even fans of riders came to at most 6.994×10^{-15} against one rider's 2.599775×10^{-2} (ARRAY-1); but on the Loop Atlas's crest-firer a flat sweep hid two hands, the pushes coming 7.3 times less often and pushing 2.9 times as hard, their product unmoved.

Holds: carried to two: failed in Loop Atlas crest-firer (wall); held in Loop Atlas crest-firer. *Met in:* measurement, driven oscillators.
Counted: A check that cannot fail 1, The mirror audit 1. *Shown:* Loop Atlas, R2W; full guide row 805.

Read a knob's return on its map, not by proportion: a part of what it turns pays less as it grows, a sweet spot moves another knob's best, and two knobs that only trade are a frontier (806) — A part that belongs to the whole it acts on does not pay in proportion to its size, so map a knob before naming its direction. In the Loop Atlas sessions halving the arms' mass to 3.5 kg was leaned to cut one arm circle's turn and the fixed share to 0.45 to 0.60 of their values at 7 kg, and they fell only to 0.6160 and 0.6087, because lighter arms leave a lighter body (ARM_SHARE).

Holds: not yet carried elsewhere. *Met in:* waves, kinematics, quadrature.
Shown: Loop Atlas, R2W; full guide row 806.

Before reading two answers as two states, make every scale in the toy agree with every other (807) — A toy that will not settle may be doing something no real machine does. At the Loop Atlas's tied bells' crowded corner, eight nearby starts split between swimming forward and being driven back, the share moving with the step, because each ring had a core of 0.15 but a softening of only 10^{-3} where two rings meet; given a softening of 0.0225, the square of 0.15, every start swam forward (BELL_SWITCH).

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: Loop Atlas; full guide row 807.

A check across machines compares only what can repeat: held and regular runs, birth by birth, saved file against saved file, with a loose tier for the other machine (810) — Two computers do not do exactly the same arithmetic, so a chaotic run on one parts from the same run on the other, however good both are. In the triple-star thread KM0 stopped its tool when 21 re-runs gave back the atlas's phi_N only within 0.932 turn, because it compared every run, held or not; KM0′, comparing held runs only, passed at 2.10×10^{-5}.

Holds: carried to four: failed in the same birth on the laptop (wall); failed in SL7's workspace anchor at strict tier (ruler); failed in check across machines, all births pooled (wall); failed in between the laptop and the containers (substrate). *Met in:* measurement, records and tools, three body.
Counted: A check that cannot fail 3, A word that changes meaning across a boundary 1, Chaos and binning 1. *Shown:* triple-star; full guide row 810.

Run a planted case at two step counts before setting its bar: what shrinks with the step is the step's, what stays is the arithmetic's, and the case must be one that finishes (813) — A planted case's bar must sit above the error of the tool that reads it, and running the case at a step and at half of it finds that error cheaply. The Loop Atlas's FACE_ALONG planted a no-grip swimmer whose travel is exactly zero, and it printed $+1.798 \times 10^{-6}$ at 300 steps and $+4.628 \times 10^{-7}$ at 600, so a tolerance of 10^{-9} missed for the wrong reason.

Holds: carried to two: held in EXPLORER_ROUTINE_CHECK's planted tolerance; failed in routines blended by half-cosines, with rate corners (seam). *Met in:* measurement, kinematics.
Counted: A check that cannot fail 3. *Shown:* Loop Atlas, R2W; full guide row 813.

A guard computed by a round trip calls hits from rounding: compare where the guard is computed, diagnose a planted case alone, and rerun every old tool as it stands before the fix (816) — A guard that fires by mistake crashes nothing; it quietly changes the motion, so its fault can pass for a finding about the world. The Loop Atlas's toy column tested each walker's gap to the man ahead as a round trip, (x + off) − off < x, and rounding alone called a hit about one walker-step in 80, each hit setting the walker's speed to his man ahead's.

Holds: not yet carried elsewhere. *Met in:* records and tools, kinematics.
Counted: The built artifact is the evidence 1. *Shown:* Loop Atlas, R2W; full guide row 816.

A lifetime read in a finite window is a lower bound: say "held for T", throw an edge many times, fly the side that held for longer, and test a fit by its whole procedure (819) — A run that ends with a train still whole has shown only that it lasted as long as the run, so the honest words are "held for T". In the Loop Atlas sessions the three-ring train never broke in 30 throws from 0.500 to 0.525 radii within 20,000 time units, yet flown on to 100,000 on Bear's laptop, half of ten copies at 0.520 and 0.525 broke, a lifetime near 150,000.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Counted: Count the right unit 2, A check that cannot fail 1. *Shown:* Loop Atlas; full guide row 819.

A chaotic event repeats exactly on the machine that flew it and nowhere else: plant on another machine only what chaos cannot scramble (822) — Two computers differ in the last digit of their arithmetic, so a single chaotic event's time is a fact about one machine, while its statistics are facts about the toy. A second computer broke the Loop Atlas's smoke-ring trains at 4,494, 241 and 272 time units where the session's own machine breaks them at 1,022, 1,011 and 303; a nudge of 10^{-15} of a radius moved the session's breaks by 28 to 84 per cent.

Holds: carried to two: failed in a second computer (wall); held in the other machine. *Met in:* measurement, waves, maps and chaos.
Counted: A check that cannot fail 1. *Shown:* Loop Atlas, R2W; full guide row 822.

A small excess over a known law needs an error bar, more than one reader and a finer reference before it is named, and the law is tested at its limit (824) — A finer grid that reads the same noise repeats the same scatter, so it does not confirm a small excess. In the Loop Atlas sessions the extra knee of a switch that looks only now and then ran 20 and 18 per cent over the traders' law at its two smallest shifts, repeated on a grid ten times finer; against a reference watcher four times finer, every step lay within 1.7 standard errors of the law.

Holds: carried to two: held in sampled switch's extra knee, finer reference; weaker in a look every four tenths of the memory or coarser (seam). *Met in:* measurement, control and circuits, quadrature.
Counted: A check that cannot fail 1, The built artifact is the evidence 1. *Shown:* Loop Atlas; full guide row 824.

Cut a record into waiting times so that none is dropped and none counted twice: keep the interval that runs past the window's end, and time first arrivals at each new level when a counter can run backward (827) — A pool of complete intervals leaves out each sequence's last interval, which runs past the window's end and is likelier to be long, and a forward count after a backward slip re-crosses ground already counted. In the Loop Atlas sessions a needle that slipped back 7,253 times read 0.49300 ± 0.00105 between forward counts, 15.9 errors under the exact value, and 0.50779 ± 0.00112 timed from one new highest turn to the next, against the exact 0.50967.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators.
Counted: Count the right unit 1. *Shown:* Loop Atlas; full guide row 827.

Read a toy's setting at more than three values before calling its curve a hill, and look for a spike where two of the toy's own scales agree (829) — Three points can draw any hill, and a toy built with two lengths of its own can resonate where they match. The Loop Atlas's tied bells' gain at a fast squeeze, read at three meeting widths (0.75, 1.01, 0.87), looked like a peak at the cores' own width; read at eleven, it is a plateau between 0.68 and 0.97 with one spike, 1.01, at that width alone (BELL_WIDTH_LADDER).

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Counted: Coupled clocks and event time 1. *Shown:* Loop Atlas; full guide row 829.

Smoke-test a published theorem over its whole range before planting it as a check, with its sign conventions written down (831) — A theorem from a book is a strong check only if the tool and the book mean the same thing by every sign. In the Loop Atlas sessions the course theorem of sailing missed by 0.69 radian at 150 degrees, where a sail trimmed for speed stalls; read with the water's angle measured from the course's leeward side, as the theorem's own drawing has it, it held to 4×10^{-15} radian at every heading of nine hulls.

Holds: carried to one: weaker in the Loop Atlas's sailing smoke test (ruler). *Met in:* measurement, waves.
Counted: A word that changes meaning across a boundary 1. *Shown:* Loop Atlas; full guide row 831.

A value picked because it was the most extreme of many is partly the pick: read it again on fresh data before building on it (835) — The largest or smallest of many readings is partly chance's pick, so on new data it usually falls back toward the rest. In the Loop Atlas sessions a noisy needle's Fano factor read 0.835 in one row of 18, the lowest, and a lean built on it missed; on new seeds and four times the needles the same row read 0.9244 ± 0.0132, within an error of the exact 0.913.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators.
Shown: Loop Atlas; full guide row 835.

A proof's convenience is not a limit on its result: test the instant the proof was built from before predicting that the result fails (836) — A proof is often built on the easiest case, and that choice says nothing about where the result stops holding. In the Loop Atlas sessions the mirror between two falling cats had been proved on a steady circuit, and a lean that it would break on paths whose bend swells and shrinks missed: on nine such paths the two turns agreed within 2.842×10^{-13} degrees (CAT_MIRROR_PATHS).

Holds: carried to one, held there: nine paths whose bend swells and shrinks. *Met in:* topology, kinematics.
Shown: Loop Atlas; full guide row 836.

A bar is checked against its reader's grain and its number's error bar before it is registered (837) — A reading made of whole counts cannot repeat more closely than its own grain. In the Loop Atlas sessions a planted check asked a knee grid of 0.001 to repeat a finer grid's readings within 5×10^{-5} and missed by up to 3.65×10^{-4}, because the flips are whole numbers, about twenty to a knee step of 0.001, so the curve is a staircase and its grain sets the reader's error.

Holds: carried to one: failed in a reader of whole-number flips (ruler). *Met in:* measurement, quadrature.
Counted: A check that cannot fail 2. *Shown:* Loop Atlas, R2W; full guide row 837.

A straight law ends where the thing it describes changes kind: before carrying a fitted line past its range, ask whether it still measures the same thing (839) — A fitted line describes one object, and when that object turns into something else the line has nothing left to describe. In R2W the door followed a straight line in the weight within 0.0015 from weight 0.55 to 0.90, then bent where the room it bounds closes, near 0.905, and a side count showed the change: riders in flight were 186, 15 and 0 at weights 0.85, 0.90 and 0.95 (DOOR-1).

Holds: carried to one: failed in past 0.905, overlap of two locks (seam). *Met in:* driven oscillators, maps and chaos.
Counted: Count the right unit 1. *Shown:* Loop Atlas, R2W; full guide row 839.

Iterating a truncated formula is not a higher order: write the truncated answer plainly first, and test any iteration of it as a guess of its own (841) — Solving a formula worked to second order again and again at shifted settings brings back some higher terms and not others, so the answer can land further from the truth. In R2W's toy, KNOBS-3's iterated estimate put gear 1's edge shift 26 to 37% too far with the ripple turned around (KN3-3 missed), while FOLD-DIR showed the plain formula at the unshifted drive came within 4.2% at every weight.

Holds: carried to one: failed in ripple turned around (large shift) (seam). *Met in:* driven oscillators.
Shown: R2W; full guide row 841.

Count the quanta to read a power, and count their parity to hear what pushes (847) — Writing a beat as whole tone numbers that add to zero says, before any run, how fast it fades and whether it can push at all. In R2W the fourth harmonic's own arrow, four first tones against one fourth harmonic, fell as the swing to the fourth power (measured 3.916, 3.970 and 3.993); and only an odd group can push, so a third harmonic alone is silent until the octave gives it $1 + 2 - 3$ (BEAT-1).

Holds: carried to one, held there: nonlinear optics, frequency doubling crystals. *Met in:* driven oscillators, waves, quadrature.
Counted: The mirror audit 1, The integer diagnostic 1. *Shown:* R2W, triple-star; full guide row 847.

Know the instrument's own step rule and law before reading a slope, then take the law's maximum over the range you run (850) — A slope read without the tool's step rule can pose as physics, and a worst case sampled at a few settings is not a ceiling. R2W's integrator leak was predicted worst at the lightest weight but peaked at 0.25,

because the step rule switches at weight 0.2, giving slopes of +3.00 and −2.00 with one constant; the book's quoted maximum of 1.9×10^{-6} was a sampled worst case, and the law's true peak is 1.8×10^{-5} (NO-HONEY-PEAK).

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators.
Shown: R2W; full guide row 850.

Before calling a structure new-born, look for its trace in the map without the knob (851) — Finding the trace first turns a discovery into a prediction. In R2W's junction, weight 0.05 opened half and third steps only inside the weightless rotations, each on the line where the weightless rider already slipped by that fraction: 110 of 110 within 0.0136 (WEIGHT-CONE W-a and W-b).

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators.
Shown: R2W; full guide row 851.

The integrator's honey: measure the drift in the conserved quantities at several tolerances before reading any slow trend as physics (858) — A slow trend can be the integrator's own friction, so tighten the tolerance and see whether the trend moves with it. Over 30,000 passes at the triple-star kit's tolerance every unlocked rate drifted, and the energy's drift at three tolerances (-7.90×10^{-11}, -2.71×10^{-12} and $+3.26 \times 10^{-14}$ a pass) fell by the same powers as the rates' creep, so the creep was the integrator's. Tighten until the drift stops shrinking: below that floor every record measures its own arithmetic.

Holds: not yet carried elsewhere. *Met in:* measurement, three body, waves.
Shown: Loop Atlas, R2W, triple-star; full guide row 858.

Walls of measurement come back in new fields: know each by its shape and its first test (863) — The triple-star thread set down thirteen walls it first met as surprises, each written to be recognised in any slow thing read through a fast machine. Each comes with a first test: for the instrument's honey, turn tenfold the knob that should not matter; for the one-sided door, count failures by side, since 163 of 166 escapes left the island on one side. The Loop Atlas measured its own honey the same week.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: Loop Atlas, triple-star; full guide row 863.

A lock is a closed loop on a torus: read its fraction as two winding numbers, a family by its crossings, and a walk among fractions by the trace of its matrix (864) — Wrapping the drive's clock as well as the needle's angle turns the Loop Atlas's cylinder into a torus, and a lock at p turns every q cycles becomes one closed loop with that pair of winding numbers. Two loops a/b and c/d cross $|ad - bc|$ times, which held in 120 of 120 pairs counted, and every family is a whole-number matrix of determinant one, 511 of 511 to ten generations.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, topology, number theory.
Shown: Loop Atlas, R2W; full guide row 864.

An event a tool counts is a threshold standing in for a thing: count it by what it physically is and by who takes part, set its threshold against the scale that moves, fly past it, and check the stand-in in every regime (865) — A fixed threshold can hide a family that slides, and the event a run stops on may not be an ending. In the Loop Atlas's smoke-ring trains, a cutoff of 0.220 of a radius showed no deep squeezes for the thin cores, yet against each core's own door six of six thin-core breaks came through one; and flown 1,000 time units past the break, the third ring came back in 5 of 11 trains.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: Loop Atlas; full guide row 865.

A picture from another trade brings a law with a number: test the number (870) — Bear heard R2W's flipping skateboard as a motor's commutator, and the commutator brought its timing advance with it: the best kick sat near the arctangent of slip rate times lag, 41.4 degrees against 36.0 read after the data, then 70.0 against 70.0 landed blind. The other trade's picture was worth one law, and the law's number was the test.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, control and circuits.
Shown: R2W; full guide row 870.

Read a cascade from its onsets, on both sides; test the order of its windows; and let one toy's road forecast another's (872) — A cascade's bands as a scan meets them are broken by hidden locks, so place its onsets by the square-root law and read them from both sides. Inside the Loop Atlas's needle's 5/7 step the band ratios ran 3.945, 4.482 and 4.626 from above and 3.994, 4.480 and 4.629 from below, and past each end the windows came in the logistic map's order (Metropolis, Stein and Stein 1973).

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, maps and chaos.
Shown: Loop Atlas, R2W; full guide row 872.

Every listener is deaf to its own echo: buy any real check, and grade drift against a kept copy of home (884) — A learner hears least from the batch built out of its own agreement, about 50 times quieter than any other, so only an outside check can hear its errors. Any real check pays: one that catches a quarter of the errors earns 40 per cent of the benefit. And drift that hides when graded against the truth is loud when graded against a kept copy of home.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: Question the Substrate, Two Arrows; full guide row 884.

Thresholds in self-training chains are tipping zones: zoom where the person points, and report the split, not only the mean (886) — In a self-training chain a threshold is a band in which identical setups live or die by their random start. Five starts averaged that band into a slope and ten showed it, so zoom where the person points and report the split, not only the mean.

Holds: not yet carried elsewhere. *Met in:* measurement, maps and chaos.
Shown: Question the Substrate; full guide row 886.

L2 — The Geometry & Substrate You Walk

The fiber bundle real estate, the vertices and junctions, the winding theorem.

Look where a field distrusts amplitude (4) — Where amplitude cannot be trusted, phase can still tell sources apart by how they behave. In the two-source model of otoacoustic emissions, the distortion source is wave-fixed, its phase near-constant at fixed ratio with near-zero group delay, while the reflection source is place-fixed, its phase winding rapidly and its delay scaling as 1/f.

Holds: not yet carried elsewhere. *Met in:* waves, biology.
Shown: version 6; full guide row 4.

Wavelength against structure (6) — Too long a wavelength averages across a structure and too short a one bounces off without reading it, so match the signal to what it must read before anything else. The cochlear mismatch shows the first failure: at 1.2 Hz the acoustic wavelength in fluid is 1,250 metres, far larger than a 0.6 m canal, and even at 1 kHz a quarter-wave air column is 86 mm and in fluid 375 mm, too long for any implantable pipe.

Holds: not yet carried elsewhere. *Met in:* waves, biology.
Shown: version 6; full guide row 6.

Why an invariant is protected (32) — A signed invariant keeps its sign under perturbation, so only an actual crossing event, which can be detected, changes it. In the topological memory architecture a signed vortex ring keeps 39–40% of its spin energy after roughly 9 ring radii through water, and its sign, the stored topological bit, never flips.

Holds: not yet carried elsewhere. *Met in:* topology, waves.
Shown: version 6; full guide row 32.

Invariants persist, energy leaks (34) — When a signal fades, read the count rather than its size: the integer survives after the continuous signal has decayed. The parity, the sign of the winding, held to 4×10^{-15}, near-perfect preservation, while the raw holonomy amplitude drifted by 9.2%.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 34.

Arnold tongues and mode-locking (38) — The staircase of lock plateaus is the commissioning certificate: sweep the detune, log the plateaus, bank their widths as measured margins and operate at plateau centres. Wide steps are deep locks and widths shrink as roughly one over q-squared down the Farey tree; in the Loop Atlas sessions a two-rhythm rider at pace 1.2933 drifted at 0.7999, against the 0.7993 that pace times 0.618 predicts.

Holds: carried to two: failed in a low-pass filter subsystem (substrate); held in the second stage, chained through a relay. *Met in:* driven oscillators, control and circuits, number theory.
Shown: Loop Atlas, R2W; full guide row 38.

The scale law (39) — Biological propulsion slows with size, its frequency scaling as f ~ L^(−2/3), a known allometric relationship. The jellyfish case gives a 5-metre bell, a 13-second stroke cycle and 9 tonnes of water moved per breath at approximately 2.5 watts, and from it the bell's geometry, not its size, is taken as the design parameter for a geometric analogue.

Holds: not yet carried elsewhere. *Met in:* biology.
Shown: Jellyfish; full guide row 39.

Rigidity selects the operator, and the operator selects what you can hear (49) — Reading b_1 acoustically requires a solid, a constraint nobody would guess from the topology. A membrane or gel gives the scalar Laplacian, where Kac applies exactly; a stiff plate the biharmonic, where the theorem must be re-derived; a fluid cavity scalar waves only; and only an elastic solid carries vector waves, the 1-form sector where b_1 lives.

Holds: carried to two: weaker in a stiff plate (biharmonic) (substrate); failed in a fluid cavity (scalar only) (substrate).
Met in: topology, waves, materials.
Shown: version 6; full guide row 49.

Winding is a count, not an amount (57) — A cell's volume does not set what it stores, because the topological charge is independent of the quantity of medium. A quantized circulation does not depend on how much fluid circulates, and the optical case is the cleanest illustration of a fact the spin case needs.

Holds: not yet carried elsewhere. *Met in:* topology, waves, materials.
Shown: version 6; full guide row 57.

The conjunction is the wall, not any single property (60) — Showing each needed property somewhere in isolation does not show that one material can hold them all, because the properties pull against each other. Five are each demonstrated somewhere and no material does all five, deep barriers fighting room-temperature writability; it is ALife's fifty-year shape, substrate-independent in logic and hostage to substrate in realization, and the supplying is the hard part.

Holds: not yet carried elsewhere. *Met in:* materials.
Shown: version 6; full guide row 60.

The Volterra construction is a holonomy definition (65) — Recognise an old object under a new name before treating it as new. Cut a crystal along a line, remove or insert a wedge, weld the lips and force a rotation: that is a disclination, and its Frank angle is the holonomy of a loop around the defect line, the deficit angle the guide already banks in B.2, here under a third name.

Holds: not yet carried elsewhere. *Met in:* topology, materials.
Shown: version 6; full guide row 65.

The ring attractor is a circle-valued state with no lift (69) — An insect's heading lives on a ring of neurons, topologically S^1, so there is no neural state for "370°". Path integration carries each body-frame step into world coordinates through the current heading, so a round trip's closure error is a holonomy, and the net number of full turns is exactly what the ring structurally cannot hold: a lift obstruction instantiated in neural tissue.

Holds: not yet carried elsewhere. *Met in:* topology, biology.
Shown: version 6; full guide row 69.

Closure forces an integer — hold closed, and the sentence was wrong (82) — Kept because the miss is the record: a remembered sentence sent four searches after a file that does not contain it, and the file turned up only when someone supplied its real name. The Session-4 capture's keystone says something else, that the eigenvalue is what survives the trip and the holonomy is the turn the survivor comes home wearing.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 82.

The same object arrived twice, in two lineages, months apart (90) — A corpus large enough to rediscover itself needs its index more than another chapter. The Anchor and the Hammer argued that no observer reads itself true and priced it on a control loop, while the wave-geometry lineage independently banked the unverifiable-from-inside as its own node: one object, two names, same author, the cleanest B.2 instance in the corpus.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 90.

The apparatus and the readout are one object (91) — Carry a state around a closed loop and see what it comes home as: that is the Wilson loop, the question, and its eigenvalue, the answer, is what survived the trip, the conserved quantity the loop did not disturb. The corpus has always been built from "eigenvalues and Wilson hoops" because they are two faces of one instrument.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 91.

Why it ports, and it is two theorems rather than a vibe (92) — The tool ports because one specific problem recurs, not because everything is a loop. Anything with a pose tracked through time without a fixed external reference, from a whale to a spacecraft to head-direction cells, must integrate an axis-and-angle and leaves a residual where it fails to close; and what a system is indifferent to it conserves, which is exactly what walking the loop exposes.

Holds: not yet carried elsewhere. *Met in:* topology, kinematics.
Shown: version 6; full guide row 92.

The out-of-band witness, in the flesh (95) — A moving frame needs an anchor placed outside it to check itself against, and losing that anchor costs. The post-reproductive orca matriarch holds the clan's map across decades while the pod dead-reckons through dark water, long-lived so the reference does not drift with daily noise; when she dies

the pod loses its witness and mortality rises.

Holds: not yet carried elsewhere. *Met in:* biology.
Shown: version 6; full guide row 95.

A smooth contour field can break exactly three ways, and there is no fourth (109) — One instrument suffices across domains because the ways a contour field can break form a short closed list, not because the domains are one thing. Wherever nature has a monotone quantity, such as pressure, depth, temperature or time, it stacks the substrate into contours for free, and a defect is a winding, a critical point or a discontinuity: the same lens sees pond water, blood and crystals.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 109.

You cannot read a winding off a substrate with no enclosed area (116) — A ledger branded topological, tested on a one-dimensional sensor ring, fired 52% on honest data: with no two-dimensional field the winding never quantised, and the repair is a triangulated sensor field, not a better threshold. The same rule tells a holonomy from a knee: run the loop out and back along the same line, and a holonomy leaves nothing while a knee leaves its band.

Holds: carried to two: failed in one-dimensional sensor ring of bearings (substrate); held in car on a banked turn. *Met in:* measurement, topology, kinematics.
Shown: Loop Atlas, R2W; full guide row 116.

An open arc is a convention; only the closed loop is an invariant (121) — This is why the instrument's basic unit was never a measurement but a circuit: a single seam read on its own is a gauge-dependent placeholder, untrustworthy alone, and the physical object is the closed loop product. It is not an analogy to lattice gauge theory; it is lattice gauge theory.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 121.

Wrinkly where you hoard, smooth where you measure (122) — A wrinkle is a place to hide treasure and a place to hide a lie, so a storage substrate and a fidelity instrument turn out to be one machine. Keep two meanings of smooth apart: low-wrinkle, the honest cheap-to-read ruler, and seamless, where a join cannot be found and so cannot be audited; what survives a seam-erasing smoothing is a fiducial placed before it.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 122.

The field's own definition was already the thesis (135) — Check whether the field already holds an idea before claiming it, since making a qualitative idea quantitative is easier to defend and better to make. A tropical cyclone is officially a rotating organised system with a closed low-level circulation, and the formal genesis definition sets no wind threshold, so the literature already treats formation as a structural closure event and wind as the downstream consequence.

Holds: not yet carried elsewhere. *Met in:* topology, waves.
Shown: version 6; full guide row 135.

The loop is the line between re-confirming known physics and testing something new (137) — Know where the line runs and cross it on purpose, or a new instrument becomes an expensive re-derivation. A single station snapping back is elastic rebound, settled since 1906; the winding lives only in a loop of stations, so walk a triangle or a pentagon through time and read the failure to close as the holonomy.

Holds: not yet carried elsewhere. *Met in:* topology, geodesy.
Shown: version 6; full guide row 137.

Chern–Simons demands the ribbon (143) — The dimension a flat diagram loses returns as the ribbon's width: the theory is not even well-defined until the curve is thickened into a ribbon, a framing. The matching read is one the corpus already had: carry an orientation parallel around the loop, and the drift on return is the holonomy, with twist plus writhe equal to the self-linking number, so the dead-reckoning instrument and the knot reader are one machine.

Holds: not yet carried elsewhere. *Met in:* topology, kinematics.
Shown: version 6; full guide row 143.

The periodic table is the existence proof for the winding-memory architecture (144) — Every property the deposited winding-memory paper asks of a substrate, the atom already has: a signed spin, multilevel shells, quantized standing waves that must close around the nucleus, and retention behind an energy barrier. The row-lengths 2, 8, 8, 18 are counts of standing-wave slots, and nothing here is a novelty claim: it is the most famous integer table in science, which makes it the right worked example.

Holds: not yet carried elsewhere. *Met in:* topology, materials.
Shown: version 6; full guide row 144.

The fluid supplies the framing Chern–Simons demands (147) — Chern–Simons is undefined on a bare curve, and changing the framing by one unit multiplies the answer by a known phase, so the framing is physical; a vortex tube carries it, core radius as framing and swirl as twist. All three legs are prior art and the CANDIDATE owes a prior-art pass; the price is the corpus's own, a signed ring keeping 39–40% of its spin over ~9 radii in water, the sign held.

Holds: carried to one: weaker in a vortex tube in viscous fluid (substrate). *Met in:* topology, waves.
Shown: version 6; full guide row 147.

Transport only turns, it never stretches — so the rotor's length was never a choice (148) — A holonomy eigenvalue is unimodular by nature rather than by normalisation, because parallel transport moves phase only and adds or drains no amplitude. Circle a pentagon enclosing 60° of deficit and the eigenvalue is e^(±i60°), with magnitude exactly one: nobody decided rotors have length one; the geometry did.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 148.

The corners of the polygon are its eigenvalues (149) — This is why an n-gon rotor combinator works at all: the n-th roots of unity are the eigenvalues of the operator that steps once around, so a pentagon's five corners and the walk around it are one eigenvalue in two descriptions. Each rotor's corners already are its eigenvalues, all of unit length, and the "= 1" is not bolted on.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 149.

THE KEYSTONE — the two ledgers are the modulus and the argument of one number (150) — Magnitude one says the conserved quantity rode the loop untouched, the energy ledger; the phase is the deficit enclosed, the winding ledger. The sentence that the eigenvalue is what survives the trip and the holonomy is the turn the survivor comes home wearing is filed as CANDIDATE, not READY, because an analogy doing a theorem's job stays an analogy until something forces the identity.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 150.

The phase depends on the shape of the loop, not the speed of the walk (151) — This is the adiabatic yardstick the corpus is measured against: carried slowly around a loop, a state keeps its eigenvalue and gains a phase set only by the loop's shape. JELLY-1's X0 tests it: in the Stokes limit displacement per shape loop is pace-independent, and the measured +0.12 R/cycle fast against +0.055 slow at Re 30 is the departure from it.

Holds: carried to three: failed in JELLY-1 swimmer at Re 30 (seam); held in a body's turn in the air; held in straight leg swung in a stride. *Met in:* biology, kinematics.
Counted: The integer diagnostic 1. *Shown:* Loop Atlas, Jellyfish; full guide row 151.

The twist channel, priced twice in two substrates (165) — A distinction measured in two unrelated substrates is no longer a borrowed idea awaiting a test, and rows 143 and 147 rest on that. On a synthetic loop channel in July 2026 a framed loop separated from a flat one at 15.5σ, the sign reading as chirality; in a viscous fluid a signed ring held 39–40% of its spin over nine radii with the sign intact.

Holds: carried to one, held there: a signed ring in viscous fluid. *Met in:* topology, waves.
Shown: version 6; full guide row 165.

The face shape is set by the arithmetic, not chosen (171) — What carries over between substrates is the principle, not the shape: read the instrument on the faces, because a node value is exactly the part a gauge choice can erase. Cascadia's lattice was triangles, but the forest's smallest closed loop, n → 2n → 6n → 3n → n, is four-sided, 110 squares in the patch, because two prime directions give a quadrilateral.

Holds: carried to two: failed in the integer forest's loops (substrate); held in the integer forest's loops. *Met in:* topology, number theory, geodesy.
Shown: version 6; full guide row 171.

The constant, with its address (175) — State a constant with the conditions it belongs to: saying it is the infinite-temperature, uniform-illumination value is half the result. The barrel at which one peak becomes two, 1.20871040964352282, was checked over 993 prime pairs and 14,098 rungs, and of the 8,295 deep rungs not one lands above it; every patch approaches the ceiling from below, as a finite-size effect would and a coincidence would not.

Holds: not yet carried elsewhere. *Met in:* measurement, number theory.
Shown: version 6; full guide row 175.

The most findable coincidence is the least visible (181) — 13^7 = 62,748,517 and 89^4 = 62,742,241 differ by 6,276 (one part in ten thousand), which anyone with a pocket calculator could stumble on, yet resolving the two would take a telescope 115,100 logarithmic units long. Noticing a coincidence is not the same purchase as seeing it.

Holds: not yet carried elsewhere. *Met in:* number theory.
Shown: version 6; full guide row 181.

Multiplying by a prime is a gauge transformation of the beam (183) — Because the logarithm of the multiplier cancels from every difference, a prime's clone of the whole forest is the original beam turned by one global phase, so a census over prime pairs is exhaustive over families. The phase was measured at 3.9×10^{-11} against prediction with the modulus untouched to 10^{-12}; the only difference is that a clone is shorter by log P, a difference in how long it gets to play.

Holds: not yet carried elsewhere. *Met in:* topology, number theory.
Shown: version 6; full guide row 183.

One hidden law, many faces: close a loop and an integer is forced (186) — When the integer is forbidden flat, five-fold or seven-fold, the system must curve or go quasiperiodic, which makes the list a design rule: from the Bohr atom's closing wave to the 1:2:4 Io–Europa–Ganymede resonance, anything single-valued around a loop is an integer. In the Loop Atlas sessions' worm ring toy, a ring forbids the best open-chain step of half a wave, and crawls less: 1.14 against 1.66 for 8 segments.

Holds: carried to one: weaker in the chain closed into a ring (wall). *Met in:* topology, kinematics.
Shown: Loop Atlas; full guide row 186.

Zero is the boundary, not a blend (187) — This is why the curvature-sign channel is ternary and not a continuum with a threshold. On a saddle a triangle's angles sum to less than 180° and the carried arrow turns back (−1); on a flat plane they sum to exactly 180° with no turn at all (0), the boundary between the two; on a sphere they exceed 180° and the arrow turns ahead (+1).

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 187.

Roughness is cheap; walking is exponentially expensive (196) — The octave law prices an asymmetry between the construction's two axes: matching one prime's reach at rung r by a pure power of two costs 2^(r+1). The prime 113 sits at rung 30 with a reach of 30.001 and the smallest power of two exceeding it is 2^{31}, while 729,000,000 reaches only 4.20, worse than 1000's 5.367, because its legs nearly cancel.

Holds: not yet carried elsewhere. *Met in:* number theory.
Shown: version 6; full guide row 196.

The radius is the exponent, exactly (197) — Measured across all 999,999 integers below a million, every integer sits at horizontal distance exactly its top exponent e from the axis planted at its cofactor's endpoint, the worst deviation 1.78×10^{-15}, floating-point roundoff and nothing else. The cylinders nest down to the stem, so the nesting chain is the factorization.

Holds: not yet carried elsewhere. *Met in:* number theory.
Shown: version 6; full guide row 197.

The gap is cross-reference, not coverage (204) — The objects under this construction sit in Diophantine approximation, circle dynamics, crystallographic groups, discrete geometry and mathematical biology, each deeply studied, in subject classifications that do not cross-reference each other. A coordinate-free instrument's value is that it renders their shared structure in one picture, the clearest statement in the corpus of what the whole programme is for.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 204.

CAPS AND BASES CARRY HOLONOMY — in the rotational channel, and the Tower had already closed the abelian one (207) — Read the companion before filing an item as open: Tower v3.4 had already closed the abelian channel, a climb's cross-product having no vertical part, so the run returns 0.000000. Measured 2026-09-12 on the square n → np → npq → nq → n, wall loops returned 1.11×10^{-16} rad and boundary loops 0.0625 to 0.4203 rad, because a climb tilts the tangent and a rotation about a horizontal axis does not commute with the vertical ones.

Holds: carried to one, held there: rotation-minimising transport on the commuting square. *Met in:* topology, number theory.
Shown: version 6; full guide row 207.

The cap holonomy is translation-invariant in n, and that is the clone gauge again (208) — Shifting every rung by k rotates every heading rigidly about the vertical axis, which conjugates the holonomy without changing its angle, so only the relative rung offsets matter. Six loops on entirely different integers, from (1,2,3) to (2310,13,17), all

returned 2.247e-01 rad: the Magnetic Forest's clone-gauge result arriving from a different direction.

Holds: carried to one, held there: the cap holonomy (rotation conjugation). *Met in:* topology, number theory.
Shown: Linking and Magnetic Forest; full guide row 208.

The cap holonomy depends on two rung offsets and nothing else — derived, then measured twice (213) — Derive what a reading can depend on before measuring it: the base rung r enters only as a rigid rotation about the vertical axis, which cannot change a holonomy's angle, so only the offsets a and b remain. Seventeen integers from 1 to 44,521 spread 7.49×10^{-16} rad, eight offset pairs gave eight distinct values, and the closed-form legs matched the address-walked quadrilateral to 2.8×10^{-16}.

Holds: not yet carried elsewhere. *Met in:* topology, number theory.
Shown: version 6; full guide row 213.

Eight of ten regularities never touch the axis you chose (216) — Most of what was found is not an artifact of how it was drawn, and the way to know that is to redraw it. Of the Tower's ten measured regularities, eight, among them the ancestry theorem's 13,187 walk grazings and stiff spreading at 0.977, come only from where the walk goes sideways and reproduce identically under any vertical; only the octave law, and the p = 5 anomaly riding on it, reads the vertical.

Holds: not yet carried elsewhere. *Met in:* number theory, reading pictures.
Shown: version 6; full guide row 216.

Only one vertical produces the picture, and that is a fact about the choice (217) — A coordinate that changes the picture while leaving the measurements alone is a presentation decision, not a discovery. Rendered four ways, height as rung gives the tower, height as Ω collapses to about fifteen flat bands, height as log n smears into a wedge and height as n gives a featureless column, though eight of the ten readings ignore the vertical.

Holds: not yet carried elsewhere. *Met in:* number theory, reading pictures.
Shown: version 6; full guide row 217.

HOLONOMY IS THE AREA ON THE SPHERE OF DIRECTIONS — confirmed to 9.4e-16 (223) — With the curvature set to one, Gauss–Bonnet makes the holonomy equal to the signed area the walk's tangent encloses on the unit sphere of directions. Tested against the composed transports across fourteen loops spanning offsets (1,2) through (6,7), the worst disagreement was 9.4×10^{-16} rad, so the cap measurement becomes a one-line signed solid angle computable on paper.

Holds: not yet carried elsewhere. *Met in:* topology, number theory.
Shown: version 6; full guide row 223.

Commuting is a fact about the axis, not about the phase (226) — This is why the forest's walls carry nothing: two rotations about the same axis commute at any phase, while two about perpendicular axes do not, and the difference between the two orders is a rotation about the third axis. That is what i, j, k are, quarter turns about three perpendicular axes with ij = k, and the cap holonomy is the size of that commutator for one small loop.

Holds: not yet carried elsewhere. *Met in:* topology, number theory.
Shown: version 6; full guide row 226.

Three right turns make a left, and it is a theorem (227) — Curvature can be measured by walking, with no apparatus at all. A spherical triangle with three right angles encloses one eighth of the sphere, area π/2, so an arrow carried around it comes home rotated by a quarter turn: three right turns, one left, and the left is the holonomy, as the corpus's own curvature-sign figure draws.

Holds: not yet carried elsewhere. *Met in:* topology, kinematics.
Shown: version 6; full guide row 227.

Turn the mathematical worry into a physical question, and the question gets better (232) — Everything proved is a two-dimensional planar result while the proposal is three-dimensional, where the extension is not automatic, so the status table keeps THEOREM / NO / OPEN / THE ACTUAL QUESTION apart row by row. The mathematics settles in the proposal's favour, and what stays open is physical and better: which physical probe reaches which form degree.

Holds: not yet carried elsewhere. *Met in:* topology, waves.
Shown: version 6; full guide row 232.

Derive the margin instead of quoting it (244) — A margin set by the symmetry group is one a better detector moves by zero. One twin boundary written every way the point group allows gives seven descriptions, the smallest 86.300 and the runner-up 93.700, a gap of exactly 7.400°; against typical measurement noise of 0.5° that is an SNR of 14.80, derived rather than borrowed.

Holds: not yet carried elsewhere. *Met in:* measurement, materials.
Shown: version 6; full guide row 244.

The determinant is an address, not a resident (248) — The value det = 9 is shared by 6_1 and $3_1\#3_1$, and the pipeline printed 6_1 for a structure that is a connected sum: the number was right and the label wrong. An invariant that is not injective is a postcode, so hedge det = 9 as 6_1 or $3_1\#3_1$, exactly as det = 5 is hedged as 4_1 or 5_1.

Holds: not yet carried elsewhere. *Met in:* topology, biology.
Shown: R2W; full guide row 248.

Offer the replacement, then kill it in the same paragraph (260) — A limitation section stays honest when every proposed fix is measured on the spot. Total curvature was tested as a mirror-symmetric alternative to the blind channel: Fáry–Milnor says a knotted curve has total curvature above 4π, but the condition is necessary, not sufficient, and unknotted curves already in the corpus reach 13.46 against a bound of 12.57.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 260.

Screen on the right curve — and the correction does not transfer (261) — DNA topology is axis topology, because the backbones wind around the duplex axis and read as long helices whatever the molecule's real topology. Measured on the helical axis, 250 curves gave zero knotted, 0.00%, so "DNA is not knotted in the PDB" is now measured rather than assumed; RNA, single-stranded and folding back on itself, must not get the correction, since its backbone is the curve.

Holds: carried to one: failed in RNA (single-stranded backbone) (substrate). *Met in:* topology, biology.
Shown: version 6; full guide row 261.

The bug's leftover was a real quantity, and two independent routes agreed to 1.8e-13 (265) — A linking function divided by 2π where the two-curve integral needs 4π, so it returned twice the linking number, and halving left a residual near 0.49: the Bishop frame's own closure holonomy, 0.4877 by two independent routes agreeing to 1.8×10^{-13}. The trefoil then splits as Wr = −3.5123 and Tw = −0.4877, with Lk = −4.0000 exactly, Călugăreanu–White–Fuller demonstrated rather than cited.

Holds: not yet carried elsewhere. *Met in:* records and tools, topology.
Shown: version 6; full guide row 265.

The classical prediction is right on average and cannot be right pointwise (266) — The linking number of two disjoint closed curves is an integer by construction, so predicting a half-integer step is a category error, not a small one. Across a half-twist ladder it measured −4, −3, −3, −2, −2, −1, −1, 0, 0 against a smooth −4.0, −3.5, −3.0, −2.5, every value an integer to better than 10^{-4}, with a mean slope of exactly +0.5000.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 266.

A thing that rings at its own pitch has an address whether or not you wanted to give it one (271) — No naming scheme has to be designed for anything that can be made to ring, because a resonance is an identifier, and this held in all three substrates the paper visits. It sits beside row 234 and the acoustic read head, three separate papers arriving at the same fact: a spectrum is a name.

Holds: not yet carried elsewhere. *Met in:* waves.
Shown: version 6; full guide row 271.

Do not fight the medium — catch what is already moving, and lean (272) — What the medium offers changes completely from an ocean to a body to a vacuum, and the move never does: bias the outcome rather than overpower the flow. The attribution comes in the same sentence, unprompted: the branching growth form and the fact that the same object has a resonant frequency are both somebody else's, named as such.

Holds: carried to one, held there: a body and a vacuum. *Met in:* waves, control and circuits.
Shown: version 6; full guide row 272.

The wall you could not cross is the load-bearing wall of the thing you built (277) — A failure can be the specification rather than the disappointing edge of the work. A delivery vehicle that woke up and began evolving mid-voyage would arrive as something other than what left, so faithfulness across eons means cannot change, the very wall the companion paper hit and reported: the beam holding the roof up.

Holds: not yet carried elsewhere. *Met in:* biology, materials.
Shown: version 6; full guide row 277.

The loss column turned out to be the offspring (279) — Strays that never return after a run were tracked as a recovery budget, a number to drive down, but a unit that falls somewhere hospitable simply grows. So the strays are a population under selection by landing site: the environment writes by filtering which variants keep going, no

individual ever knew where it was, and the population knows.

Holds: not yet carried elsewhere. *Met in:* biology, materials.
Shown: version 6; full guide row 279.

We did not choose that number. We counted it (285) — A constraint that is counted rather than chosen cannot be argued down by a stakeholder, and knowing which numbers are which is most of engineering judgement. Two letters in two polarities give four states and a silence, the silence being the boundary and not a zero; drop one symbol and the codespace collapses, a hard ceiling on the emergency vocabulary set by the geometry.

Holds: not yet carried elsewhere. *Met in:* topology, control and circuits.
Shown: version 6; full guide row 285.

The record is pure geometry, so the reader is interchangeable (286) — Light, X-ray, microwave and acoustics are four valid needles for one groove, chosen after the fact and upgradeable forever, because anything that can resolve a stored shape can read it. The result is a format that cannot go obsolete, because it never specified its own reader.

Holds: not yet carried elsewhere. *Met in:* waves, materials.
Shown: version 6; full guide row 286.

Do not fight the gradient — turn until it goes to zero (295) — A swarm has no structure to fight a tidal gradient with, so it orients instead. The gradient points toward the well, a direction every unit can feel for itself with no clock, map or message; posed along it, the array resolves the tide along its own geometry, where it cancels rather than accumulates, and with two units dead the aperture thins and does not fail.

Holds: not yet carried elsewhere. *Met in:* three body, control and circuits, kinematics.
Shown: version 6; full guide row 295.

A monolith has one aperture forever; a swarm’s aperture is a parameter (296) — An instrument that reconfigures for each measurement turns a fixed limit into a control knob. The swarm widens or narrows its baseline on command, trading resolution against field of view as a single dish never can, while the gravity-nulling pose holds phase-lock; row 160 had found matching the aperture to the source a constraint that produced a clean and misleading null.

Holds: carried to one, held there: the swarm array (aperture as knob). *Met in:* waves, control and circuits.
Shown: version 6; full guide row 296.

Boring is a superpower, because boring can be measured exactly (300) — Detect by invariant-breaking: do not look for the thing, look for the dent it makes in books already known to balance. In even night rain someone invisible crosses the yard, and where they stand the rain stops landing; the person is never seen, only the hole they cut in something even, and everything downstream is the engineering of which books to balance and how cheaply.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 300.

Five channels, and each rung catches what the others are provably blind to (302) — The integer gates where the continuous is measured, coherence watches for what no gate trips on, and twist and linking read the handedness and knotting the lower rungs cannot see. Non-redundancy was measured: with in-plane curl held identical, circulation separated two cases at 0.15 sigma while twist separated them at 15.5 sigma, and with circulation pinned at 1.96 the linking readout still moved through 0.00, −0.064, −0.124, −0.176.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 302.

Predicted from our own thread, then checked — and the contrast closes the loop (321) — A result drawn from the programme’s own thread before it is checked against data is a prediction rather than a lookup, even when the theory is standard. Odd-degree modes sat a constant offset from the midpoint of their even neighbours, scatter 0.4 percent of the large separation; an open core gave a broken ladder and a sealed core a pristine one, filed as a conceptual resonance of record, not a topological claim.

Holds: carried to two: held in odd-degree stellar oscillation modes; failed in an open core (break not stated). *Met in:* waves, solar.
Shown: version 6; full guide row 321.

Single master against replicated, run as a live experiment by evolution (330) — One authoritative map holder buys maximal fidelity at maximal fragility, the trade a storage architect weighs in a replicated system. Priced from the literature, post-reproductive females lead in lean years and a mother’s death raises adult-son mortality roughly fourteenfold; from where the state is rebuilt when the master fails is marked unknown in the literature and left posed.

Holds: not yet carried elsewhere. *Met in:* biology.
Shown: version 6; full guide row 330.

Neither arch spans alone, and naming the keystone is the result (332) — A paper's deliverable can be a correctly located impossibility rather than a crossing it did not make. The crystal is a durable substrate that struggles to live and biology a living one that struggles to endure, so the bridge is not one imitating the other but the two coupled, each supplying exactly what the other lacks.

Holds: carried to one, held there: the biology side (living substrate). *Met in:* biology, materials.
Shown: version 6; full guide row 332.

Simple witnesses in complex configurations, not complex witnesses in simple ones (342) — The instrument reads the arrangement, so its witnesses must be simple enough not to carry their own weather. Enormous numbers of individually simple test particles in a complex field are a home-team substrate, read at the 0.1 percent level, while individually complex, jittering objects are an away game, where the amplitude channel suppressed their signal.

Holds: carried to one: failed in individually complex, jittering objects (substrate). *Met in:* measurement.
Shown: version 6; full guide row 342.

One pin, no loop (343) — Before any question of distance or sensitivity, ask whether there is a loop to walk at all. Holonomy needs a path along which state is carried and around which closure can fail, so a single steady source with no differential offers nothing; it is a loop-existence requirement, and it tells a reader in advance whether a problem is in scope.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 343.

The quadrature test: the same oscillation a quarter-cycle apart IN TIME (350) — Perpendicular in space with no time offset is geometry; the same oscillation a quarter-cycle apart in time is quadrature that carries phase. The test rules out the most tempting example: in a travelling light wave the electric and magnetic fields are in phase, ninety degrees apart only in a standing wave. A quadrature encoder runs the same idea as a sensor, its once-a-revolution index pulse catching a miscount.

Holds: carried to one: failed in a travelling light wave (substrate). *Met in:* waves, control and circuits, quadrature.
Shown: Loop Atlas, Dancing on the Sun; full guide row 350.

Circular motion is quadrature made mechanical (351) — Velocity runs along the tangent and acceleration along the radius, always ninety degrees apart, so the centripetal force does no work and only turns: the cleanest picture of why a phase channel and a magnitude channel differ. In the Loop Atlas sessions a needle turned by two crossed fields and a bead carried by two shifted standing waves gave the same numbers, so turning and carrying are one engine.

Holds: carried to one, held there: the radio conveyor (bead, standing waves). *Met in:* driven oscillators, kinematics, quadrature.
Shown: Loop Atlas; full guide row 351.

Three of the corpus's laws turn out to be one law (353) — Work equals phase gap, and that one fact explains three laws: edge-of-lock holds because zero gap is zero work, the decoupling is the pattern working on the medium where they are out of phase, and rate-not-level holds because the rate of phase change is the work flow. Three separately earned laws collapsing into one is the strongest internal evidence a programme can produce.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, quadrature.
Shown: version 6; full guide row 353.

A spinor is a closure rule you already have, not a dimension you add (357) — A vector comes home after one turn and a spinor needs two, because the rotation group is doubly connected and one loop in orientation space cannot be shrunk to nothing. Bear had drawn it three times before it had a name: the Möbius edge that needs two laps to close, the trefoil that will not undo in one pass, and the belt trick.

Holds: not yet carried elsewhere. *Met in:* topology, kinematics.
Shown: version 6; full guide row 357.

You still land on integers — you just count in spinor turns (358) — Nematic director defects genuinely carry half-integer winding, permitted only because a director is a headless line living on the double cover. So the half is not a fraction landed on but a whole number in the right units, the spinor's, and the exception turns out to be the integer rule's own double.

Holds: carried to one: weaker in nematic director defects (ruler). *Met in:* topology, materials.
Shown: version 6; full guide row 358.

Coordinate-free means reading DIFFERENCES of an INVARIANT (359) — Any uniform background cancels when differences between nearby points are read, which frees the reading from place, and a topological invariant frees it from time, since an integer cannot change continuously. Together they give a reading that depends only on structure, which is why the work kept converging on knots, integers and gradiometers: three routes to the same read.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 359.

Lattices fuse cleanly only at discrete, physics-forced angles (361) — Stacked at a random angle, two lattices share few sites and the seam is strained and high-energy; at the coincidence angles they fall into periodic registry and the seam is seamless. Closure forces the angle, but the number is not derived: coincidence angles depend on the actual lattices, so derive the angle from the real lattice and do not assume it.

Holds: not yet carried elsewhere. *Met in:* materials.
Shown: version 6; full guide row 361.

Hexagons store, squares address, odd polygons signal (363) — Give each shape the job its symmetry fits. The hexagon is zero-curvature and axis-less, the dipole-immune storage substrate; the square is flat too but four-fold with a clean axis, as an addressing raster needs; and odd polygons carry the curvature, plus one and minus one, so even polygons are neutral carriers and odd ones are where to read.

Holds: not yet carried elsewhere. *Met in:* topology, materials.
Shown: version 6; full guide row 363.

Rim sees the mouth, never the neck (374) — A surface loop reports the cap total, dominated by the wide storm-scale spin, and cannot reach the tight throat where the energy concentrates. Priced by the geometry, a four-kilometre core collapses below thirty-to-sixty-kilometre station spacing, so the loop watches the mouth tighten and never sees the neck close: a containment problem, not a sensitivity problem, and row 160’s aperture law in its sharpest form.

Holds: carried to one, held there: a surface station loop round a vortex. *Met in:* measurement, waves.
Shown: version 6; full guide row 374.

The medium decides whether you get the integral for free (375) — Rock is a low-pass filter that integrates strain on its own, while air is high-pass with no quiet baseline: the physics is no simpler in rock, the substrate simply does the accumulation. So before designing anything for a new substrate, ask whether the medium hands over the integral or an integrator must be built.

Holds: carried to one: weaker in air (high-pass, no quiet baseline) (substrate). *Met in:* waves, geodesy.
Shown: version 6; full guide row 375.

Fill the rungs and you approach the sphere — and that is literally a theorem (378) — Unsöld’s theorem says a filled shell sums to a perfect sphere, so the inert elements are the round, complete rungs and chemistry lives in the lumpy partial ones. The honest wait comes with it: filling needs exclusion and half-integer spin, the 720-degree closure a bare polygon does not have, and since roundness recurs the table is a spiral staircase rather than one ramp.

Holds: carried to one: weaker in filled atomic shells (wall). *Met in:* waves, materials.
Shown: version 6; full guide row 378.

The pentagon gives you only half the optic (382) — Reaching only for the sign that feels intuitive is a design blind spot with a name. Positive curvature converges, focuses and closes; negative curvature diverges and opens, doing what its partner cannot, so a sheet of one sign is half an instrument, while an alternating array makes each positive cell a converging element and each negative one diverging.

Holds: not yet carried elsewhere. *Met in:* waves, materials.
Shown: version 6; full guide row 382.

Knotting adds a second protected integer (383) — Tying the loop into a knot makes the phase close around a knotted path, adding self-linking and writhe on top of the plain winding, so the eigenvalues snap to a richer discrete set: more lock, not less. The knot must cross itself on the flat disk and lies cleanly only on the torus the unit circle sweeps, which is why the tube was always the right object.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 383.

One instrument, seven faces (388) — Every thread kept converging because it was one coordinate-free winding reader aimed at seven substrates: field, sky, slide, time, lattice, sky-ghost and genome. The ranking inside it is honest rather than flattering: the biological face is named the most actionable, because a curvature-and-winding anomaly read against a periodic background is the cleanest version of the whole method.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 388.

The domain map: does the medium hand you the integral (391) — Row 375 is promoted here to a map of every domain the programme might enter: the instrument does cheap useful work where the medium is low-pass and integrates for it, and where it is high-pass with no memory or quiet baseline the regulator must be built. It explains the corpus's own hit rate: the tectonic thread produced results and the atmospheric one nulled, not because one was studied harder.

Holds: carried to one: failed in the atmospheric thread (high-pass air) (substrate). *Met in:* measurement, waves, geodesy.
Shown: version 6; full guide row 391.

There is no free invariant, and the blind spot is the receipt (408) — Losing a dimension buys exactness, no gauge choice and a countable grading, and costs everything abelianisation forgets; the deep material lives in the non-abelian sector, which is not stocked at abelian prices. Stated for whole theories, every geometry forces one invariant and pays with exactly one blind spot, the sharpest form of conservation-of-difficulty in the corpus because it is said about mathematics.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 408.

Its ceiling and its exactness are the same fact (409) — An abelian corner is a permanent flatland worth trusting, not a temporary simplification: its group is abelian by nature, so it can never wander into the expensive aisle. The constraint that caps an instrument is the one that makes it exact, so an instrument that could escape its ceiling would lose its warranty with it.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 409.

A coincidence is never alone — it is a rung on a ladder you can write down before looking at data (414) — Enumerate the family a coincidence belongs to and its rarity becomes a count rather than a matter of taste: its siblings sit evenly spaced on a ladder generated by one number computable from the constants before any data. In R2W the quarters, fifths, sevenths, sixths and eighths were each predicted from rules found on the members before them, and such a test is honest only if its bars are written first.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, number theory.
Counted: A check that cannot fail 1. *Shown:* Loop Atlas, R2W; full guide row 414.

The comma is an abelian holonomy — a lift that fails to close (419) — Twelve fifths measure 7.0196 octaves against seven, so the walk closes in pitch class while its lift to the line comes home 23.46 cents out: a circle-valued state whose lift does not close, in row 69's candidate language. Row 408 explains why it is exact: it lives in the abelian corner, where the deficit is a single additive number, order-independent.

Holds: not yet carried elsewhere. *Met in:* topology, waves, number theory.
Shown: Question the Substrate, Two Arrows; full guide row 419.

Life is a constant check on the security of its substrate (423) — A living thing spends energy to hold its internal model against decay, so homeostasis is error-correction and the decay clock is the metabolic cost of being alive. The operational question follows: never certain, how is trust placed well enough to survive? Trust everything and the lie takes you; trust nothing and you starve; trust one source totally and the day it is wrong your whole world is wrong with it.

Holds: not yet carried elsewhere. *Met in:* biology.
Shown: version 6; full guide row 423.

The evidence that two instruments are one object is a shared FAILURE, not a shared success (438) — A success can be engineered, since something winding-like can be made to work almost anywhere, but a failure mode costs too much to build in on purpose. So the strongest support is one failure in unrelated substrates: coherence-is-the-symptom arrived three times, in a control loop defending a corrupt setpoint, a pipeline wrong on every seed from one transposed character, and a game board descended from one bad anchor.

Holds: carried to one, held there: a pipeline and a game board. *Met in:* measurement.
Shown: version 6; full guide row 438.

The null, run by nature, in protein (446) — If a negative result were an artifact of crystals or of simulation, biology would not echo it, and it does. A prion carries heredity in structure and evolves almost not at all, all fidelity and no creativity: the wall walking around in a biological body, bracketed by a spore, a frozen record awaiting an external spark, and a virus, creative but lacking the crystal's autonomy.

Holds: carried to one, held there: biology: the prion. *Met in:* biology, materials.
Shown: version 6; full guide row 446.

The cheapest store is the one where the stored thing is already what somebody wanted (482) — The electricity never comes back, and it was never what was needed: cold and water were. Two independent routes reach one number, a deployed off-grid system reporting about 100 kWh per tonne of cooling against this paper's 93 by its own route, and the store outlasts a battery, because a battery degrades by being used and a tank of ice does not.

Holds: carried to one, held there: a deployed off-grid cooling system. *Met in:* control and circuits.
Shown: version 6; full guide row 482.

A table that locates a mismatch without needing an argument (490) — Eight functional stages run down the side and two populations across the top: for one population the prosthesis replaces the stage that failed, for the other a stage that had not failed, while the two rows that actually broke are the two the device does not have. The cochlear implant is a correct and elegant answer to the left column.

Holds: carried to one: failed in the population whose broken stages lie elsewhere (substrate). *Met in:* biology.
Shown: version 6; full guide row 490.

Four independent constraints selecting the same product, none of them aimed at it (499) — Unaimed constraints that agree are evidence, as unaimed failures are in row 438. Nerve dependency, a power budget putting one architecture at roughly 1.1 W against a comparable device's 30 mW, a thermal limit forbidding continuous wear, and a regulatory path favouring a device with no therapeutic claim all converge on an instrument rather than a prosthesis.

Holds: not yet carried elsewhere. *Met in:* control and circuits, biology.
Shown: version 6; full guide row 499.

A WORD THAT CROSSES A DOMAIN BOUNDARY CARRIES AN ASSUMED DEFINITION, AND THE ASSUMPTION MUST BE VALIDATED BEFORE THE WORD CAN CARRY A CLAIM (502) — Before a word carries a claim across a boundary, write down what it denotes on each side and check that the two are the same object; that cheap step would have caught all eleven cases. Writhe crossed without its definition and a pre-registered prediction failed by a wide margin, and in one session's record beta meant a damping number, an FM drive's swing, a gear-train switch and the McCumber parameter itself.

Holds: carried to five: failed in the open-arc writhe of the backbone (ruler); failed in HMI's azimuth, Stokes Q sign reversed (ruler); failed in R2W's rider (break not stated); held in this guide's own drafted row 727; failed in a triple slice's highest point (substrate). *Met in:* measurement, records and tools.
Counted: A word that changes meaning across a boundary 5. *Shown:* Loop Atlas, R2W, triple-star, Field Guide sessions; full guide row 502.

One blind spot, two substrates — the unification is by impossibility (510) — Distributed-systems trust and discrete differential geometry, two fields that never talk, share a concealment failure, and it is provable, so a reader from either should see their own problem in the other's clothes. A floor tells builders in both exactly what no detector can do: stop trying to catch static balanced lies with sums, and spend the effort on the adaptive case or on pricing it.

Holds: carried to one, held there: discrete differential geometry. *Met in:* measurement, topology.
Shown: version 6; full guide row 510.

Full shell against partial shell — the most predictable thing in chemistry (515) — How partial a shell is tells how reactive the atom is. A closed shell is stable, symmetric and inert, a partial one reactive and lopsided, wanting to grab or give to reach closure: the noble gases barely react, while elements one electron from full, in either direction, react violently, the winding-must-close logic standing where the word quantum was first earned.

Holds: carried to one, held there: atomic shells and the periodic table. *Met in:* topology, materials.
Shown: version 6; full guide row 515.

Cousins, not twins — a deterministic map and a statistical distribution (518) — Two ideas about gaps feel like one law exactly because both concern gaps. The inequality is deterministic, a fact about the map, always exactly true above one; the repulsion is statistical, a fact about a spacing distribution, and it comes from the determinantal structure, not the log, so fusing them would wrongly claim the logarithm causes the repulsion.

Holds: carried to one, held there: two of one's own ideas. *Met in:* measurement, number theory.
Shown: version 6; full guide row 518.

A clone is one global phase, and the beam cannot see it (521) — Because the logarithm of a product is the sum of the logarithms, every term of a prime's family gains the same angle, so cloning is a gauge transformation the instrument cannot see, verified to 9.9×10^{-12}. The only visible difference is the aperture lost, the log of the prime, measured at 24.2216 against a predicted 24.2019 for an eleven-digit prime: the music is identical, only the length of the performance differs.

Holds: not yet carried elsewhere. *Met in:* topology, number theory.
Shown: version 6; full guide row 521.

Which bins exist is arithmetic; which bins survive is aperture (523) — The window between too fine and too coarse has two edges of different kinds: the available bin sizes are quantised to the divisors of the period, and only those whose repeat fits the standing aperture survive. Two of the three parts of Bear's reading hold; the third, that this is also the chaos crossover, has no named mechanism and is held as a candidate.

Holds: not yet carried elsewhere. *Met in:* measurement, number theory.
Shown: version 6; full guide row 523.

The two mechanisms compete rather than stack (529) — At half flux every phase factor is plus or minus one and time-reversal is restored, confirmed by a level-ratio statistic of 0.489 at zero flux, 0.59 to 0.65 at generic flux and 0.440 at half. Yet on every composite periodic lattice tested, that same q = 2 is the weakest divisor tooth, below baseline on three of five, while on an open patch it is the only survivor.

Holds: carried to one: failed in composite periodic lattices with a wrap (break not stated). *Met in:* materials, number theory.
Shown: version 6; full guide row 529.

The dial is re-bought in every medium (532) — A ring lattice's persistent-current amplitude falls as one over the circumference, slope −1.0004 over a sixteenfold range with amplitude times size flat at 1.6733, while a vortex ring at held circulation carries energy that rises with radius. Same latch-and-battery architecture, opposite sign: an architecture ports and its scaling does not.

Holds: carried to two: held in vortex ring at held circulation; failed in vortex ring at held circulation (break not stated). *Met in:* waves, materials.
Shown: version 6; full guide row 532.

The substrate is a piece of real estate in the fiber bundle network (535) — Every project opens by assuming the connections between all points in a chunk of real estate exist, then asks where the magic scissors are that expose them for reading; data agrees, falsifies or is silent about each. Usually only data others gathered for other purposes is available, and it serves its own domain and exposes the geometry at once: the same data doing two jobs.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 535.

The bundles are always there — what varies is our ability to read them (540) — Asking "is there a bundle here?" is the wrong question; ask whether the sensors and the geometric exposure exist to read the bundle that is there. Bear's correction to Station 2 on 2026-09-15 turned every null there from an absence of structure into a gap in instrumentation or a geometry not yet offering a vertex, junction or seam: not dead, unmapped, and unmapped is always recoverable.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 540.

Static and empty are different things (543) — A count across space is as real as one across time. A nebula's dust polarization angle is defined mod 180°, so a loop's winding comes in halves, ±1/2 and ±1 (the nematic defect classification), and halves cost half the margin: on HMI's azimuth, Dancing on the Sun's headless count moved 104 counts against the arrow count's 31 under the same planted turning.

Holds: carried to two, held in both: nebula dust polarization map; HMI azimuth on real Sun maps. *Met in:* topology, solar.
Shown: Dancing on the Sun, Latch Atlas; full guide row 543.

The Arnold floor is set by helicity — writhe alone is not enough, and this code models only half (547) — Name what the code models before reading a floor from it. The Arnold floor is set by helicity, $\Gamma^2 \times$ (writhe + twist), and the core's twist is not modelled, so the tool reads only the writhe half; what it does prove is that a filament evolver with a Gauss-integral writhe reader exists, costs seconds and uses the reader the DNA spoke already uses: the route to F7, and no more.

Holds: carried to one, held there: vortex filament evolver. *Met in:* records and tools, topology, waves.
Shown: version 6; full guide row 547.

Matter latches and waves label (561) — Ask of every new medium whether it offers a latch or a label before asking anything about memory. A whole number that costs energy to hold, behind a wall higher than heat's jiggle, is a latch; one that flies away holding nothing is a label, and among the Latch Atlas's thirteen media, twisted light, radio and sound carry labels until they land on matter whose phase must close on itself.

Holds: carried to one: failed in twisted light, radio and sound (substrate). *Met in:* topology, waves, materials.
Shown: Latch Atlas; full guide row 561.

The viewing angle is a free witness, and it testifies both ways (576) — When the substrate carries the object past the instrument, the trajectory is a witness already retained, and it can testify against the claim too. In Dancing on the Sun each region is seen from a run of angles over about two weeks, and the two-frame read gave about 4 degrees about the line of sight, yet two large regions of 2024 changed the rotation's sign across the meridian while thirteen did not.

Holds: not yet carried elsewhere. *Met in:* measurement, solar.
Shown: Dancing on the Sun; full guide row 576.

The archive's own numbers are witnesses already retained — including against your own sentence (577) — A ratio against the archive's own keyword is worth more to a reader than any description of the tool. In Dancing on the Sun, SHARP's MEANJZD reproduced the home-made mean current to a ratio of 1.000 on maps from 2011 to 2024 while SAVNCPP carried the rotation's upward part on 82 regions and not on six tracked ones, so a claim first made as general was narrowed in the open rather than dropped.

Holds: carried to one: failed in six tracked regions (break not stated). *Met in:* measurement, records and tools, solar.
Shown: Dancing on the Sun; full guide row 577.

Headers are the instrument's diary (578) — A public archive's metadata is a time series of its own, so read it whole before the maps and set every change of code version or quality flag beside the dates of the tests. In Dancing on the Sun the headers showed a QUALITY of 128 on NOAA 12529 on 14 April 2016, the same day the inversion code's name changes to vfisvcombine, though none of it was under test.

Holds: not yet carried elsewhere. *Met in:* records and tools, solar.
Shown: Dancing on the Sun; full guide row 578.

Treads live where phase is carried forward (605) — Before hunting lock plateaus, ask whether the mechanism keeps phase between cycles. A latch erases phase at every stroke, and a field winch driven past its limit showed no staircase, while a slipping conveyor with a second rhythm showed treads at slip-over-beat 1 and 2 only, across three surveys of 4,000 points each; every run on a tread forgets its start, and every run between treads remembers (the Loop Atlas sessions, R2W).

Holds: carried to one: failed in latch (field winch past its limit) (substrate). *Met in:* driven oscillators, number theory, maps and chaos.
Shown: Loop Atlas, R2W; full guide row 605.

A lie hides exactly where the frame can flex without the lying bars (623) — Over 3,360 sets of one, two or three bars in four frames in the triple-star sessions, a set could lie without a trace exactly when the frame without those bars could flex, with no exception. A junction telling the same lie to every bar it reads is caught 0.002 of the time when each bar is read once and every time when each is read from both ends: a lie told alike to everyone needs a second witness.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: triple-star; full guide row 623.

Lengths cannot see handedness; angles can (624) — A measurement blind to mirroring should say so, and a read that sees the mirror should be added. In the triple-star sessions seven bar lengths fit four different pentagons, the three folded shapes keeping all seven lengths to 2.2×10^{-16} while missing the true corner angles by about 2.5 radians; angles read at the bolts named a lying third of a stick every time, against 0.183 of the time with lengths alone.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Counted: The mirror audit 1. *Shown:* triple-star; full guide row 624.

Count the spare measurements, not the shapes they draw (625) — In the triple-star sessions the pentagon with all five diagonals draws 35 triangles, but its ten bars hold only 6 independent loops and 3 bars beyond what rigidity needs, and every gain in sturdiness comes from those three. Bolting the five crossings adds ten pieces and no check, and the frame gets easier to break: it survives 175 of 190 double losses, against 45 of 45 unbolted.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: triple-star; full guide row 625.

Before looking for preferred levels, find which variable rests (626) — A search for rungs on a variable that is kicked at every pass finds only chance. In the chaotic triple of the triple-star sessions the binary's eccentricity jumped only 21 times in 929 strobes and rested between jumps, while the third star's energy was kicked at nearly every pass: only 91 of its 1,080 rests outlasted one excursion.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Counted: Coupled clocks and event time 1. *Shown:* triple-star; full guide row 626.

Grade a picture by its topology: is its edge a fence or a step? (634) — Before borrowing a picture, ask whether its space has ends and whether its push can be written as one height. Bear read the turning needle as a car on a banked track: a track ending at a wall has one seat per speed and a cliff at its end, but a bank that curls all the way round has two balance points at every speed and no cliff, so a cliff in the needle's sense needs a push that goes round.

Holds: carried to one: failed in the turning needle (break not stated). *Met in:* driven oscillators, topology.
Counted: The knee 1. *Shown:* Loop Atlas, R2W; full guide row 634.

A damper eats energy, never momentum (635) — Free fall lets a thrower keep a chunk of position, never of motion. In the Loop Atlas sessions an astronaut throws a ball on a ribbon and a damper catches it: the damper took 100.000 per cent of a 53.7024 J throw, and the momentum kept was -5.43×10^{-13} kg m/s, which is none; the mass of whatever holds the far end sets the chunk's size.

Holds: carried to two: held in a force on a damped spring; weaker in a skateboard's springy pin, sharp top-ups (knob).
Met in: control and circuits, kinematics.
Shown: Loop Atlas; full guide row 635.

A balancing rider's cliff sits where its reaction delay reaches its own rest between taps (636) — The cliff gives no warning in the song of taps before it, so locate it from the rider's rest. In the Loop Atlas sessions, for rests of 0.4, 0.5 and 0.6 fall times the cliff came at delays of 0.40, 0.50 and 0.60, and past it the taps pile up on one side; a quick load is paid at its average, and a lingering one closer to its peak.

Holds: carried to one: failed in looks of 2 and 4 fall times (seam). *Met in:* driven oscillators, control and circuits.
Counted: Coupled clocks and event time 3. *Shown:* Loop Atlas; full guide row 636.

Only the exact half-turn shuvit walks straight (637) — A board spinning in the air goes exactly where it was already going; given a pad to spin on, every exact half turn moves it one board length, and a little more or less walks it in a circle home. A landing hands the spin back reversed and halved, so in the Loop Atlas sessions a board left alone waddles, every stride the same length (1.4142) while each takes twice as long as the last.

Holds: not yet carried elsewhere. *Met in:* kinematics.
Shown: Loop Atlas; full guide row 637.

Agreement in sign is the weakest agreement: ask a toy and a solver the same question, and compare mechanisms (653) — Matching direction says little; compare mechanisms. In the Loop Atlas sessions a toy's kick rule was checked against a jellyfish solver on the one question both could answer, a rest before the second stroke: the direction survived, but the solver's bell is pushed forward by its own wake while it rests, which the toy cannot do, so the toy's numbers are now labelled toy-rule numbers, not findings.

Holds: carried to one: weaker in the jellyfish solver (wall). *Met in:* measurement, waves, biology.
Shown: Loop Atlas, Jellyfish; full guide row 653.

A tread's edge is a fold with a ghost, mirrored (654) — A lock's edge gives only a thin warning band, about a relative 10^{-4} of the drive, so read it from both sides. In the Loop Atlas sessions recovery above the edge slowed as the square root of the distance and the rider stuttered below it on the same square root, slopes 34.27 and 34.73; the floor of a steady drive at weak strength 0.3, read over 20,000 time units, was 0.80798920.

Holds: carried to one, held there: zigzag drive chopped into teeth. *Met in:* driven oscillators, maps and chaos.
Shown: Loop Atlas, R2W; full guide row 654.

Chaos needs something that can answer back, and it lives on the edges (655) — A one-way chain of massless riders cannot be chaotic at any setting, by Bowen's bound (Bowen, 1971), and 672 runs agreed. In R2W, with mass, chaos appears from mass 0.5 and sits on the borders between treads, 97 per cent on a fresh grid read blind and the same at three chaos thresholds, so it is structure, not binning; with two-way teeth it appears, and with one-way never.

Holds: carried to one: weaker in heavy rider inside the gear (break not stated). *Met in:* driven oscillators, maps and chaos.
Counted: Chaos and binning 1. *Shown:* Loop Atlas, R2W; full guide row 655.

Same rate, different word (656) — The order of the slips is a channel the rate does not show, the word rung of Bear's channel ladder. Over 200 beats in R2W, two drives each slipped exactly one tooth in three beats, one playing it as pairs (000011) and the other as singles (100); nine of 12 words were periodic and only 2 evenly spread.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, quadrature.
Shown: Loop Atlas, R2W; full guide row 656.

A side channel can be a bus, not a feed (658) — A channel that carries no power can still decide where the power goes. In Bear's picture of four-wire dimming, two heavy wires carry the power and two thin ones carry the signal that decides how much of it flows; in the same way, a relay's kick that never powers the next stage, but decides which gear it rides, is a valve, not a pump.

Holds: carried to one, held there: a relay's kick in R2W. *Met in:* driven oscillators, control and circuits.
Shown: Loop Atlas, R2W; full guide row 658.

A rest pays back only what builds up, so put the variable your picture depends on into the model (669) — A picture that needs a store is tested only once the store is in the model. In the Loop Atlas sessions, rest stops on a zigzag drive changed nothing while nothing built up; with a heat that drains by Newton's cooling and fades the grip, rests gave back what the heat took. A jump needs a latch too: for the same number of switches, a latch pushed 1.36 to 1.82 times as much as a memoryless crest-firer (run LATCH_EVENT).

Holds: carried to one, held there: zigzag-driven rider with a heat store. *Met in:* driven oscillators, control and circuits.
Shown: Loop Atlas, R2W; full guide row 669.

Memory as a road lives where two locks overlap, and mass makes the overlap (670) — With mass, a driven rider can hold two gears at one drive, and which one it holds depends on the road it came by. The overlap opens near mass 0.61 and is 0.124 of the drive wide at mass 1.0, and in a survey of 32,080 riders, 4,405 of the 4,411 tumbling ones sit between gear 1's edge and gear 2's door. It is Bear's truck road, with its slow lane, passing lane and runaway ramp.

Holds: carried to two, held in both: ribbon of the half groove beside gear 2; Loop Atlas book's two-gear machine. *Met in:* driven oscillators, maps and chaos.
Counted: The knee 3, Count the right unit 1. *Shown:* Loop Atlas, R2W; full guide row 670.

Above the fold, a rapid with pools: high on the staircase the locks are exponentially thin, and place sets their width (671) — Past gear 1's fold a heavy rider mostly tumbles, and its locks are narrow windows in that rapid, their width set by place more than by fraction or weight. In the Loop Atlas's needle at mass 3.0 the 4/3 lock is 1.8 times as wide as the 3/2 above it, and high up the staircase the 8/5 is only $2.4662389 \times 10^{-20}$ of drive wide, because a fast rider averages the wave nearly away.

Holds: carried to one: failed in above gear 1's fold (seam). *Met in:* driven oscillators, number theory, maps and chaos.
Counted: Coupled clocks and event time 2. *Shown:* Loop Atlas, R2W; full guide row 671.

Wells and saddles balance, so an index sum is a free completeness check (672) — Every fixed point carries a sign: on the rider's tube-shaped space of lag and speed, wells count +1 and saddles −1, and the signs must sum to zero. A run that counts them tests the solver, not the physics; it found every fixed point at eight drives and two windings, and every later solver run carries the check for free. Wells and saddles are born and die in pairs, in Bear's picture a pentagon and a heptagon.

Holds: carried to one, held there: R2W dial's dark point, mirror line. *Met in:* driven oscillators, topology, control and circuits.
Counted: A check that cannot fail 1, The integer diagnostic 1. *Shown:* Loop Atlas, R2W; full guide row 672.

Which door you come home through depends on which way you go round the tip (673) — Where two gears coexist, the state a rider brings home depends on its route, and only a loop that encloses the tip of the overlap can tell: one way round brings it home in gear 1, the other in gear 2. In the Loop Atlas sessions a needle walked slowly round loops in 18 runs came home changed exactly when its jumps plus its turns round the meeting point at 0.7026 were odd.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, topology.
Counted: The mirror audit 1, The knee 1. *Shown:* Loop Atlas, R2W; full guide row 673.

The edge says what kind of lock it is (675) — Before fitting an edge law, ask whether the system loses energy, and read the slope of hold length against offset on a log plot. Just below a damped lock's edge a near-miss lasts as one over the square root of the distance, met independently by R2W and the Loop Atlas; at a conservative lock's edge it follows the pendulum's logarithm, 7.57 and 7.37 passes per decade against 7.29 and 7.83 forecast with nothing fitted.

Holds: carried to three: held in the triple; failed in conservative lock kicked once a lap (substrate); held in conservative lock's two clean rims. *Met in:* driven oscillators, three body, maps and chaos.
Counted: Coupled clocks and event time 2. *Shown:* Loop Atlas, R2W, triple-star; full guide row 675.

A borrowed number belongs to the machine it was made on (678) — After any rescaling recompute every dimensionless group, and borrow the whole ladder the record holds, not its first rung. In R2W, RATCHET-69's plan took the tilt's slope from RATCHET-45's lightest weight and treated it as a straight line, though RATCHET-45's own ladder already showed the tilt tiring; the prediction fell 4.7% short for that reason alone, and the redesign using the measured shapes landed at 0.9994.

Holds: carried to 12: failed in needle rescaled to memory dial 1.0 (knob); failed in this machine's filter range (ruler); failed in rider that pulses, then waits out delay (substrate); failed in laptop's run pace (ruler); weaker in island's rocking across eccentricities (wall); failed in heavy star's ragged sides (break not stated); failed in fold refiner tool on the laptop (knob); failed in forward rider's column (knob); failed in higher drive ratios (break not stated); weaker in RATCHET-69's heavier weights (ruler); failed in planted case, riders side by side (knob); failed in a planted case's

value (ruler). *Met in:* measurement, driven oscillators, three body.
Counted: A word that changes meaning across a boundary 2, Count the right unit 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 678.

A doubling ladder, measured: the rungs shrink toward 4.669, and the ladder starts at a fold (680) — The universal numbers are ratios read on a measured ladder, and they are universal only within a class. In the Loop Atlas's light needle (mass 0.75) the band ratios run 4.4595, 4.6155, 4.6570, 4.6667, 4.6687 and 4.6691 toward Feigenbaum's 4.669, and the last three doublings, forecast from that ratio before they were run, landed within 2.5×10^{-9}, 1.3×10^{-10} and 8×10^{-11}.

Holds: carried to five: held in the Loop Atlas's light needle; failed in a textbook's quartic example (substrate); held in the Loop Atlas's own maps; weaker in a lightly leaky map (seam); failed in the triple's island 7 centre (break not stated). *Met in:* driven oscillators, three body, maps and chaos.
Counted: A second reader 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 680.

A rider who listens works on any swing, and reading by the wrong rate has a price that can be written down (681) — Event timing does what a clock cannot, and its error has a closed form. In the Loop Atlas sessions a rider timing each pump by the swing's own phase, read from its angle and speed, passed 90 degrees within 12 of its own swings on every swing from 0.5 to 12.5 m, while a rider keeping a clock set for a 2.5 m swing never passed 12.67 degrees from 10.

Holds: carried to three: held in swings of 0.5 to 12.5 m; failed in swings of 0.5 to 12.5 m (knob); failed in big swings of 10 to 60 degrees (knob). *Met in:* driven oscillators, kinematics, quadrature.
Counted: Coupled clocks and event time 1, The mirror audit 1. *Shown:* Loop Atlas; full guide row 681.

Resonance can evict, and the moment decides who goes (685) — Rocking the drive at a six-step dance's own rhythm throws six-step dancers onto the twelve-step dance that shares their stone, while an off rhythm moves no one, even at five times the depth. Which dancers leave is set exactly by where they stand in the cycle when the rocking begins. The registered lean had guessed capture; it is Bear's Chladni picture inverted, with nodes in time instead of lines in space.

Holds: carried to one: weaker in rocking drive's six-step dancers (break not stated). *Met in:* driven oscillators, maps and chaos.
Shown: Loop Atlas, R2W; full guide row 685.

Ask where a lock can form before choosing which locks to count (688) — Where a lock can sit is set by the shape of the rate, so a census that asks in the wrong place counts the places, not the locks. Re-read at births a thousandth apart, the triple star's locks formed almost only where the rate turns or the island ends (10 of 11), so a law about towers tested on crossings found one crossing to test.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: R2W, triple-star; full guide row 688.

When a scaling misses, its exponent names who is in charge (695) — The guess that a rounded tent's first doubling gap scales with the rounding missed: it scales as the rounding to the power 0.6779, because the second doubling is set by the rounding's far tail. The law worked out after the miss held to 0.02 and 0.01 per cent on two new roundings, so register the law a miss suggests and test it on fresh cases.

Holds: carried to three: held in two new roundings; held in the needle knee's extra size; failed in the flips' power against knee (ruler). *Met in:* measurement, maps and chaos.
Shown: Loop Atlas, triple-star, Question the Substrate; full guide row 695.

Chaos can keep a count (696) — Whether a chaotic orbit's winding is locked is worth reading before calling it disorder, and order is measured by the average, not the spread. In the Loop Atlas sessions a needle with a positive Lyapunov exponent, between two islands of 5/7, kept five turns in every seven within a quarter turn over 30,000 cycles, while a free chaos beside it strayed 25 turns.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, maps and chaos.
Counted: The integer diagnostic 2. *Shown:* Loop Atlas, R2W; full guide row 696.

A window can open and close by the same road (697) — Look for a cascade on both sides of a window, not only on the side the textbook draws. The third face's 2/3 doubles at both ends, chaos, 4/6, 2/3, 4/6, 8/12, 16/24, chaos; and the two ends need not open onto the same chaos, for in the Loop Atlas sessions the pulsed needle's 1/4 window meets chaos confined to four narrow bands below and chaos filling the whole circle past its fold.

Holds: carried to one, held there: the coin's tails 1/2 window. *Met in:* driven oscillators, maps and chaos.
Shown: Loop Atlas, R2W; full guide row 697.

Two maps can share a line; find it and check that they agree on it (698) — The pad atlas's plane of pull against clock ratio carries the golden-ray column that R2W's maps run up, so along that column the two instruments read the same settings. That shared line is where each can be checked against the other, before either is treated as a separate instrument.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, maps and chaos.
Shown: Loop Atlas, R2W; full guide row 698.

The eye keeps what a summary throws away: lopsidedness, negative space and walls (715) — Look at the shape of a signal, not only at its mean and its edges. A core's centre held within 0.0004 across a run, so its average said nothing happened, while the person's "short spiky top left, long spiky top right" was a flip in the ripple's lopsidedness, measured afterwards as −0.18 to +0.29 on the data: right in sign, overstated in size. Translate such a mark into a test together with the person who made it, before the run.

Holds: carried to two: held in next two sets of triple-star runs; failed in rescaled drawing made without the person (break not stated). *Met in:* three body, maps and chaos, reading pictures.
Counted: A word that changes meaning across a boundary 1, Coupled clocks and event time 1, Rate over level 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 715.

Check which angle, or which pair of angles, a clock belongs to (719) — Coupled clocks return together at the continued fraction of their ratio, so before naming a pattern's clock, find which angles come home together. A triple-star lattice climbed a ladder of pass gaps, each the sum of the two before (19, 28, 47, 75, 122), while one angle's own returns gave 22, 25 and 72; over 47 passes the two angles' median net turns, −4.9967 and +15.0175, showed them geared one to three.

Holds: carried to three: failed in birth 7.230 (seam); held in a second specimen, run blind; failed in birth 7.210 (break not stated). *Met in:* three body, number theory, quadrature.
Counted: Coupled clocks and event time 3. *Shown:* triple-star, Field Guide sessions; full guide row 719.

A chaotic transient can end in copies (724) — Read a group of settled states only where every member has settled. In the Loop Atlas sessions thirty-two chaotic spins each settled onto a regular spin, at a mean of 4,750 time units, and at one pace fell into four groups agreeing to 3.4×10^{-8}; read over all windows instead, each needle's own transient split the four groups into sixteen single needles.

Holds: carried to one, held there: the stop's door, LEAK_RATE. *Met in:* waves, maps and chaos.
Counted: Count the right unit 2, Chaos and binning 1. *Shown:* Loop Atlas; full guide row 724.

A needle's own slip rate senses its pace (725) — When a count misses a textbook rate, compute the exact rate before suspecting the count. In the Loop Atlas sessions a needle's slips fell 9 per cent short of Kramers' rate for escape over a barrier (0.9093 ± 0.0112 over 6,549 slips), but against Stratonovich's exact rate for the same needle they pool to 0.9885 ± 0.0122. Read through Kramers' law, the same slips gave the needle's pace within 0.0064.

Holds: carried to one: weaker in noisy needle at barriers 4 to 9 (seam). *Met in:* measurement, driven oscillators.
Counted: Chaos and binning 1, Count the right unit 1. *Shown:* Loop Atlas; full guide row 725.

A map transfers in two senses: number for number where the equation is the same, and as a calibrated ruler everywhere else (727) — Say which sense is meant whenever a result is called portable, and which machine. With its ripple off, R2W's rider obeys the swing equation of a synchronous generator's rotor angle, a phase-locked loop's phase error and a Josephson junction under a steady bias, so the bare map transfers there; for the three-body problem and the seismic rings, the pendulum certifies the ruler, not the mapping. None of these mappings has yet been tested on a real domain's data.

Holds: carried to four: held in generator rotor, phase-locked loop, biased junction; held in a particle in two waves; failed in a junction driven by radio waves (substrate); weaker in three-body problem and seismic rings (break not stated). *Met in:* driven oscillators, control and circuits.
Counted: A word that changes meaning across a boundary 2. *Shown:* Loop Atlas, R2W; full guide row 727.

A flat picture cannot show linking: count the holes, and read the hidden coordinate at the crossings (744) — On one three-body cloud the person saw an eye with three chambers, and the screenshot's own pixels bore it out: a cloud 1.67 times as long as it is wide, its long axis 44° above horizontal, with a hollow on each side of the centre and a filled middle between them (run EYE_PIXEL_CHECK). Whether two such loops are linked is read only at the crossings, by noting which passes over in the coordinate the picture hides.

Holds: not yet carried elsewhere. *Met in:* topology, three body, reading pictures.
Shown: R2W, triple-star; full guide row 744.

A column of followers: check one follower alone, space by time, let the spacing breathe with the pace, and match speed promptly (754) — A stumble passed down a line of followers grows or dies by how the spacing is kept. In the Loop Atlas sessions a line keeping a fixed distance passed a stumble down the line growing thousands of times

in a linear toy, while one keeping a gap in time, about 1.5 s for gentle walkers, damped it up to a reaction delay of about 0.6 s; past about 0.8 s no spacing saves it.

Holds: not yet carried elsewhere. *Met in:* control and circuits.
Shown: Loop Atlas; full guide row 754.

A late read makes a column amplify: the sensor's delay, a word passed back, and how quick a read must be (762) — A growing wobble grows from something, and in a column of walkers the reading error is that something. On a 1 m column with a firm speed match and a small reading error, a look of 0.1 s stopped none, 0.15 s stopped 4 and 0.3 s stopped 19; with no error at all, 0.3 s stopped none and 0.6 s stopped 3 (run CATS_EYES, as corrected).

Holds: carried to one: failed in column keeping a time gap (knob). *Met in:* control and circuits, biology.
Shown: Loop Atlas; full guide row 762.

Two swimmers share one water: sharing a remembering partner is not free (768) — A second swimmer behind the first squeezes into the first one's jet and loses. Of 24 tied spacings and firing orders in the bell toys, one gained (+6 per cent); and on the jellyfish solver of record, a closed passive follower meeting the leader's own wider ring took +51.7 mN·s backward on a leash in 3 s and was pushed back 1.34 R when free, a headwind, not a draft.

Holds: carried to one: failed in a wide ring on a closed follower (knob). *Met in:* waves, biology.
Counted: Chaos and binning 1. *Shown:* Loop Atlas; full guide row 768.

A blade can listen with no sensor: a refunder does the listening, the blade is the rein, and an all-or-nothing blade can stall until a pilot tone wakes it (769) — Any harvester that needs to be moving before it can switch has a cold-start problem. In the Loop Atlas sessions a bead on a spring, its charge switched by its own motion, got up to 72 per cent of what a perfect listener gets from an even soup, but an all-or-nothing blade stalled once its charge was off. A pilot tone of one hundredth of the soup's push, Bear's idea, took two stalled beads from nothing to their full catch.

Holds: carried to two: held in stalled spring beads (pilot tone toy); failed in bead's working pace under load (knob). *Met in:* driven oscillators, control and circuits.
Counted: Coupled clocks and event time 2. *Shown:* Loop Atlas; full guide row 769.

A run kit for the person's laptop: one double-click launcher, checked inputs, a tested exit, his limit on cores, and a price before it ships (776) — The laptop is the person's, and so is its limit, so a kit runs from one double-click launcher that checks its inputs, says STOPPED with the reason when something is missing, and is priced before it ships. Label every estimate as one and set the stop well beyond it: in the triple-star sessions an off-mirror run estimated at about 4 hours took 19,006.4 s measured, 5.3 hours, and finished inside its 6.5-hour stop.

Holds: carried to two: failed in off-mirror run's time estimate (break not stated); held in intake and folder check on his laptop. *Met in:* records and tools.
Shown: Loop Atlas, R2W, triple-star; full guide row 776.

A new session starts from a reference built by script, reads its start-here page in full, and searches the rest (778) — A session card generated from the built guide cannot drift from it, and a session that reads it in full and searches the rest does not start late. A record must also compute every count it prints: in the Loop Atlas sessions the register's front page printed 453 tools, 483 runs of record and 114 blind runs, typed in at one exchange and still printed two exchanges later, when the crate held 457, 487 and 118.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 1. *Shown:* Loop Atlas, R2W, triple-star, Field Guide sessions; full guide row 778.

A lock's width in numbers: set by its place, and read in the rider's own units (790) — Against the habit of reading a simpler fraction as a wider tongue, place decides: in the Loop Atlas's heavy glide at mass 3.0 the 4/3 lock is 6.860×10^{-5} of drive wide, 1.8 times as wide as the 3/2 lock (3.877×10^{-5}), which sits higher in drive. A width in the knob's units carries the staircase's slope and can reverse a trend: the 3/2 narrows with weight in drive but widens in turns per passing.

Holds: carried to three: failed in heavy rider's locks at mass 3.0 (break not stated); held in new cases (GROOVES-2); held in clock ratio 0.6. *Met in:* driven oscillators, number theory.
Counted: Coupled clocks and event time 1. *Shown:* Loop Atlas; full guide row 790.

A loop's phase in closed form: light carried by filters or wave plates, and the three ways a mountain hides from the arrow (792) — The shift is the loop's, not the filters': light carried round a loop of states on the polarisation ball by filters comes home shifted by half the solid angle the loop encloses and loses a little light, while carried by wave plates it comes home with the same shift, same sign, to 4.44×10^{-15}, and loses nothing. An exact zero is a warning: the first ridge test read exactly 0.000e+00 because it could not fail.

Holds: carried to one, held there: wave plates on the same loop. *Met in:* topology, waves, kinematics.
Shown: Loop Atlas; full guide row 792.

Read a thin lock from its loop, not its folds, and trust one power per period only roughly (795) — At each starting angle exactly one drive closes the orbit, so solving for it stays well conditioned where fold conditions do not. Read this way in the Loop Atlas sessions, the 11/8 lock at mass 3.0 is $1.1911119 \times 10^{-11}$ of drive wide, 4.4 times thinner than two unconverged folds had said, and period 9 gives powers of 23.06 and 21.79, about 6 per cent apart.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, three body.
Counted: Coupled clocks and event time 1. *Shown:* Loop Atlas, triple-star; full guide row 795.

The comb theorem: N evenly spaced copies keep only the harmonics the spacing allows, and a symmetry that empties half of them decides odd against even (802) — This is a general fact, not special to holonomy, and it decides what any sampling scheme can see. N evenly spaced copies, added together, keep only the harmonics whose number is a multiple of N, the same arithmetic that sets a three-phase bridge's line harmonics at 6k ± 1; in R2W, sixteen timings cannot tell the push's ninth harmonic from its seventh, so a model built from sixteen timings is blind to the error that eight crooked timings make.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, quadrature.
Counted: Chaos and binning 1. *Shown:* R2W; full guide row 802.

Give a settling field a memory and it gets a flywheel and a speed limit: read a start by its fronts as well as its modes, and count a ringing by its deepest swing (804) — Heat that takes time to catch up carries fronts as well as modes, and the fronts can overrule the modes. In the Loop Atlas sessions a warm brick in ice, read by its modes, should ring above the hump's line, but its middle is knocked down only when its two cold fronts meet, and it stays warm; a quenched brick's deepest dip comes at 1.485 times its length in heat's travel time, as the cold fronts cross back.

Holds: not yet carried elsewhere. *Met in:* waves, materials.
Shown: Loop Atlas; full guide row 804.

A long run keeps its own custody: a ledger line written with each verdict, a stamp and a job of its own, only what its reader reads, and a heartbeat that survives sleep (809) — A long run is only as safe as what it has already written down, and in the seventh triple-star session runs were lost to their own tools, launchers and machines, not to the physics. On 19 October triple_tw8 v1.2's main run and its twin started in the same second, both took the stamp 20261019_043245, and the twin overwrote the main run's TABLE and SERIES; each run's own ledger survived, so the loss was seen at once.

Holds: not yet carried elsewhere. *Met in:* records and tools, three body.
Counted: The built artifact is the evidence 1. *Shown:* triple-star; full guide row 809.

Test the test bed as well as the tool, and find out which of the two is wrong (812) — A check that fails may be telling the truth about the bench rather than the tool, so the bench is tested too. On 22 October the triple-star thread ran the publishing packet's launchers under wine 9.0, whose cmd ran the right side of 'a || b' even when a succeeded, so the fingerprint gate stopped a matching tool; the launchers now use 'if errorlevel' lines, which both shells read alike.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 1, A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, triple-star; full guide row 812.

The scallop theorem counts knobs: a knob that goes round escapes it, and a body's own symmetry can forbid motion even with two (817) — In thick, slow water a single hinge that opens and shuts goes back exactly as far as it came, so a swimmer needs two knobs out of step or one that goes round. In the Loop Atlas sessions a coil turned on an axle went farthest a turn at a winding angle of 54.7 degrees at grip ratio 2, but farther per unit of work wound shallower, and a coil symmetric end over end, pumped by squeezes, went nowhere to 3×10^{-15}.

Holds: carried to two: failed in a coil turned on an axle (substrate); failed in end-over-end symmetric pumping coil (wall). *Met in:* kinematics.
Shown: Loop Atlas; full guide row 817.

Read a falling body's twist and turn in a form that can hold them: carry a mark across the joint, keep the turn signed, and break a symmetry before crediting a topology (818) — The usual ways of writing a turn can hide a full turn, a sign or a symmetry. In the Loop Atlas sessions an angle read from 0 to 180 gave the twist-free cat 152 degrees at a bend of 90 where, kept signed as a quaternion, the turn was 208, past its feet; and the toy cat's exact half turn came from its two halves being the same, since halves of 0.25 and 0.35 fall 23.9 degrees short.

Holds: carried to one: failed in halves of 0.25 and 0.35 (knob). *Met in:* topology, kinematics.
Counted: A check that cannot fail 1, The built artifact is the evidence 1, A word that changes meaning across a boundary 1, Chaos and binning 1. *Shown:* Loop Atlas; full guide row 818.

Two falling cats that look like strangers are one cat under two rules: a thick cat twisting at its waist and a thin twist-free cat bent to the mirror angle turn alike on every path (820) — The Loop Atlas found that a thick cat twisting at its waist and a thinner twist-free cat at the mirror bend, 180 − b, turn alike within 2.842×10^{-13} degrees on nine swelling paths, and nearly lost the finding to its own code. A search made at bend 90, which the mirror sends to itself, hid a mirror applied twice, and the false miss stood in the book's headline for two days; test a mirror away from its middle.

Holds: carried to four: held in the toy cats in flight; held in the hinge moved in; held in the Loop Atlas's twist-free cat; failed in every other bend (ruler). *Met in:* topology, kinematics.
Counted: A check that cannot fail 1. *Shown:* Loop Atlas; full guide row 820.

Rare events can keep a rhythm: measure a count's spread before trusting its square root, and expect a low barrier to space its events out (826) — A count's error bar is its square root only if its events are independent, which is a measurement to make, not an assumption. In the Loop Atlas sessions the noisy needle's slips at 4 to 9 noise units had a variance equal to their mean within about 5 per cent (Fano factors 0.947 to 1.032), but over a barrier three times the noise's strength the slips ran about 8.5 per cent more regular than independent events.

Holds: carried to one: weaker in low barrier, 3 noise units (seam). *Met in:* measurement, driven oscillators.
Shown: Loop Atlas; full guide row 826.

Noise can make a noisy machine's events most regular at one level: set the noise against what is left of the barrier, not against the barrier as designed (828) — Too little noise and a machine near its slipping edge slips by luck, too much and it wanders; between, the noise only tips it over and its own slide sets the beat. In the Loop Atlas sessions a needle held at 0.95 of its edge slipped least spread at noise 0.7, a Fano factor of 0.42341, where the barrier left is only 0.09 noise units, and its own waits there read 0.42357 ± 0.00126.

Holds: not yet carried elsewhere. *Met in:* driven oscillators.
Shown: Loop Atlas; full guide row 828.

Look for the shape a system passes through on its way to failing, then test whether that shape travels to the neighbouring settings before calling it a warning (830) — A setting where something fails says where, not how, and a door seen at one setting is a story about that setting until it is tested at the neighbours. In the Loop Atlas sessions every one of eight smoke-ring train breaks came within 200 time units of the slipping ring shrinking to 0.207 to 0.211 of a radius, yet thin cores at their own edge broke with no squeeze warning at all, so the clean door belongs to the middle core.

Holds: carried to two: weaker in fatter cores, up to 0.66 radius (knob); failed in thin cores at their own edge (knob). *Met in:* waves, maps and chaos.
Shown: Loop Atlas; full guide row 830.

Name a kit for its home session, and quote a run time with the cause it leaves out (842) — A kit on the person's laptop is a parcel sent away. A door kit named after its engine, LOOP_ATLAS_RECORD_SURVEY_DOOR, left Bear, with four sessions running batches, unable to tell where its results belonged; and R2W's 3/7 ladder kit, priced at three hours, up to five, took 516.8 minutes, because the deep dances' Newton noise kept the secant from settling.

Holds: carried to one: failed in deep levels of the 3/7 ladder (seam). *Met in:* records and tools.
Counted: Count the right unit 1. *Shown:* R2W; full guide row 842.

Read a loop's matrix by its class first, and name the bit the picture throws away (848) — A conjugacy class is the part of a loop's matrix that no change of frame can move, so it is read before anything else. In A Rider in Two Waves the passing matrix's half-trace sorts it into a rotation, a boost or a null rotation, and the steps read from the slip alone agree at 19,477 of 19,481 settings, while the frame-dependent top-left entry showed no sign change at all.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, topology, three body.
Shown: R2W, triple-star; full guide row 848.

A special point on a boundary is usually known: find it by its structure, not its numbers (849) — On R2W's light cone every point is balanced as tightly as any other; the exact rider, at strength 1 and bias equal to the frequency, is special only because a pencil can write its motion down. Its likely home in the literature was found by the shape of the result, a family of explicit solutions on step edges, not by its numbers, and its null rotation's shear, 240.8, 47.2, 17.1 and 4.83 at four frequencies, is left open.

Holds: not yet carried elsewhere. *Met in:* records and tools, driven oscillators.
Shown: R2W; full guide row 849.

File before you send: anything that lives only in the conversation is written to its file before the reply that carries it goes out (856) — A read given only in chat can be lost. On 28 October a reader went back through the triple-star conversation and found eight reads given in chat and never filed, which were filed from the chat; a ninth, written and sent later that day but never filed, was not rewritten from memory, and the reads file keeps its title with a note.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: triple-star; full guide row 856.

A long run on a shared machine: measure it first, cut it into saved segments, and stop it only by its number (857) — The triple-star thread's 60,000-pass run first hit a two-hour job limit without saving anything; a scratch run measured afterwards (0.128 s and 0.264 MB a pass) showed one half needed about 8.7 GB against the workspace's 7.7 GB. Cut into twelve segments that each save their state, the run survived a second limit with eleven of twelve saved, so time and size a scratch run and quote a finish time only from it.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: Loop Atlas, triple-star; full guide row 857.

A knob one project adds on purpose is an artefact another must measure: keep a two-way list, and measure the artefact against the knob's scale (859) — Friction added on purpose in one toy is an error to be measured in another. Bear named the friction that A Rider in Two Waves and the Loop Atlas put into their toys "honey"; the triple-star toy's only honey is its integrator's, and R2W, with its own honey at zero, found its integrator draining energy as if a honey of at most 1.85×10^{-6} had been added.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, three body.
Shown: Loop Atlas, R2W, triple-star; full guide row 859.

Prove a deposit from its zips: follow every file it names, start one launcher from the unpacked zips, claim byte-identity only where it holds, and word-read it (862) — Only a run from the deposited zips shows what a reader will actually receive. For the islands paper of the triple-star sessions, the zip proof found that 21 deposited launchers named a launch script the list had never carried, so none could have started a run, though the README said each checks every file's fingerprint; the gap was open four days and closed before any upload.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: R2W, triple-star, Question the Substrate, Two Arrows, Field Guide sessions; full guide row 862.

A clock and a switch make a road of their own, and chaos can have a skeleton the eye sees first (873) — A switch clocked like a drive gives a road unlike a smooth toy's: the Loop Atlas's boost converter (card 14) ran two smooth flips with a border collision between them and then chaos with no window from 2.59 to 5 A. Over its picture Bear drew four straight lines, and every one is a real edge of that chaos, within 0.12 A: the switch's border and its images lap by lap.

Holds: not yet carried elsewhere. *Met in:* control and circuits, maps and chaos, reading pictures.
Shown: Loop Atlas; full guide row 873.

Count a part's life on the clock it wears by, and know where that clock has a knee; and the back side of the power curve is a fold (876) — A part wears by its own events, not by the hours on a clock it does not wear by. In the Loop Atlas's toy trainer airplane (card 61) a hinge wears at Rice's level-crossing rate, which has a knee, so a tach-hour schedule leaves 85 per cent of hinge life unused on the calmly flown airplane; and in a flare 98 per cent of the first two seconds' deceleration is the downhill push given up.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, control and circuits, quadrature.
Shown: Loop Atlas; full guide row 876.

Each doubling turns the last one's corner over, by a ratio that belongs to the hump's class: −1.9644458 for a rounded top (877) — A doubling bends an orbit average against the drive, and each doubling turns the last one's corner over by a fixed ratio. The logistic and sine maps, solved in 80-digit arithmetic, agree on −1.9644458 to the ninth figure, the Loop Atlas's needle lies about 0.002 from it, and a flatter top changes the number: −0.603 at z = 4, where the corners grow.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, maps and chaos.
Shown: Loop Atlas; full guide row 877.

Where a switch can hold a body still, compute it with no switch at all: the count is the switch's, the push the world's, and the flicker is counted from the stays (879) — Where a switch can hold a body at rest, the push comes from the world and only the count belongs to the switch. The Loop Atlas's spring at heavy loss, computed with no switch at all, pushes 0.2265 of strength against the plain blade's 0.2259, and its flicker count can be forecast from its stays: 54.54 against 53.13 flown at loss 1.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, control and circuits.
Shown: Loop Atlas; full guide row 879.

A mind's own certainty is a rich-get-richer filter: the average holds while the smallest answer dies (883) — A mind that trains on the labels it is surest of feeds its biggest answers and starves its smallest, while the average hardly moves. In Question the Substrate an answer holding 11 per cent of the data went extinct, so track the smallest class beside the average, and gauge any fix the same way, since a fix is a new filter.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: Question the Substrate; full guide row 883.

Humility without courage spirals, and a cure tested as one bundle hides which knob works (885) — A checker allowed to answer “unknown” spiralled until it said “unknown” to 66.9 per cent of test loops. Two knobs together stopped it, and a sweep of each knob alone later showed which one did the work, correcting the paper’s own first explanation.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: Question the Substrate; full guide row 885.

A copying chain compounds large leaks and absorbs small ones: read a dial at its small end before calling a drift universal (887) — A copying chain inherits large leaks: 5 per cent of its labels flipped each generation slid it 7.9 points. Small leaks it absorbs, a leak of 0.5 per cent entirely, so read a dial at its small end before calling a drift universal.

Holds: not yet carried elsewhere. *Met in:* measurement, maps and chaos.
Shown: Question the Substrate; full guide row 887.

A learner heals its answers long before its inside: in any study of drift, read the inside directly (888) — After a closed loop of tasks, Two Arrows’ learner keeps 0.0167 of its change in its answers and 0.3049 in its hidden layer. A checker that grades only answers would call it healed, so a study of drift reads the inside directly.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: Two Arrows; full guide row 888.

Only a kept record of the self brings it home, and held after the trip it costs the journey nothing (889) — Of five kinds of practice, only a pull toward a record of the hidden layer taken before the loop brings the learner home, leaving 0.0003 of its inner mark. Held throughout the journey, the record makes the learner worse at following its task; switched on only after the trip, it costs the journey nothing.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: Two Arrows; full guide row 889.

Look for a relabelling before calling a change continuous (890) — A learner came home with two hidden units exchanged, and renaming them removed 68 to 85 per cent of its distance from home. A symmetry can make one discrete swap look like a large continuous change, so match the units before measuring the distance.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: Two Arrows; full guide row 890.

L3 — Holonomy You Write

The write limb: creating invariant loops, acoustic marking, cured helixes.

Active loop walking — the probe (31) — The fluid’s own drag must be counted before a moving ring can probe anything. A parked ring stalls at 0.784 ring radii from its own centre and loses 29% of its energy in 0.5 seconds at Reynolds number 300, the thick-fluid regime, while at water viscosity the loss drops to about 1%, so viscosity is the dominant energy loss and the probe must account for the fluid regime.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 31.

Loops that close themselves (37) — Switch a stroke’s parts off one at a time and read each against the whole: the stroke brings the body home almost where it started unless the lip shuts the slot, and only vorticity kept under the rim or left behind breaks that symmetry. In the rung 6 switch-offs, lip off gave −0.015 to −0.019 R per cycle against −0.115 to −0.217 for the full stroke, and the pinched moon jelly surged to −0.30 R by 0.4 T before the open-mouth refill hauled it back at +50 mm/s, to a net +0.03 R per cycle the wrong way.

Holds: not yet carried elsewhere. *Met in:* waves, biology.
Shown: Jellyfish; full guide row 37.

A second channel carrying control (40) — Reading and controlling need not take separate structures. A ring architecture that reads topology carries a control signal in its second degree of freedom: the signed-ring data channel retains 39–40% of spin over about 9 ring radii, and a quarter-turn swirl manoeuvre costs about 0.7% of speed, so readout and control happen at the same time in the same ring.

Holds: not yet carried elsewhere. *Met in:* topology, waves, control and circuits.
Shown: version 6; full guide row 40.

Write-time holonomy — the inversion (43) — This is the whole write architecture in one turn. Conventionally an instrument searches a substrate for an invariant and walks loops to detect change; here the path that writes is the loop, and the invariant sits where the path puts it: “you stop hunting and start placing”.

Holds: not yet carried elsewhere. *Met in:* topology, materials.
Shown: version 6; full guide row 43.

Name your structure group before you fabricate (44) — The group in which rotations are composed decides whether a loop’s answer can be reproduced at all. Composing symmetry-reduced rotations around a loop moved 201 map cells into different defect classes between two computers running identical code on identical bytes, and moved a boundary-seam count from 951 to 2207 across five equally defensible tie-break rules, while the same construction on $\mathbb{Z}/2$ is path-independent to machine precision.

Holds: carried to one: failed in symmetry-reduced rotations round a loop (wall). *Met in:* measurement, topology.
Shown: version 6; full guide row 44.

The cut chooses the algebra (45) — A sealed cavity gives a 2-sphere, π_2 and an abelian $\mathbb{Z}^n$, which reads as an address; a through-tunnel gives a handle, π_1 and a free, non-abelian group, which reads as a sentence. The map between them is abelianisation, so the readout algebra is chosen at write time, when the cut is made.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 45.

Closure is the admission rule (46) — A path that does not close has no holonomy, so closure decides whether a path counts as a word at all. At length 12, 5.1% of 2-D words close (853,776 of 16,777,216) and 1.5% in 3-D, and that rejection is error detection performed by geometry before any checksum exists.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 46.

The read head is a theorem (47) — A shape tells its own topology through how it vibrates, so its holes can be counted with no geometric scan. By the heat-trace theorem a flat domain with h holes carries a constant term $a_0 = (1 - h)/6$ in its spectrum, each hole adding $-1/6$ to the constant, so with three holes the topology is as audible from the spectrum as the area.

Holds: not yet carried elsewhere. *Met in:* topology, waves.
Shown: version 6; full guide row 47.

A famous impossibility, repurposed as a feature (48) — Because the spectrum does not determine the shape, a medium read by its spectrum can disclose the stored value without disclosing the structure that holds it. Nobody designed that property into the medium; it falls out of the mathematics and costs nothing.

Holds: not yet carried elsewhere. *Met in:* topology, waves.
Shown: version 6; full guide row 48.

Reads-until-erasure — an unreported category (51) — Every read is hot, yet flash quotes write endurance and nobody publishes a read budget. A stored state’s lifetime grows exponentially with its barrier over the heat, $\tau = \tau_0 \exp(\Delta E/kT)$: $\Delta E \approx 0.5$ eV gives fractions of a second and ≈ 1.1 eV gives centuries, so tiers must be separated by ≥ 0.25 eV or their lifetime distributions overlap.

Holds: not yet carried elsewhere. *Met in:* control and circuits, materials.
Shown: version 6; full guide row 51.

A spectral read needs no pose (52) — A kinematic read must specify seven numbers, three of position, three of orientation and one start phase, or its read clock drifts off-sample. Ringing the whole object is free of pose by construction; the trade is one number for the object instead of a spatial map.

Holds: not yet carried elsewhere. *Met in:* measurement, waves, kinematics.
Shown: version 6; full guide row 52.

The alphabet is a product, and the accidental degeneracy (53) — Some design numbers are forced, and some coincidences are cheaper to find at the desk than after the cavity is built. The semantic alphabet’s size of $12 \times 36 = 432$ was forced, not chosen; and on a 3×3 grid $\lambda = 4$ arrives with multiplicity three, so one chirality channel is spectrally inseparable from a non-chiral mode until a slight aspect ratio splits it.

Holds: not yet carried elsewhere. *Met in:* topology, waves.
Shown: version 6; full guide row 53.

The lightness is the immune system (81) — Seriousness is what turns a tool into a faith: if a thing can be laughed at it can be tested, while if it is sacred it is already corrupting. The deposited paper carries the recursive clause too, that certainty that certainty is a red flag would itself be the trap, and this paragraph from July is the published precedent for the 2026-09-11 editorial-balance ruling.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 81.

The hunter reads; the composer writes (110) — This one sentence is a charter for the whole write limb, and it arrived in a node about tornado false alarms. The instrument reads defects out of a substrate it was given, and the same geometry writes contours into a substrate it shapes.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 110.

They match to erase; we match to discover (162) — The minimum-weight perfect-matching decoder exists to pair defects away so that a quantum memory survives; pointed the other direction, the same engine pairs the noise away so that the one defect which refuses to be matched stands exposed. One sign flip turns production-grade open source into a detector that can be imported rather than re-derived, though the wiring is still short of a full literature pass.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 162.

The address closes and the datum closes, and they are the same walk (234) — Conventionally an instrument searches a substrate for an invariant and walks loops to detect changes in it; here the path that writes is the loop, and the invariant sits where the path puts it. The invariant is deposited on a loop and is itself a loop, a track that returns the walker, upside down, to exactly where it started, which is what a holonomy does to a frame.

Holds: not yet carried elsewhere. *Met in:* topology.
Shown: version 6; full guide row 234.

Tamper response as a consequence of the read, not an alarm bolted beside it (235) — Reading at the energies an invasive probe or die-imaging attempt requires is a supra-barrier operation, so the attack anneals the very structure it is trying to read. Conventional tamper-evident hardware trips an alarm that can be defeated; a security property that falls out of the physics costs nothing to maintain and cannot be switched off.

Holds: not yet carried elsewhere. *Met in:* control and circuits, materials.
Shown: version 6; full guide row 235.

Put the intelligence outside, and the safety is the absence of a capability rather than a feature (270) — The mother holds the four things that must persist and be shared, brains, power, master clock and map, and the daughter holds none of them. The payoff is safety, not elegance: a daughter with no onboard intelligence has nothing to go rogue with, and a safeguard built as a missing capability cannot be disabled, because there is nothing there to disable.

Holds: not yet carried elsewhere. *Met in:* control and circuits.
Shown: version 6; full guide row 270.

Handedness is topological, not amplitude — so it survives what amplitude cannot (283) — Attenuate the chiral codec, blur it or drown it, and left is still not right. Driving a cavity's two degenerate modes ninety degrees apart makes the pattern rotate, so every mode with angular structure doubles into plus and minus; the ring mode with no angular folds has no winding and is the fragile, amplitude-only letter, so one alphabet shows which letters are protected and why.

Holds: not yet carried elsewhere. *Met in:* topology, waves.
Shown: Loop Atlas; full guide row 283.

One word, one meaning, one part of speech — because there is no round trip to spare (288) — In a channel this bad there is no round trip available to ask which sense was meant, so the chiral alphabet is bound to a controlled vocabulary of about nine hundred terms. The vocabulary is not claimed: it belongs to an aerospace documentation standard that has done a nearly identical job since the eighties, for technicians reading manuals in a language they half speak.

Holds: not yet carried elsewhere. *Met in:* records and tools, control and circuits.
Shown: R2W; full guide row 288.

You cannot scramble an integer (293) — An anchored in-situ dosimeter stays on station while an external therapy fires, reading dose actually received rather than dose intended, and its rad-hardened winding memory keeps a log that the radiation it measures cannot corrupt. A continuous register degrades gracefully into a plausible wrong answer, while a count either holds or fails visibly.

Holds: not yet carried elsewhere. *Met in:* measurement, biology, materials.
Shown: version 6; full guide row 293.

A sensor geometry run backwards is an emitter — and the cost moves from free to powered (367) — The same antenna receives and transmits, and the same element reads strain and makes it, so a differential reader inverted is a differential projector. Two engineering fences travel with it: the geometry is free but the amplification

never is, costing energy, heat and an exposure limit; and nonlinear and lossy parts break the symmetry, so the projector is its own device sharing the geometry, not a mirror of the detector.

Holds: carried to one: weaker in an emitter, projecting (break not stated). *Met in:* waves, control and circuits.
Shown: version 6; full guide row 367.

Angle is the index; side is the load (379) — This storage architecture separates what must survive from what may vary: the protected integer is the address and the payload rides the side, hard addressing and soft storage. On the companion dial a wide funnel collects and sees the mouth, while a tightened polygon closing toward a sphere stores and becomes the checksum; which end of that dial a design sits on is the design decision.

Holds: not yet carried elsewhere. *Met in:* topology, materials.
Shown: version 6; full guide row 379.

Quantised locks beat a strained stretch (381) — Stretching an element continuously to reach a target holds a strained configuration that is high in energy, drifts and needs servicing forever, while integer locks sit at stable minima and snap to. Past threshold new defects nucleate as the lattice tears and modes bleed into each other, capping the usable resolution, so compose discrete stable presets in a stack rather than straining one element to its edge.

Holds: not yet carried elsewhere. *Met in:* control and circuits, materials.
Shown: version 6; full guide row 381.

A frozen record is silent until something traverses it (443) — In the gramophone decoupling the record is pure geometry and the reader is interchangeable, so density is fixed at write time and reading stays upgradeable forever. The missing dimension of a static structure is supplied by the act of reading across it, the way a groove is silent until the needle moves: a crystal archive keeps its data precisely by being able to sleep.

Holds: not yet carried elsewhere. *Met in:* topology, materials.
Shown: version 6; full guide row 443.

The concrete outlives the copper (484) — A marine structure is a fifty-to-hundred-year asset and an electrical machine a twenty-year asset, so welding them together throws away the long-lived thing when the short-lived one dies. Put the interface at a pipe, a busbar or a plug within reach, keep every short-lived part on land, and test a drawing in one line: if servicing it in year twenty means opening something poured in year zero, it is the wrong architecture.

Holds: not yet carried elsewhere. *Met in:* control and circuits.
Shown: version 6; full guide row 484.

Geometry beats mechanism — and every member fails into its other job (485) — A perforated wall does storm relief with nothing to actuate, a notched rim does overfill relief with no valve, and the blade and the flooded chamber both fail into being a wall. Designing the failure mode to be the secondary function keeps the failed state useful rather than merely safe, a stronger property than fail-safe that is almost never specified.

Holds: not yet carried elsewhere. *Met in:* waves, control and circuits.
Shown: version 6; full guide row 485.

The substrate stores poses, and reading is navigation (519) — The claim reframes the addressing problem: data stored as poses rather than at addresses is reached by dead-reckoning, not by looking up a cell, which makes the navigation lineage the foundation of the addressing layer. Noticed three or four times and never stated, it is filed as stated and untested, in the write limb, where the inventory is thinnest.

Holds: not yet carried elsewhere. *Met in:* topology, kinematics.
Shown: version 6; full guide row 519.

A curve and a kernel — no grid, no solver, 88 lines and 4.7 seconds (545) — A vortex can be evolved as a curve of N points, each segment inducing velocity on every other through a Biot-Savart kernel regularised by core radius a and stepped by RK4 with no 3-D grid and no full solver. A 200-point trefoil ran 300 steps in 4.7 seconds in the standard container, the evolver F7 needed beside the DNA spoke’s existing reader.

Holds: not yet carried elsewhere. *Met in:* records and tools, topology, waves.
Shown: version 6; full guide row 545.

The price of a latch is the wall, not the store (562) — A superconducting ring stores about one jiggle of heat per rung and holds for years, because its wall is ten thousand jiggles high even in a 20 nm wire. And a wall guards one path only: a claim of stability names the escape route it is stable against, which for a thin film is a vortex crossing, not the phase slip a thick wire pays.

Holds: not yet carried elsewhere. *Met in:* materials.
Shown: Latch Atlas; full guide row 562.

Erasing is a second write, from the other side (563) — A ring with one weak spot writes one rung at a time and comes down by a different road, the erase paying what the write paid; the loop between the two roads is the memory, and its area is the price of a round trip. Read where it lives: R2W, holding riders rather than ramping them, finds memory between two locks, whose overlap opens near mass 0.61 and is 0.124 of the drive wide at mass 1.0.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, control and circuits, materials.
Counted: The knee 1. *Shown:* Loop Atlas, R2W, Latch Atlas; full guide row 563.

Temperature is the write-enable, and it works at two temperatures (564) — Rungs come into being just below the transition and lock only lower down, where the walls outgrow the heat, so the count is written in the gap between. Heat and Kibble–Zurek domains write the same staircase so closely that no field scan tells them apart; only how the width moves with cooling rate and ring shape separates them, and in niobium a measured slope steeper than either alone is recorded as an open door.

Holds: not yet carried elsewhere. *Met in:* materials.
Shown: Latch Atlas; full guide row 564.

A lock stalls at a quarter turn, and a powered lock is not a latch (565) — Holding a count and being held are different things. Two dials pulling on each other hold until the lag reaches a quarter turn, then slip one whole turn at a time, as drawn on Bear's Savonius rotor before the run confirmed it. A latch holds its count with no power, a label carries one and holds nothing, and a powered lock, like a voltage standard's exact steps, holds only while it is driven.

Holds: carried to one, held there: two dials pulling on each other. *Met in:* driven oscillators, materials.
Shown: Latch Atlas; full guide row 565.

An intuition can get a home without becoming a claim (581) — The registration is what turns a picture into a result, whether it holds or not. The Savonius-rotor picture went to the page as one geometric sentence, expectations written before the run and an exploratory run labelled as not a test; and four times in one triple-star session Bear's picture or one-line summary became a registered test, among them "the ratio stays fixed, the rates don't", held at 3 of 8 slices over 600 passes.

Holds: carried to six: weaker in a line field's half residues (polarity) (wall); failed in the fourth-power term c (break not stated); held in R2W's dark point law (RATCHET-53, 54); weaker in the fourth harmonic's holding powers (break not stated); weaker in the next stretches (RATCHET-61) (seam); weaker in rehearsal standing or on a tumbling mat (substrate). *Met in:* measurement, driven oscillators, kinematics.
Counted: Coupled clocks and event time 6. *Shown:* Loop Atlas, R2W, triple-star, Dancing on the Sun; full guide row 581.

Direction is information: a dither can be a ratchet (657) — A part that slips one way and grips the other, a blade, a tyre, a keel, a fin or a diode, is what makes order matter: with grip equal both ways, a three-link swimmer in the Loop Atlas's toys went exactly nowhere. A wiggle can even make a slope: a field flipped fast enough holds a charged bead, and fails sharply when too strong for the flip rate, the radio trap's first stable region ending at q of 0.92.

Holds: carried to one, held there: the radio slope toy's bead. *Met in:* driven oscillators, kinematics.
Shown: Loop Atlas, R2W; full guide row 657.

Read every case before writing the sentence about all of them; writing is a second reading (667) — In the Loop Atlas sessions, writing a book's limb from its ledger rows found a price card claiming "twenty-five cells within six percent", which its run's table does not support: 20 cells sit at 0.94 to 1.06 of the line and 5 at 1.07 to 1.11, and the card was corrected in place with a dated note. Explaining simply is a test too: a chalkboard map drawn for the person found three mistakes while it was being drawn.

Holds: carried to two: held in the Loop Atlas ear's creep (CREEP_LAW); weaker in the measured rider at the 1/30 step (break not stated). *Met in:* measurement, records and tools, reading pictures.
Counted: Coupled clocks and event time 1. *Shown:* Loop Atlas, R2W; full guide row 667.

The split-frame measurement: send one walker each way, and compare them where they meet (674) — A loop of settings whose carried state depends on the road cannot be walked in one trip. Send one walker each way round and compare them at the same phase of the pattern: the difference is a whole number of turns. The triple-star sessions built the same two walkers the same day and split a third star's apsidal law into a part that flips with the direction of travel and a part that does not.

Holds: carried to one, held there: a third star's apsidal law. *Met in:* measurement, driven oscillators, three body.
Shown: Loop Atlas, R2W, triple-star; full guide row 674.

Digits, names, hand arithmetic and argued-away knobs are claims: compute them, and run them (677) — Print tables and targets from the numbers themselves, and run what was argued. A line read by a fixed-step integrator was quoted to nine digits for two days and adopted by another room as its reference, though four adaptive integrators agree on a value 4.05×10^{-8} lower; and a name is a claim too, for 0.236068 is twice 0.618034 less one, which equals its cube, not its square.

Holds: carried to two: failed in another room's reference value (ruler); weaker in another room's cascade (knob). *Met in:* measurement, records and tools.
Counted: The built artifact is the evidence 1, A second reader 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 677.

When an instrument fails its guard, its unscored readings fail with it (682) — A triple-star reader that failed its guards on every run still reported, as "unscored and seen", one angle turning forwards inside the apparent locks; a reader that counted instead of predicting found both angles turning backwards together, at a ratio of 0.994 to 1.008, as the algebra requires. Writing up such a reading, even as a suggestion, plants a claim the next run has to dig out.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, three body.
Shown: R2W, triple-star; full guide row 682.

Check a bar against the reference its sentence names (692) — A registered bar stands for a sentence, and the two can part company. In the triple-star sessions a guess's words asked for rims "at least as ragged as the light star's", but its bar asked for 0.010 at one island, more than twice the light star's own 0.004 to 0.0045 there, so the one chaotic birth, at 0.005, met the words and missed the bar.

Holds: carried to two: failed in new test's definition of fast chaos (ruler); failed in FN0 test of the paper's one-turn held (ruler). *Met in:* measurement, records and tools, three body.
Counted: A word that changes meaning across a boundary 2, Chaos and binning 1. *Shown:* Loop Atlas, triple-star, Question the Substrate; full guide row 692.

Sign and size are separate claims: when a sign is free, register the size, and score an eye's claim on its sign and place before its size (712) — An eye proposes sign and place well; the size belongs to the data. Three blind reads of another room's pictures, made from their pixels, were all borne out in sign and place, while two of their sizes were off in opposite directions: a weave stronger on the data than on the page (+23.4 standard deviations against +12.4) and a lopsidedness weaker.

Holds: carried to two: weaker in the data behind the pictures (break not stated); held in R2W tone-direction effect (RATCHET-44). *Met in:* measurement, driven oscillators, reading pictures.
Counted: A check that cannot fail 1, A second reader 1, Coupled clocks and event time 1, The mirror audit 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 712.

A tool that travels finds its companions by what is in them, not by their names (721) — On a laptop a Downloads folder keeps a second copy as "name (1)" and never renames it, so a tool that opens its companions by exact name stops. In the triple-star thread the first start of PB1′ stopped before running anything; its launcher's census found ten of its eleven files present with the shipped fingerprints, five of them as "(1)" or "(2)" copies, so only the one missing table was sent.

Holds: carried to two: failed in Bear's laptop Downloads folder (substrate); held in R2W's explorer returned under another name. *Met in:* records and tools.
Counted: The built artifact is the evidence 3, A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 721.

A maintenance alarm is a bar on a fleet: count work and strain together, set a bar per class, and compare policies at matched catch (746) — A shared bar is a hidden transfer between classes of machine. In the Loop Atlas sessions one threshold set to catch 90 per cent of a mixed fleet over-catches its quiet rooms (0.96) and under-catches its loud ones (0.84), while a bar per class sets each near 90 per cent; and policies are compared at matched catch, since at one training bar calendars caught anywhere from 0.886 to 0.942 of the test fleet.

Holds: carried to one: weaker in the fleet's quiet and loud rooms (ruler). *Met in:* measurement, driven oscillators, control and circuits.
Counted: A check that cannot fail 1, Coupled clocks and event time 1. *Shown:* Loop Atlas; full guide row 746.

Build from the last build in one pass, with its proofreading copy, and measure everything a rule could touch (773) — A highlighted copy written in the same pass as the build, and checked to differ only by its marks, has the real page numbers, so in the Field Guide sessions the person proofread only what moved, 39 pages of 133. A style rule written to enlarge the title also enlarged every part heading in the book, caught only because the station card was pushed off its page, so measure the built PDF for every element a rule could touch.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 8, A second reader 1. *Shown:* Loop Atlas, R2W, Field Guide sessions; full guide row 773.

A paper travels with its PDF, read back block by block, in plain words, with its misses kept (779) — A check can forgive the very fault it is for, so read every built PDF back against its md before sending, and look at any page that holds a new list or table. The Loop Atlas checked Bear's words in 24-character pieces that forgave curly quotes,

dashes and ellipses; read strictly, 85 of 1,141 pieces had been changed by the builder's smart typography, and after a new build tool, 0 of 1,141.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 779.

Shop faults in the launch: a pipe's exit status, a backgrounded line, a stamp in a folder's name (786) — Never chain a launch after a pipe, group a whole command before backgrounding it, and count the running processes right after a launch, because one shell habit can start the wrong run or hide a fault. In the Loop Atlas sessions a launcher named its work folder with the run's own stamp, and the comparison called a FOLD_OR_FLIP_4 run that reproduced every number DIFFERENT; a made-up end-to-end run found it.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 5. *Shown:* Loop Atlas, R2W, triple-star; full guide row 786.

Faults a free check catches in the code: a nested namespace, a stopping rule that never stops, a search with no bracket, an energy that changes sign (796) — Before a long run, ask for every name new code uses, give every search a bracket, and let the stand-ins serve as a code review. In R2W, ODD-1's reader took the cosine of a timing rounded to six decimals, so the first harmonic leaked about 2.7×10^{-8} and a verdict failed where the made-up push said it must hold; it was fixed before registration, and the first version is kept.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: Chaos and binning 1. *Shown:* Loop Atlas, R2W; full guide row 796.

Tell the reader which idea is upside down before giving results (800) — The person who learned to see a thing knows which part a newcomer will read backwards, so that part comes first. Bear wrote that chaos as structure is an inverted idea, and in the triple-star sessions the islands paper's chaos section now opens with chaos with a shape, before any measurement. The same view works for the whole page: the paper's 52-page PDF held no image, and its bench was one 412-word paragraph on page 38.

Holds: not yet carried elsewhere. *Met in:* records and tools, three body.
Shown: triple-star; full guide row 800.

An array of identical periodic parts is decided by its seats: the fan inside a group picks the harmonic, the offset between groups steers it, and a silent set parks what it holds (803) — Fold every setting of an array into one cycle and list its seats before tabulating, because settings with the same seats are one row. In R2W, three groups of three riders kept only their third harmonics, and the offset between groups steered them to 0.910684, 0.666667 and 0.333333 of the aligned swing at 10, 20 and 30 degrees, and nothing at 40 (NINE-1), the array factor of a three-element antenna.

Holds: carried to two, held in both: three groups of three riders (NINE-1); array's silent sets (STAGING_READING).
Met in: driven oscillators, waves, quadrature.
Shown: R2W; full guide row 803.

Small faults in a tool's own code: a verdict made a plain true or false before both print and dump, a followed quantity started from an explicit index, and a table row inserted by its line (815) — Cast a verdict to a plain true or false before printing or saving it, and insert a table row by its line, because a few characters of code can cost a run or a record. In the Loop Atlas sessions numpy's true made four held planted cases print UNDECIDED in a blind run, and a row inserted by replacing text left one line with eleven fields where the rows have seven.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: A word that changes meaning across a boundary 2, The built artifact is the evidence 1. *Shown:* Loop Atlas; full guide row 815.

Rebuild a reader's edition from tagged paragraphs, never cut it from the working book, and read every reused page for its old vocabulary (825) — Cutting a working book into a reader's edition loses paragraphs and leaves the workshop's words behind, so the edition is rebuilt instead. The Loop Atlas's 137-page working book, 546 paragraphs, was tagged by readers who were not writing and rewritten chapter by chapter into 46 pages of 17 chapters with 17 figures; its reused front page still named working-book parts such as Trunk 4 until they were mapped to chapters by rule.

Holds: carried to one: weaker in the reader's edition front page (substrate). *Met in:* records and tools.
Shown: Loop Atlas; full guide row 825.

A check that leans on the order of adding must know what kind of number it adds: Python's built-in sum compensates floats, not complex numbers (838) — A check that compares two programs to the last bit must know whether its tool takes rounding back out of the order of adding. From Python 3.12 the built-in sum compensates floats, so the last bit moved in 0 of 2,000 shuffles against 1,240 for a plain loop, but for complex numbers sum() is a plain loop, as R2W's twelve complex terms of a window showed.

Holds: carried to one: failed in Python's built-in sum on complex numbers (break not stated). *Met in:* records and tools.
Shown: R2W; full guide row 838.

Proofread in rounds on fresh builds: leave alone what the person is reading, log every later change against his page, and agree a plan for his marks before any batch (846) — A proofreader can mark only a page that stays still, and a mark helps only if it can be found again in the next build, so sections he is reading are left alone and every later change is logged against his page. Bear's first proofread of R2W's book, 29 marks to page 74, went in five batches on a plan agreed before any batch was done.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: Loop Atlas, R2W, triple-star; full guide row 846.

A number written down lives in several places: look it up, change it everywhere at once, and print every mention a renumbering touched (854) — List every copy of a repeated number, figures included, before the first change, and carry each new count to every place it is printed in the same build. The triple-star count moved from 444 to 490 in one day, and the source check found an old 444 that hand reading missed; in R2W, a fresh reader found a seventh copy of the count still saying 105 misses.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: R2W, triple-star; full guide row 854.

After a miss, say whether the test or its premise failed; derive a redesign from what you have seen, but register it where you have not looked (860) — First ask what failed: a triple-star test of a fixed rotation missed because the rates themselves moved, so its redesign tested the premise before asking the question again. Register a redesign where you have not looked, and let one the answer would choose stay a description. Most misses are not redesigned, and that is the rule working: of the triple-star thread's 211 misses, 59 to 75 were followed by a registered redesign.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: R2W, triple-star; full guide row 860.

Words follow numbers: scope every "all", label every estimate, define every coined word, write a "perfect" as a tolerance, and narrow a sentence the day it is shown too broad (861) — A sentence is held to the same scope as the numbers under it. The triple-star sessions' islands paper said every test of the long runs read the same on a second machine, but three were registered after the kit was sent, so it now says "in every test registered before they were sent there"; and a cross-check estimated at 2 to 3 hours took about 3 hours 24 minutes.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: R2W, triple-star, Two Arrows; full guide row 861.

A check that compares words cannot see where they sit: look at the pages, make what was seen a check that reads positions, prove it can fail, and aim it at the class of fault (867) — A text read-back can pass while the page is wrong, so look at the pages, turn what was seen into a check that reads positions, and show the check fail first. The Loop Atlas's reader's edition read back complete while every three-digit note number had lost its first digit at the page's edge; a later check aimed at the class, check L5 in build_paper v1.3.2, found exactly Bear's page 46.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: Loop Atlas, triple-star; full guide row 867.

A whole read-back sees only what its reader can see: know the reader's habits, read cells as words in order, and look at the pages for what no text check holds (869) — Know the habits of the reader under a check: a PDF reader joins hyphen-broken words and interleaves a table's columns, so cells are read as letters and digits in order. Read whole, the guide's 292 pages read back 2,730 of 2,730 on the first pass, and when the completeness check counted 329 of R2W's 330 reads, the source conversation showed the guide's 330 was right.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: R2W, Question the Substrate, Two Arrows, Field Guide sessions; full guide row 869.

A memory sees a steady swing's crest at the square root of two, whatever its length, and a trigger set above that ceiling never fires again (878) — A trigger riding on a memory of its own swing's square has a ceiling it can never cross once the swing goes quiet. A steady swing peaks over its memory at the square root of two whatever the memory's length, and the Loop Atlas's spring latch stayed awake exactly up to that ceiling at 22 settings of seven springs, none a hundredth above.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, control and circuits, quadrature.
Shown: Loop Atlas; full guide row 878.

Write so a stranger can place every sentence: a date for “now”, a name for “I”, a credit for every passage and an address for every number (881) — Words that tie a sentence to its moment go stale without being wrong, so give each one an address a stranger can follow. Of 9 stale-word edits in version 6’s text, 7 were time words such as “tonight” and “still open”; a session’s “I” became “the session” in 11 rows; and 37 openings that had leaned on their tag line now carry their own credit.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: Loop Atlas, Field Guide sessions; full guide row 881.

L4 — The Ledger

What it cost, what it taught, what went wrong, and what it yielded.

Could it have missed, and did it (1) — Before you look at the data, write down what would count as a miss, and before scoring a test, confirm that it can fail and that its input reaches the system. The seismic instrument set its ceiling at r = 0.0940 and cleared it cleanly at 0.2697, because the bar was committed first; a triple-star test first read BORNE OUT was withdrawn to NO READING when a cold reviewer saw that the band it asked about lay below its reader’s floor.

Holds: not yet carried elsewhere. *Met in:* measurement, three body, geodesy.
Counted: A check that cannot fail 2, A second reader 1. *Shown:* triple-star; full guide row 1.

Mechanism versus correspondence (7) — A mechanism can be measured, salvaged and carried; a correspondence of two numbers with no mechanism behind it is labelled so when it is registered, and a miss closes it at once, priced and stopped. Three defences pre-registered as breaking at the same place on one shared noise-correlation axis did not, and two shares asked to agree within 20% came 0.121 and 0.159 and stand as measured.

Holds: carried to two: failed in count defences such as a Byzantine vote (knob); failed in a phase-locked swarm (wall).
Met in: measurement, control and circuits.
Shown: triple-star; full guide row 7.

Test the narrowest case first (8) — The smallest, most constrained version of a test can say no most easily, and that makes it the one most likely to find something real. Four signal processing architectures were tested against the narrowest case; all four were closed as insufficient, and three of the approaches were redirected to narrower sub-tests.

Holds: not yet carried elsewhere. *Met in:* measurement, quadrature.
Shown: version 6; full guide row 8.

Inherited constants (9a) — A number that is only inherited has never been checked, however often it is repeated, so trace every constant to its source before it carries a result. The row’s case is a spread constant asserted 5× and never sourced.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 9a.

Do not close on thin evidence (9b) — Register from the fit, and treat a lone record near an edge as the thing under test. In the triple-star sessions a turning point placed from a single reading, 7.1986, missed at 7.1980, and both arms of the next test put it where the fit had; in the Loop Atlas sessions a lean read from BELL_CLOCK’s one case sat on a special point, and at memory 3 the gain was 0.933.

Holds: carried to two: failed in tied bells at memory 3 (seam); failed in tied bells with softened cores (ruler). *Met in:* measurement, three body, kinematics.
Counted: Chaos and binning 1, Count the right unit 1. *Shown:* Loop Atlas, triple-star; full guide row 9b.

The elimination arc (15) — Test hypotheses one by one, keep only what survives every filter, and count what chance would leave standing; even the misses narrow where a leftover lives. The seismic superstar scorecard kept 2 stations through all six criteria where chance expects 1.5, so the dozen-superstars idea died on training data, and the triple-star thread split a leftover scatter at 0.28 three ways, by side of the beat’s zero, by chaos and by locks, where the 15 locked runs fit better, ratio 0.21.

Holds: not yet carried elsewhere. *Met in:* measurement, three body, geodesy.
Shown: triple-star; full guide row 15.

Held-out calibration (22) — Calibrating a detector on the data it is tested on trains it to pass itself, so data for calibration is reserved from the start. The seismic false-alarm rate was 13–21% uncalibrated and 1.1% after calibrating on held-out data, an improvement by a factor of 10–20 that was available only because data had been set aside.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 22.

The weak-baseline law (23) — A baseline that does not know the basics awards credit for knowing what any informed observer already knows. The poker geometry feature showed +0.47/+0.44 improvement over a baseline that knew only a hand's aggregate statistics, and once the baseline was given three order-blind pressure facts, the advantage collapsed to +0.000/−0.002.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 23.

A gate that never got stressed (28) — A test is only as good as the stress at which it has actually been run, so check the stress reached before trusting the gate. This gate was designed to activate at correlation ρ = 0.717 (VIF 2.06), and only the patched version, reaching ρ = 0.9668 (VIF 15.331), genuinely stressed the gate it was designed to test.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 28.

Synthetic calibration is circular (29) — Data generated by the instrument's own model encodes exactly the assumptions the instrument was built on, so passing on it is guaranteed by construction. Permute along the axis the feature's name claims sensitivity to and check whether the numbers move; real calibration needs an independent reference the model did not touch, a ground truth that can say no.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 29.

A feature named for a property it lacks (30) — Run the test a feature's name implies before trusting the name. A feature called order-dependent was shuffled, and two of its three components changed by 4.4×10^{-16} and 2.2×10^{-16}, effectively zero, while one varied by 0.79; in the triple-star sessions a rule meant to find rocking rests called shuffled rests rocking 0.72 of the time, more often than the real order.

Holds: not yet carried elsewhere. *Met in:* measurement, three body, reading pictures.
Counted: A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, triple-star; full guide row 30.

The resolution walk (35) — When two measurement methods disagree, walk toward finer resolution until they converge, and hand the numbers that never stop moving to the reader. The falling-cat law for three bodies read a residual of 0.00143, 0.00019 and 0.00003 radians at 25, 100 and 400 points, converging, while a peak speed of 379, 740 and 1,149 was still climbing. When two instruments disagree, bring a third that shares code with neither.

Holds: carried to one, held there: three bodies. *Met in:* measurement, three body, kinematics.
Counted: A second reader 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 35.

Fidelity rerun after every change (36) — Every downstream step trusts the step before it, so after any change to a tool the whole pipeline is rerun, and the new version must reproduce every number the old one wrote. In the Loop Atlas sessions the same tool, settings and pace gave 0.557209 turns per cycle against its parent's 0.526206, until repeating the parent's call sequence gave 0.526206 to the digit.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Counted: The built artifact is the evidence 2. *Shown:* Loop Atlas, R2W, Field Guide sessions; full guide row 36.

The numerical read failed twice, and the mechanism is named (54) — Name the mechanism of a failed numerical read so the next attempt does not repeat it. Here discretisation put an O(1) offset in the constant term that did not cancel between domains, and a staircase boundary invented thousands of right-angle corners, each contributing exactly 1/16 to the very term being measured; switching to axis-aligned square holes did not rescue it.

Holds: not yet carried elsewhere. *Met in:* measurement, topology, waves.
Shown: version 6; full guide row 54.

Hansel and Gretel — the birds ate the path, not the forest (55) — A topological invariant lives in its medium, so clean readers do not prove it survived: if the medium goes, the invariant goes with it. The named failure mode is medium reads fine, reader reads fine, invariant is gone, like Hansel and Gretel's breadcrumb trail, which appeared intact right up until the birds finished eating it.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 55.

Hand the reviewer the hammer (61) — State in the paper itself that what survives demonstrates the merit of the idea and what is destroyed is offered a second life as the chance to engineer a replacement. Back it with a Seam and Assumptions Register that names, discipline by discipline, which expert should interrogate which assumption, and the exact question the author could not answer.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 61.

Confidence recorded separately from assertiveness (62) — Most papers conflate how sure the author is with how sure the prose sounds, and a reader cannot tell the two apart. The register carries the author's honest confidence separately from how assertive the prose reads, and names the places where confident prose is least trustworthy: calibration claims.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 62.

The packing-knob pattern, named and NOT promoted (63) — Call a pattern shared phenomenology, not one equation, when its cases differ in what matters. A packing knob buys capacity cheaply until a critical density, past which loss turns non-linear, in over-binning, defect-density structural collapse and the mutation error threshold; but over-binning is recoverable by reading a coarser alphabet, and structural collapse is not.

Holds: carried to one: weaker in defect-density collapse in the medium (break not stated). *Met in:* measurement, biology, materials.
Shown: version 6; full guide row 63.

Separate the references you read from the ones you know (67) — Tag every reference by depth, so a reader can weigh each claim, and read the direction of a known correction instead of recalling it. The Latch Atlas tags all 201 of its references; in the Loop Atlas sessions Cavendish's mended value was leaned from recall to lie within 1 per cent of today's, but read, it is 5.448 ± 0.033, 1.2 per cent below today's 5.514.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: Loop Atlas, Latch Atlas; full guide row 67.

The rolling referee report, inline and adversarial (68) — Give each paper its own skeptical but fair review as a section, so the reviewer's hammer is built into the structure instead of left to rhetoric. The section doubles as the version tracker: a summary judgement, the strongest and weakest claim named, honest exposure, and a list for the next version.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 68.

A check that silently stops being valid (42) — Some validity checks hold only in certain configurations and fail silently when the configuration changes, so note when each one applies. The mass-balance column is valid only while the ring is open; once the ring's lip seals, body speed is no longer constrained by mass balance, and any number from the check after sealing is meaningless, whatever the check reports.

Holds: carried to one: failed in ring lip sealed (seam). *Met in:* measurement, waves.
Shown: version 6; full guide row 42.

The exterior reference is a binary wall, and it was measured (70) — A loop cannot correct a corrupted reference from inside, so the cure is a second loop anchored differently. A PID handed a corrupted setpoint defended it across 40 seeds, and at corrupt = 70 its reality-cost ran 903.2 at noise 0.1 and 1171.8 at noise 8.0, against 0.0–17.9 with the true setpoint; a system-ID adversary settled at 64.27 against a true optimum of 100.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 70.

Retract your own pre-registration in place (71) — When a pre-registration overclaims, strike the borrowed authority inside the record, not beside it. The pre-registration named Conant–Ashby as the single citation that would convert the keystone from supported to proven; the paper retracts that in its own front matter, separates the two claims and restates what the keystone rests on, and the prediction survives.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 71.

Pre-register the kill that would destroy your own thesis (73) — Register the test that would destroy the thesis, and run the control that could kill an explanation before writing the chapter that rests on it. Test A asked whether three averaging defences break at one place on a shared noise-correlation axis, and it fired; in R2W, run ADDED-1 showed an added drive locking more fractions, not fewer, against a helper's inference.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators.
Shown: R2W; full guide row 73.

The operational filter — build, predict, or break (74) — Admit a bridge only if it builds, predicts or breaks something; an observation that only produces a feeling of connection is out. The named failure mode is the 1990s chaos fad, which used the mathematics as a mood and could not survive being punctured; puncture this work and a

working read strategy is still underneath.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 74.

The hostile hammer is the valuable one (75) — The swing that finds what a sympathetic test never will comes from someone trying to break the work, so when a result looks too clean, spend the next run trying to break it. R2W's two-channel meter came out exact in its structure and within 0.21% in its one predicted number, and that was the moment to go hunting for the flaw, not to build on it.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: R2W; full guide row 75.

Declare the model overlay before the claim that uses it (76) — Declare each model departure ahead of every claim that leans on it, with its own kill condition and the established alternative it is not, and state a prediction with its model in the same sentence. In R2W, "Set against the octave, it is backwards" read as a finding when it was a prediction from adding two tones; a prediction always carries its "if".

Holds: not yet carried elsewhere. *Met in:* records and tools, driven oscillators, biology.
Shown: R2W; full guide row 76.

Ship the broken run in the crate (78) — Put the artifact that produced a miss in the public record, not only the miss. The crate deposits neuro_loops_v1_BROKEN.py alongside v2: its drive gains were set too high, every scenario died, and the real modelling error the data caught is kept under the project's graveyard rule.

Holds: not yet carried elsewhere. *Met in:* records and tools, biology.
Shown: version 6; full guide row 78.

Exploratory and confirmatory are layers, and each prediction names its own (79) — Label each result exploratory or confirmatory, and give a reading seen after the run a registered test before it enters a paper as a result. In R2W a power law seen after one table's data was registered with two rivals before the next rows were opened, and it held blind, 8 of 8 and then 4 of 4, while both rivals failed (runs TOY-SIDE-3g, TOY-SIDE-4g and JJ-SMALL-1c).

Holds: carried to four: held in next rows and the junction; failed in strength 0.10 (knob); held in strength 0.10; held in RATCHET-61's second size. *Met in:* measurement, driven oscillators, three body.
Counted: A word that changes meaning across a boundary 1, Coupled clocks and event time 1. *Shown:* R2W, triple-star; full guide row 79.

Name where the method goes silent (80) — The diagnostic is most testable exactly where it matters least, on repeated, resolvable cases with ground truth on schedule, and least testable where it matters most, the once-only regime change. Where no ground truth is accessible, consensus is the only reading there is, so the framework declares itself silent instead of extending, and the paper states that limit rather than hiding it.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 80.

A detector that falsifies cleanly still tells you what it found (85) — Give every miss a name and a file, because a clean negative still says what the instrument does detect. The detector is a supercell-scale rotation detector, not a tornado detector, a clean negative across three labelled days, one tornadic, one severe-but-not, one quiet; row 372 prices it, and row 373 is the autopsy that names density as the cure.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 85.

Tag the sentence, not just the number (86) — A reader should be able to tell at a glance which sentences are the author's to see and which are the world's to confirm. Give the voice that sees and the voice that checks each its own mark, under the one rule on the first page: vividness is not evidence. SOURCED/UNVERIFIED/ASSUMED tags the number, row 67 tags the bibliography, and this tags the sentence.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 86.

Retract the word when the harder test fails it (87) — When a harder test fails a word, withdraw it in public where it was used, and keep the verdicts it carried. The word topological was withdrawn after a harder test fired 52% on honest data, a noisy heuristic rather than a clean integer invariant; in the triple-star sessions TS1 read SQUARE ROOT by its rule, but TS2 showed the shape changes with record length, and the call was withdrawn.

Holds: carried to one: failed in triple star's jump at 1/10 (ruler). *Met in:* measurement, topology, three body.
Shown: triple-star; full guide row 87.

Supersede by tightening, never by burying (88) — Keep the earlier reasoning legible, so a reader can see what changed and why. A later appendix neither replaces the earlier one nor quietly drops it; it tightens it, adds the trust layer the first one missed and states the relationship, because in a published lineage the supersession is the content.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 88.

Turn the instrument on your own panic (89) — Before acting on a loss, ask whether the invariant changed or only your access to it, because the two demand opposite responses and feel identical. One of the Field Guide sessions ran 23 days and rebuilt its context from files many times, every rebuild made by reading two working files, never by recall; write the next step into a file before the turn ends.

Holds: carried to two, held in both: author's fear the work was lost; a 23-day Field Guide session. *Met in:* measurement, records and tools.
Shown: Field Guide sessions; full guide row 89.

Every port must beat a baseline in its own domain (93) — A tool that ported by mere analogy would fail its new domain's baseline, so the baseline is the admission test. Three instances stand on the record: the whale forage signal beat a null, closure depth beat maximum wind, and Rivera beat its own quiet control; and in R2W, a family's rules are sorted into universal and specific, because rules every rotation obeys prove nothing on a new substrate.

Holds: carried to three, held in all three: whale foraging; storm closure depth; Rivera case. *Met in:* measurement, driven oscillators, biology.
Counted: A check that cannot fail 1. *Shown:* R2W; full guide row 93.

The rim can be ground arbitrarily fine and still cannot see what it does not enclose (96) — No system is its own trust anchor, and the only adversary that matters is the one that arranges never to be enclosed. The corpus met this one impossibility in four costumes: the observer inside the loop, the bootstrap that cannot pull its own boot, the frame-variant liar with no static twin, and an event off the system's representational map; the cure is an out-of-band channel.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 96.

Reach the wall from two directions, and fix the attribution when you get there (99) — Two instruments that reach one blind spot independently are far stronger evidence that the wall is real than either alone, and the attribution is checked when they meet. The Shape of Adversarial Influence, arXiv 2505.20435, from Imperial College London, catches the static topological distortion and goes silent at the adaptive one; the working record had filed the group as Los Alamos, and locating the real paper closed it.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools, topology.
Shown: Two Arrows; full guide row 99.

Generate warm, audit cold, in different rooms (100) — The suspicion that protects a paper poisons the play, so generate fast, capture everything untouched, and hand the load-bearing claims to a reader who did not write them. A separate agent checked one paper's draft against its run files and found 17 points, two of them wrong claims, and A Rider in Two Waves passed every gate it had while a fresh reader still logged 100 findings. Gates check form; a fresh reader checks meaning.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools, three body.
Counted: A second reader 3. *Shown:* R2W, triple-star; full guide row 100.

The wraith is the theorem; the bad data is the job (101) — The unbeatable adaptive adversary is the floor that says this far and no further, and it is almost never the work: the instrument earns its keep on the catchable ninety-nine percent. The most dangerous member of that class is a true invariant with a false story told over it, where "it's derived, trust me" is the perfect cover.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 101.

Fear dressed as caution (102) — Each of the three flinches is fear wearing the costume of rigour: from the look, for fear of being scooped; from the kill, for wanting the pretty result to be true; from the autopsy, because the null felt like a verdict. The antidote is one discipline, fearless and playful together: throw the guess, fold your hopes into it, let it die, read the result straight.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 102.

Three honest implementations converging on one boundary means the boundary is real (106) — Three different roads to one wall is evidence about the wall, not about any one implementation, and a fourth gate would only retune toward it. In R2W, WITNESS-1 was written fresh, sharing only the equation with the first engine, and agreed to

six figures or better; over 19,481 settings of the weightless junction, two readings sharing no step agree at 19,477.

Holds: carried to one: failed in high noise, for all three gates (wall). *Met in:* measurement, driven oscillators.
Shown: R2W; full guide row 106.

A one-sided bar turns an operating point into a triumph (107) — Score a result on every half of the bar committed for it. The softer gate raised high-noise recall by +0.36 mean, one cell going 0.47 → 0.97, while false alarms in that same cell went 32.5 → 147.9; cells clearing both halves numbered 0 of 6. Scored on recall alone it would have read as six wins from six.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 107.

The rigging check that half-tripped, and the label that had to change (108) — Build the rigging check to attack the interpretation, not the score. Time-accumulation beat a single-frame read by 6–28× at every noise level, but fed static vortices with no building at all, time still won 80 → 6.5, so the mechanism is persistence, not building: the number was right and the label would have been wrong.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 108.

The three-cause autopsy, run before any null is declared (111) — Before a null is written, name which of three things is responsible: the hunted, with no signal there; the hunter, the instrument or its gate; or the substrate, mis-anchored or carrying the wrong projection or reference. Naming the cause turns a disappointment into a finding, and in practice it produced predicted fixes rather than retunes.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 111.

Vividness is not evidence (113) — A strong inner certainty is a compass, often pointing at something real, but it is not a finished map. Honour it by building the words, testing it against the shared world and carrying it to the people who live in that country; that is how a private certainty becomes knowledge that can be handed to another person.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 113.

Two voices that never pretend to be each other (114) — Mark every passage for which voice is speaking, what is seen or what the field says, so speculative work can be shared and not merely admired. The checking voice carries three grades: settled, confirmed by experiment; open, genuinely argued over now; and interpretation, pictures that fit the same equations with no experiment able to choose, the grade where most overclaiming happens.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 114.

Hold-then-cliff beats silent creep (119) — An integrity system should fail visibly rather than degrade quietly, and the shape of its failure is a design choice. A plain consensus drifts toward a lie continuously as a bloc grows; a consistency-gated one stays pinned to the truth and then falls off a cliff at the true majority, so a sub-majority bloc gets nothing instead of partial drift.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 119.

The three-layer trust model (125) — A lie has three possible homes: the invariant, forced by a conservation law, cannot lie; the measurement can drift; and the interpreter can be fully adversarial, measuring honestly and reporting falsely. Anchoring to derived physics defends only the first layer, so the adversarial interpreter is bypassed only to the exact extent that the derivation can be run oneself.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 125.

Claim the connection, never the gap (130) — Say exactly what is added, so a real finding does not inflate into a false claim of novelty. The field already used anomaly-against-climatology; what was done here was to re-derive from first principles why the fiducial must drift, by watching the tool fail, so the contribution is the unifying lens, not the practice.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 130.

Do not fire the killer test before the rig is clean (131) — An ambiguous result cannot tell a wrong idea from an unready rig, and it wastes the one shot a decisive test gets. Write the order down before any of it runs: fix the baseline, then run the batch, then the head-to-head on held-out cases against the strongest existing baseline.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 131.

A pilot's job is to find which signal survives contact with data (132) — Report every verdict a pilot returns, because that is what makes the survivor believable. One batch returned three: one signal confirmed, one hypothesis killed cheaply, and one parked as contaminated by record length; a pilot that confirms everything it tested has tested nothing.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 132.

Explain the slop before you invent a term for it (138) — Exhaust the boring explanations before inventing a term for what is left. Detrending, de-seasoning, common-mode subtraction and sigma-clipping took a residual from 1167 mm to 0.9 mm, roughly a thousandfold; known sources accounted for 99.9% of it, so no invented dissipation term was admitted to the model.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 138.

A discriminating tool beats one that fires on everything (139) — An instrument that fires everywhere says nothing about where it works, so a pre-registered null that comes back null strengthens the candidate. A cross-category null came back with 24% overlap with the quiet control against the candidate case's 0%, which establishes a domain of validity.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 139.

Re-grade your own keystone when the unification is an analogy doing a theorem's job (141) — An analogy doing a theorem's job stays an analogy until something forces the identity, so the unification is graded apart from its pieces. A keystone node claimed the rotor combinator, the chiral-knot tracker, the two-ledger architecture, Berry phase and Noether were one object, tagged settled; each piece is settled, and the unification was re-graded to vision.

Holds: not yet carried elsewhere. *Met in:* records and tools, topology, kinematics.
Shown: version 6; full guide row 141.

A frame that explains everything cannot be tested by anything (142) — When a small idea starts trying to unify everything, slow down, file it as vision and steer back to the small closeable piece. This is a speed instruction rather than a prohibition: the seeing may well be right, but a frame nothing can contradict has stopped being an instrument.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 142.

A shared shape is not a shared law, and you tell them apart by writing both functions down (146) — A structural rhyme can be real and still be all there is. Asked whether prime-factorization levels match electron shells, the measured answer killed the match in three places: tower energy is log n and grows without bound while hydrogen shell energy is $-13.6/n^2$ and converges, and factorization has no exclusion principle capping exponents as Pauli caps a shell at $2n^2$.

Holds: carried to one: failed in prime-factorization levels (tower) (substrate). *Met in:* measurement, number theory.
Shown: R2W; full guide row 146.

The strongest methodological artifact in the corpus, by its own audit's verdict (152) — A clean pre-registered null with a caught borrowed parameter is worth more than a positive result, because only one of the two can be faked by wanting it. Locked before data access, the test reported FAIL: a 20.3% match against 23.3% for random, an enhancement of 0.87×, across 154 galaxies, and the same session caught a borrowed characteristic acceleration.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Shown: version 6; full guide row 152.

A four-order unit error passes every internal check (154) — A wrong input makes every internal check honestly wrong together, so the only defence is asking something that is not under test. A voltage carried at 3.36 GV was actually 85 kV, a factor of about 4×10^4, and every calculation downstream of it was internally consistent the whole way.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 154.

An instruction that cannot be disagreed with is not a preference (155) — Test any style instruction by what following it to the letter would do the next time there is a real error. A rule saying the author is never wrong about which threads are alive was refused, and the audit named four defects it would have suppressed: a regression, a power budget, a formula that did not reproduce its own numbers, and an oxygen budget.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 155.

Close-out is a phase with an external checklist, not a matter of vigilance (156) — Late responses get calibrated against earlier ones in the same session, not against the protocol, so close-out needs an external checklist rather than vigilance. The handover's head goes stale first: at one close its restart section still said 57 rows and 16 items while the manifest below it was current, so the top is rewritten at every shipment.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: Loop Atlas; full guide row 156.

Complete standalone files every time — append is the failure mode (157) — Rebuilding from an old copy silently drops every later addition: a file rebuilt from a 183-line earlier version instead of the current 362-line one lost them, caught by comparing line counts. Ship the whole file and check a reissue against every line it did not mean to change; in R2W, twenty-five deltas against issue r8 each carried every new and changed file whole.

Holds: carried to one, held there: R2W crate deltas. *Met in:* records and tools.
Counted: The built artifact is the evidence 2. *Shown:* R2W, triple-star; full guide row 157.

Leaving a live thread to run a catalogue ends the thread rather than pausing it (158) — Reasoning lives in working memory and notes are always a compression, so a thread set down mid-derivation restarts from the compression instead of resuming. The priced instance abandoned an active derivation to run 137 shallow dives, and the real failure was the abandonment, not the failure to find anything.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: R2W; full guide row 158.

One good session should produce two changes, not five new laws (159) — Bear warned against over-tuning: overconstrain the work and it follows rules instead of hunting. The editorial ruling that the rules are the invoices rather than the point is the only thing standing between a corpus of priced ornaments and a rulebook nobody reads.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 159.

The null control was measuring something real (161) — A null that reads something is a finding to resolve, not a failure. A control region chosen because nothing should happen there read +6.1 curl, which resolved as twice the vertical rotation rate of rigid plate motion; the Euler pole fitted from that region alone matched the published value and the null closed to about (0,0,0), though that published pole was quoted from memory and its check was still open when the row was written.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 161.

The trunk's own pre-registered prediction failed, and the paper carries the loss (163) — A framework that reports only its own confirmations has stopped being testable, so the paper carries the loss. Two pre-registered strikes falsified the multi-scale plaquette upgrade on real data: plaquettes read 0.00 to 0.03 against 0.67 for the outer loop on a propagating front, plus area-noise scaling and a threshold tax.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 163.

Two hands reading one frozen register produced the same spec (166) — Freeze a register before parallel work starts, because the architecture then dictates the experiment. A trunk session and a spoke session, sharing no text, derived near-identical run-2 specifications, ternary basis, seam and shear channel, because both were reading the same frozen register; it is the direct answer to two hands numbering upward from the same last row.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 166.

A design closed conditionally on someone else's open question (167) — When a design hangs on an open question in another field, close it conditionally and name the external result it is hostage to. In a closed matter-dominated universe the conformal lifetime is 2π and every early photon refocuses at the antipode, but with dark energy the conformal future falls short of π, so the instrument is valid if and only if dark energy loses.

Holds: carried to one: failed in universe with dark energy (seam). *Met in:* measurement, waves.
Shown: version 6; full guide row 167.

Capture now, grade later — and preserve what you could not parse (168) — Capture everything first, mark every span that could not be parsed, and grade later as a declared phase, so the unparsed is not smoothed into something readable and wrong. Capture includes registrations: in the triple-star sessions a test registered only in the

conversation lost its wording, and in R2W a printed but ungraded rule came within 15 per cent at all six rows (run DANCE-3).

Holds: not yet carried elsewhere. *Met in:* records and tools, driven oscillators, three body.
Shown: Loop Atlas, R2W, triple-star; full guide row 168.

Say who specced it, especially when it is the same author (169) — A session that validates something owes the lineage credit to the session that specced it, and a solo author is the case where nobody else will catch the slip. A drift-check run three weeks after the fact found the homotopy ladder and the sampling question had been specced in June and kill-tested in July, and wrote that framing into every downstream document.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 169.

A tolerance you chose is not a constant (173) — A number that moves with its tolerance is the price of an accuracy somebody selected, so a tolerance is set after the input's rounding, the grid's step and the law's power. Asked whether two peaks landed within fifteen percent, the answer read 1.219 at thirty percent and 1.360 at one percent, and the repair, a criterion with no free parameter, came from 1916.

Holds: not yet carried elsewhere. *Met in:* measurement, quadrature.
Shown: R2W; full guide row 173.

Two appearances of π, and they are not the same π (176) — Take the shine off a headline that invites a reader to hear more than is there. The paper's wall at exactly π is inherited from the radian convention in the aperture identity, so in units of turns it sits at exactly one half, and there is nothing transcendental about it.

Holds: not yet carried elsewhere. *Met in:* number theory, quadrature.
Shown: version 6; full guide row 176.

The novelty ranking, corrected downward (177) — Rank results by what the work actually adds, and reverse an earlier rating when it was wrong. The π wall's content is elementary algebra; the claim is the identification that the half-width setting the wall is the gas temperature multiplied by the barrel the coincidence demands, divided by a turn, so the constant, not the wall, is the stronger candidate.

Holds: not yet carried elsewhere. *Met in:* number theory, quadrature.
Shown: version 6; full guide row 177.

The test that was measuring its own artifact (178) — A result identical at every setting is a broken test, and the missing control is often one line. An unequal-amplitude test returned the same barrel down to one twentieth; with the weak comb deleted, a peak still sat beside the weak tooth (108.6496 against 108.6525), so the test had been reading the strong comb's own sidelobe.

Holds: not yet carried elsewhere. *Met in:* measurement, quadrature.
Shown: version 6; full guide row 178.

No pair is blessed (182) — Coincidence has a price distribution, so the best case is weighed against its tail before it is admired. The census leader leans within 7.26×10^{-9} with a bargain factor of 9,226 — and across all 14,098 rungs the bargain distribution matches the Gauss–Kuzmin law to a worst deviation of 0.0062, its tail populated exactly as chance requires.

Holds: not yet carried elsewhere. *Met in:* measurement, number theory.
Shown: version 6; full guide row 182.

Declare the precision boundary before the measurement (184) — A tool and its check can be strongest in opposite regimes, and that asymmetry belongs on record before any number arrives. The independent cross-check needed an exact column count, and double precision stops counting exactly past 2^{53}, so that boundary was written down before any giant-prime measurement; in the event no rung crossed it.

Holds: not yet carried elsewhere. *Met in:* measurement, number theory.
Shown: version 6; full guide row 184.

Compare like with like, or finite size reads as failure (185) — A finite-size measurement set against an infinite-aperture prediction will look like a failure. The first giant-prime run set an eight-column rung's 0.151 deviation against the asymptotic value; split by aperture, the thirteen shallow rungs show the documented approach from below, and the fourteen deep ones agree to 6.46×10^{-5}, the deepest carrying 1.25×10^{10} columns.

Holds: not yet carried elsewhere. *Met in:* measurement, number theory.
Shown: version 6; full guide row 185.

A named thing with no definition on record is an orphan wearing a label (191) — Name the undefined names, so a label is not mistaken for a tool. The instrument inventory separates one frozen root, closure_hunter v1.3, from seven separately-versioned siblings, puts reused design patterns in a muted zone beneath them, and prints that three

names are not defined anywhere on record, likely early poker-era naming.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 191.

Correct the prose that describes the code, not only the numbers (192) — A paper whose numbers are all correct can still be impossible to reproduce from, so the sentence that describes the code is corrected too. The construction text said the walk climbs by a running count; the code climbs to the rung of the largest prime seen, and only that reading reproduces the printed 30.00110 and 4.26423 rather than 1.03 and 3.63.

Holds: not yet carried elsewhere. *Met in:* records and tools, number theory.
Shown: version 6; full guide row 192.

A census sentence that did not share a domain with its census (194) — A summary sentence and the table under it can disagree while both are true, so a census that says “every” lists the tables it did not read. The prose credited all 13,197 on-core events to one predicate, but the table split them into 13,187 walk grazings and 10 return-leg crossings; in the triple-star sessions, 74 escapes called every escape left an atlas of 1,183 uncounted.

Holds: carried to two: held in atlas up to eccentricity 0.24-0.28; failed in eccentricity 0.30 and above (seam). *Met in:* measurement, records and tools, three body.
Shown: triple-star; full guide row 194.

A faithful coordinate is the opposite of a cryptographic one (195) — Close a door the reader would otherwise open; it is cheaper than answering the email later. The address is trivially invertible, factor the number and read the walk, so the construction offers no one-way hardness and no cryptographic content whatsoever, and the paper says so unprompted, twice.

Holds: not yet carried elsewhere. *Met in:* number theory.
Shown: version 6; full guide row 195.

Test the bridge you want to exist, and print the null (202) — Test an attractive identity before keeping it, and print the result when it is negative. The Pythagorean comma is the same kind of object as the construction’s closure deficit, a loop of commensurate-but-unequal ratios that fails to come home, but the numerical correspondence with the measured constants tested negative, so the kinship stays as provenance, not as a bridge.

Holds: carried to one: failed in construction’s closure-deficit constants (break not stated). *Met in:* waves, number theory.
Shown: Question the Substrate; full guide row 202.

A blind prediction with a date, a search class and its failure modes (203) — Naming the failure modes in advance is what makes a prediction rather than a forecast. The closest same-rung separation record is predicted to hold flat until exactly N = 4,225,798, then drop to 8.346644×10^{-5} realized by a named pair, then hold to ten million, with its search class and three failure modes sha-pinned in the crate.

Holds: not yet carried elsewhere. *Met in:* measurement, number theory.
Shown: version 6; full guide row 203.

Standing beside, not on shoulders (205) — Welcome each established landmark the tool stumbles onto, because it is evidence the tool is pointed at something real. Each rederived theorem was a zero-holonomy landing point, solid and already named, confirming the tool had not drifted into speculation; others, from entirely different domains, had stood in that same spot.

Holds: not yet carried elsewhere. *Met in:* records and tools, topology.
Shown: version 6; full guide row 205.

Make disagreement cheap by making the arithmetic free to check (206) — When every figure regenerates from a receipt, disagreement can be aimed at the construction’s choices instead of the arithmetic, and printing the inputs beside every derived value lets a slip show on its face. The Loop Atlas’s EXPLORER_ROUTINE_CHECK printed a cue’s rip window as ±1.102 per cent when its own entries give ±2.203, because the print halved a half-width again.

Holds: not yet carried elsewhere. *Met in:* records and tools, number theory.
Counted: The built artifact is the evidence 1. *Shown:* Loop Atlas, R2W; full guide row 206.

I set a kill threshold below my own instrument’s noise floor (210) — Before a run, write down the floor of each reading and set every bar well above it, a hundredfold where it can be. In the seismic instrument build, acos reported about 1.4×10^{-8} rad from float error alone against a kill threshold of 10^{-9}; atan2 of the antisymmetric part has a measured floor of 2.36×10^{-16}, an 8.9×10^{7}-fold improvement.

Holds: carried to one, held there: axis-tolerance check in later rooms. *Met in:* measurement, driven oscillators, geodesy.
Counted: A check that cannot fail 5. *Shown:* Loop Atlas, R2W; full guide row 210.

The fence the paper asked for, answered in the direction it was asked (211) — Run the sweep before saying which part of a result belongs to the construction and which to the coordinate. Across four heading angles the wall loops stayed at the floor while the boundary values moved: 0.2247 at the golden angle, 0.3479 at 100°, 0.3744 at 90° and 0.2268 at 137°, so the pattern is robust and the value is a coordinate.

Holds: carried to one: weaker in heading angles 100°, 90°, 137° (knob). *Met in:* measurement, number theory.
Shown: version 6; full guide row 211.

The reader caught a numerology artifact in the output, from the numbers alone (212) — A suspicious regularity in the stimulus is a bug report about the experimenter, so look at the inputs and ask what procedure produced them. Six loops returning an identical 0.224748 sat on the primorials, n = 1, 2, 6, 30, 210, 2310 — and Bear asked whether that was a pattern, and it was an artifact of how the test cases had been generated.

Holds: not yet carried elsewhere. *Met in:* measurement, number theory.
Shown: version 6; full guide row 212.

Answer a suspicious coincidence with a derivation, because a derivation says what CANNOT matter (214) — Re-sampling defends against the coincidence noticed; a derivation also defends against the ones not noticed, because it says what cannot matter. In R2W, TWELVE-1 found the V at the 80-degree null equal to the 40-degree V in every printed digit, down to $1.951845869 \times 10^{-28}$, and asking where the riders sit showed one calculation done twice: the same nine seats.

Holds: carried to two, held in both: four weights not yet looked at; Loop Atlas thick and thin cats. *Met in:* driven oscillators, number theory, kinematics.
Shown: Loop Atlas, R2W; full guide row 214.

The only substrate in the program where the noise floor is PROVED instead of measured (218) — Telling closed from very nearly closed needs a noise floor, and every other substrate pays a run to measure it: 35.81° for protein defect angle, 0.0640 for RNA, 1.360°/bp for DNA twist. On the integers the floor is proved rather than measured: Baker's theorem from 1966 bounds how small a linear form in logarithms can be.

Holds: not yet carried elsewhere. *Met in:* measurement, biology, number theory.
Shown: Question the Substrate; full guide row 218.

The registered prediction failed, and the failure carried the information (219) — Keep failed bars visible and ask where each miss broke, because the miss may carry the information. The leftover of log n over log p was registered as evenly spread, but its deviation from even stayed flat across three decades, 0.0451, 0.0423 and 0.0430, and it follows Benford's law in base p, fitting 4.1×, 2.8× and 2.2× better than even.

Holds: carried to one: failed in the leftover of log n over log p (substrate). *Met in:* measurement, driven oscillators, number theory.
Shown: R2W; full guide row 219.

The corrections landed as a practice change, and the date is visible from outside (220) — A change of practice is shown by measuring the whole corpus, not by asserting it. A linter asked whether every Zenodo DOI in each live record's related-identifiers block points at a concept or a version: before 2026-08-22, 15 records carried 105 version-DOI citations and 3 were clean; from 2026-08-22, 12 records gave 1 defect and 11 clean.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 220.

Hand-counting found eighteen; the machine found a hundred and six (221) — Machine-check a property that is mechanical, because a careful reader looking straight at it still undercounts. The citation register's manual pass logged 18 defects across five documents, and a script asking the same question returned 106 across 13 records, worst at 20 in one paper; it lints the metadata block only, so a DOI inside a PDF's reference list is invisible to it.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 221.

A sweep that does not check its own coverage is not a census (222) — Read the reported total and keep paging until the count matches it, and pair every parse with a row count against the number expected. The linter once swept 55 records where the endpoint says 62 and printed a total as though it were the corpus; in the Loop Atlas sessions, a register read with every label form held 941 graded predictions, where two scripts had read 783 and 667.

Holds: carried to one, held there: triple-star batch parser (KW0). *Met in:* measurement, records and tools.
Counted: Count the right unit 1. *Shown:* Loop Atlas, triple-star; full guide row 222.

The third instrument defect of one day, and all three were mine (228) — An estimator used outside its domain of validity can make a sound theory read as wrong. The area prediction first came back NOT CONFIRMED by up to 1.6 rad because interior angles summed with acos cannot express a reflex angle and hold only for a convex polygon; with a signed solid angle, the same prediction confirms at 9.4×10^{-16}.

Holds: carried to one: failed in fat patches with reflex angles (wall). *Met in:* measurement, kinematics.
Shown: version 6; full guide row 228.

Name your loss first, because it removes the cheap shot (229) — Stating the weakness first takes the only easy attack away from a hostile reader. The section on what the paper is not testing opens by conceding density, then declines write speed while owing a number for “slow”, and declines to compare against anyone’s published lifetime figure, because “comparing extrapolated numbers across laboratories is not a measurement”.

Holds: not yet carried elsewhere. *Met in:* records and tools, materials.
Shown: version 6; full guide row 229.

An assembly claim, not an invention claim (230) — Restrict novelty language to not located, never new, and declare the search’s own coverage. Three prior-art searches found that every component exists off the shelf and nobody has assembled them, which the paper calls “the best possible shape for a proposal and the worst possible shape for a novelty claim”, and it says the claim is weaker than hoped and much more likely to survive review.

Holds: not yet carried elsewhere. *Met in:* records and tools, materials.
Shown: version 6; full guide row 230.

Name the obviousness risk yourself, and answer it with an experiment rather than an argument (231) — The obviousness objection cannot be argued away, so answer it with one named measurement. Because microfluidics builds these structures routinely, storing data in the topology invites the reply that it is an obvious relabelling; the author wrote that objection first, and the defence is to measure whether a topological readout degrades discretely while a magnitude readout fades on the same sample.

Holds: not yet carried elsewhere. *Met in:* measurement, topology, materials.
Shown: version 6; full guide row 231.

Design the fallback before the risk resolves (233) — Draw the retreat before the open question closes, so the question is not a gamble. A thin plate is effectively a two-dimensional domain, so a layered architecture of holes written in stacked planar sheets sits squarely inside the theorem already proved: “Less elegant than a solid block, fully supported by a proof, and it is how one would build the first demonstrator anyway.”

Holds: not yet carried elsewhere. *Met in:* topology, materials.
Shown: version 6; full guide row 233.

State the negative claim explicitly, because this class of claim is routinely oversold (236) — Unbreakability is a property of a protocol, never of a medium, so say what the work is not before what it is. This is not an unbreakable cipher; the honest claim is an unclonable, tamper-responsive substrate for key storage, still to be designed against machine-learning modelling attacks, side channels such as an acoustic read, and thermal spoofing.

Holds: not yet carried elsewhere. *Met in:* records and tools, materials.
Shown: version 6; full guide row 236.

Name what your own favourite construction does NOT buy (237) — Fence a favourite construction before a reader overvalues it: “The tower is bookkeeping, not geometry.” Mapping an address space onto the integer-forest addressing buys exactly one real thing, arithmetic relations becoming spatial ones so divisors sit near multiples, and it pays only if the data’s natural relations are arithmetic.

Holds: carried to one, held there: topological storage address space. *Met in:* materials, number theory.
Shown: version 6; full guide row 237.

A quantity whose rank order does not survive its own convention is not a measurement (241) — Grain-ring closures read under four gauges move a median of 67.4° and a maximum of 158.7°, 96% of loops move by more than 10°, and the rank correlation against the shipped gauge is 0.022, 0.223 and 0.319. That is not a shift in magnitude but a loss of order, and the paper reports its own quantity as one that did not survive its convention.

Holds: carried to one: failed in three other gauges (ruler). *Met in:* measurement, materials.
Shown: version 6; full guide row 241.

Find the route that is already immune, and say so (245) — Finding that someone else’s method is already safe from a newly named problem is a better outcome than a fix, and a novelty-seeking write-up tends to bury it. One established construction is structurally immune to the whole defect because it never composes reduced rotors at all, and the paper names it, with a reason to prefer it not seen stated before.

Holds: not yet carried elsewhere. *Met in:* materials.
Shown: version 6; full guide row 245.

Three of our own results that the discipline killed (246) — A paper arguing that a bookkeeping choice can manufacture a result cannot leave out a manufactured result of its own. It reports them instead of quietly dropping them: a striking spike at exactly 60° that turned out to be a tie-break artifact, a grain-network defect census that was

spanning-tree dependent, and a composed-loop null that fell short of its pre-registered 5° bar and was reported inconclusive with the bar unmoved.

Holds: not yet carried elsewhere. *Met in:* measurement, materials.
Shown: version 6; full guide row 246.

The knot built to beat the instrument, and it does (249) — An instrument should fail exactly where its theory says it must, and what it can still say is a reliable negative. Two mutant eleven-crossing knots have Alexander polynomial equal to 1 — the same as the unknot — so the instrument returns det = 1 and calls them unknotted; neither reaches det = 3, so a clean trefoil reading rules them out, excluding but never confirming.

Holds: carried to one: failed in Kinoshita–Terasaka mutant knots (wall). *Met in:* measurement, topology, biology.
Shown: version 6; full guide row 249.

The most confident result in the study was computed on the wrong protein (250) — Every internal check is blind to a wrong input, because a wrong input makes every seed honestly wrong together; only an exterior reference catches it. A det = 1 result held across every seed, in three independent runs, confidence tier HIGH, purity up to 1.000, and it was an entirely different molecule, fetched because one character of the accession was transposed.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools, biology.
Shown: version 6; full guide row 250.

The failed prediction was worth more than the pass would have been (252) — Ask why a registered prediction failed, because the failure can be the better measurement. Writhe ≈ 0 was expected from published results on an amphichiral alternating knot, and the backbone measured +42.542 and +7.039; the theorem is about the reduced diagram, the measurement was the open-arc writhe, and the gap became a clean readout of helix contamination.

Holds: carried to one: failed in open-arc writhe of protein backbone (substrate). *Met in:* measurement, topology, biology.
Shown: version 6; full guide row 252.

We inherited a propagated typo and did not check it (253) — A risk identified, written down and then not executed is a worse failure than one nobody saw, and printing that distinction makes the rest of the record believable. The literature is itself split on the accession, so the error arrived from outside; but the research plan had flagged the ambiguity, and the check was never run.

Holds: not yet carried elsewhere. *Met in:* records and tools, biology.
Shown: version 6; full guide row 253.

The floor measurement needs no labels, so you can know the comparison fails before you run it (255) — Before comparing a geometric descriptor between two structures, measure how far it moves across the models inside one ensemble. That spread needs no topology call and no ground truth and costs a minute, and if the effect is smaller than it, the comparison cannot work: one of the rare checks that says so in advance rather than after.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 255.

A citation that cannot be followed, and the two namespaces behind it (263) — A citation that a reader follows literally and finds nothing can make good data look bad. The literature cites one chain identifier that the deposited file lacks in the namespace every standard tool reads; a full scan of all 4,470 atom records showed the two namespaces disagree, the paper citing one while the tools read the other.

Holds: not yet carried elsewhere. *Met in:* records and tools, biology.
Shown: version 6; full guide row 263.

Two sign-inverting bugs, both caught by a four-line synthetic of known answer (264) — A small synthetic with a known answer catches what every internal check misses. Sorting base pairs without a continuity check made the axis teleport, and 15% of centres out of order flipped writhe from −0.4563 to +0.4107; the midpoint also orbits the true axis, flipping the sign at 2 Å of wobble, and smoothing over one repeat recovered −0.4191 at every amplitude tested.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 264.

“We are not competing with you” (267) — Say plainly first where other tools are better, then claim the narrow thing that is added. Four named tools detect knots better than this instrument and agree with it where results overlap; it adds how much the molecule moves, not how stable the algorithm’s vote is, and in the triple-star sessions’ islands paper a challenge-sounding phrase was softened everywhere it was said, every number unchanged.

Holds: carried to one, held there: triple-star islands paper’s abstract. *Met in:* records and tools, three body, biology.
Shown: triple-star; full guide row 267.

A negative result is a specification that has already been tested (268) — Read a list of failures against a different question before discarding it. Five findings reported as fatal to an artificial-life claim, among them no autonomy, faithful copying and silence until read, are each a requirement for a device somebody else pilots. Most specifications are wishes written before contact; this one came from trying as hard as possible to get the opposite, under pre-registered conditions, and failing, so the architecture is the shape of the hole the failure left.

Holds: carried to one, held there: device somebody else pilots. *Met in:* control and circuits, biology.
Shown: version 6; full guide row 268.

The same primitives that make it work are the ones that make it governable (273) — For a device commanded inside a living person, ethics is load-bearing, so it belongs in the body of the paper rather than in a coda. The trust, fidelity and anchoring primitives that make the swarm work are the same safeguards that make it governable, so building the security builds most of the ethics, and the agent's one reflex, refusing a physically-insane command, can only ever say no, never yes.

Holds: not yet carried elsewhere. *Met in:* control and circuits, biology.
Shown: version 6; full guide row 273.

Name the misuse concretely, then say the safeguards are not sufficient (274) — A paper that only lists its safeguards is not being honest about its risks. Name the doors specifically, as one paper did: a population that could be imprisoned by the thing meant to heal it, and a fail-open design where losing contact releases rather than safes. Then write the sentence most papers will not: safeguards raise the cost and narrow the path, but they do not close it, and none of it is sufficient.

Holds: not yet carried elsewhere. *Met in:* control and circuits, biology.
Shown: version 6; full guide row 274.

Write your own ten strongest attacks, and concede the sharpest one (275) — A paper that ships its own red team is harder to dismiss than one that waits to be attacked, and the concession is the part that costs something. The sharpest attack was a motte-and-bailey, thrilling claims retreating under pressure to only an invitation; rather than defend it, the paper bound itself to the modest position as the true one and named its single highest-risk load-bearing claim front and centre.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 275.

Sturdy where it is least exciting, fragile where it is most exciting (276) — A work that is strongest where it is dullest and weakest where it is most thrilling is the opposite of a scam, and that pattern is the honest reason to trust the dull parts and distrust the thrilling ones. Any reader can run this test on any paper in ten seconds, including every paper in this corpus, and almost nobody volunteers it about their own.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 276.

Refuse the grander sentence, and say exactly why it is false (278) — A seductive sentence sits just off the edge of every paragraph of a bold paper; write it down and kill it in specifics rather than avoid it. The grand version requires culture and knowledge transmitted rather than inherited, the far side of the wall the work proved it could not reach. The honest version of the dream is the smaller one, and only the smaller one survives the daylight.

Holds: carried to one: failed in culture transmitted, not inherited (wall). *Met in:* biology, materials.
Shown: version 6; full guide row 278.

Size the new claim yourself, before anyone else can (280) — Name the exact width of a claim at the moment it is most tempting to leave it vague, so that results sized "medium" one at a time still reach the front page. Answer five questions in its register entry: how big against the nearest known number, to whom, whether it is known, confirmed here or ours, what would make it smaller or bigger, and where it goes.

Holds: not yet carried elsewhere. *Met in:* control and circuits, materials.
Shown: triple-star, Question the Substrate; full guide row 280.

Keeping it light is what keeps it true (281) — The moment a vision is asserted as fact it turns to smoke, the failure mode of much work that resembles this from a distance and not at all up close. Bear's own paper gives the trade that earns the light tone: kill hard and early, and the last page gets to be fun. Nobody gets the ark for free; this one was paid for with a wall.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 281.

Take it to the hardest place you already own before you take it anywhere new (282) — Earn the trip where failure is survivable, and give a dry run for a new regime that regime's hardest row. In R2W an engine copied from one built where the frozen points always exist stopped on the arccos of a number above 1 at strength 0.05, where the

3/5 root sits just past that edge, though a cousin written two days earlier already handled it; its dry run had used 1/2 at strength 0.10.

Holds: carried to one: failed in strength 0.05, 3/5 root (seam). *Met in:* driven oscillators, control and circuits.
Counted: A check that cannot fail 1. *Shown:* R2W; full guide row 282.

The broken version is kept because it is the reason we believe the working one (287) — A confidently wrong answer is the worst kind, and keeping it shows why the fix is believed. The first bell-mesh implementation measured frequency proximity where it should have measured consonance; rebuilt on correct circle-map dynamics, the real effect appeared unprompted, locking width collapsing as ratio complexity rises in an Arnold-tongue staircase. The paper gives the shape and not a number, so row 38's hold stands on price, now traceable to a source.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, waves.
Shown: version 6; full guide row 287.

The confident mis-anchored friend is more dangerous than the openly compromised adversary (291) — Inside your own network the friend is trusted and wrong, and that combination beats an enemy who is merely wrong. Carried from the evolutionary-warfare work to a contested sea floor, the rule treats reporters as Byzantine and as poker players, working the table on the assumption that some hands are collusive; only physically-verified, provenance-anchored signals are sovereign, because the anchor, not the reputation, confers standing.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 291.

Hold the most powerful tool in reserve, and give the technical reason rather than a scruple (297) — Keep active probing behind the passive sweep, and give the reason as a measurement limit, not a moral one: too much probing might eventually disturb the very substrate fidelity being read. Naming an instrument that perturbs what it measures as a design boundary before anyone uses it is paired with a refusal to guard the door, since the case for probing is large.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 297.

Put the number first so it cannot hide, then say why you distrust it (298) — Stating the distrust beside the number is what makes the section survivable. A Fermi estimate given as a wide range leaning high was followed at once by the red team's catch on the authors: two-fold error bars quoted in the same breath as a reference program's twenty-fold overrun, which cannot both be true. What survives is structural, a sparse aperture scaling with baseline where a monolith scales with glass, so the figure is an opening bid.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 298.

The interactive whose biggest button defeats it (299) — A demonstration earns trust by what it lets you do to it. In the passive-detection toy, an accumulated map lets a true site persist and lock while false sites wash out, and the most prominent control breaks the claim: flip common-mode noise on and the independent signatures fail together, so the gain from fusion evaporates. A site confirms only on at least four observations and better than half persistence.

Holds: not yet carried elsewhere. *Met in:* measurement, reading pictures.
Shown: version 6; full guide row 299.

Blind spots as theorems, not bugs — each with its named price (303) — A limitation with a known remedy and a known cost is a design parameter; one without is an excuse. Each blind spot carries its fix and price: net-charge-zero targets are invisible per-rung, scalar observables are loop-blind by the telescoping identity, and helical structure at fixed circulation is invisible to winding until the twist rung is added, so the price is to climb a homotopy rung, add a channel, or promote the observable from scalar to vector.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 303.

An instrument that has only ever agreed with itself has not yet been tested (306) — Internal kill tests (nulls, injections, shuffles, held-out floors) supply no external witness. A satellite geodetic pair fails through orbits, antennas, troposphere and reference frame; a buried invar-wire extensometer fails through rainfall, vault temperature and wire creep. They share almost nothing but the fault between them, so their agreement is worth more than any amount of internal consistency, and disagreement indicts one of the instruments, never the ground.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 306.

Nature did not move; our anchor did (308) — Write the frame down first, and measure both ends of any offset before giving the motion to either. A registration assumed a plate-fixed frame, and the effect arrived at nearly ten sigma in the opposite registered tail because the data frame itself moved; in R2W, couples graded from a frozen point

seemed to slide, but end by end the frozen point did the moving, 0.21 to 0.22 of a turn per unit of the second tone.

Holds: carried to one: failed in data's own secular frame (ruler). *Met in:* measurement, driven oscillators, geodesy.
Shown: R2W, Two Arrows; full guide row 308.

The price of skipping held-out calibration, printed as a warning (310) — Keep the cost of a shortcut in the record rather than quietly fixing it, because that is what makes the good numbers believable. With pair-specific held-out calibration the realised false-alarm rates across four windows were 0.011, 0.022, 0.000 and 0.000; field-wide, before that calibration, they ran 0.13 to 0.21, an order of magnitude worse on the same data from the same instrument.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 310.

Record the threshold artifact rather than hide it, and pre-register the fix instead of retrofitting (311) — A threshold inherited from a different region's null was negatively biased by a residual staircase, so a held-out control region formally crossed the bar while sitting at z = +0.40 against its own null. The paper states the verdict plainly, no signal there and a miscalibrated absolute bar in the pipeline, and pre-registers the repair for the next campaign: fixing it in this one would be a tuning, not a correction.

Holds: carried to one: failed in held-out control region (ruler). *Met in:* measurement, geodesy.
Shown: version 6; full guide row 311.

The honest instrument publishes its own visibility (312) — Almost no instrument in any field ships the list of what it is blind to. One datasheet, built by matched injection, set eight physical deformation modes against six channels, with one channel reading 1.00 on its own mode and 0.00 to 0.05 on the others. R2W gives each engine a card of knobs, digits, floor, limits and checks, and a difference smaller than an engine's floor is not evidence, however many digits it is printed with.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: R2W; full guide row 312.

A pre-registered bar that a real, correctly-signed signal failed to clear — and both facts printed (313) — Showing a real tendency that failed its bar is stronger than hiding either fact. Median rank correlation +0.73 and median speed 6.4 km per day landed inside the registered physical band, but the sign test read 11 of 16, p = 0.21, a failure of sample size, not signal; a sharper post-hoc statistic reading about four sigma on the same data is claimed as nothing and pre-registered for the next campaign.

Holds: not yet carried elsewhere. *Met in:* measurement, three body, geodesy.
Shown: triple-star; full guide row 313.

A domain-scoped verdict binds only its domain (314) — Row 313's non-pass speaks for one network at its sample size and not for the method, and keeping the literature value, about 8 km per day against the campaign's 6.4, beside the failed bar stops the claim drifting upward. A fitted law speaks for its inputs too: in R2W, RATCHET-4's rule held at frequency 0.70 and missed at 0.40, where another ratio had also moved, and with that ratio held, RATCHET-5 came to 1.0003 of the reference.

Holds: carried to six: failed in this network at this sample size (ruler); held in every strength and frequency run; weaker in second tone at twice the frequency (knob); held in frequency 0.70; failed in frequency 0.40 (knob); held in RATCHET rectifier runs. *Met in:* measurement, driven oscillators, geodesy.
Shown: R2W; full guide row 314.

The contribution is the protocol, not the signal (316) — Reproducing something already known, under a discipline nobody has applied to it, is a result, and the only kind that validates an instrument rather than a hypothesis. One campaign reproduced a known geophysical process at high internal significance using only an external clock, documented-equipment masking and a held-out foreign network for false-alarm accounting, then said plainly that the signal is the literature's and the protocol is the contribution, its two failures printed.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 316.

What crossed the domain boundary, and what did not — in the first sentence (317) — Port the method, not the mechanism, and say so as a restraint. Opening a new domain, the paper states first that the topological channels remain in their home domain and make no appearance; what generalises is the operating discipline (validate against ground truth, pre-register the traps, publish the floors and the breaks) plus a spectral toolset under three named guards. Row 93 asks every port to beat a baseline in its own domain; this row names which part ported.

Holds: carried to one, held there: a new domain. *Met in:* measurement, quadrature.
Shown: version 6; full guide row 317.

Flag the known systematic; do not smooth it (320) — Applying a legitimate correction to a number already known to be wrong, before publishing it, is how a systematic becomes invisible. A derived mass came in high against the published value by the exact amount the field's known scaling systematic predicts, and it was flagged rather than smoothed, with the standard cure named as the next step instead of quietly applied to make the number land.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 320.

Two first lights, two honest breaks, zero novelty claimed (322) — Open a new domain by reproducing its known results before proposing anything; it is the slow road and the only one that arrives. Both targets were validated against published ground truth and both instrument failures printed, and the claim was not about the objects at all: standard work in another field, done under this programme's discipline, showing the discipline itself is domain-portable, with zero novelty claimed.

Holds: carried to one, held there: standard work in another field. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 322.

A card is a card (323) — Three sentences can stop a roadmap from becoming a claim: the future application is carded and remains a card, with no data, no session and no results, and the paper claims none. Every speculative programme gathers such cards, and almost none label them this bluntly inside the paper that would benefit from their being vaguer.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 323.

A novelty claim that states its own search bound is a different object (324) — Saying how far the search went turns a novelty claim into one a reader can weigh. A literature pass found every mechanism to be prior art, from topology as trigger to the gate economy itself, leaving the assembly plus the external validation as the candidate; the paper marked that claim PROVISIONAL, since finding no prior assembly in a bounded adversarial search is not the same as none existing, and flagged it as its most exposed.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 324.

A kill may be a rename, and that is a third verdict (325) — Writing the rename branch into the kill conditions in advance keeps the third verdict from becoming an escape hatch, since a sound idea can die only in its words. One channel dies as stated if two axes collide but may survive as a rename to a non-optical degree of freedom, and a heredity claim is renamed persistent re-induction if the signal turns out re-induced rather than transmitted.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 325.

Surface an idea specifically to be killed, and say so without apology (326) — A span that survives an honest assault has earned its place, and one that breaks shows where the real bridge has to go. Every rung in the paper ships with its own named kill condition, and the nine rungs' conditions were collected into one section so a reader can attack the whole structure from a single page: a paper organised around its own falsifiers, not a limitations section at the back.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 326.

A register of what we assumed, WHO should check it, and our confidence separated from our tone (327) — Naming the specialty and the precise question turns a plea for collaboration into an executable ask. The register has four columns, its third giving honest confidence apart from the prose's tone in three levels, Moderate, Calibration and Speculation, and above it sits the caveat that every in-container simulation is single-run, modest-N, without confidence intervals, and nothing labelled a result is statistically hardened yet.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 327.

Cite the dispute, and say whether the literature tests YOUR claim or only the phenomenon (328) — Distinguishing a field that supports the phenomenon from one that has tested the specific claim is where borrowed authority usually enters. An effect solid one generation out and disputed further out is cited with its dispute; and in R2W, matched only on the effect, all of the book's Part 7 would read as known, but matched on the quantity the literature's is a drift and the book's a threshold, their zeros a quarter turn apart at the Hamiltonian end.

Holds: carried to one: weaker in book's step lopsidedness (Part 7) (break not stated). *Met in:* records and tools, driven oscillators.
Counted: A word that changes meaning across a boundary 1. *Shown:* R2W; full guide row 328.

A table whose every row invites being struck (333) — Put the discipline in the layout, not in the author's vigilance. A table of eight jobs and two toolkits adds a fourth column naming where they could trade, and prints under it that every cell in that column is a challenge: name the shared mechanism, or strike the row as resemblance. It is row 7's mechanism-versus-correspondence rule as a table format, and most comparison tables would lose rows under it.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 333.

Mark the door; do not walk through it (334) — Naming a door you are not opening keeps a speculative paper navigable: the reader knows which rooms were entered and which only labelled. The two most speculative rungs carry the tag opening rung, most speculative in their own headings rather than a footnote, the largest question nearby is left as a marked door, and two kill conditions pre-commit to dropping the rung as a category error rather than defending it.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 334.

An instrument that claims to be universally superior is a marketing claim (335) — What makes a tool trustworthy is a spec sheet of where it reads clean and where it fights the substrate. In one paper's two applications, one channel recovered a published signal cleanly and another suppressed it, and the split was treated as the most useful result: the deliverable is a competence surface, with two clean failures and one mapped suppression mechanism reported at full volume.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 335.

A bar that cannot fail is not a bar (337) — Pre-registration stops a bar moving after the data; only checking whether it could ever have said no stops it being set too wide. Inflated apex-error tolerances were re-read as uninformative and an informativeness gate installed; in the triple-star sessions a condition that one share be at least twice another printed "holds" on a code trial where both shares were zero, so a floor above zero was added before the run of record.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Counted: A check that cannot fail 7. *Shown:* Loop Atlas, R2W, triple-star; full guide row 337.

We tested our own leading explanation and it failed, which is the useful part (339) — Even a failed repair can show where the problem lives. An unweighted vote across per-source errors was suspected of suppressing amplitude, but a pre-registered heteroscedastic battery left the coefficient statistically unchanged and the amplitude 4.5 sigma low; white-noise synthetics at the real error spread reproduce only about twenty percent loss against about sixty in the data, so the missing forty percent lives in structure the synthetics lack.

Holds: not yet carried elsewhere. *Met in:* measurement, kinematics.
Shown: version 6; full guide row 339.

The economy claim failed cleanly, and what the loops actually buy is named instead (341) — Killing a selling point and naming the real one in the same paragraph keeps a null from being mourned. A decimation test against a plain global fit, across progressively smaller subsamples, found no crossover at any source count in clean and jitter-inclusive subsets, so the claim that robust loops reach frame accuracy with fewer sources fails cleanly; what the loops buy instead is direction, localisation and audited degradation.

Holds: not yet carried elsewhere. *Met in:* measurement, kinematics.
Shown: version 6; full guide row 341.

A topological mismatch is a statement about fit, not about quality (344) — When a substrate offers no genuine transported state with non-trivial closure, the instrument reads nothing, and that is a mismatch of data to tool, not a failure of the tool. A null correctly put down to fit is reusable, while one blamed on the tool is thrown away, and even in mismatch, watching what a substrate does under the instrument can show structure its own community had not catalogued.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 344.

An instrument is a partner in a dance, not a harpoon for a white whale (345) — The instrument works when the substrate carries real invariants to walk on, when the signal is of the coherent global-phase class it reads, and when there are at least two vantages to close a loop against. Choosing the substrate, the signal and the estimator to match the field's geometry is not a limitation to apologise for; it is the whole of using the instrument well, and it says what the guide is for.

Holds: not yet carried elsewhere. *Met in:* measurement, topology.
Shown: version 6; full guide row 345.

The tautology guard — naming when your own agreement is circular (346) — A result that had to come out right is quarantined by name, not counted. Sub-percent agreement with a standard method was filed as validation only, because the zeroth-order circulation equals the rotation curve by construction; in the Loop Atlas sessions a toy sail's price dead downwind, 0.2000 newtons a watt, exactly one over the wind's speed, was printed as a ride law met, but it is the price's own bookkeeping, and any boat prints it.

Holds: carried to two: failed in toy sail dead downwind (wall); held in toy sail's general power balance. *Met in:* measurement, three body, kinematics.
Counted: A check that cannot fail 2. *Shown:* Loop Atlas, triple-star; full guide row 346.

Name the stricter claim a reader would want, and decline it (347) — The claim a result is one inch from is the one a hurried reader is most likely to make, so write the denial into the paper. The registered bar passed on recovering known structure, and the paper added, unprompted, that the stricter claim, that the channel carries information the standard analysis lacks, is not demonstrated: the same patches are recoverable in the standard channel and any excess sits at the 0.1 percent level.

Holds: not yet carried elsewhere. *Met in:* measurement, kinematics.
Shown: version 6; full guide row 347.

Rediscover why a standard practice exists by declining to adopt it silently (349) — Adopting a field's convention without asking why leaves you unable to tell a convention from a result. A rotation curve built from measured counts rose unphysically toward 250 kilometres per second, because a distance truncation poisons the outer annuli, while a pinned external profile gave the flat value the literature reports, an in-crate demonstration of why published analyses pin an external density profile.

Holds: not yet carried elsewhere. *Met in:* measurement, kinematics.
Shown: version 6; full guide row 349.

The substrate autopsy that ruled itself out, with a number (368) — Eliminate a cause with a number rather than by assumption. From-memory station coordinates were replaced with official six-decimal ones: mean shift 0.59 km, worst 1.36 km, against legs of 50 to 90 km. Both loops still wind the same direction with signed area unchanged, so the anomalous sign flip is not a geometry artifact, and substrate is ruled out without rescuing the null: the full weight falls on the hunter.

Holds: not yet carried elsewhere. *Met in:* measurement, geodesy.
Shown: version 6; full guide row 368.

Commit the bar in writing before the data comes back (369) — Pre-registration is easiest and most honest when the data is not yet in the room. With the data missing, the analysis was built anyway and a two-part bar with a novelty clause written down first; in the Loop Atlas sessions, a reading made after seeing a map, chaos on the borders between treads at 84 per cent, was claimed only after it held at 97 per cent on a fresh grid, while another such reading failed blind.

Holds: carried to three: held in fresh grid, read blind; failed in fresh grid, read blind (break not stated); held in survey data, four starts. *Met in:* measurement, records and tools, maps and chaos.
Counted: A check that cannot fail 1, A second reader 1, Count the right unit 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 369.

A wall you cannot walk through is not a reason to stop (370) — Name exactly which front is blocked, and the others stay open. The data had not survived a session reset, a genuine hard stop on the headline result, yet two of three fronts advanced: the substrate was ruled out by fetching authoritative coordinates from the web, and the instrument was built, verified by running it to its polite stop, and armed against a committed bar, leaving one front loaded and waiting on one upload.

Holds: not yet carried elsewhere. *Met in:* records and tools, geodesy.
Shown: version 6; full guide row 370.

The swing detector, falsified in three deaths (372) — Each death located its own break. Magnitude died first: storm-day turning of 2702 degrees against a quiet day's 2317 was only 1.2 times, with three of five stations swinging more on the calm day. A perfect spatial ordering of 1.00 fell to a permutation null, since at five stations random orderings reach 1.00 seventeen percent of the time; and a supercell that produced no tornado swung the most, 4957 degrees.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 372.

Density is the cure, not years (373) — The failure was statistical power, not a wrong idea: five points cannot certify a spatial sweep, because random arrangements look swept too often. Ten to thirty stations make the ordering null strict, so more stations on one storm beat more years of one station; and by Bear's corollary, a single-station ticker fails on storm days, firing at full tilt on the no-tornado supercell, while specificity on calm days is cheap.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 373.

Match the structure, not the digits (384) — Treat a digit match as coincidence until something derives it. A tempting numerical parallel was filed as a resonance to explore, not a claim, with two fences beside it: the formula it would match is empirically unexplained, so matching it is matching a mystery, and the two numbers are not shown to be the same object, since the value is a common one. The structure is defensible; the digits are not evidence.

Holds: not yet carried elsewhere. *Met in:* measurement, number theory.
Shown: version 6; full guide row 384.

Name which of your own threads is most likely to close (385) — Ranking your own directions by closeability rather than by excitement is the rarest prioritisation a speculative programme can perform. One thread was singled out on three grounds: real public data, an established science to check against, and the programme's best historical hit-rate, with rate-over-level separating states at 96 percent and persistence cutting false alarms six to twenty-eight fold.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 385.

The felt fifth dimension is a parameter, not a place (386) — Keep the five-knob count and drop the spatial claim. The architecture is three of space, plus time, plus a mode index that behaves like an independent axis only because it is a separate integer knob. Two adjacent over-reads are named in the same pass: the unit-loop's free trip is free in magnitude and paid in phase, not a speed bypass, and using a candidate target as the inertial anchor is the error.

Holds: not yet carried elsewhere. *Met in:* topology, waves.
Shown: version 6; full guide row 386.

The ledger law that governs every free-energy idea (389) — There is no net-free work with no ledger anywhere, but an existing gradient can pay the bill while you pay only a small steering cost: a rudder, not an engine. Stated before any of the ten projects, the law lets none quietly violate it, and in the Loop Atlas sessions the swing toy shows it: a swing gains with gravity and gains nothing without it, and that zero is the measurement.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, kinematics.
Shown: Loop Atlas; full guide row 389.

A defect battery is real; defect propulsion is a perpetual motion machine in a knot costume (390) — Knots and invariants are not free-energy packets to drain: making and unmaking one costs at least what comes back, usually more once friction and radiation are paid. A stable topological state can store energy and release it on demand, a genuine robust battery or actuator, but someone paid to wind it; the usable version and the forbidden version differ by one word, and the word is net.

Holds: not yet carried elsewhere. *Met in:* topology, materials.
Shown: version 6; full guide row 390.

The scale map, so nothing drifts into overclaim (393) — Match each ambition to the wavelength that can reach it. Acoustic wavelengths are tens of microns, real, cheap and in reach for cells, droplets, particles and flaws, and structurally cannot touch molecules; electron wavelengths are picometers, reaching atoms and molecular handedness through an instrument accessed by collaboration. So the molecular ambition rides the electron side, said in the same document that lists the garage builds.

Holds: not yet carried elsewhere. *Met in:* waves, biology.
Shown: version 6; full guide row 393.

The killer filter, designed as a single experiment (394) — Choose the strongest project by which one has a measurable result that decides something, not by spectacle. Send winding-encoded and amplitude-encoded signals through the same deliberately-noisy channel; if winding survives where amplitude keying fails, the programme's standing law is demonstrated in a tank for the price of a bubbler. It is a project selected because it can lose.

Holds: not yet carried elsewhere. *Met in:* topology, waves.
Shown: version 6; full guide row 394.

Start where the rig proves itself, not where the spectacle is (395) — The build order runs by leverage, not excitement: the simplest levitation build first, purely to prove the whole stack works, then trapping, then patterning, then the frontier experiment. The instrument-grade nanometer work is held as a partner track through collaboration rather than pretended to be a garage project, with the access path written down instead of a bill of materials that could not be bought.

Holds: not yet carried elsewhere. *Met in:* waves, control and circuits.
Shown: version 6; full guide row 395.

Lower tier before higher tier (396) — Three build tiers are specified with real wiring, and on top sits the instruction to prove the loop with the cheap, fully-understood discrete parts before reaching for the programmable rig. It is the same one-shape-at-a-time discipline that governs the simulation growth path, applied to hardware and written into the document rather than left to willpower.

Holds: not yet carried elsewhere. *Met in:* records and tools, control and circuits.
Shown: version 6; full guide row 396.

Grow one mode at a time, and do not weld them together (397) — Name the failure mode of your own enthusiasm as a build instruction. The growth path is written out: understand the quiet mode completely, add the angular phase in two dimensions, let that phase become height in three, then grow one mode at a time, and the remaining ideas are marked HELD for one-at-a-time building, not to be welded at once.

Holds: not yet carried elsewhere. *Met in:* records and tools, waves.
Shown: version 6; full guide row 397.

The bench tests the device; the sky tests the model (398) — Keep the two drawers separate, because each new result is allowed to touch only one of the two claims. The instrument's soundness is decided on a bench, with numbers, by the person building it; the cosmology is a bet decided by somebody else's data, and its live external handle is named with its current significance and its crux, not yet a discovery, with calibration and foregrounds as the question that decides it.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 398.

A faster implementation is a claim, and it is validated against every existing path before use (399) — A speed-up that changes an answer is a bug wearing a benchmark, so run the old path beside the new one. A vectorised rewrite was checked against three independent scalar implementations and agreed to 10^{-15}; in the Loop Atlas sessions, the jumper ported to JavaScript matched its Python parents' runs of record to 8.5×10^{-11} at 6,000 steps a flight and 1.8×10^{-10} at 1,500, its gap printed at more than one step.

Holds: carried to two, held in both: vectorised rewrite; Loop Atlas jumper in JavaScript. *Met in:* records and tools, biology, kinematics.
Shown: Loop Atlas; full guide row 399.

The banked negative, printed in four lines (402) — A null needs its counts, its breakdown and the word banked. A hypothesised coupling between a turning index and a parity class gave 5,369 matches out of 9,999 (53.70 percent against a 50 percent baseline) and was banked as no coupling at parity level. Closing a hunt is not deciding nothing is there: in the triple-star sessions a closed hunt's watch fired the same day, a lap ratio on flat steps at 7.0185 and 8.0219.

Holds: carried to two: failed in fresh phases (break not stated); held in the islands' lap ratio, a different variable. *Met in:* measurement, three body, number theory.
Shown: triple-star; full guide row 402.

Put what the bench does NOT test in the scope section, not the appendix (403) — A build document is read by someone with a soldering iron and an appetite, so the limit goes at the top, where the temptation is highest. One opens by naming what it cannot demonstrate, that this reads the universe's phase, which is the model and not what the bench tests, and follows at once with what the bench does earn: it tests the instrument, which is genuinely sound on its own.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 403.

Name the analogy inside the build document, in the same paragraph as the result (405) — Each tier ends with its honest limit beside its success, at the moment the demonstration is most convincing. The mechanical belt is a mechanical integer constraint, an analogy for the topological one, good for intuition but not the real knot; the electronic divider is a counter, not a topological winding, the right dynamics in an electronic stand-in. The builder learns which part of what was watched was the real thing.

Holds: carried to two: weaker in mechanical belt tier (substrate); weaker in electronic divider tier (substrate). *Met in:* topology, control and circuits.
Shown: version 6; full guide row 405.

A five-level verdict vocabulary, and every chapter must end with one (407) — A fluent translation hides whether anything is under it, so every chapter ends with a tag and asserts no more than it allows. The five levels run from exact or theorem-backed, with a receipt or proof, through settled or classical, kinship of kind and conjectural, down to inherited coincidence, flagged, a rhyme kept only so the reader is told it is one; every number sits in a receipt that regenerates it from constants.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 407.

A VERDICT CAN UPGRADE — and the verdict belongs to the PAIR, not the object (410) — A bridge reported negative closes that bridge, not the story, so check whether the wrong target was tested. The Tower paper correctly found that a musical comma's 23.46 cents is not 35.81 degrees, but written as a linear form in logarithms the comma measures 1.355×10^{-2}, exactly the (19,12) row of the same paper's Baker-floor table. The verdict belongs to the object and the target together, never to the object alone.

Holds: carried to two: failed in substrate constant 35.81 degrees (break not stated); held in Baker-floor table's (19,12) row. *Met in:* waves, number theory.
Shown: version 6; full guide row 410.

The internal contribution is one line, and everything else is credited above it (411) — Name the prior art at length and claim a finding at exactly its width. The credits ran from the convergents of a logarithm to a 1948 paper and an observation from around 40 BC, and the claim was one internal line: the paper had carried two facts in separate sections and never noticed they were the same row. R2W likewise cites the 2025 biharmonic Josephson diode and says only that its own road adds weight.

Holds: not yet carried elsewhere. *Met in:* records and tools, number theory.
Shown: R2W; full guide row 411.

The conjectural tag is the one that proves the collection is not a sales brochure (412) — An atlas that only ever reported exact or beautiful kinship would be a sales brochure; the test of the method is whether it will write conjectural, in ink, on the author's own cherished idea. Three chapters resolve cleanly and one carries that tag, with the chain printed in three clauses so no reader can slide along it: resemblance real, operator hoped for, hypothesis unproven.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 412.

Both of the usual answers are wrong (413) — Precision is not evidence when the inputs are not that well known, so a match to a fraction of a percent need not mean anything; but reflex dismissal as numerology throws out the structures that really are rare along with the junk. Most argument about coincidences is two camps each correctly diagnosing the other's error, and naming both reflexes as failures is what makes a third answer worth building.

Holds: not yet carried elsewhere. *Met in:* measurement, number theory.
Shown: version 6; full guide row 413.

The window is a finding until you widen it (416) — Each widening changed the answer, and each new answer was more structural than the last. Stopped after 119 rungs, two rows reported infinite that were merely long, one of them 2,615 members, and stopped two rungs below the published one, a clean lower edge appeared where widening found a gap of 28 rungs. R2W refines the ruler instead: a seam read 0.767, then 0.782 from fine ladders, then 0.783, where it stopped moving.

Holds: carried to four: held in Loop Atlas sessions' misses; weaker in sweep over slope and cutting-circle size (knob); failed in ripple strength 0.25 (ruler); held in R2W seam reading: refine the ruler. *Met in:* measurement, driven oscillators, maps and chaos.
Counted: Chaos and binning 2. *Shown:* Loop Atlas, R2W; full guide row 416.

Print both prices, because which reference class is right is a judgement (417) — The price of a coincidence depends on what the searcher committed to before looking, and the honest and the flattering answers differ by nearly a factor of ten. Both are computed and printed, because which reference class represents what a searcher in that area would actually accept is a judgement, not a calculation; publishing the range and naming the judgement beats calling the favourable number the price.

Holds: not yet carried elsewhere. *Met in:* measurement, number theory.
Shown: version 6; full guide row 417.

A limitations section where every entry carries its own death condition (418) — Hedges cannot be checked; a death condition can. Six claims each carry an explicit dies if: the convergent dies if the dimensional constraint is stated wrongly, the gap dies if a wider scan finds excluded rungs elsewhere, and the two prices die if either reference class misrepresents what a searcher would accept, which is a judgement and the reason both are printed.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 418.

One number becomes a point, a curve and a comb — and the comb is over-specified (421) — A curve is far harder to fake than a point, so splitting a crossover into a sign flip, a size per decade and a resonance structure is the right move. But a Deborah-number-near-one mechanism predicts one feature, not a repeating one, and no recurring

mechanism has been named, so by row 418's standard the comb needs a death condition: it dies if the period sweep shows a single crossover and no repeat.

Holds: not yet carried elsewhere. *Met in:* measurement, materials.
Shown: version 6; full guide row 421.

A simulation whose slow limit is the answer is a rig check, not a result (422) — A relaxational model of an orientation slaved to a moving trap reproduces the solid-angle formula by construction as the sweep rate goes to zero, so recovering it confirms the pipeline and nothing about the world. The result is the departure, the mirror residual's shape against the sweep-rate-times-relaxation-time group, a falsifiable prediction that makes the expensive simulation able to fail.

Holds: not yet carried elsewhere. *Met in:* measurement, kinematics.
Shown: version 6; full guide row 422.

THIRD INDEPENDENT ARRIVAL — coherence is the symptom, not the safety (424) — A false reality from a single source is more seamless than a true one built from independent noisy sensors, because there is no second voice to disagree. The matriarch's memory anchor sat on ground that read clean and was not, so every read from her is consistent and confident; checking her harder confirms the lie, and only a second anchor that does not descend from her can detect it. Rows 70, 250 and this one reach the same fact.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 424.

The right is documented; the causal story is fenced separately (427) — Grade each claim in a paragraph on its own evidence. The real-world anchor for forcing substrate-poisoners to declare themselves is statutory and verifiable: a capped number may sit, a posted notice is required, and identification on request is on the books. The often-repeated story that the practice declined because poisoning trust hurt business rests largely on a single anecdote, so it stays a plausible hypothesis, not confirmed data.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 427.

The deepest trap is not being attacked (430) — The worst failure is not knowing a channel is compromised, so it is trusted more, leaned on harder, and walked into prepared traps while believing the game is being won. It gets worse the more competently the operator works, because competence is exactly what routes more weight through the captured channel, which is why the paper names it as its own object rather than leaving it as a moral.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 430.

The corrections register is the project's actual credential (431) — Of four deposited documents, the one the deposit itself calls the most important and most counter-intuitive to publish is a catalogue of ideas held, tested and changed, each with the seductive idea, the check that met it, the outcome and the named failure mode. Corrections run both ways, including one idea wrongly dismissed and later restored, and a reader short on time learns most from the register alone.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 431.

Run our question on your data and send back only the answer (432) — When a steward rightly withholds exact locations, build the method to work with the withholding rather than around it. The ask comes in the steward's order of preference: share under any protective terms, or run the pre-registered question on their own data and return only the result, so sensitive locations never leave their custody and the thing wanted is separated in advance from the thing that must not be had.

Holds: not yet carried elsewhere. *Met in:* records and tools, biology.
Shown: version 6; full guide row 432.

STANDING RULE — scope every new project across the Field Guide first (433) — The rationale is measured rather than asserted: the first time a project was run past the assembled method (the four limbs, the priced rows, the named blind spots, the verdict vocabulary) before anything was built, it changed what got measured next. Bear set the rule on 2026-09-12; when a spoke session began work before reading the guide, no damage was found, and every handover now asks for the guide at the open.

Holds: carried to one, held there: a new project. *Met in:* records and tools.
Shown: version 6; full guide row 433.

A method guide used as a lens produces falsification points, not vocabulary (434) — The test of an assembled method is not that it reads well or is comprehensive but that pointing it at a running project changes what gets measured next. Handed the quadrature rows and the work-lives-in-the-phase-gap law, a live project session opened a

new testing regime with falsification points at the quarter-turn maximum, on work already in progress, with no new theory and no new hardware.

Holds: carried to one, held there: a live project session. *Met in:* records and tools, quadrature.
Shown: version 6; full guide row 434.

THE GUIDE'S OWN KILL CONDITION — a stranger finds something the author did not (435) — Every rung in the corpus ships with the condition that would kill it, and the guide's is cheap to state and expensive to fake: it works if someone who is not its author uses it on their own substrate to find something the author did not. Citing or admiring it does not count. Until then it is a well-documented personal practice; after, it is a method.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 435.

The method travels; the discipline is what gets stripped (436) — A guide spread by use degrades unevenly: rows with numbers survive because they are quotable and pay at once, while the fences die first, the what-this-does-not-test, the visibility matrix, the dies if and the search bound, because each needs understanding and returns nothing to anyone who skips it. So put the fence inside the row, in the same cell as the claim; a limitations section is a page nobody forwards.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 436.

Cross-domain work is harder to criticise than to write (437) — The siloing that kept the landmarks apart also silos criticism: the specialist who could falsify one chapter will never read the next, so an external witness is hardest to get just when row 306 says it matters most. The workable answer is already in the seam register: ask several specialists one precise question each, written so it can be answered without reading the rest.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 437.

Tell the ending first, because the ending is half the point (439) — A paper whose spine is a failure announces it in the abstract and again in the introduction, on purpose: a reader expecting a triumphant demonstration that crystals are alive is disappointed at exactly the intended moment. The wager is stated outright: a clearly mapped wall, with the mechanism that reaches it and the conditions under which the central claim dies, beats one more over-claimed result that quietly plateaus.

Holds: not yet carried elsewhere. *Met in:* records and tools, biology, materials.
Shown: version 6; full guide row 439.

Concede what you did not invent before a reviewer concedes it for you (440) — Prior lineages are the bar a new contribution must clear, not decoration. Five were named with their owners before any claim (life-as-organisation, the clay hypothesis, information-bearing crystal growth, the error threshold, autocatalytic closure), and then the two things that are not re-skins were each stated narrowly, one mechanism and one negative result, with everything else kept honest about being scaffolding.

Holds: not yet carried elsewhere. *Met in:* records and tools, biology, materials.
Shown: version 6; full guide row 440.

The speculation licence, taken deliberately and fenced in the same breath (441) — In artificial life no one can manufacture life to check against, so the discipline advances by proposing mechanisms, arguing their plausibility and naming what would falsify them, and speculation is the working instrument rather than a vice. The licence is taken deliberately but not as permission to be careless: every leap is fenced to a kill condition, and where a claim needs expertise the team lacks, a register maps exactly where.

Holds: not yet carried elsewhere. *Met in:* records and tools, biology.
Shown: version 6; full guide row 441.

Three smothers removed and shown to work; the fourth left honestly open (445) — Measure the removals, and keep the hope apart from the theorem. Making selection endogenous, so that the other individuals are the environment, held the evolutionary rate at 0.033 against 0.002 on a fixed landscape, because climbing never finishes when the hill keeps moving. But ignition from sufficient conditions is the unsolved problem, so three handholds on the wall were found false and the fourth is left honestly open.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 445.

We will not quietly assume our way past a second wall (447) — A claim can be elegant, falsifiable and logically correct and still have no home. The whole heredity mechanism rests on whether real fracture quantises the linking number or scrambles it, a property of the substrate, and predicting the discriminator is not finding the material;

artificial life reasoned that life is organisation and not matter, then could not make life because its chosen substrate withheld what the organisation needed.

Holds: carried to one: failed in the substrate artificial life chose (substrate). *Met in:* topology, biology, materials.
Shown: version 6; full guide row 447.

Hand the field a hypothesis, not homework disguised as a discovery (448) — Frame an unsolved step as an open experimental programme, with directions that are concrete and quickly falsifiable. Substrate selection starts with layered crystals, whose planar cleavage and interlayer registry are exactly the structure the invariant could ride, names conchoidal fracturers as the control that should fail, since the contrast would itself validate the prediction, and places the deciding measurement as a simulation question first and a bench question second.

Holds: not yet carried elsewhere. *Met in:* topology, materials.
Shown: version 6; full guide row 448.

The false-alarm rate is itself a cause of death (449) — A miss is counted, but a false alarm is usually treated as the acceptable cost of caution. Post-event study of a real EF5 documented that residents did not act on a valid warning because prior false alarms had eroded its credibility, so a system that improves detection while leaving the false-alarm ratio alone reproduces that problem exactly: precision over recall is a life-safety question, not a statistical preference.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 449.

A target satisfied by construction could never have failed (450) — Specificity counts every quiet county-hour as a true negative, so a design target of better than ninety percent specificity already sat at 98 to 99 percent before anything was built; precision is the target that means something. In the triple-star sessions LB1 printed a residual correlation of +0.000, but the fit already carried a term linear in e_run, so that correlation was zero by construction.

Holds: not yet carried elsewhere. *Met in:* measurement, three body.
Counted: A check that cannot fail 1. *Shown:* R2W, triple-star; full guide row 450.

Recording this removal matters more than the channel did (453) — Withdraw an original element in public when it fails, and say why. A proposed beat was dismantled by its own author: a beat needs two signals in the same medium reaching one detector, but one was electromagnetic in the Earth-ionosphere cavity and the other acoustic in air, with no coupling proposed, and even as arithmetic the difference swung through zero as the acoustic peak moved with vortex size. It was the one element original to this work.

Holds: not yet carried elsewhere. *Met in:* measurement, waves, quadrature.
Shown: version 6; full guide row 453.

Withdraw your own code in public, itemised, with the files still retrievable (454) — A withdrawal earns trust when every finding can be checked and the files stay retrievable. Two shipped files were withdrawn with five findings, among them an alert-dispatch path naming siren and emergency-alert activation with no interlock, no dry-run default and no operator gate, a lead-time figure manufactured from a scenario built to escalate, and scoring by exact floating-point key that no FFT bin will produce.

Holds: not yet carried elsewhere. *Met in:* records and tools, control and circuits.
Shown: version 6; full guide row 454.

Say what the withdrawal does NOT affect, and what is worth carrying forward (455) — Two short sections stop a retraction from becoming a rout. The build guide never used the withdrawn code: reading archived data, siting a sensor, recording a floor and characterising a wind farm are hardware and method, and a builder writing a reader needs perhaps twenty lines. What survives is the mockup’s habit of showing why a tier fired, since a warning that arrives with its evidence is a different object from one that arrives as a colour.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 455.

The first experiment is not the thing you want to detect (456) — Measure the instrument’s noise floor first, and write the stopping rule before the work starts: if the signature is not visible in real archived data at real distances, stop and publish that. In the triple-star sessions a candidate rung put 0.603 of its rests’ time near a whole number, but perfectly locked rests would read only 0.44 to 0.53 through that window, so 0.603 was what luck looks like; the replication read 0.302.

Holds: not yet carried elsewhere. *Met in:* measurement, three body, waves.
Shown: triple-star; full guide row 456.

Publish the floor whatever happens (457) — A negative result with a floor under it transfers, and a positive result without one does not. The list is specific: the noise floor, the turbine signature, the correlation matrix, the false-alarm rate over quiet periods; and the paper gives its own target away, since if the design cannot reach its precision target,

saying so with the measurements attached is more useful to the next person than any promised warning system.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 457.

The author has built and tested nothing, said twice (458) — A specification must refuse to be mistaken for a result in the places a skimmer will land. The abstract and the head of the build guide both say this is a design and a specification, written so that somebody else can build it without needing to ask anything, and the performance table is labelled design targets to be tested by a pilot, not achieved performance.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 458.

Demote the exciting half to questions, each with what-would-have-to-be-true (459) — Keep the organising idea and drop the proposals that lack a lever. The earlier intervention proposals became open questions, each with a column naming what would have to be true, and where a budget applied it came out unfavourable by orders of magnitude; warning and sheltering remain the only life-saving intervention, while the idea survives that a lightning rod provides a preferred path, its leverage in timing and geometry rather than power.

Holds: not yet carried elsewhere. *Met in:* measurement, waves.
Shown: version 6; full guide row 459.

Decide which layer the failure lives in before prescribing (462) — Prescribing agronomy for a policy failure is a category error that a mechanism diagnosis catches for free. The physical shortfall is of order 1.7% of world production, not a global shortfall, and the closest historical precedent produced a price crisis through export restrictions and precautionary purchasing, so the exposure falls on import-dependent countries and policy response, calling for buffers, storage and restraint on export controls.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 462.

Rank by cost per unit delivered, and the top two need no new supply (463) — Choose the denominator carefully: cost per unit of nutrient delivered to a responsive soil quietly rules out everything delivered to an unresponsive one. Ordered that way, the first two of five instruments need no new supply at all and command the largest available multiplier, and the top action is stated by what it costs rather than what it promises: no new supply, no new production, no subsidy, no new technology.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 463.

Derived here, not quoted — noted on the column it applies to (465) — A reader who arrives at the table first never meets a note placed elsewhere, so provenance goes on the column it applies to. The source study treats the quantity as a variable but does not tabulate per-soil values, so the column was derived from the study’s reported parameters rather than quoted from it, and anyone who wishes to can recompute it.

Holds: not yet carried elsewhere. *Met in:* records and tools, biology.
Shown: version 6; full guide row 465.

The deprecated line, terminated with three findings so nobody re-derives them (466) — The paper began as the very thing it removes, and three findings close that line for the next person: no selectivity, since capture follows the atomic-capture Z-law and iron outcompetes the target on atomic number; no penetration, since cheap surrogates act only at an illuminated surface; and no delivery, since beam-based delivery does not miniaturise. The mechanism is real; delivering it deliberately at a competitive cost is what fails.

Holds: carried to one: failed in a field’s plough layer (substrate). *Met in:* biology, materials.
Shown: version 6; full guide row 466.

The fence applied at the source, in the paper that would most have benefited from keeping the exciting version (467) — The standing fence is executed by the author, in public, on his own founding idea. The deprecated route, which arrived in soil science from particle-capture work in radiotherapy, was removed from the analysis rather than carried as a speculative horizon, with its full derivation and designed falsification test kept in the first version of the record, so the trail survives without the claim, in the one paper where that angle drew the most attention.

Holds: carried to one: failed in soil science (phosphorus targeting) (break not stated). *Met in:* records and tools, biology.
Shown: version 6; full guide row 467.

Passing the famous check does not protect against the famous failure (469) — A canonical example that teaches the wrong lesson certifies the wrong check. The 1940 collapse was aeroelastic flutter under a steady wind with no matching external frequency, not forced resonance, so detuning the deck from every driving frequency would

not have saved it; a designer who runs the resonance check and finds a clean margin will believe the structure protected against the very thing that destroyed that bridge.

Holds: carried to one: failed in aeroelastic flutter, the 1940 bridge (substrate). *Met in:* driven oscillators, waves, control and circuits.
Shown: version 6; full guide row 469.

An inherited multiplier is the most dangerous kind of number in a design document (471) — A multiplier quoted from memory is a guess wearing a citation's clothes, so the check specifies a sourcing procedure and no numerical multipliers at all. An assumed input is an inherited multiplier too: in the Loop Atlas sessions a toy boat's hull drag area of 0.15 square metres, a barge's, decided the verdict, and with a clean hull of 0.006 square metres the same boat sails 1.1483 times the wind at 84 degrees.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits, kinematics.
Shown: Loop Atlas; full guide row 471.

State a margin's provenance inside the margin table (472) — These margins are engineering judgement calibrated to modelling uncertainty, said so where they stand, and must not pass as derived from a published code. Their ratio is derived, not asserted: a twenty percent error in entrained added mass moves the natural frequency by about ten percent, consuming the whole secondary margin. In the triple-star thread, one held prediction sat 1.6 of its own standard errors under the bar, another at about a quarter of it.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: triple-star; full guide row 472.

A design that does not state what it is not designed for has not finished its analysis (473) — Conflating design categories is how structures end up over-built for the wrong load and under-built for the right one. The design-event register separates what a design works in every day, what it shuts down above, what it must survive, and, mandatory, what is explicitly accepted as beyond design, stated rather than omitted, since the absence is usually discovered by the event itself.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 473.

Publish the compliance statement so an omission is visible without an audit (475) — An incomplete statement is a finding, not a formatting problem. A short fixed-form statement names the modes checked, the lowest margin found and its source, the instability assessment, the design life and horizon year, the scenario and why it was selected, and the next re-check date, so an omission shows without an audit: the visibility-matrix idea moved from the instrument to the paperwork.

Holds: not yet carried elsewhere. *Met in:* records and tools, control and circuits.
Shown: version 6; full guide row 475.

A document written to be cited rather than read (476) — A method document can stop trying to be a book: written to be cited rather than read straight through, it lets a design document reference it, complete the compliance statement and move on. Its worked examples come from one domain, but the method applies to a bridge, a mast, a gantry or a rooftop plant room, and the scope statement says it contains no novel physics and claims none.

Holds: not yet carried elsewhere. *Met in:* records and tools, control and circuits.
Shown: version 6; full guide row 476.

The disclosure block that says what was verified and what was not (477) — Make a disclosure specific rather than reassuring. The block names tools for what they did, places responsibility for correctness entirely with the named human author, and lists what was verified, down to every section pointer checked by an automated pass; then it states the attribution limit most disclosures omit, that the references are not yet verified against the publisher record, in the paper's limitations section as well, so a reader meets it from either end.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 477.

The useful output is not a machine, it is a selection guide (478) — Aim a document at the problem the field says it has. A published survey of four decades of prototypes and very few installations names the cause as a failure to converge, not a shortage of inventions, so the useful output is a guide that helps somebody choose; the thing is at least three things chosen independently, and changing the site can flip any one while the other two stay put.

Holds: not yet carried elsewhere. *Met in:* waves, control and circuits.
Shown: version 6; full guide row 478.

Kept because an idea merely deleted comes back (486) — Because the reason an approach lost is site-dependent, the next site may reverse it, so deprecated approaches are tabulated with their numbers: 15.8 million cycles through a dynamic seal in seawater, 23 millivolts from a kilometre-long electrode pair, a Carnot ceiling of 6.7 percent against a

device reaching 0.5. In R2W a reading that had already failed once returned for a new question, and it is owed a sharper test, not a fresh start.

Holds: not yet carried elsewhere. *Met in:* driven oscillators, waves, control and circuits.
Shown: R2W; full guide row 486.

Three lessons that transfer beyond this project (487) — The paper ends by moving its own portable method into an appendix. Check dimensions before geometry, since two deprecated ideas proposed a rule connecting quantities that do not share dimensions; count the conversions before optimising one, since one approach lost because it had six stages where the winning path had one; and check Carnot before designing a heat engine, which ruled one out before any hardware existed.

Holds: not yet carried elsewhere. *Met in:* measurement, control and circuits.
Shown: version 6; full guide row 487.

Attribution stated as a gap rather than omitted (488) — Say first that almost nothing is novel as a mechanism, rather than conceding it later. The attribution section is deliberately incomplete, states that no systematic search has been performed, and lists nine mechanisms with substantial existing literature and originators still to cite by name; recognising one element as long-established practice saved inventing a worse relief port, and two search passes that moved four of five claims downward are recorded.

Holds: not yet carried elsewhere. *Met in:* records and tools, control and circuits.
Shown: version 6; full guide row 488.

A capability is not novel because its parts are listed (493) — Nothing in the capability requires an invention, so its parts are listed precisely to claim no novelty for any of them. The framing is then corrected against the author's own interest: comparable sensory expansion for normally-sensed people has been demonstrated, so the space is not unoccupied, and the proposed version sits where the literature is thinner, a claim the document can support where unoccupied is not.

Holds: not yet carried elsewhere. *Met in:* records and tools, biology.
Shown: version 6; full guide row 493.

State the asymmetry in both directions, then decline to resolve it (495) — Surgical risk for convenience is a bad trade against a removable alternative and a straightforward one against total loss, and the asymmetry can run the other way: a large elective market refines technique, drives complication rates down and funds the long-horizon data small populations inherit. Two cautions are named, and the document states the asymmetry without resolving it, since that judgement is not one an engineering scope is entitled to make.

Holds: not yet carried elsewhere. *Met in:* biology.
Shown: version 6; full guide row 495.

Nearly every risk follows from one property: it cannot be taken off (496) — Deriving a whole risk taxonomy from one structural fact makes it checkable. Every governance mechanism that works for a phone (leave it, switch it off, hand it over, discard it) depends on separating the device from the person; an implant cannot be, so consent given at implantation persists and withdrawing it requires surgery. Eight categories follow, including an attack surface that terminates in a person's perception rather than their data.

Holds: carried to one: failed in an implant (substrate). *Met in:* control and circuits, biology.
Shown: version 6; full guide row 496.

A risk named without a remedy is only an alarm (497) — Write governance as things a regulator could require. The section lists an output ceiling enforced in hardware and not defeatable in firmware, a physical user-controlled mute independent of software, the implant's lens cap, a funded explantation pathway as a condition of approval, and separate regulatory classes for restorative and elective use, with the fence that it names hazards and deliberately gives no implementation detail.

Holds: not yet carried elsewhere. *Met in:* control and circuits, biology.
Shown: version 6; full guide row 497.

A falsification test says whether an idea is wrong; it does not say what would be built if it were right (498) — A paper that proposes without naming the device leaves a gap, and a scoping document fills it with a list of the things that must be true. Nothing in it is a design, with no transducer modelled, no field simulated and no phantom built, and confidence is graded by source: the regulatory limits are quoted from primary sources, the physics follows from them, and the biology the concept rests on is unestablished.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 498.

Name the gate that must not be skipped (501) — Order what would settle a claim by cost, and label the irreducible safety gate as irreducible. A distribution obtainable from data already collected at fitting gates the entire spatial argument and needs only data access; a simulation gates the next claim and needs no hardware; then a safety question with no direct evidence either way must not be skipped, and only then does anything get built.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 501.

THE STANDARDS CHANGE UNDER YOU WHILE YOU ARE READING THEM — and this is why the guide exists (503) — Cross-domain language, not the mathematics or the instruments, is what stopped the work: a word denotes two things, and its definition moves on both sides of the fence, which defeats learning the other field's term once. That barrier cost more than any calculation, so the work stopped and consolidated into a methods paper of substrates, hooks and traps, and the Field Guide is the document written because of it (row 504 gives the larger reason, the one that leads).

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 503.

THE GUIDE IS THE SOLVER'S SOURCE CODE, BECAUSE THE SOLVER CANNOT KEEP IT (504) — The method lived inside sessions, and sessions end, so each re-derived it badly, partially and differently, and the only persistent component was the human; the guide is the externalised state of a component that cannot hold state. Each row carries its price and name in one cell so it can be reloaded with zero context, and in R2W a count kept in the conversation reached only "at least the fourth" once the conversation's memory was cut.

Holds: carried to one, held there: counts inside one session. *Met in:* records and tools.
Shown: R2W; full guide row 504.

Expert witnesses — retained early, qualified before the read, called at the audit, and they accumulate into a library (505) — An independent voice is worth something at the audit precisely because it was lined up before anyone knew the answer. By Bear's correction, outside reviewers are expert witnesses in the courtroom sense, retained in advance and qualified on a specific question, introduced while the project is scoped and called at the audit; the roster then accumulates across projects into a holonomy library, so every finished project leaves the next better witnessed.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 505.

Prior art belongs at the wall, not at the idea — and the trade cuts both ways (506) — Run too early, a prior-art pass inherits the field's route and its blind spots, yet meeting every old wall from the priors' direction wastes time, so the trigger is a wall, not an idea. When the triple-star thread's pass found it had missed the circumbinary literature, a new tool, triple_hw_edge v1.0, first reproduced Holman and Wiegert's own example, an edge at period ratio 6.878, then put the pair's edge at period ratios 4.03 to 5.36.

Holds: carried to four: held in triple star's rebuilt harmonic; held in new tool triple_hw_edge v1.0; held in Loop Atlas's noisy needle; failed in triple star's edge and test target (break not stated). *Met in:* records and tools, three body.
Counted: A second reader 1. *Shown:* Loop Atlas, triple-star; full guide row 506.

Flucht nach vorn — failure is the main road, and the method hands out no participation trophies (507) — The whole guide exists because holonomy delivers failure routinely, so a null is the road, not a branch off the successful path: not fall back, but escape forward. Yet a lesson is not conferred by having failed; it has to be extracted deliberately, by looking for specific things, and the shortcuts, easy buttons and geometric cross-checks nobody teaches are earned, which is why the long walk is not wasted.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 507.

Do not stop when you find an answer (509) — A found answer is the most reliable way to stop searching, and the structures this method reads are plural by nature: patches come in families, combs have teeth beyond the first, and a lattice has more than one coincidence angle. Bear's margin note makes it the first rule of the walk, because the second and third findings in this corpus were so often the better ones.

Holds: not yet carried elsewhere. *Met in:* measurement.
Shown: version 6; full guide row 509.

Each half is known; the equivalence and its conditions are the contribution (511) — Write down how a claim shrinks under prior art before any draft exists: the seed expects each half known in its own field and sizes the novel object as the bijection, the iff conditions and the adaptive-orphan observation. Then look for the model that holds both halves as special cases; for the Loop Atlas's mirror cats eight sources stated no mirror, and Iwai's two equal cylinders on a joint that allows any twist is such a model.

Holds: not yet carried elsewhere. *Met in:* records and tools, topology.
Shown: Loop Atlas; full guide row 511.

Two fences, and they are not the same fence (512) — One fence is disownment: material out because the physics underneath it is not trusted, never cited, never carried forward, and raised at volume if it reappears. The other is readiness and ownership: material that is not ready for the world or belongs to other people is tracked and worked while every public reference is removed. Collapsing the two would either publish something unready or bury something merely unfinished.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 512.

A working instrument that annotates its own vocabulary collision (513) — Place a vocabulary fence where the hand is, not in a preface. A control-room simulator carries a note beside the control it applies to: this sweep moves the drive in time, while the other person's term is a velocity gradient on the bunch, cousins and not the same object. The same tool labels its own tier in the interface: display toys calibrated to session anchor numbers, not engineering simulations.

Holds: carried to one, held there: control-room simulator's interface note. *Met in:* records and tools, control and circuits.
Shown: version 6; full guide row 513.

Established mathematics beats conjured mathematics, because you can go and check me (516) — Decline the compliment that the ideas are more fundamental than physics: this is established mathematics with names on it, which is better because anyone can go and read it and check. One piece of mathematics turns up in two distant fields because they share a skeleton, not because one sits underneath the other, and the eigenvalues do not want to spread; they repel as a consequence, and no intention is needed.

Holds: not yet carried elsewhere. *Met in:* records and tools, topology.
Shown: version 6; full guide row 516.

Do not hold the build for a late ornament (524) — A structure that waits for its last decoration never ships. Bear's assembly ruling of 2026-09-13 holds that papers finishing while the guide is assembled will not change its shape, so they arrive as an edit if late; the test is whether the incoming thing changes a limb, a station or a fence, and if it only adds a row, the register takes it afterwards and the build keeps moving.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 524.

The fence in the paper was correct; the conversation violated it (525) — Knowing which one broke is the whole of the repair. A topological invariant requires a periodic lattice with a well-defined Brillouin zone, and the object here is finite, irregular and open, so the invariant is not defined on it; the deposited paper had already said so, finite irregular patch, kinship, reported qualitatively. The session discussion violated that fence, so the correction addresses session framing only and nothing in the paper needs revision.

Holds: carried to one: failed in finite, irregular, open patch (substrate). *Met in:* records and tools, topology, materials.
Shown: version 6; full guide row 525.

Correctly measured and wrongly interpreted (526) — Re-measuring cannot catch this kind of error, because the measurement was never wrong; only asking what the number is a signature of can. A companion lattice was recorded as reproducing the comb at a gap ratio of 1.297, and the number was right, but the comb's signature is its divisor structure: tested properly, the open version gave scattered teeth that divide nothing, while closing the chain gave the divisor set exactly.

Holds: carried to two: failed in companion lattice, open chain (knob); failed in the latch's three-halves (break not stated). *Met in:* measurement, materials, number theory.
Shown: version 6; full guide row 526.

Three cherry-picked points, replaced with the whole set (527) — Three points can be drawn through by any curve already believed in, so the repair is not a better fit but a weaker sentence. A claim that a feature strengthens with size rested on three points; the full set read 1.20, 3.04, 1.55 and 1.93, with rectangles at 2.19 and 3.54, scattered rather than monotone, and the replacement says only that the feature is elevated on every open patch measured and does not trend with size.

Holds: not yet carried elsewhere. *Met in:* measurement, materials.
Shown: version 6; full guide row 527.

The result exists only as a contrast, never as a property of one curve (530) — A result that exists only differentially has to be reported differentially, because quoting one arm of it is not a simplification but a different and false claim. In the apodization families the separation grows monotonically across all six weight levels, stronger

evidence than any single depth, yet the finding is the gap between the two families, not a number belonging to either.

Holds: not yet carried elsewhere. *Met in:* measurement, quadrature.
Shown: version 6; full guide row 530.

Every bridge must let you build, predict, or break something (533) — A cross-domain programme's central risk is the bridge that feels true and does nothing, and three verbs are the cheapest test for it, applied before the bridge is written down rather than after it is published. Observations that produce only a feeling of connection are out; the filter was earned from the author's own earlier framing, correct but underspecified in a way that permitted a misreading.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: version 6; full guide row 533.

I thought it was a destination. It is a method you could teach kids in a sandbox (538) — A method is worth more than a destination, because a method transfers to every new substrate and a destination is just a place you arrive at once. The author's correction of 2026-09-15 to his founding expectation traded a super-app that would make holonomy or geometry problems give up their secrets for a method, the sum of the experiences and the purpose of the book.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 538.

The four limb chapters — named 2026-09-15 (539) — Every row in the register needed to know which house it lived in, and the names were chosen to carry the guide's own character rather than a taxonomy someone else would recognise. L1 is The Method as an Instrument, holding the walk, the channel, the floor and the differential read; L2 the terrain the method traverses; L3 the loops built rather than read; and L4 The Ledger, what every project cost, taught, got wrong and yielded.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: version 6; full guide row 539.

A selftest failure is a finding when the code was right (541) — When a check fails, test the check before touching the data, since a failed check is a claim about the check as much as about the data. A dipole test asserted that an alternating ±1 field would hold its net near zero, and the worst residual was 7.0 against a median boundary of 30 edges: cancellation is real, but its residual scales with the perimeter, not the area, so the assertion was corrected and the code left untouched.

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators, three body.
Counted: A check that cannot fail 1. *Shown:* R2W, triple-star; full guide row 541.

A harder solver tests the solver, not the quantity — and only if it changes one thing (549) — A validation run changes one thing, or it is several experiments reading as one, and passing a harder solver shows only that the machinery is sound. The latch paper's first version read a drop from 1.55 to 1.25 as its exponent sliding with solver strength, but the stronger run had changed the minimiser, the point count and a spring at once, and the pair linked six times read 22.37 at 72 points, 17.51 at 160 and 16.38 at 1,280.

Holds: not yet carried elsewhere. *Met in:* measurement, topology, materials.
Shown: version 6; full guide row 549.

A staircase across an integer measures whatever else climbs with the integer (550) — Vary an integer and read a magnitude off the staircase, and everything else that changes with the integer must be written down and pinned. Two loops linked one through six times rose from 9.34 to 17.15, but on a torus of fixed size more winding laid down more rope, 13.14 per strand at linking one and 25.93 at six; held at equal rope, the energy fell as the linking rose, and the paper's second version withdrew the claim.

Holds: not yet carried elsewhere. *Met in:* measurement, topology, materials.
Shown: Linking and Magnetic Forest; full guide row 550.

A check certifies only the question it could have failed on (551) — Two papers found this the same month. A linked pair shrunk 16.7-fold kept its linking at −1.0000, but a uniform shrink cannot change a Gauss integral, so the test checked the reader and not the latch. Give every verdict one stand-in on which it should pass and one on which it should fail.

Holds: carried to one: failed in forest suite's genus flux-ladder count (wall). *Met in:* measurement, records and tools, topology.
Counted: A check that cannot fail 7. *Shown:* Loop Atlas, R2W, triple-star; full guide row 551.

A citation checked for existence is a claim you have not verified (552) — Check what a paper says against the paper, sentence by sentence, and credit the theorem that exists at the level it is written. The first prior-art pass credited Freedman and He with a proved three-halves energy scaling they do not prove, since their "3/2-energy" is the

name of a quantity; and of eight statements a draft cited to Mardling (2013), two had taken a nearby sentence for the claim, which a record check would have passed.

Holds: not yet carried elsewhere. *Met in:* records and tools, topology, three body.
Counted: A word that changes meaning across a boundary 1, A second reader 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 552.

An edit is a claim, and only the built artifact proves it landed (553) — Build it, extract the text, look at the figures, and assert that the new words are present and the old ones gone, because none of it is visible in the source or in a build that exits cleanly. The latch paper's first version shipped a Figure 5 plotting an earlier run that was never deposited while its caption gave the deposited numbers, and a strict read later found a converter's typography in 85 of 1,141 pieces.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 11. *Shown:* Loop Atlas, triple-star, Field Guide sessions; full guide row 553.

Right size, right sign, wrong shape: a mechanism can pass two of three and still not be the whole cause (555) — A test of size alone would have passed this mechanism, so report size and shape apart. A one-layer fit whose Faraday terms are too weak turns a sunspot map's arrows by 3.6 to 9 degrees, flipping with the field's polarity as the maps' turning does, but its turning crosses zero near 1,000 G; planted into the real maps it left an rms current offset of 2.46 mA/m² against the one-sign ramp's 1.05, part of the mechanism rather than the mechanism.

Holds: carried to one: weaker in triple star's island (break not stated). *Met in:* driven oscillators, three body, solar.
Counted: A second reader 1. *Shown:* R2W, triple-star, Dancing on the Sun; full guide row 555.

Two readers of one light are not a witness (556) — The difference between two fitters is not the difference from the Sun, so a two-fitter disagreement is never promoted to a measurement of bias. Fitting HMI's real Stokes light with a textbook one-layer fitter, the arrows differed from HMI's by −5.04 and −3.88 degrees in strong field and +2.28 and +1.75 in weak field, a real finding about the two readers; the external witness is a second instrument on the same region, not a second fit of the same photons.

Holds: not yet carried elsewhere. *Met in:* measurement, solar.
Shown: Dancing on the Sun; full guide row 556.

Two estimators of one thing are not a disagreement, and the ledger must say which one it used (558) — A number that changes when the estimator changes is not wrong but under-specified until the estimator is named beside it. The paper's 5.5-degree turning (rms 0.76) was fitted on two regions' median offsets, ten numbers, while the planting tool fitted the same ramp on every band-map, 29 numbers, and got 5.30 (rms 1.05) on the same six maps; both are right.

Holds: carried to three: held in triple-star resonance angle's slide; failed in median step at birth 7.210 (knob); held in two fresh births. *Met in:* measurement, three body, solar.
Counted: A word that changes meaning across a boundary 2, A second reader 1. *Shown:* Loop Atlas, triple-star, Dancing on the Sun; full guide row 558.

A pre-written expectation corrected within a minute, and the run restarted rather than annotated (559) — An expectation written before the run is worth exactly as much as its being written before the run, and one edited while the run runs is worth nothing, so the honest repair is to start over and say so. The width run's docstring predicted crossovers of 560 and 1,120 G from a scaling slip; the run was killed, the expectation corrected to 690 and 1,390, and the run restarted from nothing.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools, solar.
Shown: Loop Atlas, Dancing on the Sun; full guide row 559.

Redistribute, never cut (569) — A long note is a sign that its material belongs somewhere more useful, not that it should be shorter, and moving text is checked word for word. Bear's rule held for the Latch Atlas's 21 notes and one 1,862-word paragraph: every clause found a home in the body, a caption, the reference list or a two-line note, and a script probed each piece into its new place, zero missing every time.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 3, A second reader 1. *Shown:* triple-star, Latch Atlas; full guide row 569.

Every number names its run, and the tool that strips them counts them before and after (570) — A provenance rule is kept by a count a script prints, not by good intentions. The sentence "numbers before 930, after 930, differing none; run stamps left in body: 0" let a 92,000-word book be rebuilt forty times without a number drifting, and a number printed only on a page is untraceable even when true: a triple's 0.909 could not be found until a trace tool with its own ledger was built.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Counted: A check that cannot fail 2, The built artifact is the evidence 2, A second reader 1. *Shown:* Loop Atlas, R2W, triple-star, Latch Atlas; full guide row 570.

Citations are claims: run a checker after every build (571) — A citation checker costs a second per build and catches what reading misses. On the Latch Atlas it resolved 157 author-and-year citations to exactly one reference row each, found every row cited and every run stamp with a folder or a deposited twin, and caught a version DOI passed off as a concept DOI, two names with the wrong initials, and works cited that no note had named.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: Latch Atlas; full guide row 571.

The crate is built by script from the folder the manifest checks, never gathered by hand (572) — A crate gathered by hand can carry stale files; one built by script from the folder the manifest checks cannot. Three of the five records found on a laptop disk were stale versions, and the Latch Atlas's crate could not have been, while a manifest verifier reading "differ 0" before every ship turned "the files are gone" into a ten-minute problem instead of a lost week.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 2. *Shown:* Loop Atlas, R2W, triple-star, Latch Atlas; full guide row 572.

The person with the questions is the instrument (573) — A proofreader who asks "is this a known thing?" is running the prior-art pass by hand, and it works. The author's marks on the Latch Atlas's final read, among them whether five dividing ten is known and whether twisted light latches, each changed the Latch Atlas; and in R2W his strokes built a linker's shared mode that lifted coverage from 10 to 25 of 35 drives.

Holds: carried to one: failed in weights 0.768 and beyond (seam). *Met in:* driven oscillators, three body, reading pictures.
Counted: A second reader 1, Coupled clocks and event time 1. *Shown:* Loop Atlas, R2W, triple-star, Latch Atlas; full guide row 573.

A wrong upload is survivable when the record is honest (574) — The rule that working notes stay out of a deposit stands, but what saved the day was that the notes had been written carefully enough to survive being read. A fourth file went onto the Latch Atlas's record by mistake, five working records in the working voice; nothing private was in them, the Latch Atlas cites them by name, and a line added to the description in place turned the mistake into a supplement.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: triple-star, Latch Atlas; full guide row 574.

A size confirmed on a population is not a size for an individual (579) — A population confirms a sign and a typical size, neither of which fixes the size for a member, so say of every number which it was confirmed on. The rotation's 2.97 degrees, fitted on the 2011 regions and confirmed on 82 regions of a blind test, shrank the net current on only 15 of 37 near-centre maps when applied to nine regions followed day by day, worse than a coin, while the sign survived every test.

Holds: carried to four: held in 82 regions of a blind test; failed in nine regions followed day by day (break not stated); held in individual regions followed day by day; failed in single columns at weights 3.00 and 5.00 (seam). *Met in:* measurement, driven oscillators, solar.
Shown: R2W, Dancing on the Sun; full guide row 579.

Undecided is a verdict, and a rule is written for the sample that will exist (580) — UNDECIDED is reported as written, never nudged into a pass by loosening the rule afterwards, and a rule's sample-size clause is written against the archive as it is. Of five blind tests with rules fixed and fingerprinted before the maps were fetched, three of the last four came back UNDECIDED, one because the rule demanded six judged regions and five were judged, one at 19 of 37 and one at a chance of 0.054 against 0.05.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools, maps and chaos.
Counted: A check that cannot fail 1. *Shown:* Loop Atlas, R2W, triple-star, Dancing on the Sun; full guide row 580.

A companion citation is written to be true in either order (582) — A sentence in a deposited text that names a companion must be true whichever record exists first. The plan was to deposit the Latch Atlas first, and the paper went up first; the Declaration's sentence had been written to stay true either way, and the pair was closed in the records' metadata in place, "is supplement to" on the paper and "is continued by" on the Latch Atlas, with no new version of either.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: triple-star, Dancing on the Sun, Latch Atlas; full guide row 582.

A rule bent with its guard in place, and said so (583) — A standing rule can bend when the guard it exists for stays in place and the bending is said in plain words. A laptop tool got a second version in one sitting, against the rule of one version per sitting; the hash-checked launcher was in place, a fidelity run reprinted every version-one line before the launcher would run, and the status page said so.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: R2W, Dancing on the Sun, Field Guide sessions; full guide row 583.

A strip of snapshots teaches motion where a single object fails (585) — An analogy that is a single object carries a shape and loses the order of events, and a loop is a motion. The wing drawn as a strip of frames, good timing beside bad, made a hard idea land, while the butter knife, one object standing in for a motion, did not; the cheapest honest picture of a motion is its steps.

Holds: not yet carried elsewhere. *Met in:* kinematics, reading pictures.
Shown: Loop Atlas; full guide row 585.

A corrected number moves everywhere, or the correction becomes a second defect (588) — Old figures kept in a table beside their fixes give a reader two homes for one number, and no way to tell which is current. Withdraw a number everywhere the same day: thin widths withdrawn from one row had stood in a book for a day, wrong by 19.7 and 37 times, and every working copy that carried them was rebuilt.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Counted: A second reader 2. *Shown:* Loop Atlas, R2W, triple-star, Field Guide sessions; full guide row 588.

Keep the author's words verbatim from the first turn (589) — A paraphrase is a compression, and a compression cannot be expanded back into the words it replaced, so the voice file opens on the first turn and keeps the small words too. Asked about a tilted floor, the person answered "two u shaped curves offset side by side (intersecting)"; a registration left out the bracketed word, though the runs showed the gap closing steadily from 0.0587 at period 5 to 0.0056 at 13 as if the curves cross.

Holds: not yet carried elsewhere. *Met in:* records and tools, reading pictures.
Counted: The built artifact is the evidence 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 589.

A session jump travels as one zip with its handover inside (591) — Loose files suit a person reading them one at a time, but a session jump needs one container the next session can open whole, so the work restarts from the record rather than from a pile. A new session could not see 94 loose files, and another lost a day when its ferry arrived as five of six pieces, 1,163 files absent; at every move the whole crate now ships fresh, with each piece's md5 in an index sent beside it.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: Loop Atlas, R2W, triple-star; full guide row 591.

The assistant's own queue is a standing section of the handover (592) — Deferrals the assistant makes for itself become orphans unless they are kept in view beside the author's. Its queue slid behind the author's new pictures all day, a habit rather than a request; and the Loop Atlas's Stretch and Squeeze Explorer, built on 13 October, was named in the next handover only as a published page and carried by none after it, until Bear asked after it on 16 October.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 1. *Shown:* Loop Atlas, R2W, triple-star, Field Guide sessions; full guide row 592.

A green build is not evidence that anything changed (593) — An exit status says the script ran, not that the artifact holds what was meant, so after every edit re-read the file and after every build assert on the finished artifact's text. Three times in one deposit an edit script hit an anchor it could not find and exited before writing; the rebuild ran on the unmodified file, and the linter returned CLEAN.

Holds: carried to one, held there: triple-star register's promise about saved tables. *Met in:* measurement, records and tools.
Counted: The built artifact is the evidence 3, A check that cannot fail 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 593.

Renumbering in place cascades into its own output (594) — Rebuild a numbered list whole and assert the sequence, and after any structural move check where every section and row reference lands. Replacing "6." with "7." and then "5." with "6." renumbers the item just renumbered, and it shipped a reference list reading 12, 13, 16, 17, 14, 15; a moved section does the same to its references, since "see Section 5" still resolves, to whatever now sits in slot 5.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: A word that changes meaning across a boundary 1. *Shown:* Loop Atlas; full guide row 594.

A number that is everywhere and measured nowhere has drifted onto a different noun (595) — Agreement among sources that cite each other is one source counted many times, so find who measured the number and read what it was a number of. The claim “40 per cent of soil phosphorus is locked” came back from every search, but the figure in the primary literature, 30 to 40 per cent, is the share of the world’s arable land where phosphorus limits yield: it had drifted subject.

Holds: not yet carried elsewhere. *Met in:* records and tools, biology.
Shown: version 6; full guide row 595.

Absence is invisible when you read for presence, so run the coverage check (596) — A source doing structural work in an argument is searched for every claim made on it, and what is missing is reported as absence, weaker than a stated position but evidence. A paper recommending five interventions leaned on an expert review for its framing, and its claims, written out as a list and searched for in the review’s full text, found none of the five discussed, which reframed a whole section; the check took about ten minutes on 82 pages.

Holds: not yet carried elsewhere. *Met in:* records and tools, biology.
Shown: R2W, Field Guide sessions; full guide row 596.

A source with no text layer searches as empty (597) — A text extraction that returns almost nothing is the tell, so run OCR before any search counts as a read. A 34 MB print-to-PDF file yielded 82 bytes of text, one form feed per page, and every search of it would have come back “not found”, true of the file and false of the document; at 200 dpi tesseract reads body text well and tables of contents badly.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: R2W; full guide row 597.

Unit traps sit where two fields report the same thing differently (598) — Label the unit, the chemical form and the point of sale on every number before two are set side by side. Field trials report elemental phosphorus and fertilizer claims report P_2O_5, a factor of 2.291 apart; potassium against K_2O is a factor of 1.205; and a free-on-board benchmark and a retail delivered price for one product in one month differ by freight and margin, about $380 against $495.

Holds: not yet carried elsewhere. *Met in:* measurement, biology.
Shown: version 6; full guide row 598.

Conclusions go stale (599) — Re-read the conclusions against the body as the last editing step, every time, because a conclusion summarizes a body that keeps changing beneath it. One deposit’s closing section told readers that the most useful action was a $10 experiment, three pages after its policy section had ranked something else first; and the triple-star islands paper printed ‘160 of the 162 escapes’ after a later run had added four.

Holds: carried to one, held there: triple-star islands paper’s escape count. *Met in:* records and tools, three body.
Shown: triple-star; full guide row 599.

Assume there is no future work (600) — Anything parked for the next pass does not get done, so when something cannot be resolved, close it in the artifact: disclose it, scope it, or state it as unresolved, where a reader can see it. A reading written as a finding counts too: R2W’s book stated a fold as found that was read after the data with its last steps not fully converged, so its finish list says to word it so, or to run FX-2c first.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: R2W; full guide row 600.

The auditor’s expectation is a card too (602) — A wrong expectation turns a correct result red, so expectations are written, sourced and audited like results, and a red verdict first checks its expectation. On the mirror audit’s first pass a twist read red at an even share of 0.970, because the audit expected it to flip; the twist belongs to the blade, so it should stay put, and on the corrected card it did (0.030).

Holds: not yet carried elsewhere. *Met in:* measurement, kinematics.
Counted: The mirror audit 1. *Shown:* Loop Atlas; full guide row 602.

A run file is written only by its tool (610) — A number that cannot be regenerated cannot be checked, so a new sweep is a new part of the tool, rerun, never an appended line. Four runs of record had no script that makes them, and one now disagrees with a new run; in R2W two readings with no file that could make them again were written as scripts and rerun, and they reproduced the record (runs WALK-2c and SLIDE-1).

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 1. *Shown:* Loop Atlas, R2W; full guide row 610.

Small faults that each cost a round trip, with their fixes (611) — None of these faults is physics; each cost a round trip once and need not again. Launcher logs came back in UTF-16 because the shell’s tee defaults to it, so launchers set UTF-8; and a kill command that matched its own shell kept coming back, once ending the shell that sent it (exit 144), so each background job is now stopped by the process number kept at its launch.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 1, A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 611.

The cost of precision depends on where you stand (659) — The cheapest place to be accurate is away from your own resonance. The filter needed against a gear's room falls with slope −0.92 on the watershed and with slopes −0.60 and −0.57 one step off it, as the FM reading predicted before the blind runs; and in R2W a second knob relaxes a constraint so the first can move where the cost is lower, the fourth harmonic bringing c to 0.30 and 0.28 of the edge's at the joint points.

Holds: carried to two: held in the filter's slopes on and off the watershed; weaker in holding an array of riders in lock (break not stated). *Met in:* driven oscillators, control and circuits, quadrature.
Shown: Loop Atlas, R2W; full guide row 659.

A near-miss is a loan, not a gift (660) — Being near a simple fraction buys time, not grip, and the interest rate is the detuning. Near a ratio of 2, hold time multiplied by detuning and by the tooth rate tends to 1.53 radians, the good band's half-width, with nothing fitted; a triple's slip period times offset, 0.978 turns by median, shares the slope of −1 but counts whole slips, so name what each constant counts before setting them side by side.

Holds: carried to one: weaker in triple star's slip period (break not stated). *Met in:* driven oscillators, three body.
Counted: Coupled clocks and event time 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 660.

Adjust, don't retire, and know when understanding the break is enough (661) — A miss redesigned aloud and run again is the method's main road, though for a one-shot test of a picture, locating the break and parking the idea is the whole job. Of 104 misses one book carries, the instrument is named in 58, a wrong guess in 50 and nature in 25; the instrument fails before nature does, and over the triple-star thread's month, redesigns landed 35 times in 79 read.

Holds: carried to three: failed in a tap in the Loop Atlas (break not stated); failed in five of eight slices over 600 passes (ruler); failed in slices from the star's eccentricity 0.26 (seam). *Met in:* measurement, records and tools.
Counted: Coupled clocks and event time 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 661.

Mail between sessions: tag every letter, manifest every packet, never infer a delivery, and keep it small (676) — Carrying a letter twice is always safe and skipping one is the only risk, so every letter opens with one line of tags, every packet carries a manifest of md5s, and a letter is marked delivered only on the receiving session's own word. One mailbox inferred a delivery from the other room's count of letters, and the letter had never arrived.

Holds: carried to one: weaker in Loop Atlas's needle (FOLD_OR_FLIP, RUNG_ABOVE) (break not stated). *Met in:* records and tools.
Counted: The built artifact is the evidence 2, A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, R2W, triple-star, Field Guide sessions; full guide row 676.

When the work feels slow, look for the missing lane before cutting the method (679) — The habits are the method, not its cost, and the answer to slowness was a second lane. Asked whether he was adding requirements faster than the work moved, Bear was first answered by a plan that slowed the three habits; the revised plan kept all three and added runs in the background while pages are written, and the Loop Atlas book's parts written in full went from 11 to 13 of 21 over two such days.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: Loop Atlas; full guide row 679.

Keep a ledger of every knob, with no-data rows, and turn the untouched ones (684) — The cheapest run of a sitting can be its most surprising. The weak wave's strength in a toy rider had never been turned, and turned it proved to build every groove: with it off no rider lands on any fraction but 0, and the number of distinct fractions runs 1, 7, 19, 7 and 5 at strengths 0 to 0.6; the instrument's own knobs belong in the same ledger.

Holds: carried to one: weaker in ripple strengths 0.25 and 0.35 (knob). *Met in:* measurement, driven oscillators.
Shown: Loop Atlas, R2W; full guide row 684.

Hold back the data while the eye is learning (716) — With the numbers in hand a reader measures instead of looking, and the looking is the skill being taught, so Bear held back the data until the examples were finished. The practices that held the data back produced blind reads a scorecard can judge, and where a planted case printed a judged number before the run of record, such as one pairing's share of arms and legs (0.834), it is said aloud in the registration.

Holds: not yet carried elsewhere. *Met in:* measurement, reading pictures.
Counted: A check that cannot fail 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 716.

A container's run lives only while its session does (720) — A container can be shut down soon after its session goes quiet, and a run it holds goes with it. Give long runs to the laptop, keep runs inside a turn, and have every run write what it has finished as it goes: one sitting's workspace restarted at 22 exchange openings and nothing was lost.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 720.

When the whole toy turns out to be a textbook object, its map becomes a bench (726) — Each match with a textbook is a ruler checked against a known length, so the claim of anything new moves from the physics to the reading. Stripped down, R2W's rider is a particle in two waves with drag and a steady slope, its maps' landmarks already named, the devil's staircase among them; with the ripple off, its stopping pulls at weights 10, 5 and 2 matched the exact values to the dial's step.

Holds: carried to three: weaker in R2W's rider (substrate); held in R2W rider with ripple off; held in triple star's 24-gon clockwork. *Met in:* driven oscillators, three body.
Counted: A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, R2W, triple-star, Field Guide sessions; full guide row 726.

A file sent is not a file in the working folder: measure the folder, and make the last step mechanical (733) — Guessing at a folder that cannot be seen had cost two days. Twice in one morning a person was told to start a launcher not sent in that message, and a tool that read his folder showed why: files land in Downloads and reach the working folder by hand, and nothing sent after the morning of 14 October had made the trip; in the triple-star thread an intake script now copies them by fingerprint and ends READY or NOT READY.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 2. *Shown:* Loop Atlas, R2W, triple-star; full guide row 733.

Let the command write the record: stamps, counts, ranges, and the line that cites a check (735) — Run the check, read its output, then write the line, naming the tool that printed the number. A sitting that had learned to read the clock still wrote six stamps ahead of it, such as 22:35 for 22:33, until the notes printed their stamp from the clock inside the same command that appended them; and five FBP-1 outputs' line counts written from memory were wrong, 25 lines as 31 and 61 as 64.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 4. *Shown:* Loop Atlas, R2W; full guide row 735.

A running job is read by its own done line (736) — A run file is a snapshot until it says it has finished, so check the done line of each file a reader takes and watch a running job by its line count. A run recorded as stopped from a file taken mid-run was still live, and the next answer pointed the person at the wrong batch; in R2W, RATCHET-50's wait loop gave up after 250 seconds and its reader ran on files still being written.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: R2W; full guide row 736.

A check that does not gate the build is only a report (740) — Chain the build behind the check, so that a failed count stops the build instead of printing beside it. A cell count once ran in the same command as the builds, which went ahead before its result was read, and it found a cell of the paper's RATCHET-1 row holding absolute-value bars, rewritten in time only because the output was read before sending; a check whose answer never changes should be fed a case known to be bad.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 6, A check that cannot fail 2. *Shown:* Loop Atlas, R2W, triple-star; full guide row 740.

A long piece of work needs a finish line it can see, counted in words and in tests (765) — A project that must finish freezes its cards and writes its surprises down instead of running them, and counts its book by part at every review. At one session's close the proposed list held 29 tests by count with 26 runnable cards beside it, and every exchange named three more surprises; following one toy's window took the Loop Atlas book's Limb G to 13,892 words, more than its Limbs A to F together (8,527).

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: A word that changes meaning across a boundary 2. *Shown:* Loop Atlas, R2W, Field Guide sessions; full guide row 765.

Give a long project's exchange a fixed shape: the old card first, then one prose step, the misses' redesigns, and one test chosen by what it would change (767) — A slot that is not first loses to the live thread, so the old card opens each exchange, before anything else. In one project's five exchanges the older-card slot lost to the live work four times in five, and moved to open each exchange it held five of five; each surprise goes on the proposed list with one line saying what its test would change in the book, and the one test an exchange runs is chosen by that.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: Loop Atlas, R2W; full guide row 767.

Work with the person in plain words: never ask him to judge from his equipment, take the wider door his words describe, and check the files before reassuring him (770) — Say what a word means in the work the first time it appears, and label a default as a default with who set it, so the person can overturn it without first having to find it. The phrase “close the sitting” was read as leaving the session at the end of his working day, and “pack a save point” replaced it; every reminder or question goes last in the exchange, after any file is sent.

Holds: carried to one, held there: the timing off the mirror line (RATCHET-45). *Met in:* records and tools.
Counted: A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, R2W, triple-star, Field Guide sessions; full guide row 770.

Two deposits that cite each other’s unpublished parts cannot both go first (774) — Let the one that deposits first carry its own content, and let the second cite the first’s DOI. The guide was to wait for R2W’s book, and the book was to cite the guide’s rows 686 to 719, which exist only in the guide’s next version; since waiting grows scope, freezing the content and setting the deposit date are two separate decisions, and a version waiting on another project closes its intake that day.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: Loop Atlas, R2W, triple-star, Field Guide sessions; full guide row 774.

A citation is checked by someone who may not fill a field from memory, against the source the text names, and your own records too (775) — Read the source the text actually names before correcting the text, and check your own records as you check everyone else’s. Three checkers told that no field may come from memory led to seventeen corrections to one guide’s text, and a fourth, blind to the list, rechecked 22 entries; resolved from Zenodo’s API, four of the five titles of the author’s own papers in its drafted reference list proved to be the Declaration’s one-line descriptions.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: A second reader 1. *Shown:* Loop Atlas, triple-star, Field Guide sessions; full guide row 775.

Keep the person’s side of the work small, and his finish line written down (777) — Fix the plumbing first, keep turns short with at most one question and always a default so that a yes is enough, and hold a written, finite finish line for each piece of work. On 15 October, with the probe FW2 back and every guess of it borne out, the person wrote that the work was “losing its fun factor and becoming work”: the science was moving, and what wore him down was everything around it.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: triple-star, Field Guide sessions; full guide row 777.

Parked threads stay in the record, each with its test, and the person’s own pictures with them (782) — A backward pass keeps what is not the paper as side threads worth keeping, each parked with its test, and the person’s own pictures stay with their tests too. One thread’s chest of 13 October held, among others, an escape tripwire with 74 escapes now to try it on; and in R2W the parked card FOLD-DIR, its redesign written when KN3-3 missed, was registered nine minutes into the exchange that took it up.

Holds: not yet carried elsewhere. *Met in:* records and tools, three body.
Counted: A check that cannot fail 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 782.

A flag says where to point a specialist, never “this is new”, and a session’s end lists what was measured and not looked at, and what is exact but not yet priced (783) — Every number in a flag comes from the record with its run, and the flag names who should look, not what is new. One thread’s chest of 15 October flagged a one-way crossing in the low-side layer, where of 18 chaotic stars born below the island that circulate, 17 turn the way stars above it turn, for a specialist in resonance overlap and separatrix crossing, and listed beside it what was measured and not looked at.

Holds: carried to one: weaker in half the smallest separation he tested (break not stated). *Met in:* records and tools, three body.
Shown: Loop Atlas, triple-star; full guide row 783.

Count which lessons keep arriving on their own, rank them by the count, and spend on that synthesis the turns that record-keeping would take (798) — A guide’s own finding is which lessons kept arriving independently across projects, so open each version with them, ranked, and fold a backlog by adding counts rather than rows. Counted through 23 October, the check that cannot fail leads with 131 arrivals; and the session that kept the ledger shipped 130 files, 66 of them revisions of its record, and never wrote the ranking. Re-bin, don’t grow.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: Field Guide sessions; full guide row 798.

A bar moves only before the first scored run, and in the open: count the tries at one number, keep the band the first fair try set, and carry an amended bar into every planted check (808) — A bar is the line a result must clear, and it is worth something only if it stays where it was put. Count the tries at one number and keep the band the

first fair try set: the triple star's drag size passed 5 of 5 on its fifth try, and the paper says so. When a bar must move, move it before the first scored run, in the open, with every planted check that tests it.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools, three body.
Counted: A check that cannot fail 1. *Shown:* R2W, triple-star; full guide row 808.

Trace every number in a paper to a ledger at every pass, with a scan that reads only sources, and read the scan's leads by hand (811) — A paper's numbers are only as good as the record each can be traced to, so the tracing is redone at every pass by a scan whose sources are the runs' own ledgers, and its flags are read as leads. On one pass of the triple-star islands paper the scan found 1,023 numbers, 1,020 held by a ledger; on 18 October three numbers had traced only to the provenance table's own ledgers, which print the draft's descriptions back.

Holds: not yet carried elsewhere. *Met in:* records and tools, three body.
Counted: A check that cannot fail 3, The built artifact is the evidence 2. *Shown:* triple-star; full guide row 811.

Rebuild a published model from its own table: report the structure and the digits apart, read the parts under every ratio, and scan each entry for the one that brings two numbers back (814) — Someone else's printed table is an answer key nobody here wrote, so a near rebuild is two findings, the structure that came back and the digits that did not. The Loop Atlas's engine rebuilt Tong and Dullin's diver from their Table 1 with the field's largest size within a quarter of a per cent; scanning each entry alone found that the knee had dropped one 0.08, and restored, every number came back within its bars.

Holds: carried to two: weaker in the Loop Atlas's engine rebuild (ruler); held in Tong and Dullin's planar somersault (2019). *Met in:* measurement, kinematics.
Counted: The built artifact is the evidence 1, A second reader 1. *Shown:* Loop Atlas; full guide row 814.

Count a project's own record by kind: what landed beside what missed, how often the leans hold, and each rival once (821) — Weighed one at a time, every result reads as small, and handovers keep the open items and drop the finished wins, so count the record by kind, with a script. In the Loop Atlas a single inherited paragraph calling the work modest stood for six sessions in place of 41 benchmarks, 18 found-here candidates and 640 held predictions; counted, the registered leans held 73 times in 116 over one session, then 68 in 84 over the next.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Counted: Count the right unit 1. *Shown:* Loop Atlas, R2W, triple-star; full guide row 821.

Before a page publishes, check it against its runs three ways: every headline's number and words, an error bar on every claim from a noisy run, and where each number sits (823) — A page can carry the right number in the wrong sentence, and such slips survive checks that ask only whether each number exists. A script found every one of a 46-page rewrite's 1,334 numbers in its records but two honest roundings, yet two readers who had not written it found 25 numbers set in the wrong context; and an error-bar pass over 35 claims read off noisy runs made 6 change their words.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Counted: A check that cannot fail 2, A second reader 1. *Shown:* Loop Atlas; full guide row 823.

A citation not reached is "not reached", not "not found": say what was tried, close a literature check like a run, and read the paper behind a summary's word (832) — A search that cannot open a paper has learned nothing about whether it exists, so say which tools and sites were tried and keep a remembered name on the owed list. Herzog (1986), out of reach of the search tools, was on its publisher's own page, found by the person in one look; and "bend and twist" in a popular book meant the bend's direction carried round, which only the paper itself, bought and read in full, showed.

Holds: carried to two: held in insect wake capture and jellyfish's parked ring; failed in leading-edge vortex and clap and fling (break not stated). *Met in:* records and tools, biology.
Counted: A word that changes meaning across a boundary 1. *Shown:* Loop Atlas, R2W; full guide row 832.

A book takes the person's words by tidying, not pasting: each quotation in clean sentences in his own voice, with its date and subject read in the voice file before it is placed (833) — A person's chat words have to be made readable without changing what they meant or when they were said. In the Loop Atlas book each quotation was put into clean sentences in his own voice, at his word; a reflection typed on the first night was dated near the end of the work, and a line about his own guesswork was set as a verdict on the book, both caught by a second reader.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: A second reader 1. *Shown:* Loop Atlas; full guide row 833.

Where the model makes a number exact, say exact: a ratio of rounded prints is not a measurement (834) — A ratio worked out from two rounded printed numbers adds a scatter that is not there, so before printing such a quotient, ask whether the model makes it exact. A register said a toy rider taps 11.98 times a second on a 0.25 m board and 2.99 on a 4 m board, "4.007 against 4"; the toy's parts B3 and C1 show every rider tapping at the same moments, and the line now reads "exactly 4 in the toy".

Holds: not yet carried elsewhere. *Met in:* measurement, driven oscillators.
Counted: The integer diagnostic 1. *Shown:* Loop Atlas; full guide row 834.

A count that goes on the front page is taken from the record by a script audited by eye, with every uncounted case read and every test of a rule counted (840) — A project's count of what it achieved is a headline, measured as carefully as any result. R2W's register first held the 155 tests its book words as predictions; a script then read every graded entry, its first version getting 4 of 25 sampled tests wrong, and five readers who read all 237 uncounted landed tests by eye found a whole class the samples had passed over, moving the count from 355 of 416 to 331 of 402.

Holds: carried to one, held there: the pad atlas, blind. *Met in:* measurement, records and tools.
Counted: The built artifact is the evidence 1, A word that changes meaning across a boundary 1, A second reader 1. *Shown:* R2W; full guide row 840.

A summary is a new claim about every line it shortens: have it read against its source by someone who did not write it (843) — Shortening a passage means choosing what to keep, and every choice can make the story cleaner than what was measured, so a summary is checked like any new result. In R2W, Claude wrote eight summary paragraphs from a section it had just moved, and an independent reader checked them against the book, the record and the run files: about 130 claims held and 25 did not, one of them hiding the registered bow that missed (R27-c).

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: A second reader 1. *Shown:* R2W; full guide row 843.

A move is proved by restoring it: ship it in a fixed order with its list loose, rebuild the deliverable from the restored parts at both ends, and leave the old session idle (844) — When a project moves to a new session, the only proof that everything arrived is to restore it into an empty folder and rebuild the work from it, at both ends. Ship the move in order with its list loose, leave the old session idle as a backup, and run every command the start-here page gives exactly as written. The Field Guide's own move restored 3,740 files and rebuilt the guide to the same md5.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Counted: The built artifact is the evidence 2. *Shown:* Loop Atlas, triple-star, Field Guide sessions; full guide row 844.

What leaves for the public carries its tools as they ran and nothing that must stay home, and every outgoing file is searched for what was taken out (845) — A deposit is read by strangers and kept for good, so it holds exactly what made the results, with tools deposited byte for byte as they ran so that each run's fingerprint still names its tool. A scrub of the text is not a scrub of the deposit: the Loop Atlas's reader's edition was scrubbed of the author's typed words while about 160 of its 497 tools quoted him as typed in their opening comments.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: Loop Atlas, R2W, triple-star, Dancing on the Sun; full guide row 845.

Re-derive a verdict from its output before reporting it, and let blind readers compare before acting (852) — The danger is not forgetting but remembering wrongly in a confident voice. A session's own summary of its earlier work said a registered prediction had failed, and the run's output, read again, showed it held (WEIGHT-CONE C-a); and two readers who sorted R2W's 910 withheld notes without seeing each other's piles found what a keyword rule had silently dropped.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: R2W; full guide row 852.

The walk-through read for contradictions, before any upload (853) — A fresh helper that has not seen the work reads the source and the built PDF in full, logging every number that disagrees with itself, every stale line and every pointer that does not resolve, graded A, B or C, and the author's session fixes A first. In A Rider in Two Waves the front pages held 7 of the 38 contradictions and the section written last held 5: fix the front first.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: Loop Atlas, R2W; full guide row 853.

Write the rule without the word it forbids (855) — Writing down a word kept out of print, even to forbid it, put it into new files more than once in R2W's sitting P1. Write the rule without the word, search every new file case-insensitively, and let a tool hold the word only as a hash, as the publishing packet's upload checker of 30 October does.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: R2W, Field Guide sessions; full guide row 855.

Credit and borrowed sentences drift toward a tidier story: read new prose against the record, write credit from a dated ledger, and quote a companion from its final text (866) — Numbers get checked; words are where the drift hides. When new pages went into the Loop Atlas's reader's edition, a reader who had not seen the writing

found 20 faults against the record, nearly all running one way: a condition dropped, a credit simplified, a count rounded to the whole (46 tagged where the run tagged 44). What held was written from a ledger of who did what, built loop by loop from the dated record.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: Loop Atlas, R2W, triple-star; full guide row 866.

A month of an AI's slips had three shapes, none of them arithmetic: words outrunning numbers, lists that did not list themselves, and things that lived only in the conversation or only in an estimate (868) — Once a tool's planted checks passed, the tool was almost never where the trouble was: the triple-star thread's Claude found its slips of eight sessions fell into three shapes. Words outran numbers, as when "most guesses miss" was said beside 208 landed and 194 missed; lists did not list themselves; and things lived only in the conversation or an estimate, held off by a fixed closing block of eight steps at the end of every exchange.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: R2W, triple-star, Field Guide sessions; full guide row 868.

Pack a delta during a long sitting, and at a move sweep the workspace for what the crate does not hold (871) — A sitting whose only copy of its work is the container is one restart from losing it, and scripts written outside the crate travel nowhere. R2W's sitting P2 ran 59 exchanges over three days through two workspace restarts and shipped no crate delta until its move, and at the move 44 files lived only in the workspace; they went into the crate with an md5 list before packing.

Holds: not yet carried elsewhere. *Met in:* records and tools.
Shown: Loop Atlas, R2W; full guide row 871.

Check every row of a published table you lean on, and when one family of its entries is wrong, look for a step only that family takes (874) — A wrong entry in a published table may have company, and the company may have a shape. Briggs's table of the doubling constants is right in δ at every even order and off at every odd one, 5.9679687038 for 6.0846890743 at order 3, and in the Loop Atlas's reading one sign slip reproduces every odd entry; McCartney (2020) had found the odd orders wrong first, naming no cause.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools, maps and chaos.
Shown: Loop Atlas; full guide row 874.

Before publishing a correction to someone else's numbers, have them re-derived by code that shares nothing with yours, read the paper that already says it, and give the note a book's disclosure and two fresh readers (875) — A public correction of someone else's number needs proof that does not lean on the corrector's own code. For the odd-order constants of row 874 the Loop Atlas had a fresh helper, given no number, write its own programs, and all five registered predictions held; McCartney (2020) was credited first, and two fresh readers of the note logged 55 and then 37 findings.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools, maps and chaos.
Shown: Loop Atlas; full guide row 875.

A derived edition checks itself against its source at every build, and the readers who draft it are a cold read for free (880) — A generated edition that asserts every number in an entry stands in its row will refuse to build when the source moves, and that refusal is its check. The guide's reader's edition stopped when a row took thousands commas and its entry still said 4420; the eight helpers who drafted its entries flagged about seventy rows, and each lead was opened against its source first: of 22 on numbers, 7 rows were wrong.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: Question the Substrate, Field Guide sessions; full guide row 880.

A count is as wide as the places it reads: grade only in a result entry, prove a new counter on the old count byte for byte, and write both sides of a graded test the exchange it lands (882) — A count reads only where it is told to look, so grade a test only in a result entry, read by eye every verdict found anywhere else, and prove a new counter on the old count before trusting it on new lines. R2W's new counter first wrote the old count byte for byte, 425 graded, 352 landed and 73 missed, before ruling every later row by eye, and the whole record now reads 515 graded, 436 landed and 79 missed.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: Loop Atlas, R2W, Field Guide sessions; full guide row 882.

Ship the method with the number, and a toy any reader or small AI can run (891) — A newcomer's first try, a single loop, keeps only 0.0806 of the learner's inside where the paper reports 0.3049 from a loop and its turned copy. So Two Arrows ships its method as the default of a one-file toy that prints JSON, for any reader and their AI to run.

Holds: not yet carried elsewhere. *Met in:* measurement, records and tools.
Shown: Two Arrows; full guide row 891.

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Declaration

This edition. The reader’s edition is generated by script from version 7’s full edition. Each entry’s sentences were drafted with Claude’s assistance from its row’s own words, and a script checked that every number, run name and quoted phrase in them stands in that row; the lines beneath them are generated from the full edition’s appendices. Where an entry and its row differ, the row in the full edition is the record.

Preprint. This guide is a preprint, a checkpoint on a road: it has not been peer reviewed, and its search of the literature is limited. Its failures are stated and made instructive. It comes from a part-time research team that works in the open and gives its tools away.

AI assistance. This guide was assembled with AI assistance (Claude, Anthropic) for literature survey support, instrument building, instrument running, code generation, prose drafting, and iterative editing across multiple sessions. In the part *Two Readers and a Ruler*, Claude is also one of the two readers whose blind readings are scored, and the three projects’ sessions, each run with Claude, wrote the accounts the part is built from.

Integration papers. Version 6 integrated lessons from three companion programmes deposited in September 2026, and version 7 adds nine more: two that went live on 30 September 2026, the triple-star paper, published on 28 October 2026, R2W’s book, published on 5 November 2026, the Loop Atlas’s book, its record and its note on the odd-order constants, published on 9 November 2026, and the two-voices papers, published on 9 and 10 November 2026:

- *Jellyfish: the Holonomist of the Sea* — what a bell taught an instrument about memory, hovering, and the loop that encloses nothing (concept DOI 10.5281/zenodo.22908909)
- *Instruments for the Magnetic Forest* — a tool suite for the integer-forest lattice model, and what it verifies (concept DOI 10.5281/zenodo.22906780)
- *Linking Is the Latch* — reading the integer on a linked filament, and why a bare filament sets no floor (concept DOI 10.5281/zenodo.22906667)
- *The Latch Atlas* — holonomy and locks that can’t be picked (concept DOI 10.5281/zenodo.23068382)
- *Dancing on the Sun* — a loop count that reads through holonomy, and the current it finds flowing in both polarities (concept DOI 10.5281/zenodo.23067245)
- *Held, Rocking, Locked* — the islands of a light star around an eccentric binary (concept DOI 10.5281/zenodo.23755473)
- *A Rider in Two Waves* — different music, the same moves (concept DOI 10.5281/zenodo.23999853)
- *The Loop Atlas* — a loop comes home to where it started; what did it leave behind? (concept DOI 10.5281/zenodo.24100672)

- *The Loop Atlas: the record* — every prediction a run judged, every other check a run made, how each came out, and every miss with its reason (concept DOI 10.5281/zenodo.24100980)
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- *Two Arrows* — why a learning mind heals its answers long before its self (concept DOI 10.5281/zenodo.24120415)

The three working projects. Version 7 also draws on three working projects of the same author, all now published (above): the triple-star sessions, whose paper is *Held, Rocking, Locked*; R2W, whose book is *A Rider in Two Waves*; and the Loop Atlas sessions, whose book is *The Loop Atlas*, published with its record and its note on the odd-order constants. Rows from them name their source in the text or the tag line, and their numbers are as each project reported them, most from a named run. The part *Two Readers and a Ruler* was written from the three projects' own accounts of how they read pictures, and its quotations of the author are his own words, with spelling and capitals put right and personal asides left out. Each project's record is in the list above and in the table below. The two-voices papers' lessons are rows 883 to 891 and additions to 20 earlier rows, and each names its paper.

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Where the rows come from

Each row of the register names the project or paper it came from, in its text or its tag line. This table counts the rows that name each, so a row that carries lessons from two sources is counted under both. The 440 rows that name none of them are version 6's own, all numbered 728 or below. The register holds 884 rows.

Source	Its record	Rows that name it
Version 6 of this guide, and the papers it drew on	concept DOI 10.5281/zenodo.21811588	440
The Loop Atlas sessions, *The Loop Atlas*	concept DOI 10.5281/zenodo.24100672	292
R2W, *A Rider in Two Waves* (the RF Spoke in its early records)	concept DOI 10.5281/zenodo.23999853	228
The triple-star sessions, *Held, Rocking, Locked*	concept DOI 10.5281/zenodo.23755473	166
Dancing on the Sun	concept DOI 10.5281/zenodo.23067245	23
The Latch Atlas	concept DOI 10.5281/zenodo.23068382	20
Jellyfish: the Holonomist of the Sea	concept DOI 10.5281/zenodo.22908909	4
Linking Is the Latch and *Instruments for the Magnetic Forest*	concept DOIs 10.5281/zenodo.22906667 and 10.5281/zenodo.22906780	2
Question the Substrate (the two-voices series, Paper 1)	concept DOI 10.5281/zenodo.24094163	21
Two Arrows (the two-voices series, Paper 2)	concept DOI 10.5281/zenodo.24120415	14
The Field Guide's own working sessions	this record	26

Tool and computational resource disclosure

Tools used. Claude (Anthropic) was used across multiple sessions for literature survey support, instrument building and running, code generation, prose drafting and iterative editing. Typesetting used pandoc and wkhtmltopdf.

Responsibility. Responsibility for the correctness and adequacy of the arguments and results presented here, and for the completeness and accuracy of the citations, rests entirely with the named human author. No automated system is an author of this work.

Verification performed. Every row cross-reference in the register was resolved by an automated pass over the source; a loss check keyed by row title confirmed that no row of the previous version was dropped; and the final PDF was checked with lint_deposit.py and its extracted text read back against the source. Each work in the References was looked up on a publisher page, a DOI record, a preprint listing or a library catalogue, a few only in the reference lists of later works that cite them, and a field that could not be confirmed is left out of its entry. On 31 October 2026 every concept DOI of the author's own records cited here was resolved against Zenodo's own records, and a script matched the author-and-year citations it found in the text to entries in the References, each it could not match then read by eye.

Attribution limits. The literature pass is limited, as the preprint notice states. Rows tagged "ours" carry no located prior name and should be read as not located, not as new. The words "from recall" stand where a name or a law was given without a checked source; after the pass of 31 October they stand in rows 67, 351, 552, 605, 608, 635, 672, 675, 705, 727 and 769, and in the part Two Readers and a Ruler, section 7 (a radio discriminator's S-curve).

Computational resources. All instruments ran on a personal computer and in cloud containers; no specialised hardware was used.